

American Community Survey and Puerto Rico Community Survey Design and Methodology

Version 4.0

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Acknowledgements (2024 Edition)

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Foreword

The American Community Survey—A Revolution in Data Collection

The American Community Survey (ACS) is the cornerstone of the U.S. Census Bureau’s effort to keep pace with the nation’s ever-increasing demands for timely and relevant data about population and housing characteristics. This survey provides current demographic, social, economic, and housing information about America’s communities every year—information that until 2010 was only available once a decade. Implementation of the ACS is viewed by many as the single most important change in the way detailed decennial census information is collected since 1940, when the Census Bureau introduced statistical sampling as a way to collect “long-form” data from a sample of households.

Beginning with the survey’s full implementation in 2005, the ACS replaced the census long-form questionnaire that was sent to about one-in-six addresses in the 2000 Census. As with the long form, information from the ACS is used to administer many kinds of government programs and to help inform how trillions of dollars in federal funds are distributed each year. Obtaining more current data throughout the decade from the ACS has long-lasting value for policy and decision-making across federal, state, local, and tribal governments, the private sector, and virtually every local community in the nation.

The Beginning

In 1994, the Census Bureau started developing what became the ACS with the idea of continuously measuring the characteristics of population and housing, instead of collecting the data only once a decade with each decennial census. Testing started in four counties across the country and expanded to 31 test sites by 1999, after seeing encouraging results.

The sample was increased from about 165,000 addresses in 1999 to about 800,000 addresses in 2000 to test the feasibility of conducting the ACS concurrent with the decennial census. The demonstration period continued through 2004, and the Census Bureau collected sufficient information to produce data for the nation, states, and most geographic areas with 250,000 or more population. Evaluations and comparisons with the results from the 2000 Census long-form data collection demonstrated the quality of ACS data.

With some changes to the sample design and other methodologies, the ACS was fully implemented in 2005 with a yearly sample of three million addresses each year, increasing to 3.5 million addresses phased in between 2011 and 2012. The ACS program was also implemented in Puerto Rico, where the survey is known as the Puerto Rico Community Survey (PRCS). In 2006, a sample of approximately 20,000 group quarters facilities was included in the ACS and PRCS so that estimates would reflect complete characteristics of all residents.

ACS and PRCS data are now available for all areas. Single-year and multi-year estimates are published every year. Detailed 1-year estimates are available for geographic areas with populations of 65,000 or more; less detailed supplemental 1-year estimates are available for areas with populations of 20,000 or more; and detailed 5-year estimates are available for all areas.

The 2024 release of the ACS and PRCS Design and Methodology Report

This ACS and PRCS Design and Methodology Report is an update of the third version that was released in 2022 (the original publication was released in 2006). This edition includes information on changes to the ACS program through 2021 unless otherwise indicated.¹ Recent developments include the addition of the internet as a response option for residents in certain types of group quarters facilities, the use of administrative data for workload optimization, the use of email reminders for partially complete internet responses, changes to content through 2024, weighting adjustments for late internet and mail returns, and results of the 2022 Content Test.

We hope that data users find this report helpful and that it will aid in improving an understanding of the ACS statistical design and the methods.

Dedicated Staff and a Cooperative Public are Essential to Success

The ACS is the largest household survey conducted by the federal government. The ACS program has been successful largely because of the innovation and dedication of many people who have worked hard to achieve the program's goals, as well as the willingness of the public to participate as survey respondents.

All of the primary survey activities are designed and managed by the staff at the Census Bureau, headquartered in Suitland, Maryland. These staff continually strive to improve the accuracy of the ACS estimates, streamline ACS operations, analyze ACS data, and conduct important research and evaluation to achieve greater efficiencies and program effectiveness. They also serve as educational resources and experts for the countless data users who come to the Census Bureau in need of technical assistance and help to use ACS data.

In addition, the Census Bureau's field partners in the six Field Regional Offices, thousands of field representatives across the country who collect ACS data, and survey managers and other staff at the Census Bureau's National Processing Center (NPC) in Jeffersonville, IN, and the Census Bureau telephone call centers in Jeffersonville, IN and Tucson, AZ make it possible to achieve a smooth and efficient running of a very complex and demanding survey operation.

¹ For ease of discussion, the term ACS is used to represent both the ACS and PRCS unless noted otherwise.

At the most fundamental level, the ACS program’s achievements are based on the willingness of the public to provide information that makes it possible for the Census Bureau to release the most current, reliable, and accessible source of local statistics on critical planning topics such as age, children, veterans, commuting, education, income, and employment. The ACS data are used every day to make important data-driven decisions that improve the quality of life for those living in communities throughout the nation. Millions of people living in the United States and Puerto Rico willingly provide the data that are collected each year by the ACS or PRCS. The Census Bureau thanks each and every respondent who takes the time and effort to participate in the ACS or PRCS.

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E. Poehler	10/16/2024	Added section on the 2024 release.	4.0

Chapter 1: Introduction

The American Community Survey (ACS) is an ongoing survey conducted by the U.S. Census Bureau that provides vital information about our nation and its people on a yearly basis. It uses a series of monthly samples to produce annually updated estimates for the same small areas (census tracts and block groups) formerly surveyed via the decennial census long-form sample. Five years of samples are required to produce these small-area data. The Census Bureau released its first 5-year estimates in December 2010; new small-area statistics are now produced annually. The Census Bureau also produces 1-year data products for larger geographic areas (i.e., areas with populations greater than 65,000) and 1-year Supplemental Estimates for areas with populations of 20,000 or more.² The ACS includes people living in both housing units (HUs) and group quarters (GQs). The ACS is conducted throughout the United States and in Puerto Rico, where it is called the Puerto Rico Community Survey (PRCS). For ease of discussion, the term ACS is used here to represent both surveys.

This document describes the basic ACS design and methodology as of the 2021 data collection year unless noted otherwise. The purpose of this document is to provide data users and other interested individuals with documentation of the methods used in the ACS. Future updates of this report will reflect additional design and methodology changes. This document is organized into 16 chapters. Each chapter includes an overview and detailed documentation. Most chapters also include a list of references.

- Chapter 2 provides a short summary of the history and evolution of the ACS, including its origins, the development of a survey prototype, results from national testing, and initial implementation.
- Chapters 3 and 4 focus on the ACS sample. Chapter 3 describes the survey frame, including methods for updating it. Chapter 4 documents the ACS sample design, including how samples are selected.
- Chapters 5 and 6 describe the content covered by the ACS and define several of its critical basic concepts. Chapter 5 provides information on the survey's content development process and addresses the process for considering changes to existing content. Chapter 6 explains the interview and residence rules used in ACS data collection and includes definitions of key concepts covered in the survey.
- Chapters 7, 8, and 9 cover data collection and data capture methods and procedures. Chapter 7 focuses on the methods used to collect data from respondents who live in HUs, while Chapter 8 focuses on methods used to collect data from those living in

² From 2007 to 2013, 3-year ACS estimates, compiling 36-months of data, were released for areas with populations of 20,000 or greater. ACS 3-year estimates have been discontinued.

GQs. Chapter 9 discusses the ACS language assistance program, which serves as a critical support for data collection.

- Chapters 10, 11, and 12 focus on ACS data processing, weighting and estimation, and variance estimation methods. Chapter 10 discusses data preparation activities, including the coding required to produce files for certain data processing activities. Chapter 11 provides a technical discussion of the process used to produce survey weights, while Chapter 12 describes the methods used to produce variance estimates.
- Chapters 13 and 14 cover the definition, production, and dissemination of ACS data products. Chapter 13 explains the process used to produce, review, and release ACS data estimates. Chapter 14 explains how to access ACS data products and provides examples of each type of data product.
- Chapter 15 documents the methods used in the ACS to control for nonsampling error and includes examples of measures of quality produced annually to accompany each data release.
- Chapter 16 describes the ACS research and evaluation program.

A glossary of terms and acronyms used in this report can be found in the appendix. Sample English and Spanish questionnaires from 1996 through the present are available on the [ACS & PRCS Forms and Instructions](#) website.

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E. Poehler	9/18/2024	Minor editorial changes. This document describes the basic ACS design and methodology as of the 2021 data collection year, unless noted otherwise.	4.0

Chapter 2: Program History

2.1 Overview

Continuous measurement has long been viewed as a possible alternative method for collecting detailed information on the characteristics of population and housing; however, it was not considered a practical alternative to the decennial census long form until the early 1990s. At that time, demands for current, nationally consistent data from a wide variety of users led federal government policymakers to consider the feasibility of collecting social, economic, and housing data continuously throughout the decade. The benefits of providing current data, along with the anticipated decennial census benefits in cost savings, planning, improved census coverage, and more efficient operations, led the Census Bureau to plan the implementation of continuous measurement, later called the American Community Survey (ACS). After years of testing, outreach to stakeholders, and an ongoing process of interaction with key data users—especially among expert statisticians and demographers—the Census Bureau expanded the ACS to the planned full sample size for housing units (HUs) and the household population in 2005 and added the group quarters (GQs) population in 2006.

The history of the ACS can be divided into five distinct stages: design, development, demonstration, full implementation, and continuous improvement. The Census Bureau first proposed the concept of continuous measurement in the 1990s. From 1990 to 1993, design proposals were drafted and vetted. In the development stage (1994 through 1999), the Census Bureau tested early prototypes of continuous measurement for a small number of sites. During the demonstration stage (2000 to 2004), the Census Bureau carried out large-scale, nationwide surveys and produced reports for the nation, the states, and large geographic areas. The full implementation stage began in January 2005, with an annual HU sample of approximately 3 million addresses throughout the United States and 36,000 addresses in Puerto Rico. In 2006, approximately 20,000 group quarters and the population within them were added to the ACS so that the data fully describe the characteristics of the population residing in geographic areas. Since the full implementation stage began in 2005 and 2006, we have been continuously improving. The continuous improvement stage has included a fundamental reexamination of the systems and processes that underlie the ACS and an exploration of new methods, techniques, and approaches designed to improve the ACS program and the Census Bureau's relationships with stakeholders and data users.

2.2 Design Origins and Early Proposals

In 1981, Leslie Kish introduced the concept of a rolling sample design in the context of the decennial census (Kish, 1981). While Kish was conducting his research, the Census Bureau also recognized the need for more frequently updated data. In 1985, Congress authorized a mid-decade census, but funds were not appropriated. In the early 1990s, Congress expressed renewed

interest in an alternative to the once-a-decade census. Based on Kish’s research, the Census Bureau began developing continuous measurement methods in the mid-1990s.

The Census Bureau developed a research proposal for continuous measurement as an alternative to the collection of detailed decennial census sample data (Alexander, 1993g), and Charles (Chip) Alexander, Jr. developed three prototypes for continuous measurement (Alexander, 1993i). Based on staff assessments of operational and technical feasibility, policy issues, cost, and benefits (Alexander, 1994e), the Census Bureau selected one prototype for further development. Designers made several decisions during prototype development. They knew that if the survey were to be cost-efficient, the Census Bureau would need to collect data through a self-response mail questionnaire. They also determined that a nonresponse follow-up would be conducted to reduce nonresponse bias. Like the decennial census, the survey would be mandatory. It was decided that the survey would use both telephone and personal visit nonresponse follow-up methods. In addition, the designers made critical decisions regarding the prototype’s key definitions and concepts (such as the residence rule), geographic makeup, sampling rates, and use of population controls.

With the objective of producing 5-year cumulations for small areas at the same level of sampling reliability as the long-form census sample, a monthly sample size of 500,000 HUs was initially suggested (Alexander, 1993i), but this sample size drove costs into an unacceptable range. When potential improvements in nonsampling error were considered, it was determined that a monthly sample size of 250,000 would generate an acceptable level of reliability.

2.3 Development

Development began with the establishment of a permanent Continuous Measurement Staff in 1994. This staff continued the development of the survey prototype and identified several design elements that proved to be the foundation of the ACS:

- Data would be collected continuously by using independent monthly samples.
- Three modes of data collection would be used: mail, telephone nonresponse follow-up, and personal visit nonresponse follow-up.³
- The survey reference date for establishing housing unit occupancy status, and for many characteristics, would be the date the data were collected. Certain data items would refer to a longer reference period (for example, “last week” or “past 12 months”) anchored by the response date.

³ The ACS introduced internet as a self-response mode in 2013. A centralized telephone nonresponse follow-up was discontinued in October 2017, although field representatives who conduct personal visit nonresponse follow-up may also call respondents and respondents may call the telephone questionnaire assistance line.

- The survey's estimates would be controlled to population and (beginning in 2000) housing estimates from the Population Estimates Program.
- All estimates would be produced by aggregating data collected in the monthly surveys over a year so that they would be reported annually based on the calendar year.

The documentation of early development took several forms. Beginning in 1993, a group of 20 reports, known as the Continuous Measurement Series (Alexander, 1992; 1993a–1993i; 1994a–1994f; and 1995a–1995b; Alexander and Wetrogan, 1994; Cresce, 1993), documented the research that led to the final prototype design. Plans for continuous measurement were introduced formally at the American Statistical Association's (ASA) Joint Statistical Meetings in 1995. Love et al. (1995) outlined the assumptions for a successful survey, while Dawson et al. (1995) reported on early feasibility studies of collecting survey information by telephone. Possible modifications of continuous measurement data also were discussed (Weidman et al., 1995).

Operational testing of the ACS began in November 1995 at four test sites: Rockland County, NY; Brevard County, FL; Multnomah County, OR; and Fulton County, PA. Testing was expanded in November 1996 to encompass areas with a variety of geographic and demographic characteristics, including Harris County, TX; Fort Bend County, TX; Douglas County, NE; Franklin County, OH; and Otero County, NM. This testing was undertaken to validate methods and procedures and to develop cost models for future implementation; it resulted in revisions to the prototype design and identified additional areas for research. Further research took place in numerous areas, including small-area estimation (Chand and Alexander, 1996), estimation methods (Alexander et al., 1997), nonresponse follow-up (Salvo and Lobo, 1997), weighting in ACS tests (Dahl, 1998), item nonresponse (Tersine, 1998), response rates (Love and Diffendal, 1998), and the quality of rural data (Kalton et al., 1998).

Operational testing continued, and in 1998 three counties were added: Kershaw County, SC; Richland County, SC; and Broward County, FL. The two counties in South Carolina were included to produce data to compare with the 1998 Census Dress Rehearsal results, and Broward County was substituted for Brevard County. In 1999, testing expanded to 36 counties in 26 states (U.S. Census Bureau, 2004e). The sites were selected to represent different combinations of county population size, difficulty of enumeration, and 1990–1995 population growth. The selection incorporated a variety of geographic areas as well as areas representing different characteristics, such as racial and ethnic backgrounds, migrant or seasonal populations, American Indian reservations, changing economic conditions, and predominant occupation or industry types. Additionally, the Census Bureau selected sites with active data users who could participate in evaluating and improving the ACS program. Based on the results of the operational tests, revisions were made to the prototype and additional areas for research were identified.

Tests of methods for the enumeration of people living in GQs were also held in 1999 and 2001. These tests focused on the methodology for visiting GQs, selecting resident samples, and conducting interviews. The tests selected GQ facilities in all 36 test counties and used the procedures developed in the prototyping stage. The results of the tests led to the modification of sampling techniques and revisions to data collection methods.

While the main objective of the development phase testing was to determine the viability of the methodologies used, it also generated usable data. Data tables and profiles were produced and released in 1999, providing data on demographic, social, economic, and housing topics.

Additionally, public use microdata sample (PUMS) files were generated for a limited number of locations during the period of 1996 through 1998. PUMS files show data for a sample of all HUs, with information on the housing and population characteristics of each selected unit. All identifying information is removed, and other disclosure avoidance techniques are used to ensure confidentiality.

2.4 Demonstration

In 2000, a large-scale demonstration was undertaken to assure Congress and other data users that the ACS was capable of producing the demographic, social, economic, and housing data previously obtained from the decennial census long-form sample.

During the demonstration stage, the ACS was called the Census 2000 Supplementary Survey (C2SS). Its primary goal was to provide critical assessments of feasibility, quality, and comparability with the 2000 Census to demonstrate the Census Bureau's ability to implement the ACS fully. Although ACS methods had been successful at the test sites, it was vital to demonstrate national implementation. Additional goals included refining procedures, improving the understanding of the cost structure, improving cost projections, exploring data quality issues, and assuring users of the reliability and usefulness of ACS data.

The C2SS was conducted in 1,239 counties, of which 36 were the ACS test counties from 1999, and 1,203 were new to the survey. It is important to note that only the 36 ACS test counties used the proposed ACS sample design. The others used a state-based primary sampling unit stratified design similar to the Current Population Survey (CPS). The sample of counties was designed to produce estimates for all states and for geographic areas of 250,000 or more. The annual sample size increased from 165,000 HUs in 1999 to 866,000 HUs in 2000. The test sites remained in the sample throughout the C2SS, and through 2004 were generally sampled at higher rates than the C2SS counties. The sampling rates for 1999-2001 were higher than the full implementation design so that the 3-year estimates for this period from the ACS in these counties were comparable to the planned 5-year period estimates of a fully implemented ACS, as well as to data from the 2000 Census.

Eleven reports issued during the demonstration stage analyzed various aspects of the program. There were two types of reports: methodology and data quality/comparability. The methodology reports reviewed the operational feasibility of the ACS. The data quality and comparability reports compared C2SS data with the data from the 2000 Census at the state and national level as well as comparisons of ACS 3-year data to the 2000 Census data for the ACS test sites.

Report 1 (U.S. Census Bureau, 2001) found that the C2SS was operationally successful, its planned tasks were completed on time and within budget, and the data collected met basic Census Bureau quality standards. However, the report also noted that certain areas needed improvement. Specifically, because they coincided with the decennial census, telephone questionnaire assistance (TQA) and failed-edit follow-up (FEFU) operations were not staffed sufficiently to handle the large workload increase. The evaluation noted that the ACS would improve planning for the 2010 Census and simplify its design and that implementing the ACS, supported by an accurate Master Address File (MAF) and Topologically Integrated Geographic Encoding and Referencing (TIGER) database, promised to improve the decennial census coverage. Report 6 (U.S. Census Bureau, 2004c) was a follow-up evaluation of the feasibility of utilizing data from 2001 and 2002. The evaluation concluded that the ACS was well-managed, was achieving the desired response rates, and had functional quality control procedures.

Report 2 (U.S. Census Bureau, 2002) concluded that the ACS would provide a reasonable alternative to the decennial census long-form sample and added that the timeliness of the data gave it advantages over the long form. This evaluation concluded that, while ACS methodology was sound, its improvement needed to be an ongoing activity.

A series of reports compared national, state, and limited substate 1-year period estimates from the C2SS and the 2000 Census. Reports 4 and 10 (U.S. Census Bureau, 2004a; 2004g) noted differences. For example, although 63 of the 86 housing characteristic estimates differed significantly between the C2SS and the 2000 Census long-form results at the 90 percent confidence level, only 23 of these differences were greater than 0.5 percentage points, and only 8 differed by more than 1.0 percentage points. At the national level, distributions of the general demographic and housing characteristics from the C2SS were very similar to those produced from the 2000 Census counts. A review of select county-level statistics confirmed most national findings but showed some count-level differences. The overall conclusion of the C2SS and the 2000 Census comparisons was that the research supported the proposal to move forward with plans for the ACS.

Report 5 (U.S. Census Bureau, 2004b) analyzed economic characteristics and concluded that estimates from the ACS and the 2000 Census long form were essentially the same. Report 9 (U.S. Census Bureau, 2004f) compared social characteristics and noted that estimates from both methods were consistent, with the exceptions of disability and ancestry. The report suggested the completion of further research on these and other issues.

A set of multiyear period estimates (1999–2001) from the ACS test sites was created to help demonstrate the usability and reliability of ACS estimates at the county and census tract geographic levels. Results can be found in Reports 7 and 8 (U.S. Census Bureau, 2004d; 2004e). These comparisons with the 2000 Census sample data further confirmed the comparability of the ACS and the 2000 Census long-form estimates and identified potential areas of research, such as variance reduction in subcounty estimates.

At the request of Congress, a voluntary response test also was conducted during the demonstration phase. The test, conducted between March and June of 2003, was designed to examine the impact that a methodological change from mandatory to voluntary response would have on mail response, survey quality, and costs. Reports 3 and 11 (U.S. Census Bureau, 2003; 2004h) examined the results. These reports identified the major impacts of instituting voluntary methods, including reductions in self-response rates by about a third (with a self-response rate in low-response areas of 27.7 percent), reductions in the reliability of estimates, and cost increases of more than \$59 million (in 2003 dollars) annually.

2.5 Full Implementation

In 2003, with the full implementation of the ACS approaching, the American Community Survey Office (ACSO) came under the direction of the Associate Director for the Decennial Census. While the Census Bureau had planned to implement the ACS fully in 2003, budget restrictions pushed back full HU implementation of the ACS and the Puerto Rico Community Survey (PRCS) to January 2005. The GQ component of the ACS was implemented fully in January 2006.

With full implementation, the ACS expanded to all 3,141 counties in the 50 states and the District of Columbia and all 78 municipios in Puerto Rico. The annual ACS sample increased from 800,000 addresses in the demonstration phase to 3 million in full implementation. Workloads for all ACS operations more than tripled. Monthly mailouts from the National Processing Center (NPC) went from approximately 67,000 to 250,000 addresses per month. Telephone nonresponse follow-up workloads, conducted from three telephone call centers, expanded from 25,000 calls per month to approximately 85,000. More than 3,500 field representatives (FRs) across the country conducted follow-up visits at 40,000 addresses a month, up from 1,200 FRs conducting follow-ups at 11,000 addresses each month in 2004. And, approximately 36,000 addresses in Puerto Rico were sampled every year, using the same three modes of data collection as the ACS. Beginning in 2006, the ACS sampled 2.5 percent of the population living in GQs. This included approximately 20,000 GQ facilities and 195,000 people in GQs in the United States and Puerto Rico.

With full implementation beginning in 2005, population and housing profiles for 2005 first became available in the summer of 2006 and have been available every year thereafter for specific geographic areas with populations of 65,000 or more. Three-year period estimates,

reflecting combined data from the 2005–2007 ACS, were available for the first time late in 2008 for geographic areas with populations of 20,000 or more. Five-year period estimates, reflecting combined data from the 2005–2009 ACS, became available late in 2010 for areas down to block groups, census tracts, and small local governments.⁴ Beginning in 2010, the nation had a 5-year period estimate, available as an alternative to the decennial census long-form sample, for nearly all geographic areas recognized by the Census Bureau, including census tracts and block groups.

2.6 Continuous Improvement

With the publication of the first 5-year estimates, the Census Bureau met its goal of replacing the decennial census long form with the ACS because those estimates were designed to be comparable to the long form estimates produced following each decennial census. Since then, the ACS program has been in a continuous improvement stage, which began with a program review. Program improvements have continued through consultation with external experts and research and implementation of updates.

The ACS Program Review

The publication of the first 5-year estimates was followed by planning for a detailed review of the systems and processes that underlie the ACS program. An initial goal of this review, which began in 2011, was to identify possible opportunities for improvements. By 2012, the review was well underway and had expanded to include other aspects of the ACS program. In 2013, as part of what was by then called the ACS Program Review, managers in several divisions that contributed to the ACS participated in a series of meetings and off-site events designed to envision the ACS program of the future (U.S. Census Bureau, 2015). Other developments associated with the program review include:

- The organization of a set of review teams (the highest level of which was comprised of division chiefs) to function as a set of program management boards. These boards, including Change Control boards and a Portfolio Management Governing Board, provide oversight to the review of technical decisions for the program.
- The incorporation of a risk review and formal work request process to acknowledge the total resources required to implement planned improvements to the program.
- The use of off-site events and organizational management techniques to solicit staff views more effectively in the day-to-day management of the program.

⁴ Beginning in 2014, the 3-year estimates were discontinued; however, a small set of 1-year Supplemental Estimate Tables were added for areas with populations of 20,000 or more. The 2005-2007, 2006-2008, 2007-2009, 2008-2010, 2009-2011, 2010-2012 and 2011-2013 ACS 3-year estimates will remain available to data users, but no new 3-year estimates will be produced.

- The adoption of more effective documentation processes.
- The recognition of the need for greater involvement with corporate infrastructure solutions such as adaptive design.
- The creation of an ACS Data Users Group to solicit information more effectively on what stakeholders need to use ACS estimates and what usability issues arise among stakeholders with similar interests in ACS data applications.

ACS Stakeholders and External Engagement

The ACS program depends heavily on engaging stakeholders in the development of the program and seeking stakeholder input as much as possible in decisions affecting ACS data products.

Consultations with stakeholders began early in the ACS development process with the goals of gaining feedback on the overall approach and identifying potential pitfalls and obstacles.

As a formal ACS testing period was under development, ACS managers started forming plans to ensure local communities were aware that they were included as test sites for the ACS. ACS testing was launched in four sites in 1995, as described earlier in this chapter. From March 1996 to November 1999, 31 town hall-style meetings were held throughout the country, with more than 600 community representatives attending the meetings. Similar meetings took place in the years to follow. A series of three regional outreach meetings in Dallas, TX; Grand Rapids, MI; and Seattle, WA, were held in mid-2004, with an overall attendance of more than 200 individuals representing data users, academicians, the media, and local governments. Other early stakeholder engagement efforts included the development of special-purpose advisory panels in partnership with the Committee on National Statistics of the National Academies of Science and a Rural Data Users Conference held in May 1998 in Alexandria, Virginia, to discuss issues of concern to representatives of small areas and populations. Annual meetings of individual State Data Center representatives and affiliate organizations have frequently featured presentations to update members on the latest ACS program developments and data products.

Changes based on stakeholder input were important in shaping the design and development of the ACS and continue to influence its form as the ACS program moves forward. A “Symposium on the ACS: Data Collectors and Disseminators” took place in September 2000. It focused on the data uses and needs of the private sector. A periodic newsletter, the ACS Alert, was established to share program information and solicit feedback.⁵ The Interagency Committee for the ACS was formed in 2000 to discuss the content and methods of the ACS and how the survey meets the needs of federal agencies. From 2003 to 2005, the Census Bureau invited federal agencies to participate in an ACS Federal Agency Information Program designed to arrange meetings at

⁵ This newsletter no longer exists. The ACS program currently sends monthly emails to subscribers on topics such as changes to ACS methodology, data releases, and data products.

federal agencies where specific questions by federal agency representatives on the ACS design, methods, and data products could be addressed by Census Bureau technical experts.

The Census Bureau has also consulted with experts convened by the Committee on National Statistics (CNSTAT). For example, in 2007, the Committee on National Statistics (CNSTAT) issued an important report, “Using the American Community Survey: Benefits and Challenges,” which reflected the input of many stakeholders and addressed the interpretation of ACS data by a wide variety of users. In 2013, the Census Bureau requested that CNSTAT convene a workshop on the benefits of the ACS to a broad array of non-federal data users. A summary of the workshop is described in the CNSTAT publication, “Benefits, Burdens, and Prospects of the American Community Survey: Summary of a Workshop.” Additional reports published by CNSTAT are included in the references (see National Academies of Sciences, Engineering, and Medicine 2016 and 2019 and National Research Council 2001, 2007, 2012, 2013, and 2015).

Meetings with the Census Advisory Committees—Bureau of the Census Scientific Advisory Committee (CSAC) and the Census Bureau National Advisory Committee (NAC) on Racial, Ethnic, and Other Populations—provide opportunities for ACS staff to receive specific advice and recommendations on the ACS design, survey methods, and data products. The Census Bureau’s Field Division and regional directors have played a prominent role in communicating the importance of participating in ACS data collection to state and local government representatives and circulating pamphlets and similar publications to explain the ACS program and its benefits to communities.⁶ Further, in 2021, the 2020 Census National Partnership Program transitioned to the Office of Strategic Alliances (OSA). The OSA will continue leveraging hundreds of national partners obtained during the 2020 Census to help with enterprise mission work, such as the ACS. This includes working with partners to help program areas expand ways to disseminate data to current and non-traditional data users and collaborate to increase responses to the Census Bureau’s economic and demographic censuses and surveys. The OSA connects program areas with diverse and influential organizations to improve data collection, data products, and data dissemination.

The ACS program leadership regularly briefs several oversight groups, including the Office of Management and Budget (OMB), the Government Accountability Office (GAO), and the Inspector General of the U.S. Department of Commerce (DOC). The Census Bureau also briefs Members of Congress regularly on multiple aspects of the ACS.

The number and scope of groups and organizations representing ACS stakeholders has expanded dramatically since the survey was implemented. Some of these organizations represent broad areas of interest, such as statistical methodology, public opinion research, demography, regional science, sociology, and geography. Others advocate for specific interests, such as housing,

⁶ An example of such a publication is the [ACS Information Guide](#).

transportation, and education; some represent population groups, such as older people, veterans, and American Indians and Alaska Natives; and some are professional groups representing a specific occupation, such as librarians.

In partnership with the U.S. Census Bureau, the Population Reference Bureau (PRB) and Sabre Systems created an ACS Data Users Group in the fall of 2012. The purpose of the ACS Data Users Group is to improve understanding of the value and utility of ACS data and to promote information sharing among data users about key ACS data issues and applications. The group is led by a Steering Committee comprised of selected external stakeholders representing a broad spectrum of data users with different interests. To facilitate communication among ACS data users, an online community forum and a biennial ACS Data Users Conference were also established. The inaugural ACS Data Users Conference was held on May 29-30, 2014, in Washington, DC.

In addition, efforts have been made to promote the international sharing of the Census Bureau's experiences with the development and implementation of the ACS. Presentations have been given to many international visitors who have come to the Census Bureau to learn about surveys and censuses. Presentations about the ACS have also been made at many international conferences' working sessions and meetings. Outreach to stakeholders has been a key component of launching and gaining support for the ACS program, and its importance and prominence continue.

Transformation: Expanded Use of Administrative Data

Title 13 of the U.S. Code, Section 6 instructs the Census Bureau to maximize the use of already available data whenever possible instead of collecting data directly from respondents.

Administrative data are microdata records collected and maintained by administrative (i.e., program) agencies and commercial entities. Government and commercial entities maintain these files to administer programs and provide services. The ACS program has made it a priority to use alternative data sources and expand the use of administrative data to meet customer needs and reduce the dependency on traditional methods of data collection (U.S. Census Bureau, 2021, 2022; Thieme, 2022).

The ACS program has long used administrative data to support the operational administration of the survey, such as frame updates from the U.S. Postal Service and phone numbers for respondents from commercial entities. More recently, the ACS has moved beyond operational uses of administrative data and began investigating the use of administrative data to produce high-quality data products while reducing respondent burden. Once data sources have been identified, ACS staff determine how to use the data to produce the required data products. See U.S. Census Bureau, 2018, for the set of guiding principles ACS staff follow to evaluate data sources.

The expanded use of administrative data in the ACS is expected to reduce data collection costs and improve operational efficiency, reduce respondent burden, and improve the quality of ACS data products. The long-term goal is to provide data users with information on ACS topics relying on an active library of data from administrative data, third-party data, and a full suite of Census Bureau-collected data rather than solely on data collected directly from the public through the ACS.

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Chapter 2 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) that was published	2.0
S. Clark & E. Poehler	11/28/2022	Addressed formatting issues and updated content.	3.0
E. Poehler	8/2/2024	Updated advisory committee names, added CNSTAT reports and URLs for several references, and added information about OSA. Added information about expanded administrative data use.	4.0

Chapter 3: Frame Development

3.1 Overview

The Master Address File (MAF) is the Census Bureau’s official inventory of known housing units (HUs), group quarters (GQs), and selected non-residential units (public, private, and commercial) in the United States and Puerto Rico. It serves as the source of addresses for the American Community Survey (ACS), other Census Bureau demographic surveys, and the decennial census. It contains mailing and location address information, geocodes, and other attribute information about each living quarter. A geocoded address is one for which state, county, census tract, and block have been identified.

The MAF is linked to the Topologically Integrated Geographic Encoding and Referencing (TIGER) system. TIGER is a database containing a digital representation of all census-required map features and related attributes. It is a resource for producing maps, tabulating data, and assigning addresses to geographic locations in geocoding. The resulting database is called the MAF/TIGER database (MTdb).

The initial MAF was created for the 2000 Census using multiple sources, including the 1990 Address Control File, the U.S. Postal Service’s (USPS’s) Delivery Sequence File (DSF), field listing operations, and addresses supplied by local governments through partnership programs. The MAF was used as the initial frame for the ACS, in its state of existence at the conclusion of the 2000 Census. Updates from frame building and enumeration operations of the 2010 Census and the 2020 Census were incorporated into the MTdb and included in the ACS sampling frame beginning in the middle of 2021. The Census Bureau continues to update the MAF using the DSF and various automated, clerical, and field operations.

The remainder of this chapter provides detailed information on the development of the 2021 ACS sampling frame. Section 3.2 provides basic information about the MAF and its contents. Sections 3.3 and 3.4 describe the MAF development and update activities for HUs in the United States and Puerto Rico. Section 3.5 describes the MAF development and update activities for GQs. Finally, Section 3.6 describes the ACS extracts from the MAF.

3.2 Master Address File Content

The MAF is the Census Bureau’s official inventory of known HUs, GQs, and selected non-residential units in the United States and Puerto Rico. Each MAF record contains some or all of the following information: geographic codes, a mailing and/or location address, the physical characteristics and/or location description of the unit or any relationships to other units, residential or commercial status, latitude and longitude coordinates, and source and history information indicating the operation(s) that added or updated the record (see Section 3.3). ACS

obtains this information from the MAF in files called MAF extracts (see Section 3.6) and uses it for sampling, data collection, and data tabulation activities.

The geographic codes in the MTdb identify a variety of areas, including states, counties, county subdivisions, places, American Indian areas, Alaska Native areas, Hawaiian Homelands, census tracts, block groups, and blocks. Two important geographic code sets are the decennial census tabulation geography set, based on the January 1 legal boundaries of the most recent decennial census year, and the current geography set, based on the January 1 legal boundaries of the most recent year (for example, MAF extracts received in July 2018 reflect legal boundaries as of January 1, 2018). Each MAF record contains geographic codes from the TIGER database. Because each record contains a variety of geographic codes, it is possible to sort MAF records according to different geographic hierarchies. ACS operations generally require sorting by state, county, census tract, and block.

The MAF contains both city-style and non-city-style mailing addresses. A city-style address is one that uses a structure number and street name format; for example, 201 Main Street, Anytown, ST 99988. Additionally, city-style addresses usually appear in a numeric sequence along a street and frequently follow parity conventions, such as all odd numbers occurring on one side of the street and even numbers on the other side. They often contain information used to uniquely identify individual units in multiple-unit structures, such as apartment buildings or rooming houses. These are known as unit designators and are part of the mailing address.

A non-city-style mailing address is one that uses a rural route (RR) and box number format or a post office (PO) box format. Examples of these types of addresses are RR 2, Box 9999, Anytown, ST 99988 and PO Box 123, Anytown, ST 99988.

In the United States, city-style addresses are most prevalent in urban and suburban areas and accounted for 98.2 percent of all residential addresses in the MAF at the conclusion of the 2010 Census. Following the 2020 Census, this percentage has continued to increase. Most city-style addresses represent both the mailing and location addresses of the unit. City-style addresses are not always mailing addresses, however. Some residents at city-style addresses receive their mail at those addresses, while others use non-city-style addresses (U.S. Census Bureau, 2000b). For example, a resident could have a location address of 77 West St. and a mailing address of P.O. Box 123. In other cases, city-style “E-911” addresses have been established so that state emergency service providers can find a house even though mail is delivered to a rural route and box number.

Non-city-style mailing addresses are prevalent in rural areas. Because these addresses do not provide specific information about the location of a unit, finding a rural route and box number address in the field can be difficult. Post Office Box addresses cannot be located in the field because they are associated with a post office location, not a structure location.

To help field staff locate non-city-style addresses in the field, the MAF often contains a location description of the unit and/or its latitude and longitude coordinates.⁷ The presence of this information in the MAF makes field follow-up operations possible.

Both city-style and non-city-style addresses can be either residential or non-residential. A residential address represents a living quarter in which a person or persons live or could live. A non-residential address represents a structure, or a unit within a structure, that is used for a purpose other than residence. While the MAF includes many non-residential addresses, it is not a comprehensive source of such addresses (U.S. Census Bureau, 2000b).

The MAF also contains some address records that are classified as incomplete because they lack a complete city-style or non-city-style mailing address. Records in this category often just describe the unit's location and, usually, its latitude and longitude.

For more information on the MAF, including a description of its content and structure, see U.S. Census Bureau (2000b).

3.3 Master Address File Development and Updating for the U.S. Housing Unit Inventory

MAF Development in the United States

For the 1990 and earlier decennial censuses, the Census Bureau compiled address lists from several sources (commercial vendors, field listings, and others). Before 1990, these lists were not maintained or updated after a census was completed. Following the 1990 Census, the Census Bureau decided to develop and maintain a master address list to support the decennial census and other Census Bureau survey programs to avoid the need to rebuild the address list before each census. The Census Bureau created the MAF by merging city-style addresses from the:

- 1990 Address Control File⁸
- Field listing operations⁹
- USPS Delivery Sequence File (DSF)

⁷ For example, "E side of St. Hwy, white house with green trim, garage on left side."

⁸ The Address Control File is the residential address list used in the 1990 Census to label questionnaires, control the mail response check-in operation, and determine the response follow-up workload (U.S. Census Bureau 2000a, p. XVII-1).

⁹ In areas where addresses were predominantly non-city-style, the Census Bureau created address lists through a door-to-door canvassing operation (U.S. Census Bureau 2000a, p. VI-2).

- Addresses supplied by local governments through partnership programs, such as the Local Update of Census Addresses (LUCA).¹⁰
- Other 2000 Census activities, including the Be Counted Campaign.¹¹

At the conclusion of the 2000 Census, the MAF contained a complete inventory of known HUs nationwide.¹²

MAF Improvement Activities and Operations

MAF maintenance is an ongoing and complex task. New HUs are built continually, older units are demolished, and the institution of addressing schemes to allow emergency response personnel to find HUs with non-city-style mailing addresses render many older addresses obsolete. Maintenance of the MAF occurs through a coordinated combination of automated, clerical, and field operations designed to improve existing MAF records and keep up with the nation's changing housing stock and associated addresses. With the completion of the 2000 Census, the Census Bureau implemented several short-term and one-time operations to improve the quality of the MAF. These operations included Count Question Resolution (CQR), MAF/TIGER reconciliation, and address corrections from rural directories. For the most part, the Census Bureau implemented these operations to improve the addresses recognized in the 2000 Census and their associated characteristics. CQR was implemented again after the 2010 Census.

The 2010 Census operations improved the coverage and quality of the MAF. The operations included several nationwide field canvassing and enumeration operations. In preparation for the 2010 Census, the Census Bureau implemented a nationwide address canvassing field operation (except for remote areas in Alaska and rural Maine) to update the housing unit inventory in the MAF. Other field operations to support the 2010 Census enumeration identified HU and GQ corrections, additions, and deletions and updated the MAF with those results. Additionally, the Census Bureau repeated the same partnership and count coverage programs used for the 2000 Census for the 2010 Census, including the LUCA and the Be Counted programs.¹³ The Census

¹⁰ The 1999 phase of the LUCA program occurred from early March through mid-May 1999 and involved thousands of local and tribal governments that reviewed more than 10 million addresses. The program was intended to cover more than 85 percent of the living quarter addresses in the United States in advance of the 2000 Census. The Census Bureau validated the results of the local or tribal changes by rechecking the 2000 Census address list for all blocks in which the participating governments questioned the number of living quarter addresses.

¹¹ The Be Counted program provided a means to include in the 2000 Census those people who may not have received a Census questionnaire or believed they were not included on one. The program also provided an opportunity for people who had no usual address on Census Day to be counted. The Be Counted forms were available in English, Spanish, Chinese, Korean, Tagalog, and Vietnamese. For more information, see Carter (2001).

¹² For details on the address list development for the 2000 Census, see U.S. Census Bureau (2000a).

¹³ The Census Bureau redesigned the LUCA program for the 2010 Census, allowing participants a choice of several methods of reviewing the census address or housing unit inventories in their jurisdictions. Participant feedback was included in the Address Canvassing field operation for verification.

Bureau determined the final 2010 Census status of each HU record in the MAF in late 2010. These operations improved the MAF extracts used for the ACS sample selection. ACS and the 2010 Census planners worked together closely to assess the impact of the decennial census operations on the ACS. For details on the 2010 Census operations, see U.S. Census Bureau (2011).

The 2020 Census used a similar set of frame-building and enumeration operations as previous censuses. Results from 2020 Census operations began updating the MAF with the start of LUCA in 2018. The Census Bureau redesigned the Address Canvassing operation to include an in-office verification of the address list using imagery and administrative records to identify blocks where costly fieldwork was unnecessary. As in 2010, the 2020 Census frame-building and enumeration operations improved the address list by identifying additions, deletions, and corrections. The final 2020 Census inventory of HUs became the main component of the ACS HU frame once the final 2020 Census results were incorporated into the MAF for ACS data year 2022. For details on the 2020 Census operations, see U.S. Census Bureau (2018).

Some ongoing improvement operations are designed to deal with MAF errors, while others aim to keep pace with post-census address development. In the remainder of this section, we discuss several of these operations, including DSF updates, the Geographic Support System Initiative, and field listing operations. Table 3-1 summarizes the development and improvement activities.

Table 3-1. Master Address File Development and Improvement

Initial Input (2000 and earlier)	Improvements (Post-2000)
1990 Decennial Census address control file	DSF updates
USPS Delivery Sequence File (DSF)	MAF Coverage Study
Local government updates	Community Address Updating System (CAUS)
Census 2000 operations	Demographic Area Address Listing (DAAL) Operations
	2010 Census field operations
	Geographic Support System Initiative
	2020 Census operations

Delivery Sequence File

The DSF is the USPS's master list of all delivery-point addresses served by postal carriers. The file contains specific data coded for each record, a standardized address and ZIP code, and codes that indicate how the address is served by mail delivery (for example, carrier route and the sequential order in which the address is serviced on that route). The DSF record for a particular address also includes a code for delivery type that indicates whether the address is a business or residential. The DSF is the primary source of new city-style addresses used to update the MAF

between decennial censuses. DSF addresses are not used to update non-city-style addresses in the MAF because those addresses might provide different and unmatchable address representations for HUs whose addresses already exist in the MAF. New versions of the DSF are shared with the Census Bureau monthly, and updates or “refreshes” to the MAF are made twice a year.

When DSF updates do not match an existing MAF record, a new record is created in the MAF. These new records, which could be new housing units, are then compared to the USPS Locatable Address Conversion Service (LACS) file, which indicates whether the new record is a change to an existing address. In this way, the process can identify duplicate records for the same address.¹⁴

Field Listing Operations

Nationwide field listing operations are undertaken between censuses to address coverage concerns. In a listing operation, a defined land area—usually a census tabulation block—is traveled in a systematic manner, while a field representative (FR) records the location and address of every structure where a person lives or could live. The Census Bureau conducts listings using laptop computers with a specially designed application called the Listing and Mapping Application (LiMA). The LiMA uses extracts from the current MTdb as inputs. Functionality in the LiMA allows FRs to edit, add, delete, and verify addresses, streets, and other map features; view a list of addresses associated with the assigned geography; and view and denote the location of HUs on the electronic map. Global Positioning System (GPS) functionality in the LiMA enables FRs to collect latitude and longitude coordinates for the structure. Compared to information once collected by paper and pencil, the LiMA allows for the standardization of data collected through edits and defined data entry fields, standardization of field procedures, efficiencies in data transfer, and timely reflection of the address and feature updates in the MTdb.

The Community Address Updating System (CAUS), Demographic Area Address Listing (DAAL), and the MAF Coverage Study are examples of nationwide field listing operations.

The Census Bureau designed the CAUS program specifically to address concerns about ACS coverage. The Census Bureau recognized that the DSF, the primary source of ACS frame updates, does not adequately account for changes in predominantly rural areas of the nation where city-style addresses are generally not used for mail delivery. An automated field data collection operation, CAUS, was designed to provide a rural counterpart to the update of city-style addresses received from the DSF. It improved coverage of the ACS by (1) adding addresses that exist but do not appear in the DSF; (2) adding non-city-style addresses in the DSF that do not appear on the MAF; (3) adding addresses in the DSF that also appear in the MAF but are

¹⁴ For additional details on the MAF update process via the DSF, see Hilts (2005).

erroneously excluded from the ACS frame; and (4) deleting addresses that appear in the MAF but are erroneously included in the ACS frame.

Implemented in September 2003, CAUS focused its efforts on census blocks with high concentrations of non-city-style addresses and suspected growth in the HU inventory. Of the approximately 8.2 million blocks nationwide, the CAUS universe comprised the 750,000 blocks where DSF updates were not used to provide adequate coverage. The Census Bureau selected CAUS blocks using a model-based method that used information gained from previous field data collection efforts and administrative records to predict where CAUS work was needed. The CAUS program was suspended from October 2007 to March 2010 until the 2010 Census Address Canvassing and field follow-up activities were completed. The CAUS program resumed listing activities again in April 2010. Approximately 36,000 blocks were listed from October 2010 until 2016, when CAUS was suspended again.¹⁵

DAAL is a combination of operations, systems, and procedures associated with coverage improvement, address list development, and automated listing for the CAUS and demographic household surveys. The objective of DAAL is to update the inventory of HUs, GQs, and street features in preparation for sample selection for the ACS and surveys such as the Current Population Survey (CPS), the American Housing Survey (AHS), and the Survey of Income and Program Participation (SIPP). DAAL was a major source of intercensal updates to the MAF until it was suspended after the current surveys changed to a MAF-based sample design.¹⁶

The MAF Coverage Study was a component of Address Canvassing for the 2020 Census. Implemented in 2016, the MAF Coverage Study listed a nationwide sample of 18,500 blocks using the LiMA. Updates in those blocks were applied to the MAF and used to produce estimates of MAF coverage.¹⁷

Geographic Support System Initiative

The Geography Division of the U.S. Census Bureau initiated a broad-based geographic support system initiative in preparation for the 2020 Census. The initiative covered many aspects of geographic support for these programs, including investigating various partnering opportunities with local governments and pursuing commercial resources and crowdsourcing to maintain the MTdb throughout the decade. This program improved coverage of the MAF by adding new HUs and updating block geocodes for existing HUs.

¹⁵ For details on the CAUS program and its block selection methodology, see Hartman (2009, 2011) and Schar (2012a, 2012b).

¹⁶ For details on DAAL, see Perrone (2005). For information about the current surveys redesign, see Kennel et al. (2011).

¹⁷ For more information on the MAF Coverage Study, see Williams (2018).

All of these MAF improvement activities and operations contribute to the overall update of the MTdb.

3.4 Master Address File Development and Updating for Puerto Rico

The Census Bureau created an initial MAF for Puerto Rico through field listing operations. This MAF did not include mailing addresses because, in Puerto Rico, the 2000 Census used an Update/Leave methodology through which a census questionnaire was delivered by an enumerator to each living quarter. The MAF update activities that took place from 2002 to 2004 were focused on developing mailing addresses, updating address information, and improving coverage through yearly updates.

MAF Development in Puerto Rico

MAF development in Puerto Rico also used the 2000 Census operations as its foundation. These operations in Puerto Rico included address listing, Update/Leave, the LUCA program, and the Be Counted Campaign.¹⁸

The Census Bureau designed the 2000 Census procedures and processing systems to capture, process, transfer, and store information for the conventional three-line mailing address. Mailing addresses in Puerto Rico generally incorporate the urbanization name (a geographic area roughly equivalent to a neighborhood), which creates a four-line address. Using the urbanization name eliminates the confusion created when street names are repeated in adjacent communities. In some instances, the urbanization name is used in lieu of the street name.

The differences between the standard three-line address and the four-line format used in Puerto Rico created problems during the early MAF building stages. The resulting file structure for the Puerto Rico MAF was the same as that used for states in the United States, so it did not contain the additional fields required to handle the more complex Puerto Rico mailing address. These processing problems did not adversely impact the 2000 Census operations in the U.S. because the record structure was designed to accommodate the standard U.S. three-line address. However, in Puerto Rico, where questionnaire mailout was originally planned as the primary means of collecting data, the three-line address format turned out to be problematic.

MAF Improvement Activities and Operations in Puerto Rico

Because of these address formatting issues, the MAF for Puerto Rico as it existed at the conclusion of the 2000 Census required significant work before it could be used to fully implement the Puerto Rico Community Survey (PRCS) starting in 2005. The Census Bureau had to revise the address information in the Puerto Rico MAF. This effort involved splitting the

¹⁸ For details on the 2000 Census for Puerto Rico, see U.S. Census Bureau (2004).

address information into the various fields required to construct a mailing address using Puerto Rico addressing conventions.

The Census Bureau contracted for updating the list of addresses in the Puerto Rico MAF. Approximately 64,000 new Puerto Rico HUs were added to the MAF, with each address geocoded to a municipio, tract, and block. The Census Bureau also worked with the USPS DSF for Puerto Rico to extract information on new HU addresses. Matching the USPS file to the existing MAF was only partially successful because of inconsistent naming conventions, missing information in the MAF, and the existence of different house numbering schemes (USPS versus local schemes). Data collection activities for the 2005 PRCS began in November 2004 with the best address information available given these shortcomings.

In preparation for the 2010 Census, the Census Bureau conducted address canvassing in Puerto Rico as it was done in the United States, updating the inventory of housing units in the MAF for Puerto Rico before the 2010 Census. Results from the 2010 Census Update/Leave and follow-up operations also updated the MAF addresses in Puerto Rico. The Census Bureau determined the final 2010 Census status of each HU record in the MAF in Puerto Rico in late 2010.

As part of the MAF/TIGER redesign effort before 2010, the Census Bureau redesigned the MAF to accommodate the Puerto Rico-specific address components that were lacking previously. The MAF now accommodates these specific address components, allowing the potential to update the MAF in Puerto Rico by census field operations and other methods. In spring 2016, the Census Bureau updated the MAF with DSF addresses in Puerto Rico for the first time.

Address Canvassing was not done in Puerto Rico before the 2020 Census. The 2020 Census was conducted by visiting every housing unit in Puerto Rico during the Update/Leave operation. Geocoded DSF addresses in Puerto Rico were included in this operation. Addresses counted in the 2020 Census comprise the majority of the addresses on the PRCS frame after the 2020 Census.

3.5 Master Address File Development and Updating for Group Quarters in the United States and Puerto Rico

MAF Development for GQs

In preparation for the 2000 Census, the Census Bureau developed an inventory of special places (SPs) and GQs. SPs are facilities such as prisons, hotels, migrant farm camps, and universities. GQs are contained within SPs and include college and university dormitories and hospital/prison wards. The SP/GQ inventory was developed using data from internal Census Bureau lists, administrative lists obtained from various federal agencies, and numerous Census 2000 operations such as address listing, block canvassing, and the SP/GQ Facility Questionnaire operation. Responses to the SP/GQ Facility Questionnaire identified GQs and any HUs

associated with the SP. Similar to the HU MAF development process, local and tribal governments had an opportunity to review the SP address list. In August 2000, after the enumeration of GQ facilities, the Census Bureau incorporated the address and identification information for each GQ into the MAF.

MAF Improvement Activities and Operations for GQs

Maintenance of the GQ universe is an ongoing and complex task. The earlier section on MAF Improvement Activities and Operations for HUs mentions short-term/one-time operations that also updated GQ information. Additionally, the Census Bureau completed a GQ geocoding correction operation to fix errors (mostly census block geocodes) associated with college dormitories in the MTdb.

The Census Bureau may identify new GQ facilities and update address information for existing ones during field listing operations. This information is used to update the MAF. Additionally, it is likely that DSF updates of city-style address areas are providing the Census Bureau with new GQ addresses; however, the DSF does not identify such an address as a GQ facility.

Prior to 2010 Census operations, the Census Bureau developed a process to supplement these activities to create an updated GQ universe from which to select the ACS sample. The Census Bureau constructed the ACS GQ universe by merging the updated SP/GQ inventory file, extracts from the MAF, and a file of those seasonal GQs that were closed on April 1, 2000 (but might have been open if visited at another time of year). To supplement the ACS GQ universe, the Census Bureau obtained a file of federal prisons and detention centers from the Bureau of Prisons (BOP) and a file from the Department of Defense (DOD) containing military bases and vessels. The Census Bureau also conducted research to identify new migrant worker locations, new state prisons, and state prisons that had closed.

ACS FRs, while administering the Group Quarters Facility Questionnaire (GQFQ), collect updated address and geographic location information. Updates collected via the GQFQ were used to provide more accurate information for subsequent facility visits and update the ACS GQ universe. For more information about the GQFQ, see the Group Quarters (Facility-Level Phase) section in Section 8.2 of Chapter 8.

The Address Canvassing operation for the 2010 Census identified records as “other living quarters” or OLQs. All OLQs and GQs were then visited during the Group Quarters Validation operation, where their final status as a HU or GQ was determined. GQs were then enumerated in the GQ Enumeration operation. The Census Bureau applied updates from all of these operations to the MAF. The Census Bureau determined the final 2010 Census status for each GQ in late 2010.

The final 2010 Census universe of GQs is the basis of the ACS GQ frame for 2012 through 2021. ACS also includes GQs that were identified as having no population on Census Day, as those GQs may contain people if visited at another time of the year. New GQs from ongoing operations are also included in the ACS GQ frame. The Census Bureau updates the ACS GQ frame and the MAF with results from ACS GQ data collection operations as well as results of state prison research using the individual state Department of Corrections websites. ACS continues to partner with the BOP to ensure the most accurate sampling frame for federal prisons.¹⁹

Instead of identifying addresses as OLQs, the 2020 Address Canvassing operation was able to convert addresses to GQs. Any known GQs after Address Canvassing were included in the 2020 Census GQ Advance Contact operation. This operation attempted to contact known GQ facilities by phone or personal visit to obtain updated information to assist with the GQ enumeration process. The universe of GQs enumerated as part of the 2020 Census became the foundation of the ACS GQ frame for the data year 2022 after the census had been completed.

3.6 American Community Survey Extracts from the Master Address File

Data from the MTdb are provided for use in the ACS in files called MAF extracts. These MAF extracts contain a subset of the data items in the MAF. The major classifications of variables included in the MAF extracts are address variables, geocode variables, and source and status variables (see Section 3.2).

As an inventory of living quarters (HUs and GQs) and some non-residential units, the MAF is a dynamic entity. It contains hundreds of millions of addresses that reflect ongoing additions, deletions, and changes; these include current addresses as well as those determined to no longer exist. Each Census Bureau program that relies on the MAF defines the set of valid addresses for its individual program.

Since the ACS frame must be as complete as possible, the Census Bureau applies filtering rules during the creation of the ACS extracts to minimize both overcoverage and undercoverage and to obtain an inclusive listing of addresses. For example, the ACS filter rules include units that represent new construction units, some of which may not exist yet. The ACS also includes other housing units that are not geocoded, meaning that the address has not yet been linked to a census tract and block. In addition, the ACS includes units that are “excluded from delivery statistics” (EDS); these units often are those under construction (i.e., the housing unit is being constructed and has an address, but the USPS is not yet delivering mail to the address).

¹⁹ For more information on the post-2010 Census ACS GQ frame, see Bates (2020).

Filtering rules change as new, vetted operations update the MAF. The post-2020 ACS frame consists of 2020 Census addresses plus any new records added to the MAF after the 2020 Census, including post-Census DSF adds; new or validated records from listing operations; new addresses from CQR and the Geographic Support System Initiative; and 2020 Census deletes that persist on the DSF.

As discussed above, the ACS attempts to create a sampling frame that is as accurate as possible by minimizing both overcoverage and undercoverage.²⁰ In the process, the ACS filter rules can lead to net overcoverage, reflecting some duplicate and nonexistent units. This overcoverage has been estimated to be approximately 1.9 to 5.2 percent from 2002–2009. See Kephart (2010) for a discussion of this issue.²¹

²⁰ Definitions of the terms “overcoverage” and “undercoverage” are provided in the Glossary. For details on how the ACS publishes estimates of overcoverage and undercoverage, see Chapter 15.

²¹ For details on the ACS requirements for MAF extracts, see La Kose and Bates (2019). For information on the ACS sample selection, see Chapter 4. For a description of data collection procedures for HUs and GQs, see Chapters 7 and 8, respectively. For details on the MAF, its coverage, and the implications of extract rules on the ACS frame, see Shapiro and Waksberg (1999) and Kephart (2010).

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Chapter 3 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) that was published	2.0
L. Bates	11/28/2022	Made several revisions to bring the current version up to date. Removed references to the ALMI since it has been replaced by the LiMA. Consolidated the sections on DAAL and CAUS into a “Field Listing Operations” section and added the MAF Coverage Study as a field operation. Added a sentence or two in the Geographic Support System Initiative section now that the program is updating the MAF. Added a sentence about the DSF now being used to update the MAF in Puerto Rico. Added information about 2020 Census operations and updated some of the references. Minor wording changes.	3.0
A. Williams	8/21/2024	Edited the sources that have updated the MAF since the 2010 Census. Removed most references to MAF address types from 2010 Census. Where possible, generalized references to 2010 Census to decennial census.	4.0

Chapter 4: Sample Design and Selection

4.1 Overview

The American Community Survey (ACS) and Puerto Rico Community Survey (PRCS) each consist of two separate samples: housing unit (HU) addresses and residents of group quarters (GQ) facilities. As described in Chapter 3, we derive the sampling frames from which we draw these samples from the Census Bureau’s Master Address File (MAF). The MAF is one component of the combined database that combines both address information from the MAF with spatial information containing the geographic boundaries from the Topologically Integrated Geographic Encoding and Referencing (TIGER) database. The combined MAF/TIGER database is used both for sampling and tabulation for the ACS as well as most Census surveys and data products. The MAF is the Census Bureau’s official inventory of known living quarters and selected nonresidential units in the United States and Puerto Rico.

We select independent HU address samples for each of the 3,143 counties and county equivalents in the U.S., including the District of Columbia, as well as for each of the 78 municipalities in Puerto Rico. In 2004, we selected samples of HU addresses for every county and county equivalent for field data collection in 2005.²² Each year from 2005–2010, we selected approximately 2.9 million HU addresses in the U.S. and 36,000 HU addresses in Puerto Rico.

Beginning in 2011, we implemented the following changes to the ACS sample design:

- We increased the HU sample in June 2011, bringing the size of the sample selected to 3.54 million addresses per year.
- We added several new HU sampling rates that allow better control of the allocation of the sample and improve estimate reliability for small areas.
- We increased the follow-up sample to 100 percent in select geographic areas.

In addition, starting in 2013, we restricted the assignment of the GQ sample for college dorms to the non-summer months (January–April, September–December).

Since 2013, we have instituted several changes to how the housing unit address sample is selected. Most notably, beginning with the 2016 sample selection that occurred in September of 2015, we moved to selecting two independent six-month samples for a given year, each consisting of roughly half of the target sample size of 3.54 million. The first sample, referred to as Period 1, is selected from the July delivery of the Master Address File extracts during September, and these addresses are assigned to January–June of the sample year. The second sample, referred to as Period 2, is selected from the addresses on the January delivery of the MAF extracts during March and is assigned to the July–December panels. All addresses on the

²² In the remainder of this chapter, the term “county” refers to counties, county equivalents, and municipalities.

July delivery and that are in the current year’s subframe²³ are assigned to one of the two sampling periods (Period 1 or Period 2). During Period 2, we use this pre-designation of period for the addresses processed during Period 1 and that remain valid, and all new addresses are randomly assigned to either Period 1 or Period 2.

Essentially, all the steps of the sampling are performed twice a year. In addition to the change to two six-month samples, all blocks are now assigned to a sampling stratum twice a year. This is a further improvement in the design, as any changes in the size of the set of geographic areas used in the sampling are now reflected every six months instead of only once a year.

Beginning in 2018, an improvement in sampling efficiency was made by introducing the concept of “Eligible for Data Collection.” Based on feedback from the Field Division and research into the make-up of certain addresses in the MAF extracts, all addresses must now have a minimum set of information to be eligible for second-stage sampling²⁴, and thus for data collection.²⁵ Addresses without this minimum set of information were difficult to contact via mail and difficult to locate in the personal visit data collection. Thus, they often were resolved as a nonresponse despite being unable to locate them, as there was no direct evidence that they were invalid housing units. These sample cases were, in essence, wasted, given that we could reasonably predict our inability to contact these addresses. To improve efficiency, therefore, only addresses deemed to be valid and now flagged as “Eligible for Data Collection” (satisfying the minimum set of information necessary to contact them) are eligible to be selected for the ACS. Since a number of the addresses flagged as “not eligible” are still flagged as “valid” or “good”, there is an adjustment to the sampling weight to account for these addresses, and the weights of the sampled addresses still represent all “valid” or “good” addresses on the frame, as they always have.

Full-implementation samples of GQ facilities and persons began in 2006 and are selected independently within each state, including the District of Columbia and Puerto Rico. In 2006 and 2007, the ACS and the PRCS included approximately 2.5 percent of the expected number of residents in GQ facilities. Beginning in 2008, we increased the sampling rates in 16 states with small GQ populations to meet publication reliability thresholds. See Chapter 8 for details on the GQ data collection methods.

This chapter details the selection of the HU address and GQ person samples. As the ACS is a continuous survey, periodic refinements are made to the sampling methodology over time. For this chapter, we focus on the methodology used to select the 2022 sample as our reference date to coincide with the last standard 1-year data release at the time that this chapter was drafted. The final section describes the differences in sampling and data collection methodologies for some

²³ This document explains the concept of subframes in detail in section 4.3

²⁴ Addresses that are valid, but not eligible for data collection, are only eligible for first-stage sampling.

²⁵ For more details regarding the minimum set of information required for addresses to be eligible for data collection, see Bates (2021).

hard-to-reach areas in Alaska (referred to as Remote Alaska). The section on Remote Alaska also details recently modified sampling and data collection procedures for these areas.

4.2 Housing Unit Sample Selection

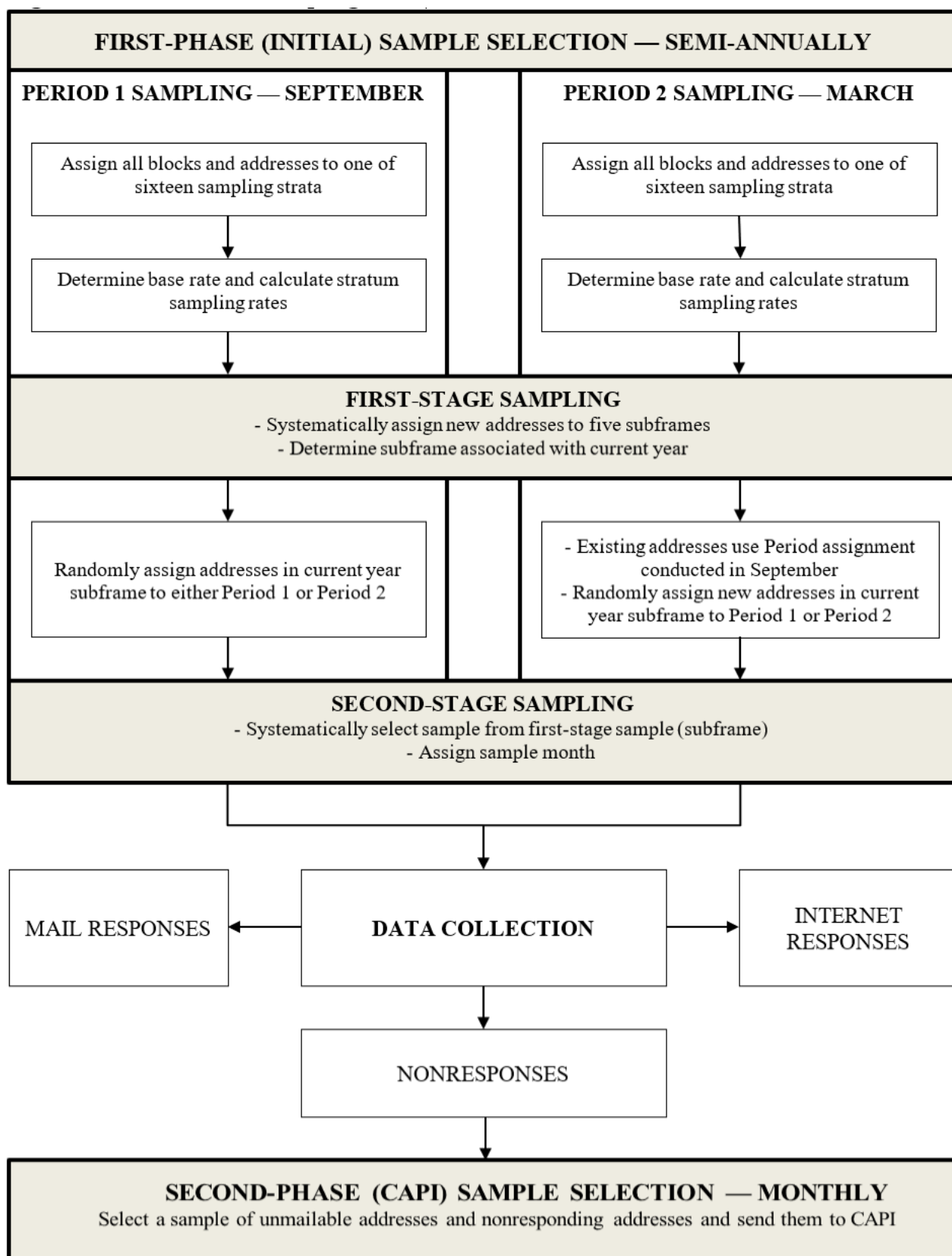
There are two phases of HU address sampling for each county.²⁶ During the first phase of sampling, we assign blocks to sampling strata, calculate sampling rates, and select the initial sample. During the second phase of sampling, we select a sample of nonresponding addresses for Computer Assisted Personal Interviewing (CAPI). This is referred to as the CAPI sample.

First-phase sampling produces the annual ACS initial sample of addresses and includes two processes—Period 1 and Period 2 sampling. The Period 1 and Period 2 samples in the first-phase sampling each include two stages. The first-stage sample selection systematically assigns new addresses to one of five subframes and identifies the appropriate subframe associated with a specific year’s sample. The second-stage sample selection systematically selects the sample from the selected subframe.

Figure 4-1 provides a visual overview of the housing unit address sampling process described in the previous paragraph.

²⁶ For first-stage HU sampling, “addresses” refers to valid ACS addresses. For second-stage HU sampling, “addresses” refers to valid ACS addresses that are eligible for data collection.

Figure 4-1. Overview of Sampling Process



4.3 First-Phase Sample

The first phase of sampling is comprised of two separate stages. The first stage of first-phase sampling maintains five distinct partitions, or subframes, of the addresses on the sampling frame within each county. Each county subframe is a representative sample of all valid addresses in the county. The subframes are assigned to specific years and are rotated annually. The subframes maintain their annual designation over time. First-stage sampling systematically sorts and assigns addresses that are new to the frame to one of the five subframes.²⁷ First-stage sampling also affects the second-stage sampling rates for each stratum for the current sample year.

The second stage of first-phase sampling selects a sample of the addresses from the current year's subframe and allocates this sample to the twelve months of the year for data collection.

Determining the First-Phase Sampling Rates Overview

Having a fixed 3.54 million annual sample, the ACS must recalculate the sampling rates each year that will achieve that fixed sample when applied to a sampling frame that generally grows over time. To that end, we define sixteen sampling strata for the first-phase sampling where three of the strata have fixed sampling rates and the remaining 13 strata have variable sampling rates that allow us to meet our sample size constraint. Each of these variable sampling rates is a multiple of a common base rate (BR). Thus, by adjusting the BR, we can adjust the size of the overall sample while maintaining a fixed relationship between the sampling rates for the variable sampling rate strata.

The first step in this process is to assign each address from the sampling frame for the period to a first-phase sampling stratum. Next, these universe counts by sampling strata are used to derive the individual stratum sampling rates that will achieve the target sample. The details for each of these steps are described in the subsequent sections.

Assigning Blocks and Addresses to the First-Phase Sampling Strata

Each year and for each Period, the sampling operation assigns each block to one of the sixteen sampling strata and, consequently, assigns each block one of sixteen sampling rates.²⁸ The ACS produces estimates for geographic areas having a wide range of population sizes. To ensure that the estimates for these areas have the desired level of reliability, we must sample areas with smaller populations at higher rates relative to those with larger populations. We base the stratum assignment for a block on information about a set of specific geographic entities—referred to as

²⁷ All existing addresses retain their previous assignment to one of the five subframes. The five subframes are maintained to meet the requirement that no address be in sample more than once in any five-year period.

²⁸ From 2005–2010, five sampling strata were used where one used a fixed rate and the other strata had a variable rate.

sampling entities—that contain the block or on information about the size of the census tract that contains the block, as discussed below.

Sampling entities are:

- Counties
- School districts
- American Indian areas/Alaska Native areas/Hawaiian Home Lands (AIANHH)
- American Indian Tribal Subdivisions with active and functioning governments
- Minor civil divisions (MCDs) with active and functioning governments in 12 states.²⁹
- Places with active and functioning governments.³⁰
- Census Designated Places (CDPs) in Hawaii only.³¹

We base the sampling stratum assignment for most blocks, at least in part, on the measure of size (MOS) for the smallest sampling entity to which any part of the block belongs. To calculate the MOS for a sampling entity, we derive block-level counts of addresses from the sampling frame. This count is converted to an estimated number of occupied HUs by multiplying it by the proportion of occupied HUs in the block in the 2020 Census. For American Indian and Alaska Native Areas (AIANA³²) including Tribal Subdivisions, we multiply the estimated number of occupied HUs by the proportion of its population that responded as American Indian or Alaska Native (either alone or in combination) in the 2020 Census.³³ For each sampling entity, we sum these estimates across all blocks in the entity to create the MOS for the entity. In AIANAs, if the sum of these estimates across all blocks is nonzero, then this sum becomes the MOS for the AIANA. If it is zero (because of a zero-census count of American Indians or Alaska Natives), the occupied HU estimate for the AIANA is the MOS for the AIANA. For more details, see the computer specifications for calculating the MOS for the ACS (Keathley, 2022a). We assign each

²⁹ The 12 states considered “strong” MCD states are Connecticut, Maine, Massachusetts, Michigan, Minnesota, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and Wisconsin.

³⁰ Functioning governments have elected officials who can provide services and raise revenue.

³¹ CDPs are used in Hawaii because the county is the principal unit of local government, with Honolulu being both a city and county.

³² AIANA is a general term used to describe American Indian and Alaska Native areas. For detailed technical information on the Census Bureau’s Alaska Native Areas Program for the 2020 Census, see the publication in the *Federal Register* (U.S. Census Bureau, 2008a). For detailed technical information on the Census Bureau’s American Indian Areas Program for the 2020 Census, see the publication in the *Federal Register* (U.S. Census Bureau, 2008b). Note that the 2010 Federal Register Notices for American Indian and Alaska Native Areas were also used for the 2020 Census (U.S. Census Bureau, 2024).

³³ 2020 Census information was used for the first time to define the measures of size in the 2022 sample selection.

block the smallest MOS of all the sampling entities in which the block is contained, and we refer to it as the Smallest Entity Measure of Size, or SEMOS.

If the SEMOS is greater than or equal to 1,200, we base the stratum assignment for the block on the MOS for the census tract that contains it as well as historical survey response rates prior to CAPI data collection. The tract measure of size (TMOS) is calculated as the sum of the estimated number of occupied HUs across all blocks contained within it. The survey response rates used measure the completed questionnaires across all modes before CAPI as a proportion of the sample that was eligible (estimated occupied mailable addresses).

This adjustment is to compensate for the increase in costs because of increasing the CAPI sampling rates in tracts predicted to have the lowest rate of completed interviews in the pre-CAPI modes. Note that the initial identification of these tracts, performed in 2004, was used in the 2005 sample selection and was revised in 2007 based on more recent data. It has been used since the 2008 sample selection. At that time, we used a combined mail and Computer Assisted Telephone Interview rate. A threshold of 60 percent was determined through research to be a tipping point between the higher and lower response rate tracts (Asiala, 2005).

Using SEMOS, TMOS, and the tract-level historical response rates, we can assign blocks to the sixteen strata defined in columns 1 and 2 in Table 4-1 on the following page.

Table 4-1. Sampling Strata Thresholds and Relationship between the Base Rate and the Sampling Rates

Stratum	Smallest Entity Measure of Size (SEMOS) and Tract Measure of Size (TMOS)	Sampling Rates
Blocks in smallest sampling entities	$0 < \text{SEMOS} < 200$	15% (fixed)
Blocks in small sampling entities	$200 \leq \text{SEMOS} < 400$	10% (fixed)
Blocks in medium sampling entities	$400 \leq \text{SEMOS} < 800$	7% (fixed)
Blocks in large sampling entities	$800 \leq \text{SEMOS} < 1,200$	$2.8 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and smallest tracts with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	$0 < \text{TMOS} \leq 400$	$3.5 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and smallest tracts with predicted levels of completed interviews prior to CAPI sampling > 60 percent		$0.92 \times 3.5 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and small tracts with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	$400 < \text{TMOS} \leq 1,000$	$2.8 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and small tracts with predicted levels of completed interviews prior to CAPI sampling > 60 percent		$0.92 \times 2.8 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and medium tracts with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	$1,000 < \text{TMOS} \leq 2,000$	$1.7 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and medium tracts with predicted levels of completed interviews prior to CAPI sampling > 60 percent		$0.92 \times 1.7 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and large tracts with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	$2,000 < \text{TMOS} \leq 4,000$	BR
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and large tracts with predicted levels of completed interviews prior to CAPI sampling > 60 percent		$0.92 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and larger tracts with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	$4,000 < \text{TMOS} \leq 6,000$	$0.6 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and larger tracts with predicted levels of completed interviews prior to CAPI sampling > 60 percent		$0.92 \times 0.6 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and largest tracts with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	$6,000 < \text{TMOS}$	$0.35 \times \text{BR}$
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and largest tracts with predicted levels of completed interviews prior to CAPI sampling > 60 percent		$0.92 \times 0.35 \times \text{BR}$

Figure 4-2 shows a Census Block that is in City A and also contained in MCD A with the surrounding County. In addition, the block is contained within census tract 1 (other tracts with City A are not shown). Therefore, it is contained in three sampling entities:

- County
- Minor Civil Division A—Minor Civil Division with an active and functioning government
- City A—A place with an active and functioning government

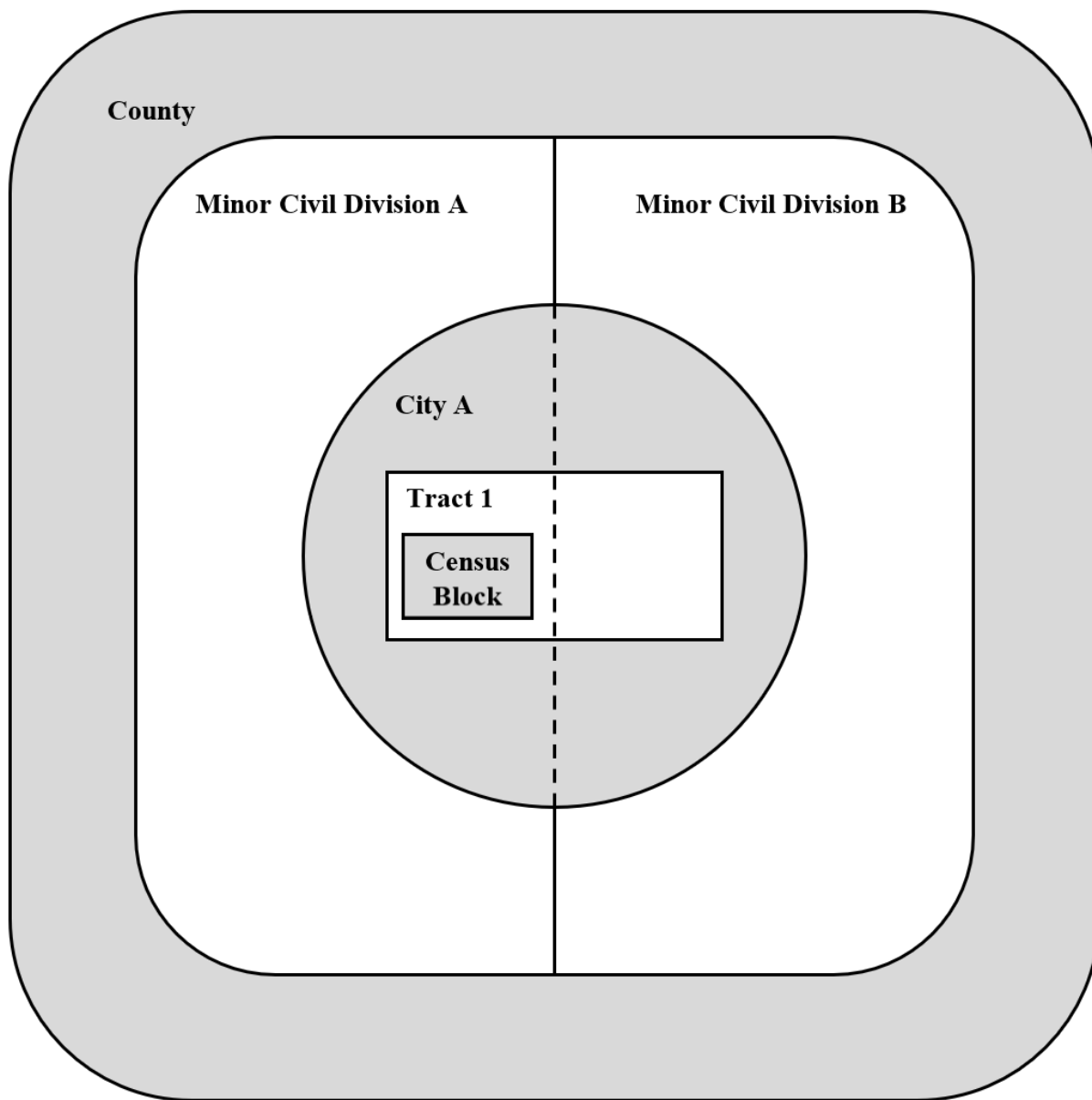
Example 1: Suppose the MOS for City A is 600 and the MOS for MCD A is 1,100. Then the SEMOS for the Census Block is 600 and it is placed in the $400 \leq \text{SEMOS} < 800$ stratum.

Example 2: Suppose the MOS for City A is 1,300, the MOS for MCD A is 300 and the MOS for MCD B is 1,000 (i.e., most of City A is contained within the more populated MCD B). Then the SEMOS for the Census Block is 300 and it is placed in the $200 \leq \text{SEMOS} < 400$ stratum. In this example, a different census block with City A that was contained in MCD B would likely fall into a different sampling stratum. Thus, this example demonstrates how the sampling rate may not be uniform across blocks within a given sampling entity.

Example 3: Suppose the MOS for City A is 1,300 and the MOS for MCD A is 2,400. Then the SEMOS for the Census Block is 1,300. Since the SEMOS for the block is greater than 1,200 the block will be assigned to one of the twelve strata with $\text{SEMOS} \geq 1,200$ depending on the size of the Tract 1 and the predicted level of completed interviews prior to CAPI sampling in the tract. In this example, suppose the TMOS is 1,800, and the predicted level of completed interviews prior to CAPI sampling is ≤ 60 percent, then the Census Block will be placed in the $1,000 < \text{TMOS} \leq 2,000$ stratum with a predicted level of completed interviews prior to CAPI sampling ≤ 60 percent.

Figure 4-2. Assignment of Blocks (and their Addresses) to First-Phase Sampling Strata

Note: The land area of a sampling entity does not necessarily correlate to its measure of size.



Determine the Base Rate and Calculating the First-Phase Sampling Stratum Rates

Each year, we must determine the specific set of sampling rates for each of the thirteen non-fixed rate sampling strata defined in Table 4-1. Before we can do this, we must perform the following two steps. The first step is to calculate a base rate (BR) for the current year. Thirteen of the sixteen sampling rates are a function of a base rate. The three fixed rate strata are 15 percent, 10 percent, and 7 percent. Column 3 of Table 4-1 shows the relationship between the base rate and the sixteen sampling rates.

After having assigned each address from the sampling frame for the period to a first-phase sampling stratum, universe counts for each stratum are tallied. These counts are then multiplied by the stratum sampling rates (which are a function of BR) and an algebraic expression for the total sample size, as a function of the BR, is obtained. Setting this expression equal to the fixed period sample size of 1.77 million (each period selects half of the annual sample) and solving determines the BR for the period. While this is done independently for each period, in practice the BRs for each period differ only slightly from one another, since the growth in the MAF from period to period is small relative to the total size of the sampling frame.

As an effect of having a fixed sample size, we reduce the sampling rates for all but the smallest sampling entity strata ($\text{SEMOS} < 800$) each year as the number of addresses in the U.S. and Puerto Rico increases. However, as shown in Table 4-1, among the strata where the rates are decreasing, the relationship between the sampling rates will remain proportionally constant. The sampling rates for the smallest sampling entities will remain at 15 percent, 10 percent, and 7 percent.

For the blocks that fall into higher response rate tracts, we multiply the sampling rates by 0.92 (reduced by 8 percent) for the six strata in which the SEMOS was greater than or equal to 1,200 to offset the costs of the increased second-phase sampling rates.

Because of this adjustment, there are sixteen sampling rates used in the U.S., and ten in Puerto Rico, as shown in columns 2 and 3 of Table 4-1. See the research report (Asiala, 2005) for a full description of the relationship between this reduction and the CAPI sampling rates. This reduction does not occur in Puerto Rico; therefore, there are ten sampling strata eligible to be used in Puerto Rico. Table 4-2 shows the final sampling rates used for 2022 after our final calculations. Note that only five strata in Puerto Rico contain valid addresses on the 2022 Period 1 sampling frame, so for 2022, we only used the five sampling rates shown in Table 4-2.

Table 4-2. 2022 ACS/PRCS Period 1 Sampling Rates

Stratum	Sampling Rate (percent) for the United States	Sampling Rate (percent) for Puerto Rico
Blocks in smallest sampling entities ($0 < \text{SEMOS} < 200$)	15.0	(NA)
Blocks in small sampling entities ($200 \leq \text{SEMOS} < 400$)	10.0	(NA)
Blocks in medium sampling entities ($400 \leq \text{SEMOS} < 800$)	7.0	7.0
Blocks in large sampling entities ($800 \leq \text{SEMOS} < 1,200$)	3.9	(NA)
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and smallest tracts ($0 < \text{TMOS} \leq 400$) with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	4.9	4.8
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and smallest tracts ($0 < \text{TMOS} \leq 400$) with predicted levels of completed interviews prior to CAPI sampling > 60 percent	4.5	
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and small tracts ($400 < \text{TMOS} \leq 1,000$) with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	3.9	3.9
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and small tracts ($400 < \text{TMOS} \leq 1,000$) with predicted levels of completed interviews prior to CAPI sampling > 60 percent	3.6	
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and medium tracts ($1,000 < \text{TMOS} \leq 2,000$) with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	2.4	2.3
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and medium tracts ($1,000 < \text{TMOS} \leq 2,000$) with predicted levels of completed interviews prior to CAPI sampling > 60 percent	2.2	
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and large tracts ($2,000 < \text{TMOS} \leq 4,000$) with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	1.4	1.4
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and large tracts ($2,000 < \text{TMOS} \leq 4,000$) with predicted levels of completed interviews prior to CAPI sampling > 60 percent	1.3	
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and larger tracts ($4,000 < \text{TMOS} \leq 6,000$) with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	0.8	(NA)
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and larger tracts ($4,000 < \text{TMOS} \leq 6,000$) with predicted levels of completed interviews prior to CAPI sampling > 60 percent	0.8	
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and largest tracts ($6,000 < \text{TMOS}$) with predicted levels of completed interviews prior to CAPI sampling ≤ 60 percent	0.5	(NA)
Blocks in large sampling entities ($\text{SEMOS} \geq 1,200$) and largest tracts ($6,000 < \text{TMOS}$) with predicted levels of completed interviews prior to CAPI sampling > 60 percent	0.5	

Note: The sampling rates in the table are the original rates rounded to one decimal place, expressed as percentages. The Period 2 sampling rates were nearly equal to the Period 1 rates within the precision shown in the table. (NA) = Not applicable.

First-Stage Sampling: Random Assignment of Addresses to a Specific Year

One of the ACS design requirements is that no HU address be in sample more than once in any five-year period in order to reduce respondent burden. To accommodate this constraint, the addresses in the frame are assigned systematically to five subframes, each containing roughly 20 percent of the frame and each being a representative sample. Addresses from only one of these subframes are eligible to be in the ACS sample each year, and each subframe is used every fifth year. For example, 2022 will have the same addresses in its subframe as did 2017, with the addition of all new addresses that we assigned to that subframe during the 2017–2022 time period. Therefore, both the Period 1 and Period 2 sample selection occur in two stages. The first stage partitions addresses new to the sampling frame into the five subframes and determines the subframe for the current year. The second stage, described in more detail below, selects addresses to be included in the ACS from the subframe eligible for the sample year (Keathley, 2022b).

Before the 2005 sample selection, there was a one-time allocation of all addresses then present on the ACS frame into the five subframes. In subsequent years, addresses new to the frame are systematically allocated to these five subframes. This is accomplished by sorting the addresses in each county by stratum and geographic order, including tract, block, street name, house number, and MAF Identification Number (MAFID). Addresses are then sequentially assigned to each of the five existing subframes. This procedure is similar to the use of systematic sampling with a sampling interval of five, in which the first address in the interval is randomly assigned to one of the subframes, the second address in the interval is assigned to the next year (in order), and so on. Specifically, during Period 1 sampling, only the addresses new to the MAF since the previous year's Period 2 sample selection are eligible for first-stage sampling and go through the process of assignment to a subframe. Similarly, during Period 2 sampling, only addresses new to the MAF since Period 1 sampling go through first-stage sampling.

Assignment of Addresses to Period 1 or Period 2

After first-stage sampling is complete for Period 1, all addresses on the current year's subframe are then allocated to Period 1 or Period 2. The addresses assigned to Period 2 will wait for further processing in the later period. Those addresses assigned to Period 1 will then move on to second-stage sampling for the period.

After first-stage sampling is complete for Period 2, all addresses new since Period 1 that are on the subframe will similarly be assigned to Period 1 or Period 2, although only those randomly assigned to Period 2 will go on to second-stage sampling for the given year. Addresses in the current subframe that existed at the time of Period 1 sampling and were assigned to Period 2 will also go on to second-stage sampling for Period 2.

The Period 1 sample will be used for the January–June monthly samples, and the Period 2 sample will be used for the July–December monthly samples.

Second-Stage Sampling: Selection of Addresses

With the first-stage sampling and period assignment complete, the second-stage sampling selects the entire first-phase sample from the current year’s subframe for the given period. Thus, the second-stage sampling rates need to account for the first-stage sampling rates to achieve our target first-phase rates. These second-stage sampling rates are approximately equal to the final sampling rate divided by 20 percent.³⁴

The second-stage sample is selected using systematic sampling after applying a geographic sort within sampling stratum. The sample selection is performed independently by county using systematic sampling from the portion of the subframe allocated to the current period. As noted earlier, the selection of the Period 1 and Period 2 samples together comprise the entire annual ACS HU sample.

Second-Stage Sampling: Sample Month Assignment for Address Samples

We must assign each sample address for a particular year to a specific data collection month. The set of all addresses assigned to a specific month is the month’s sample or panel. We sort addresses selected by stratum and geography and assign them systematically to the first six months or the last six months of the year, depending on the Period. However, we assign addresses that one of several Census Bureau household surveys have also selected to an ACS data collection month based on the interview month(s) for these other household surveys.³⁵ The goal of the assignments is to reduce the respondent burden of completing interviews for both the ACS and another survey during the same month.

4.4 Second-Phase Sampling for CAPI Follow-up

The ACS uses three modes of data collection—internet, mail, and personal visits. (see Chapter 7 for more information on HU data collection). Mailable sample addresses are eligible to complete the survey during the entire three-month time period. Cases without a completed internet or mail questionnaire are eligible for CAPI in the third month, as are the unmailable addresses (this category includes addresses in remote Alaska). An address is unmailable if the address is incomplete or directs mail to only a post office box. Table 4-3 summarizes the eligibility of addresses for CAPI sampling.

³⁴ The second-stage rate is approximately equal to the sampling rate divided by 20 percent since the first-stage sampling rate is approximately 20 percent, and the first-stage rate times the second-stage rate equals the overall sampling rate. An adjustment is made to account for uneven distributions of addresses in the county-level subframes.

³⁵ These surveys include the Survey of Income and Program Participation, the National Crime Victimization Survey, the Consumer Expenditures Quarterly and Diary Surveys, the Current Population Survey, the State Child Health Insurance Program Surveys, and the American Housing Survey (in odd-numbered years only).

Table 4-3. Addresses Eligible for CAPI Sampling

Mailable Address	Responds to Mailing (internet or questionnaire)	Eligible for CAPI
No	Not Applicable	Yes
Yes	No	Yes
Yes	Yes	No (completed)

The CAPI sample selects a systematic sample of these addresses for CAPI data collection each month using the rates shown in Table 4-4. The selection is made after sorting within county by CAPI sampling rate and geographic order within the address frame (Bullock, 2022a).

The variance of estimates of HUs and people living in them for a given area is a function of the number of interviews completed within that area. However, because of the subsampling, CAPI cases generally have larger weights than cases completed by internet or mail. The variance of the estimates for an area will tend to increase as the proportion of internet and mail responses decreases. Large differences in these proportions across areas of similar size may result in substantial differences in the reliability of their estimates. To minimize this possibility, tracts in the U.S. that are predicted to have low levels of interviews completed by internet or mail have their CAPI sampling rates adjusted upward from the default 1-in-3 rate for mailable addresses. This tends to reduce variances for the affected areas both by potentially increasing their total numbers of completed interviews and by decreasing the differences in weights between their CAPI and non-CAPI interviews.

No information was available to reliably predict the levels of completed interviews prior to second-phase sampling for CAPI follow-up in Puerto Rico before 2005, so we initially used the sampling rates of 1-in-3 for mailable and 2-in-3 for unmailable addresses. Based on early response results observed during the first months of the PRCS, we changed the CAPI sampling rate for mailable addresses in all Puerto Rico tracts to 1-in-2 beginning in June 2005.

We made several enhancements to the CAPI sampling, beginning with the 2011 sample, to increase the reliability of the ACS estimates for populations in certain well-defined geographic areas. Beginning in January of 2011, we send all Remote Alaska sample addresses to CAPI where previously they had been sampled at the rate assigned to unmailable cases (2-in-3). In addition, we do not subsample CAPI-eligible addresses that are either unmailable or that did not respond via internet or mail in the following areas:

- Hawaiian Home Lands
- Alaska Native Village Statistical Areas

- All American Indian areas with at least ten percent of the population in the 2020 Census reporting their race as American Indian or Alaska Native (alone or in combination).

The CAPI sampling rates that are used for addresses of each type are summarized in Table 4-4 below.

Table 4-4. CAPI Sampling Rates (in percent)

Area	Address and Tract Characteristics	Sampling rate
United States	Mailable and unmailable addresses in Remote Alaska, Alaska Native Village Statistical Areas, Hawaiian Home Lands, and select American Indian areas	100.0
United States	Other unmailable addresses	66.7
United States	Mailable addresses in tracts with predicted levels of completed interviews prior to CAPI subsampling between 0 percent and 35 percent	50.0
United States	Mailable addresses in tracts with predicted levels of completed interviews prior to CAPI subsampling greater than 35 percent and less than or equal to 50 percent	40.0
United States	Mailable addresses in other tracts	33.3
Puerto Rico	Unmailable addresses	66.7
Puerto Rico	Mailable addresses	50.0

Note: A 100 percent CAPI sampling rate applies to all Remote Alaska areas, Alaska Native Village Statistical Areas, and Hawaiian Home Lands. Only those American Indian Areas with 10 percent or more American Indian and Alaska Native (AIAN) population (alone or in combination) in the 2020 Census are included.

4.5 Group Quarters Sample Selection

GQ facilities include such places as college residence halls, residential treatment centers, skilled nursing facilities, group homes, military barracks, correctional facilities, workers' dormitories, and facilities for people experiencing homelessness. We classify each GQ facility according to its GQ type. (For more information on GQ facilities, see Chapter 8.) As noted previously, the 2005 ACS did not include GQ facilities, but we have included GQs since 2006. We select the GQ sample for a given year during two operations. Federal prisons managed by BOP are sampled in July of the sample year and all other GQs are sampled in October of the previous year. The MAF extract delivered in July of the previous year defines the sampling frame for the primary GQ sampling. Since 2006, the Census Bureau and the Federal Bureau of Prisons (BOP) have had an inter-agency agreement, and the BOP has delivered the most recent list of federal prisons (managed by the BOP) as well as the most recent roster of federal inmates in those prisons to the Census Bureau. From 2006 to 2015, the BOP sampling identified the number of people to sample during the same procedure as the non-BOP sample. Beginning with the 2016

sampling, a separate sampling frame was created using the facilities file delivered from the BOP, and the BOP sample was selected from this frame. The ultimate sampling units for the GQ sample are the GQ residents, not the facilities. The GQ samples are independent state-level samples.

The ACS sampling and data collection operations exclude certain GQ types, including domestic violence shelters, soup kitchens, regularly scheduled mobile food vans, targeted non-sheltered outdoor locations, commercial maritime vessels, natural disaster shelters, and dangerous encampments. There are several reasons for their exclusion, varying by GQ type. Concerns about privacy and the operational feasibility of repeated interviewing for a continuing survey, rather than once a decade for a census, led to the decision to exclude these GQ types. However, we control ACS estimates of the total population to be consistent with the Population Estimates Program estimate of the GQ resident population from all GQs, even those excluded from the ACS. Overall, the excluded GQ types contain a relatively small proportion of the total GQ population.

We classify all GQ facilities into one of two groups: (1) small GQ facilities (having 15 or fewer people according to the 2020 Census or updated information) and (2) large GQ facilities (with an expected population of more than 15 people). There are approximately 115,000 small GQ facilities and 79,000 large GQ facilities on the 2022 GQ sampling frame and 189 BOP facilities. We create two sampling strata to sample the GQ facilities. The first stratum includes both small GQ facilities and those with no available population count. The second stratum includes large facilities. In the remainder of this chapter, these strata will be referred to as the small GQ stratum and the large GQ stratum. We compute a GQ measure of size (GQMOS) for use in sampling the large GQ facilities. The GQMOS for each large GQ is the expected population count divided by 10.

The sampling procedures differ for these two strata. We sample GQs in the small GQ stratum, like addresses in the HU sample, and collect data for all people in the selected GQ facilities. Like HU addresses, small GQ facilities are eligible to be in the sample only once in a five-year period. For GQs in the large stratum, a subsample of the people living in the GQ is selected. The first phase of sampling selects groups of 10 people (“hits”) for interview from GQ facilities in the large GQ stratum, and the number of these groups selected for a large GQ facility is a function of its GQMOS. Unlike HU addresses and small GQs, large GQ facilities are eligible for sampling each year. For more detail, see the computer specifications for the GQ sampling (Bullock, 2022b).

4.6 Small Group Quarters Stratum Sample

For the small GQ stratum, a two-phase, two-stage sampling procedure is used. In the first phase, we select a GQ facility sample using a method similar to that used for the first-phase HU address sample. Just as we saw in the HU address sampling, the first phase has two stages. Stage 1 systematically assigns small GQ facilities to a subframe associated with a specific year. During

the second stage, we select a systematic sample of the small GQ facilities. In the second phase of sampling, we interview all people in the facility as long as there are 15 or fewer at the time of interview. Otherwise, we select and interview a subsample of 10 people.

First Phase of Small GQ Sampling—First Stage: Random Assignment of GQ Facilities to Subframes

The sampling procedure for 2006 assigned all of the GQ facilities in the small stratum to one of five subframes, each containing approximately 20 percent of the full frame. We sorted the GQ facilities within each state by small versus closed on Census Day, new versus previously existing, GQ type (such as skilled nursing facility, military barracks, or dormitory), and geographical order (county, tract, block, street name, and GQ identifier) in the small GQ frame. In each year after 2006, we assigned new GQ facilities systematically to the five subframes. The subframe for the 2022 GQ sample selection contains the facilities previously designated to the subframe for calendar year 2022 and 20 percent of new small GQ facilities added since 2021. The small GQ facilities in the 2022 subframe will not be eligible for sampling again until 2027 since the once-in-five-year period restriction also applies to small GQ facilities.

First Phase of Small GQ Sampling—Second Stage: Selection of Facilities

The second-stage sample is a systematic sample of the GQ facilities from the assigned subframe within each state. The GQs are sorted by the order in which they were selected during first-stage sampling. Because the range of sizes for these GQs is relatively narrow, we use the same probability of selection for all small GQs regardless of their actual size. The second-stage sampling rate combined with the 1-in-5 first-stage sampling rate yields an overall first-phase sampling rate equal to the sampling rate for each state. As an example, if the sampling rate for the state is 2.5 percent, then the second-stage sampling rate would be 1-in-8 so that the overall GQ sampling rate would be $(1\text{-in-}5) \times (1\text{-in-}8) = 1\text{-in-}40 = 2.5$ percent. Table 4-5 shows the 2022 state-level sampling rates.

Table 4-5. 2022 Group Quarters State-level Sampling Rates

State	Sampling Rate (percent)	State	Sampling Rate (percent)
Alabama	1.17	Montana	3.39
Alaska	2.89	Nebraska	2.41
Arizona	1.78	Nevada	3.37
Arkansas	2.00	New Hampshire	2.52
California	2.19	New Jersey	2.61
Colorado	1.76	New Mexico	2.61
Connecticut	2.38	New York	2.12
Delaware	4.90	North Carolina	2.00
District of Columbia	2.73	North Dakota	3.90
Florida	1.98	Ohio	2.39
Georgia	2.29	Oklahoma	2.15
Hawaii	3.09	Oregon	2.26
Idaho	2.23	Pennsylvania	2.58
Illinois	2.46	Rhode Island	2.42
Indiana	2.32	South Carolina	1.97
Iowa	2.18	South Dakota	3.46
Kansas	2.34	Tennessee	2.25
Kentucky	2.32	Texas	2.02
Louisiana	2.37	Utah	1.83
Maine	2.64	Vermont	4.25
Maryland	2.46	Virginia	2.40
Massachusetts	2.05	Washington	1.88
Michigan	2.54	West Virginia	2.14
Minnesota	2.31	Wisconsin	2.23
Mississippi	2.25	Wyoming	6.83
Missouri	2.14	Puerto Rico	2.13

Second Phase of Small GQ Sampling: Selection of Persons within Selected Facilities

Every person in the GQ facilities selected in this sample is eligible to be interviewed. If the number of people in the GQ facility exceeds 15, interviewers perform a field subsampling operation to reduce the total number of sampled people to 10, similar to the groups of 10 selected in the large GQ stratum.

4.7 Large Group Quarters Stratum Sample

Unlike the HU address and small GQ samples, we do not divide the large GQ facilities into five subframes. For field workload balance, we conduct interviews in groups of ten rather than sampling all residents of the GQ facility. Like the small GQs, we use a two-phase sampling procedure. The first phase indirectly selects the GQ facilities by selecting groups of ten within the facilities. The second phase selects the actual ten people for each facility's assigned group(s) using the roster obtained from the facility at the time of interviewing. The number of groups

eligible to be sampled from a large GQ facility is equal to its GQMOS (equal to its estimated size divided by 10). For example, if a facility had 550 people in the 2020 Census, its GQMOS is 55, and 55 groups of ten are eligible for selection in the sample.

First Phase of Large GQ Sampling: Selection of Groups of Ten (and Associated Facilities)

We sort all of the large GQ facilities in a state by GQ type and geographical order in the large GQ frame and select a systematic sample of groups of ten. For this reason, in states with a 2.5 percent sampling rate, a GQ facility with fewer than 40 groups (or roughly 400 individuals) may or may not have one of its groups selected for the sample. GQ facilities in a state with a 2.5 percent sampling rate and between 40 and 80 groups will have at least one group selected with certainty. If the GQ facility has between 80 and 120 groups, it will have at least two groups selected, and so forth.

Second Phase of Large GQ Sampling: Selection of Persons within Facilities

For the non-BOP GQs, the second phase of sampling takes place within each large GQ facility with at least one group selected in the first phase. When a field representative (FR) visits a GQ facility to conduct interviews, an automated instrument randomly selects the ten people to be included for each group of ten being interviewed. The instrument is pre-loaded with the number of expected person interviews (ten times the number of groups selected) and a random starting number. The FR then enters the actual number of people in the facility and a roster of their names. To achieve a group size of 10, the instrument computes the appropriate sampling interval based on the observed population at the time of interviewing. Then, it selects the actual people for the interviews using a pre-loaded random start and a systematic algorithm. If the large GQ has an observed population of 10 or more people, the instrument selects a group size of 10; if the observed population is less than 10, the instrument selects everyone in the GQ.

For most GQ types, if multiple groups are selected within a GQ facility, their groups of ten are assigned to different sample months for interviewing. Very large GQ facilities with more than 12 groups selected have multiple groups assigned to some sample months. In these cases, we are able to avoid selecting the same person more than once in a sample month. However, there is no attempt made to prevent the selection of someone more than once across sample months within a year. Thus, we could interview someone in a very large GQ facility in consecutive months. All GQ facilities in this stratum are eligible for selection every year, regardless of their sample status in previous years.

The person sample for BOP GQs in sample is selected during the BOP sampling operation and is drawn directly from the list of federal inmates on the roster file delivered by the BOP. Actual names are selected for each hit, and the names of the selected inmates are pre-printed on paper questionnaires, which the FR delivers to each BOP GQ in sample. The paper questionnaires are distributed to the sampled inmates by staff in each prison and, after completion, are mailed to the Census Bureau.

4.8 Final Group Quarters Sampling Steps

In 2010, we instituted a procedure that guarantees each county has at least one GQ in sample in any five-year period. This was driven by concerns about undercoverage of the GQ population in some counties as a result of the inherent clustering of the GQ population combined with a relatively small sampling interval. This final sampling step is performed after the selection of both the small and large GQ stratum sampling is complete. It will select one GQ from any county, without controlling for type, that has not had a GQ in sample over the past five years and has at least one GQ eligible in the first-stage sample of the small GQs or the large GQ sampling frame. The number of GQs selected in this manner is quite small.

Once the final first-phase GQ sample is selected, we assign the selected small GQ facilities and groups of ten for large GQ facilities to months using a procedure similar to the one used for sampled HU addresses. We combine and sort all GQ samples from a state by small versus large stratum and first-phase order of selection.

Consecutive samples are assigned to the 12 months in a pre-determined order, starting with a randomly determined month.

Because of operational and budgeting constraints, we assign the same month to all sample groups of ten within certain types of correctional GQs or military barracks. For example, we assign all samples in federal prisons to September, and data collection may take up to 4.5 months, an exception to the six weeks allowed for all other GQ types. For the samples in non-federal correctional facilities—state prisons, local jails, halfway houses, military disciplinary barracks, and other correctional institutions—or military barracks, individual GQ facilities are randomly assigned to months throughout the year.

Beginning with the 2013 GQ sample, we no longer assign college dorms to the months of May–August. This is in response to the relatively low interview rates at these GQs during the summer months. Since 2013, we only assign college dorm GQs to January–April and September–December.

4.9 Remote Alaska Sample

Remote Alaska is a set of rural areas in Alaska that are difficult to access and for which all HU addresses are treated as unmailable. In 2022, there were approximately 48,000 HU addresses and 850 GQs in Remote Alaska. Because of the difficulties in field operations during specific months of the year and the extremely seasonal population in these areas, data collection operations in Remote Alaska differ from the rest of the country. In both the Period 1 and Period 2 HU address samples, the month assigned for each Remote Alaska HU address is based on the county, place, AIANA, or block group (in that order) in which it is contained. We assign all designated addresses located in each of these geographical entities to either January or July in such a way as to balance workloads between the months and to keep groups of cases together geographically.

We sort the addresses for each month by county and geographical order in the address frame, and beginning in 2011, all sample addresses are sent directly to CAPI (bypassing mail and internet for the HU sample) in the appropriate month. We assign the GQ sample in Remote Alaska to January or July using the same procedure and allow up to six months to complete the HU and GQ data collection for each of the two data collection periods.

4.10 References

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Chapter 4 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
M. Asiala & E. Poehler	11/28/2022	Changes to content and formatting	3.0
M. Asiala	11/22/2024	Updates for data year 2022, including GQ sampling rates.	4.0

Chapter 5: Content Development Process

5.1 Overview

The American Community Survey (ACS) content is designed to meet the needs of federal government agencies and is a rich source of local area information useful to a wide variety of data users, including state and local governments, universities, and private businesses. The Census Bureau coordinates the content development and determination process for the ACS with the Office of Management and Budget (OMB) through an interagency committee comprised of more than 30 federal agencies. In addition, the Interagency Council on Statistical Policy Subcommittee on the ACS (ICSP-SACS) reviews proposals for content changes and provides input to inform the decisions made by the Chief Statistician of the United States and the Director of the Census Bureau. All requests for content changes are managed by the ACS Content Council, whose role is to provide guidance for testing and implementing new content and changes to existing ACS content. This chapter provides details on the history of content development for the ACS, survey content, and the content determination process and policy.

5.2 Initial Content Development

The ACS is part of the Decennial Census Program and is an alternative method for collecting the long-form sample data collected from the 1940 Census to the 2000 Census.³⁶ The long-form sample historically collected detailed population and housing characteristics once a decade through questions asked of a sample of the population. Since 2005, the ACS has collected this detailed information on an ongoing basis, thereby providing more accurate and timely data than was possible previously. Since 2010, the decennial census only includes a short form that collects basic information for a total count of the nation's population.³⁷

Historically, the content of the long form was constrained by including only the questions for which it was mandatory, required, or programmatic:

- There was a current federal law calling for the use of decennial census data for a particular federal program (mandatory).
- A federal law (or implementing regulation) clearly required the use of specific data, and the decennial census was the historical or only source or case law requirements imposed by the U.S. federal court system needed the data (required).

³⁶ Sampling began in the 1940 Census when a few additional questions were asked of a small sample of people. A separate long-form questionnaire was not implemented until 1960.

³⁷ In addition to counting each person at every address, the basic information collected on the 2010 Census short form included a very select set of key demographic characteristics needed for voting rights and other legislative requirements, including tenure at residence, sex, age, relationship, Hispanic origin, and race.

- The data were necessary for the Census Bureau's operational needs, and there was no explicit requirement for the use of the data, as explained for mandatory or required purposes (programmatic).

Constraining the content of the ACS was, and still is, critical because of the mandatory reporting requirement and to minimize the burden on respondents. To do this, the Census Bureau works closely with the OMB, the Federal Interagency Committee for the ACS, and the Interagency Council for Statistical Policy Subcommittee on the ACS (ICSP-SACS).

The [Federal Interagency Committee for the ACS](#) was established in July 2000 and includes representatives from more than 30 federal departments and agencies that use decennial census data. Working from the 2000 Census long-form justification, the initial focus of the committee was to verify and confirm legislative justifications for every 2003 ACS question. The members examined each question and provided for their agency justification(s) by subject matter, the legal authority for the use, the lowest geographic level required, the variables essential for cross-tabulation, and the frequency with which the data are needed. They cited the text of statutes and other legislative documentation and classified their uses of the ACS questions as “mandatory,” “required,” or “programmatic,” consistent with the constraints of the traditional long form.

In the summer of 2002, the United States (U.S.) Department of Commerce General Counsel’s Office asked each federal agency’s General Counsel to examine the justifications submitted by the committee for its agency and, if necessary, to revise the information so that the agency would be requesting only the most current material necessary to accomplish the statutory departmental missions in relation to census data. This step ensured that the highest-ranking legal officer in each agency validated its stated program requirements and data needs.

Only those questions whose subjects were classified as either “mandatory” or “required” were included on the 2003 ACS questionnaire, along with questions on two programmatic subjects (fertility and seasonal residence). The result of this review was a 2003 ACS questionnaire with content almost identical to the 2000 Census long form.

5.3 Initial Content (2003–2007)

From 2003 to 2007, the ACS consisted of 25 housing and 42 population questions (six basic and 36 detailed population questions). (See Table 5-1 for a complete list of ACS topics, listed with the questionnaire number on the paper questionnaire.) The ACS Group Quarters (GQ) questionnaire consisted of the same population questions except for the “relationship to householder” question. The ACS GQ questionnaire also included one housing question: the food stamp benefit question.

Table 5-1. 2003–2007 ACS Topics Listed by Type of Characteristic and Question Number

Housing Topics	Population Topics
H1 Type of Unit	P1 Sex
H2 Year Structure Built	P2 Date of Birth
H3 Year Householder Moved into Unit	P3 Relationship to Householder
H4 Acreage	P4 Marital Status
H5 Agriculture Sales	P5 Hispanic Origin
H6 Business on Property	P6 Race
H7 Rooms	P7 Place of Birth
H8 Bedrooms	P8 Citizenship
H9 Plumbing Facilities	P9 Year of Entry
H10 Kitchen Facilities	P10 Type of School and School Enrollment
H11 Telephone Service	P11 Educational Attainment
H12 Vehicles Available	P12 Ancestry
H13 House Heating Fuel	P13 Language Spoken at Home, Ability to Speak English
H14 Cost of Utilities	P14 Residence 1 Year Ago (Migration)
H15 Food Stamp Benefits ³⁸	P15 Disability: Sensory, Physical
H16 Condominium Status and Fee	P16 Disability: Mental, Self-Care
H17 Tenure	P17 Disability: Going out Alone, Ability to Work
H18 Monthly Rent	P18 Fertility
H19 Value of Property	P19 Grandparents as Caregivers
H20 Real Estate Taxes	P20 Veteran Status
H21 Insurance for Fire, Hazard, and Flood	P21 Period of Military Service
H22 Mortgage Status, Payment, Real Estate Taxes	P22 Years of Military Service
H23 Second or Junior Mortgage Payment or Home Equity Loan	P23 Worked Last Week
H24 Mobile Home Personal Property Taxes, Site Rent, Fees	P24 Place of Work
H25 Second residence, month, and reason	P25 Means of Transportation
	P26 Private Vehicle Occupancy
	P27 Time Leaving Home to Go to Work
	P28 Travel Time to Work
	P29 Layoff, Temporarily Absent, Informed of Recall or Return
	P30 Looking for Work
	P31 Available for Work
	P32 When Last Worked
	P33 Weeks Worked
	P34 Usual Hours Worked Per Week
	P35 Class of Worker
	P36 Name of Employer
	P37 Type of Business
	P38 Business Classification
	P39 Occupation
	P40 Primary Job Activity
	P41 Income in the Past 12 Months (by type of income)
	P42 Total Income

³⁸ In 2008, the Food Stamp Program was renamed to the Supplemental Nutrition Assistance Program (SNAP).

Puerto Rico Community Survey (PRCS) Content

The content for the PRCS is identical to that used in the United States (ACS) except for several questions worded differently to accommodate differences in the data needs and language differences. Figures 5-1 and 5-2 show examples of two ACS questions modified for the PRCS.

Figure 5-1. Example of ACS Utilities Question Modified for the PRCS

ACS Question			PRCS Question		
8 Does this house, apartment, or mobile home have –	Yes	No	8 Does this house, apartment, or mobile home have –	Yes	No
a. hot and cold running water?	☐	☐	a. running water?	☐	☐
b. a flush toilet?	☐	☐	b. a water heater?	☐	☐
c. a bathtub or shower?	☐	☐	c. a flush toilet?	☐	☐
d. a sink with a faucet?	☐	☐	d. a bathtub or shower?	☐	☐
e. a stove or range?	☐	☐	e. a sink with a faucet?	☐	☐
f. a refrigerator?	☐	☐	f. a stove or range?	☐	☐
g. telephone service from which you can both make and receive calls? Include cell phones.	☐	☐	g. a refrigerator?	☐	☐
			h. telephone service from which you can both make and receive calls? Include cell phones.	☐	☐

Figure 5-2. Example of ACS Year of Entry Question Modified for the PRCS

ACS Question	PRCS Question
9 When did this person come to live in the United States? Print numbers in boxes. Year 	9 When did this person come to live in Puerto Rico? Print numbers in boxes. Year

5.4 Content Policy and Content Change Process

In 2006, the ACS content development and change process hinged on the status of a question as *mandatory*, *required*, or *programmatic*, consistent with the constraints of the traditional long form. These statuses are defined as follows:

- **Mandatory** – a federal law explicitly calls for the use of decennial census or ACS data on that question.
- **Required** – a federal law (or implementing regulation) explicitly requires the use of data, and the decennial census or the ACS is the historical source; or the data are needed for case law requirements imposed by the U.S. federal court system.
- **Programmatic** – the data are needed for program planning, implementation, or evaluation, and there is no explicit mandate or requirement.

Since 2006, the OMB, in consultation with Congress and the Census Bureau, adopted a more flexible approach to content determinations for the ACS. In the new content determination process, the OMB, in consultation with the Census Bureau, considers issues such as frequency of

data collection, the level of geography needed to meet the required need, and other sources of data that could meet a requestor's need in lieu of ACS data. In some cases, legislation still may be needed to justify including a measure on the ACS. In other cases, OMB may approve a new measure based on an agency's justification and program needs.

The Census Bureau recognizes and appreciates the interests of federal partners and stakeholders in the collection of data for the ACS. Because participation in the ACS is mandatory, the OMB only approves necessary questions on the ACS. The OMB's responsibility under the Paperwork Reduction Act requires that new questions demonstrate the practical utility of the data and that the respondent burden be minimized (especially for the mandatory ACS collections).

The Census Bureau's [ACS Content Policy](#) is used as a basic guideline for handling all new question proposals from federal agencies, Congress, and the Census Bureau. The content change process is part of a risk management strategy to ensure that each new or modified question has been tested fully and will collect quality data without reducing overall response rates.

One vision for the ACS that emerged in the early years of the ACS program was that of a national resource to address emerging policy questions in the public sector. The idea that the ACS could be flexible, providing more frequent opportunities to add new content via new questions on the ACS itself or through follow-on or supplementary modules, was put forward in discussions with federal agencies about the program's future. The response to the survey is required by law. Therefore, the ACS has generally attained very high participation rates relative to other federal government surveys. So, the ACS estimates are attractive for users concerned about statistical bias because of nonparticipation. Also, the continuing efforts to improve the Census Bureau's Master Address File that is used for the decennial census, ACS, and other surveys the Census Bureau conducts make the ACS attractive in terms of its coverage rates.

At the same time, and serving to counterbalance this vision, the OMB and the Census Bureau recognized the need to develop priorities for including questions on the ACS and ensure that a consistent set of criteria were used to make decisions about adding or deleting questions from the ACS, using the ACS as a frame for follow-on surveys, and including a module of questions for a subsample of ACS cases. This recognition and guidance from the OMB that the respondent burden for completing the ACS would remain fixed led to the creation of a group tasked with advising the OMB and Census Bureau on issues related to content practices and policies.³⁹

The OMB and Census Bureau established the [ICSP-SACS](#) in 2012.⁴⁰ The ICSP-SACS exists to advise the Chief Statistician at OMB and the Director of the Census Bureau consistent with key governing policies and practices for the ACS from the perspective of the Federal Statistical

³⁹ Respondent burden is measured as the number of minutes each respondent requires to complete the ACS form.

⁴⁰ For more information see the [charter of the Interagency Council on Statistical Policy Subcommittee on the ACS](#).

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System. This subcommittee provides advice on how the ACS can best fulfill its role in the portfolio of federal household surveys, provide the most useful information with the least amount of burden, and address technical issues such as the deletion or addition of questions.

To implement a policy for the development of ACS content, the Census Bureau coordinates input from federal agencies, including the Interagency Committee for the ACS, to ensure broad input is acquired. The Census Bureau also coordinates the creation of subject-area subcommittee groups that include representatives from federal agencies and the Census Bureau. These groups provide expertise in designing question wording and response categories, along with research and testing objectives and measures, so that the questions will meet the needs of all agencies. Census Bureau staff review the subcommittee proposals and provide comments and internal approval of content changes.

The [ACS Content Change Process](#) provides guidance for Census Bureau testing for all new or modified questions before incorporating them into ACS instruments; this guidance is based on the standards outlined in the Census Bureau’s Statistical Quality Standards, Chapter A2: Developing Data Collection Instruments and Supporting Materials (U.S. Census Bureau, 2023a). The Census Bureau will change the content of the ACS only after the OMB gives approval.

Content Change Factors

The OMB and the Census Bureau consider several factors when new content is proposed. Federal agencies must provide specific information about the new data collection need(s). The uses of the data must be identified, along with citations to the appropriate statute or regulation, to determine the appropriateness of collecting it through a mandatory, federal survey. Other federal surveys or other sources of data are reviewed and considered. The respondent burden must also be considered.

Federal agencies interested in content changes must be able to demonstrate that they require detailed data (i.e., for small geographies and/or small populations) with the frequency of ACS data collection and that failure to obtain the information with this frequency will result in a failure to meet agency needs. Requests for new ACS content are assessed relative to the impact on the requesting agency if the data are not collected through the ACS. Federal agencies requesting new content must demonstrate that they have considered legitimate alternative data sources and why those alternatives do not meet their needs. Alternative data sources may be used to develop data products to meet federal agency needs without adding a question to the survey. Similarly, the Census Bureau may be able to take questions off the survey if it is determined that alternative data sources could meet the same need as respondent-provided survey answers, as determined by a thorough research and evaluation process.

Content Change Requirements

Federal agency or Census Bureau proposals for new content or changes to existing ACS questions are subject to the following requirements:

- ACS content can be added to or revised only once a year. Because the ACS produces estimates using one year and five years of data, the program strives to change questions infrequently to ensure questions are consistent over time. Additionally, revisions are restricted to once per year because of the annual nature of the survey and the number of operations that must be revised. New content is incorporated into the ACS only after testing, potentially including a field test, has been completed and the OMB has provided final approval.
- The requesting federal agency assists with the development of a draft question(s), works with the Census Bureau and other agencies to revise the question, and submits the proposal to the OMB and Census Bureau for further review. In addition, a plan to test new or modified content, including a field test, must be developed in accordance with the Census Bureau's Statistical Quality Standards (U.S. Census Bureau, 2023b).
- Testing is conducted to identify potential measurement error and to determine whether a change would increase or decrease a respondent's understanding of what is being asked. Alternative versions of questions are tested to identify the version most likely to be answered accurately by respondents, and then are field tested.

5.5 2014 Content Review

In 2014, the Census Bureau conducted a comprehensive content review to ensure that only the information needed is requested and that the justifications provided by federal agencies for the ACS questions are current and valid. The results of this analysis informed recommendations about which questions could be retained in their current form, which could be removed from the survey, and which could be candidates for further research.

The Census Bureau evaluated every question on the ACS questionnaire as part of this review. The questions were reviewed against 19 decision criteria developed by the OMB Interagency Council on Statistical Policy (ICSP) Sub-Committee for the American Community Survey (ACS). The decision criteria included metrics to assess the benefits associated with the usefulness of the questions and data and costs associated with respondent burden.

To assess the benefits, information from federal agencies was collected about question and data usage, such as the level of geographic detail the data are used for and whether the use was mandatory, required, or programmatic. The Census Bureau also assessed whether a data source other than ACS could be used to meet the federal agency need and if the data are used as another

federal survey's sampling frame. Additionally, coefficients of variation associated with each question's county-level estimate and interquartile ranges associated with each question's median county-level estimate were calculated.

The Census Bureau analyzed respondent burden by collecting feedback from a survey of field representatives and examining indicators of burden such as the time taken to answer the question, item nonresponse, and complaints received.

The information on the benefits and costs was used to identify which questions were candidates for removal based on being characterized as “Low Benefit and Low Cost” or “Low Benefit and High Cost.” Questions were candidates for removal if they fell into the Low Benefit/Low Cost or Low Benefit/High-Cost quadrants and had (1) No Mandatory Uses, (2) No Required Uses at the block group/tract, and (3) No Required Uses at the place/county/Metropolitan Statistical Area.

After reviewing more than 300 federal agency uses of ACS data and public comment on proposals, the Business/Medical Office on Property question (housing question 6 on the 2014 ACS) was removed from the ACS beginning in 2016.

For more information on the 2014 Content Review, see Treat, 2014.

5.6 Content Testing

The Census Bureau uses the term “content tests” to describe the testing, research, and evaluation processes used to determine the best wording, format, and placement of proposed new questions or revisions to existing questions on the ACS. Content tests are one of several kinds of tests the Census Bureau conducts as part of the ACS testing program. The primary objective of content tests is to test whether changes to question wording, response categories, and the redefinition of underlying constructs could improve the quality of the data collected. In addition to content, the ACS tests proposed improvements to ACS data collection methods and techniques. Such improvements include, for example, replacing the initial ACS mailing package with an initial pressure seal letter, using a new format to organize questions and guidance provided on the ACS questionnaire, or developing new instructions used by ACS interviewers for the Computer Assisted Personal Interviewing (CAPI) phase of data collection. This chapter will focus on content testing alone. Descriptions of other kinds of testing are included in other chapters of this report.

The Census Bureau conducts a content test for the ACS about every five years. If approved, changes are usually implemented on the ACS two years after field testing. The Census Bureau typically conducts cognitive testing to determine the final wording for a field test; field testing is used to make final recommendations about changes to implement. The methodology used for the field portion of the content test is designed to be similar to the regular ACS data collection process. A content test typically has two treatments: a control treatment, which usually has the

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current version of the question, and a test treatment, which has the alternative version of the question. Data quality may be further evaluated through a follow-up interview with respondents to either re-ask a question to measure response variance or ask more detailed questions to measure response bias. Differences between the control and test treatments are evaluated on several test metrics, typically including gross difference rates, item nonresponse rates, and measures of distributional changes. Table 5-2 summarizes changes to the ACS content on the paper questionnaire between 2010 and 2025.⁴¹

⁴¹ See the [ACS questionnaire changes](#) website for details about changes from 2005 to 2009.

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Table 5-2. Revisions, Deletions, and New ACS Questionnaire Topics Between 2010-2025

Question Type	Question Topic	2010	2011	2013	2016	2019	2020	2021	2024	2025
Roster	Household Roster									R
Housing	Year Built							R		
Housing	Business on Property				D					
Housing	Flush Toilet				D					
Housing	Public Sewer / Septic									N
Housing	Computer Use			N	R					
Housing	Internet Accessibility			N	R					
Housing	Internet Subscription			N	R					
Housing	Electric Vehicles									N
Housing	Home Heating Fuel								R	
Housing	Solar Panels									N
Housing	Condominium Fees								R	
Housing	Supplemental Nutrition Assistance Program (SNAP)	R								
Population	Race						R			
Population	Hispanic Origin						R			
Population	Educational Attainment									R
Population	Veteran Status			R						
Population	Period of Military Service			R					R	
Population	Health Insurance Coverage					R				R
Population	Health Insurance Premium and Subsidy					N				
Population	Means of Transportation					R			R	
Population	Time Leaving Home to Go to Work					R				
Population	Weeks Worked					R				R
Population	Name of Employer					R				
Population	Type of Business					R				
Population	Occupation					R				
Population	Primary Job Activity					R				
Population	Income in the Past 12 Months			R		R				
Population	Total Income					R				
Administrative	Revised page flow instructions		R		R	R	R			
Administrative	Cover Page/Front Page			R			R			

Note: R=Revised, N=New, D=Deleted

The following sections describe Content Tests conducted in 2006, 2007, 2010, 2016, and 2022 and the implementation of testing results in the content of the 2008, 2009, 2013, 2019, 2020, 2024, and 2025 ACS.⁴²

5.7 2006 and 2007 Content Tests

In 2004, planning began for the 2006 ACS Content Test so the Census Bureau could field-test potential content changes for implementation on the 2008 ACS questionnaire. Consistent with the procedures described above, the OMB and the Census Bureau asked members of the ACS Interagency Committee to review the legislative authority for current or proposed ACS questionnaire content and to identify any questions that needed to be reworded or reformatted.

The 2006 ACS Content Test was the first opportunity to test revisions to the long-form sample questions used in the 2000 Census.

- The 2006 ACS Content Test included new questions on the subjects of marital history, health insurance coverage, and veterans' service-connected disability ratings. These questions were added to the survey in 2008.
- The 2006 Content Test also tested modifications to questions on 21 other topics.
 - Housing topics tested included year built, rooms, bedrooms, plumbing facilities, kitchen facilities, telephone service, vehicles available, house heating fuel, food stamp benefits, property value and mortgage status, payments, and real estate taxes. Based on the results of the test, changes were not made to the heating fuel, vehicles available, mortgage status, payment, and real estate tax questions.
 - Population topics tested included place of birth, citizenship, year of entry, school enrollment, educational attainment, residence 1-year ago (migration), disability, military service status, period of military service, employment status, weeks worked, and industry and occupation. Based on the results of the test, changes were not made to the place of birth, year of entry, military service status, period of military service, and industry and occupation topics.

Additional content testing was conducted in the 2007 ACS Content Test on a Field of Degree question. The National Center for Education Statistics (NCES) began work with the Census Bureau and two groups of academic researchers in 2006 to develop and test alternative formats of a Field of Degree question that asked the field of degree for a person having a bachelor's degree or higher level of educational attainment. Based on the preliminary research, the Census Bureau developed and tested two alternative formats of the question in the 2007 ACS Content

⁴² [Content Test Evaluation Reports](#) are available on the Census Bureau's website.

Test, carried out in the fall of 2007. In 2009, the Census Bureau added a field of degree question with an open-ended response category to the 2009 ACS.⁴³

The 2006 Content Test and a separate test in 2007 also tested using a sequential format for the basic demographic questions rather than a grid format for the paper questionnaire (see Chesnut, 2008). Based on those results, the sequential format was adopted in 2008.

See U.S. Census Bureau (2007) for a summary of changes made in the 2008 ACS based on the results of the 2006 and 2007 Content Tests. Individual topic reports were published in 2007 and 2008 and can be found here: [Content Test Evaluation Reports \(census.gov\)](http://www.census.gov/contenttest/).

5.8 2010 Content Test

In response to federal agencies' requests for new and revised ACS questions, the Census Bureau conducted the 2010 ACS Content Test.⁴⁴ The Interagency Committee for the ACS helped identify possible changes to ACS content and additional new content that would be the subject of testing. The topics for content testing were new questions on computer ownership and internet usage and parental place of birth and revisions to questions about veteran status, period of military service, cash public assistance income, wages, property income, and the Food Stamp Program name.

- **Computer Ownership and Internet Usage** – As authorized by the Broadband Data Improvement Act of 2008, the Federal Communications Commission proposed the addition of the computer and internet usage topic. The Broadband Data Improvement Act requires that the Secretary of Commerce, in consultation with the Federal Communications Commission, expand the ACS to elicit information from residential households, including those located on native lands, to determine the following: whether persons at such households own or use computers at their address; whether persons subscribe to internet service; and, if so, whether they subscribe to dial-up or broadband internet service at that address. The additions to the questionnaire consist of three questions with a mix of fixed-choice and open-ended responses.
- **Parental Place of Birth** – Questions about a person's mother's and father's place of birth were tested to identify immigration generations (i.e., second generation or children of immigrants). These generational categories would allow policymakers and researchers to examine questions about the adaptation and integration of immigrants and their descendants over time. The results of the 2010 ACS Content Test demonstrated that questions on parental place of birth were clearly understood by

⁴³ See the [ACS questionnaire changes](http://www.census.gov/questionnairechanges/) website for details about changes from 2005 to 2009.

⁴⁴ For full details about the test, please see [reports published in 2012](http://www.census.gov/2010acs/reports/) and [reports published in 2013](http://www.census.gov/2010acs/reports/) for each topic.

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respondents and provided consistent and reliable data. Based on these results. However, ultimately, the Census Bureau decided not to pursue adding the questions to the ACS.

- **Veteran Status** – At the request of the Department of Veteran Affairs (VA), the Census Bureau revised the veteran status question to simplify the question and generate more reliable and accurate estimates of veterans. The results of the test found that the estimate of veterans from the test version was comparable to the estimate from the control version. The VA recommended the adoption of the test version of the veteran status question.
- **Period of Military Service** – The VA proposed several revisions to the wording of the period of military service question to simplify the question and to categorize period of military service categories in ranges that are meaningful for the VA. The resulting response distributions and gross difference rates in the test version are not significantly different from the response distributions and gross difference rates in the control version in the majority of the categories. The VA recommended the adoption of the new test version.
- **Cash Public Assistance Income** – The Census Bureau tested revisions to the cash public assistance income question to address an underreporting of income of this type. Researchers believe that the ACS question did not make it clear to the respondents that they should include participation on behalf of children or that they should report even single-month participation in a cash public assistance program. Several unfavorable results of the test were in opposition to the original intent of the question change, and therefore, it was recommended that changes not be made to the cash public assistance question.
- **Wages** – The Census Bureau tested a revision to the wages question in interviewer-administered modes to improve response by breaking up the question, which references wages, salary, commissions, bonuses, and tips, into shorter pieces to improve comprehension. The test found an increase in wage reciprocity for the Computer-Assisted Telephone Interview (CATI) mode. The net difference rates decreased. The test version's median estimates of wage income were significantly higher than the control. Item missing data rates for wages amounts and reciprocity increased. While the results were somewhat mixed, it was recommended to implement the test version.
- **Property Income** – The Census Bureau tested a revision to the property income question in interviewer-administered modes to improve response by breaking up the question, which references interest, dividends, net rental income, royalty income, or income from estates and trusts, into shorter pieces to improve comprehension. Because more questions were asked in the test version, there was a higher item nonresponse.

However, the test version raised the reciprocity rate and net difference rates were lowered. It was recommended to implement the test version.

- Food Stamp Program Name – At the request of the Food and Nutrition Service, the Census Bureau revised one housing question on food stamps to incorporate the program name change to the Supplemental Nutrition Assistance Program (SNAP). The test did not identify any differences in response distributions, item nonresponse, or in response reliability by incorporating the name change and recommended implementing the change.

Based on results of the 2010 Content Test, one new question topic, computer ownership and internet usage, was added to the ACS in 2013. In addition, OMB approved the modification of one housing question and four population questions for the 2013 ACS: veteran status, period of military service, wages, property income, and the name of the food stamp program.

5.9 2016 Content Test

In 2016, the Census Bureau tested changes to 10 topics on the ACS as well as a new question on health insurance premiums and subsidies.

Items Tested:

- Telephone Service – To reflect changes in how Americans use the telephone, the question on telephone service was separated from a list of questions about the physical characteristics of the housing unit and the instruction to “include calls using cell phones, land lines, or other phone devices” was added.
- Computer and Internet Use – To keep up with changes in technology, modifications to questions about computer and internet use were tested and implemented in the 2016 ACS. The test allowed the Census Bureau to evaluate the changes to the questions.
- Health Insurance Coverage – The health insurance coverage question was tested to reduce the underestimate of those with Medicaid coverage, allow respondents to better distinguish direct-purchase insurance from other types, and improve the accuracy of the estimates.
- Health Insurance Premiums and Subsidies – The Census Bureau also tested a new question about health insurance premiums and subsidies to measure subsidized Marketplace coverage (a type of health insurance coverage introduced through the Patient Protection and Affordable Care Act).
- Commuting Mode – The question on commuting mode was changed to update the terms used to describe how people get to work. For example, “Streetcar or trolley

car” was changed to “Light rail, streetcar, or trolley.” The time of departure question was also revised to address concerns raised among some respondents about privacy because the question specifically references the time a person leaves home to go to work.

- Weeks Worked – The question about the number of weeks worked was changed to an open-ended response format to better meet the needs of data users.
- Class of Worker – Several changes to the questions about the class of workers were tested to clarify definitions and improve the layout by grouping categories under general headings (e.g., private sector. Active-duty status in the military was also added as a response category.
- Industry and Occupation – Changes to the questions on industry and occupation were tested to make the question easier to understand and obtain the level of detail needed to classify these data. For example, the space for write-in responses was increased, and new, consistent examples were provided for the main occupation and description of job activities or duties.
- Retirement, Survive, and Disability Pensions – New instructions about the types of retirement, survivor, or disability pensions to include were added to elicit responses for certain types of income that were not being captured or were only partially captured, such as 401(k) accounts and Roth Individual Retirement Arrangements (IRAs).
- Relationship – To reduce errors in reports of same-sex and opposite-sex coupled households, more detailed response categories were provided for the question on household relationships and an automated consistency check was introduced in electronic modes to ask respondents to verify their responses when there are inconsistencies between sex and relationship.
- Race and Hispanic Origin – A combined version of the questions on race and Hispanic origin was tested, which included write-in response areas for every major group and a distinct response option for Middle Eastern or North African (MENA). A modified version of the separate race and Hispanic origin questions, which included the option to write in detailed origin information for the White and Black response categories, was also tested.

*Results.*⁴⁵

- The test results supported implementing the changes to the questions on telephone service, commuting mode, time of departure, number of weeks worked, class of

⁴⁵ For full details about the test, please see [reports published in 2016](#) and [report published in 2017](#) for each topic.

worker, industry and occupation, retirement income, and household relationships. The Census Bureau implemented these changes on the 2019 ACS.

- Given the rapid changes in technology, the Census Bureau chose to implement changes to the computer and internet use questions on the ACS in 2016 in conjunction with field testing. The test results further validated the decision to revise the computer and internet use questions.
- The test results showed that the changes to the health insurance coverage question did not appreciably improve the health insurance coverage estimates and did not support implementing these changes. The lessons learned in the 2016 test informed future testing of the coverage question. The test results did support implementing the new question on health insurance premiums and subsidies, which were added to the ACS in 2019.
- The Census Bureau implemented the version of the race and Hispanic origin questions used on the 2020 Census on the ACS in 2020.⁴⁶

5.10 2022 Content Test

The following topics were included in the 2022 Content Test: household roster, educational attainment, health insurance coverage, disability, income, SNAP, and labor force questions. Additionally, three new questions were tested on solar panels, electric vehicles, and sewage disposal. A summary of changes tested for each topic is as follows:

- Household Roster – The roster instructions have not changed since the 1990s while household living arrangements have increased in complexity. The revisions attempted to capture more complex living situations and improve within household coverage, especially among young children and tenuously attached residents.
- Educational Attainment – A relatively high percentage of respondents are selecting the response category, “No schooling completed.” Ongoing research suggests that this includes adults who have completed some level of schooling. The revision attempted to reduce the erroneous reports in this category through formatting and wording changes to clarify the response options.
- Health Insurance Coverage – The purpose of testing the revised health insurance coverage question was to enhance question reliability and validity. Since implementation in 2008, research has found that Medicaid and other means-tested programs were underreported in the ACS and that direct-purchase coverage was overreported, in part because of misreporting of non-comprehensive health plans

⁴⁶ The version of the race and Hispanic origin questions was not the version tested in the 2016 ACS Content Test. Results from the 2015 decennial Content Test ultimately led to the content decision.

and reporting multiple coverage types for the same plan. Moreover, revisions to the health insurance coverage question would help capture changes to the health insurance landscape that occurred with and since the passage of the Patient Protection and Affordable Care Act.

Changes to the health insurance coverage question included the following: reordering some response options and rewording response options for direct purchase, Medicaid, employer, and veteran's health care. A second version of the question tested these same changes along with a change to the format of the question that added an explicit response category for those who were uninsured.

- **Disability** – The purpose of testing revisions to the series of six disability questions was to capture population information on functioning in a manner that reflected advances in the measurement of disability and was conceptually consistent with the World Health Organization's International Classification of Functioning, Disability, and Health (ICF) disability framework. Changes included using graded response categories to reflect the continuum of functional abilities (the control questions use a dichotomous yes/no response), reordering the questions, and modifying question text. Additionally, a new test question attempted to capture difficulties related to psychosocial and cognitive disability in addition to problems with speech.
- **Income and SNAP** – The Census Bureau conducted research to determine the feasibility of using administrative sources as a replacement or supplement to the income questions currently fielded on the survey. Administrative data sources on employment, income, and public assistance benefits from the Internal Revenue Service, Social Security Administration, and state administrative offices could meet the agency needs for many types of income, transfer benefits, and employment data. If administrative data use in ACS production was implemented, it could provide far-reaching benefits for multiple ACS topics including income, SNAP, and employment.

To better align with administrative data sources on many types of income and transfer benefits, the Census Bureau tested a change in the reference period from the "past 12 months" to asking about the prior calendar year. Changes to the income and SNAP questions, aside from the reference period change, included the following: updating overall instructions along with instructions for public assistance and retirement income, changing question wording for self-employment, public assistance, and total income, and collecting rental income separately from the interest question. Two versions of the income questions were tested: one with a change in reference period and the question modifications and the other with only question modifications. The SNAP question only tested changes to the reference period; the question itself did not change.

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- Labor Force Questions – To align with the income question series, the Census Bureau tested a change in the reference period from “during the past 12 months” to asking about the prior calendar year. Changing the reference period to the prior calendar year would allow the ACS to better align with administrative records, which, if used, could improve the quality of ACS estimates. Additional changes to the labor force question series included the following: adding a question for respondents who have worked in the past five years (to implement the change in reference period) and modifying question instructions (tested in two formats).
- Solar Panels – This new question was tested to determine if the housing unit uses solar panels that generate electricity. Adding this question would provide data on operational solar panels on a housing unit level across the country. This information would help the Energy Information Administration (EIA) match energy consumption to energy production across the United States.
- Electric Vehicles – This new question was tested to determine if there are plug-in electric vehicles kept at the housing unit. Adding this question would provide the ability to project future energy sources, infrastructure, and consumer needs for the growing popularity of electric vehicles. The ACS is the only data source at the housing unit level for adequately making these projections.
- Sewage Disposal – This new question was tested to determine if the housing unit is connected to a public sewer, septic tank, or other type of sewage system. Adding this question would provide consistent data on the status of decentralized wastewater infrastructure in rural and other communities. These data are needed to protect public health and water quality and to understand and meet the country's growing infrastructure needs. The ACS is the only available survey that can provide these levels of data in a timely, consistent, and standardized manner.

The 2022 Content Test included a control and two experimental treatments to evaluate the proposed changes. The Control Treatment included the current ACS (production) version of the questions. New topics were also included in this treatment. The Test Treatment included the test question version for all topics except the household roster. Because changes to within-household coverage can impact the results of person-level questions, the household roster question(s) were the same as the Control Treatment to avoid confounding. Changes to the roster question(s) were tested in a separate treatment, called the Roster Test Treatment. Additionally, health insurance coverage, labor force, and income included a second set of experimental questions in the Roster Test Treatment.

Each topic was evaluated using a variety of metrics, including item missing data rates, response distributions, comparisons to benchmarks and administrative data, response reliability, and other topic-specific metrics. Comparisons were made between the Control Treatment and the experimental treatments. In some cases, the two experimental treatments were also compared.

*Results*⁴⁷

- The test results supported implementing the changes to the questions on household roster, educational attainment, health insurance coverage, and labor force (wording changes only) along with the addition of three new questions on solar panels, electric vehicles, and sewage disposal. The Census Bureau will implement these changes on the 2025 ACS.
 - Reference period changes to SNAP, income, and labor force questions are pending further research and implementation of administrative data use. Question changes to the income question are also pending further research.
 - The Census Bureau initially proposed to change the disability question as a result of the 2022 Content Test. However, based on public feedback on the proposal, the Census Bureau decided to retain the current ACS disability questions for the 2025 ACS and not move forward with the change (Santos, 2024).
- In addition to the topics included in the 2022 Content Test, three additional topics had changes that were cognitively tested: home heating fuel, condominium fees, and transportation to work. The Census Bureau, with input from the ICSP-SACS and approval from OMB, implemented changes to these questions on the 2024 ACS based on the cognitive testing results.
 - Home Heating Fuel – The change to the home heating fuel question updated the wording of the natural gas and bottled gas categories to aid respondents in identifying the correct category more easily by using more commonly used terminology. In Puerto Rico, the question wording also changed to indicate respondents should only include fuel that heats their homes.
 - Condominium and HOA Fees – The condominium fees question was extended to include homeowners’ association (HOA) fees. Data sources continue to show housing units that are part of HOAs outnumber housing units in condominiums. In order to provide more comprehensive and accurate costs of owning a home, the ACS needs to capture HOA fees for these homes. Adding these fees to the existing condominium fees question avoids adding a new question to the ACS and, therefore, minimizes respondent burden.
 - Transportation to Work – The transportation to work question was updated to include ride-sharing services as a mode of transportation to work to account for new and growing travel trends. The current category, “taxicab,” was updated to “taxi or ride-sharing services.” This will reduce ambiguity in the current

⁴⁷ For full details about the test, please see [reports published in 2022](#) and [reports published in 2023](#) for each topic.

question about where respondents should report ride-sharing commutes and will allow the government to monitor changes in transportation patterns for planning purposes.

- **Period of Military Service** – A change was made to the Period of Military Service question based on feedback from the VA. The VA requested that the ACS adjust the dates for the Vietnam War and Korean War to reflect the dates that they use for program evaluation (each period was adjusted by one month). The VA also requested that “Post 9/11” be added as a descriptor for the current service period, “Vietnam era” be changed to “Vietnam War,” and names of war periods be moved to the end of the date range for a uniform appearance. The VA requested that the date ranges use the word “through” instead of “to” for clarity. The updated dates for period of service match the dates that the VA uses for program evaluation as well as the official historical dates of war periods published by the Congressional Research Service. Moving names of war periods to the end of service categories creates a more uniform appearance of the question text, with dates listed first for all periods. These changes were recommended and approved for implementation on the 2024 ACS without cognitive or field testing.

5.11 2025 ACS Content

The ACS continues to provide the most detailed demographic, social, economic, and housing data for each of America’s communities. Table 5-3 provides the list of 2025 ACS Topics listed by type of characteristic and question number on the paper questionnaire.

Table 5-3. 2025 ACS Topics Listed by Type of Characteristic and Question Number

Housing Topics	Population Topics
H1 Type of Unit	P1 Name
H2 Year Structure Built	P2 Relationship to Householder
H3 Year Householder Moved into Unit	P3 Sex
H4 Acreage*	P4 Date of Birth
H5 Agriculture Sales	P5 Hispanic Origin
H6 Rooms and Bedrooms	P6 Race
H7 Plumbing and Kitchen Facilities	P7 Place of Birth
H8 Sewage Disposal	P8 Citizenship
H9 Telephone Service	P9 Year of Entry
H10 Computer Use	P10 Type of School and School Enrollment
H11 Internet Accessibility	P11 Educational Attainment
H12 Internet Subscription	P12 Field of Degree
H13 Vehicles Available	P13 Ancestry
H14 Electric Vehicles	P14 Language Spoken at Home, Ability to Speak English
H15 House Heating Fuel	P15 Residence 1 Year Ago (Migration)
H16 Solar Panels to Generate Electricity	P16 Health Insurance
H17 Cost of Utilities	P17 Health Insurance Premium and Subsidies
H18 Supplemental Nutrition Assistance Program (SNAP)	P18 Disability: Sensory
H19 Condominium and Homeowners Association	P19 Disability: Mental, Physical, Self-Care

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Housing Topics	Population Topics
(HOA) Status and Fee	P20 Disability: Doing Errands Alone
H20 Tenure	P21 Marital Status
H21 Monthly Rent	P22 Marital History
H22 Value of Property	P23 Number of Times Married
H23 Real Estate Taxes	P24 Last Year Married
H24 Insurance for Fire, Hazard, and Flood	P25 Fertility
H25 Mortgage Status, Payment, Real Estate Taxes	P26 Grandparents as Caregivers
H26 Second or Junior Mortgage Payment or Home Equity Loan	P27 Veteran Status
H27 Mobile Home Cost	P28 Period of Military Service
	P29 Veterans Disability
	P30 Worked Last Week
	P31 Place of Work
	P32 Means of Transportation
	P33 Private Vehicle Occupancy
	P34 Time Leaving Home to Go to Work
	P35 Travel Time to Work
	P36 Layoff, Temporarily Absent, Informed of Recall or Return
	P37 Looking for Work
	P38 Available for Work
	P39 When Last Worked
	P40 Weeks Worked
	P41 Usual Hours Worked Per Week
	P42 Class of Worker
	Name of Employer
	Type of Business
	Business Classification
	Occupation
	Primary Job Activity
	P43 Income in the Past 12 Months (by type of income)
	P44 Total Income

*NOTE: In 2024, the ACS began using administrative records data for acreage instead of survey responses, when possible. See Clark, et al. 2023 for more details.

5.12 References

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Chapter 5 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
D. Reese, S. Clark, & E. Poehler	11/28/2022	Expanded details of the process for changing ACS Content; Updated text to include the 2016 ACS Content Test; Added a table to summarize the topics that have undergone content testing; Added a table to list the 2020 ACS topics. Added information about the 2022 Content Test and 2022 content. Updated formatting and incorporated comments.	3.0
E. Poehler	10/17/2024	Provided more detail about the 2010 Content Test, added results of the 2022 Content Test and changes to content in 2024 and 2025.	4.0

Chapter 6: Survey Rules, Concepts, and Definitions

6.1 Overview

Interview and residence rules define the universe, or target population, for a survey and so identify the units and people eligible to be included. The American Community Survey (ACS) uses residence rules based on the concept of current residence.

Sections 6.2 and 6.3 in this chapter detail the interview and residence rules. Section 6.4 describes the full set of topics included in the ACS and is organized into four sections to parallel the organization of the ACS questionnaire: address, housing unit (HU) status, household information, basic demographic information, detailed housing information, and detailed population information.

6.2 Interview Rules

The Census Bureau classifies all living quarters as either HUs or group quarters (GQ) facilities. A HU is a house, an apartment, a group of rooms, or a single room either occupied or intended for occupancy as separate living quarters. GQ facilities are living quarters owned and managed by an entity or organization that provides housing and/or services for the residents. GQ facilities include correctional facilities, group homes, health care and treatment facilities, and college and university student housing.

Interview rules define the scope of data collection by defining the types of places included in the sample frame, as well as the people eligible. Beginning in 2006, the ACS included HUs and GQ facilities (only HUs and those living in HUs were included in the 2005 ACS). Like the decennial census, the ACS interviews the resident population without regard to legal status or citizenship and includes people residing in HUs at the residence where they meet the residence rules (see below).

6.3 Residence Rules and Reference Dates

Residence rules define who (if anyone) should be interviewed at a sample address and who is considered, for purposes of the survey or census, to be a resident. Residence rules decide the occupancy status of each HU and the people whose characteristics are to be collected.

ACS data are collected every day of the year. The survey's residence rules are applied, and its reference periods are defined as of the date the respondent completed the survey, either online, by paper questionnaire, or by telephone or personal visit interview.

Housing Units

The ACS defined the concept of current residence to determine who should be considered residents of sample HUs. This concept is a modified version of a de facto rule in which a time interval is used to determine residency.⁴⁸ The basic idea behind the ACS current residence concept is that everyone who is currently living or staying at a sample address is considered a current resident of that address, except for those staying there for only a short period of time, unless they have nowhere else to stay or live. For the purposes of the ACS, the Census Bureau defines this short period of time as two consecutive months or less (often described as the 2-month rule). Under this rule, anyone who has been or will be living for two months or less in the sample unit when the unit responded (either by internet, mail, telephone, or personal visit) is not considered a current resident unless they have no other place to stay or live. This means that their expected length of stay is two months or less, not that they have been staying in the sample unit for two months or less.

In general, people who are away from the sample unit for two months or less are considered to be current residents, even though they are not staying there when the interview is conducted. People who have been or will be away for more than two months are considered not to be current residents. The Census Bureau classifies this as a vacant HU in which no one is determined to be a current resident.

As noted earlier, residency is determined as of the date of the interview. A person who is living or staying in a sample HU on interview day and whose actual or intended length of stay is more than two months is considered a current resident of the unit. That person will be included as a current resident unless he or she, at the time of the interview, has been or intends to be away from the unit for a period of more than two months. There are a few exceptions:

- Children (below college age) who are away at boarding school or summer camp for more than two months are always considered current residents of their parents' home.
- Children who live under joint custody agreements and move between residences are always considered current residents of the sample unit where they are staying at the time of the interview.
- People who stay at a residence close to work and return regularly to another residence to be with their families are always considered current residents of the family residence and not the work residence.

⁴⁸ A de facto rule would include all people who are staying at an address when an interview is conducted, regardless of the time spent at this address. It would exclude individuals away from a regular residence even in they are away only for that one day.

- A person whose length of stay at the sample HU is for two months or less and has another place where he or she stays for more than two months is not considered a current resident at the sample HU.
- A person who is staying at a sample HU when the interview is conducted but has no place where he or she stays for more than two months, is considered to be a current resident of the sample HU.

Group Quarters

Residency in GQ facilities is determined by a purely de facto rule. All people living or staying in the sampled GQ facility when the roster of residents is made are eligible for selection to be interviewed in the ACS. Data are collected for all people sampled, regardless of their length of stay. Children (below college age) staying at a GQ facility functioning as a summer camp are not considered GQ residents.

Reference Period

As noted earlier, the survey's reference periods are defined relative to the date of the interview. Specifically, the survey questions define the reference periods and always include the date of the interview. When the question does not specify a time frame, respondents are told to refer to the situation on the interview day. When the question mentions a time frame, it refers to an interval that includes the interview day and covers a period before the interview. For example, a question that asks for information about the "past 12 months" would be referring to the previous 12 months relative to the date of the interview.

6.4 Structure of the Housing Unit Questionnaire

The ACS questionnaires and survey instruments used to collect data from the HU population are organized into four sections, with each section collecting a specific type of information. The first section verifies basic address information, determines the occupancy status of the HU, and identifies who should be interviewed as part of the ACS household. The paper questionnaire does this by stating the residence rules and asking the respondent to report the number of people living at the residence. The electronic modes (internet self-response or an interview by a Census Bureau telephone interviewer or field representative) have a more detailed set of questions to establish who is a resident. The second section of the questionnaire collects basic demographic data. The third section collects housing information, and the fourth section collects population data.

There are data collection instruments for each data collection mode (internet, mail, and in-person interviews). An automated internet instrument and paper questionnaire are used in the self-response mode. For telephone, there is a Computer-Assisted Telephone Interviewing (CATI)

instrument.⁴⁹ For personal interviews, there is a Computer-Assisted Personal Interviewing (CAPI) instrument. This section describes the basic data collection process from a personal visit perspective, but the same basic process is followed for internet, mail, and telephone modes.

Address, Housing Unit Status, and Household Information

During personal visit follow-up, the field representative (FR) must first verify that he or she has reached the sample address and then determine if the sample address identifies a HU. If a HU is not identified, the address is not eligible and is considered out of scope. An out-of-scope address includes an HU determined to be nonexistent because it has been demolished, is under construction, or is a business and not a residential unit. Interviewers use the residence rules to determine whether the sample HU is occupied (at least one person staying in the unit is a current resident) or vacant (no one qualifies as a current resident). Interviewers also apply the residence rules to create a household roster of current occupants to interview. The name of the household respondent and the telephone number are collected in case follow-up contact is needed. The terms below are key for data collection.

Housing Unit. A HU may be a house, an apartment, a mobile home or trailer, a group of rooms, or a single room that is occupied (or, if vacant, intended for occupancy) as separate living quarters.

Housing Unit Status. All sample addresses are assigned a status as either an occupied, vacant, or temporarily occupied HU or a status of delete, indicating that the address does not identify a HU. A temporarily occupied unit is a HU where at least one person is staying but where no people are current residents according to the ACS residence rules; this is considered a type of vacant unit. An example is a vacation home, which is used periodically. Deleted units are addresses representing commercial units or HUs that have been demolished, are under construction, or are nonexistent.

Household. A household is defined as all related or unrelated individuals whose current residence at the time of the ACS interview is the sample address.

Household Roster. The roster is a list of all current residents living at the sample address; all the people on the household roster will be interviewed.

Household Respondent. The household respondent is the person or persons in the household who provides the data. The respondent is knowledgeable about the household and its members. ACS field representatives try to restrict their household respondents to members who are at least 18 years old, but, if necessary, household members who are 15

⁴⁹ The ACS discontinued CATI nonresponse follow-up in 2017. The ACS program continues to conduct interviews by telephone in the Telephone Questionnaire Assistance (TQA) program when respondents contact the Census Bureau for assistance.

and older can be interviewed. If no household member can be found to provide the survey information, the interviewer must code the case as a noninterview.

Basic Demographic Information

The basic demographic data of sex, age, relationship, Hispanic origin, and race are collected at the outset of the interview and are considered the most critical data items. They are used in many of the survey's tabulations. Age defines the critical paths and skip patterns used in the instrument/questionnaire. Name is collected for all household members. One individual in the household must be identified as a reference person to define relationships within the household. The section below provides details and definitions associated with the basic demographic data.

Reference Person or Householder. One person in each household is designated as the householder. This is usually the person, or one of the people, in whose name the home is owned, being bought, or rented and who is listed as "Person 1" on the survey questionnaire. If there is no such person in the household, any adult household member 15 and older can be designated.

Sex. Each household member's sex is marked as "male" or "female."

Age and Date of Birth. The age classification is based on the age of the person in complete years at the time of interview. Both age and date of birth are used to calculate each person's age on the interview day.

Relationship. The instrument/questionnaire asks for each household member's relationship to the reference person/householder. Categories include both relatives and nonrelatives.

Hispanic Origin. The Office of Management and Budget (OMB) defines "Hispanic or Latino" as a person of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race. People who identify their origin as Spanish, Hispanic, or Latino may be of any race. Like the concept of race, Hispanic origin is based on self-identification.

Race. According to OMB, and as used by the Census Bureau, the concept of race reflects self-identification by people according to the race or races with which they most closely identify. These categories are socio-political constructs and should not be interpreted as scientific or anthropological in nature. The minimum race categories are determined by OMB and required for use in all federal information collections.

Detailed Housing Information

The ACS housing section collects data on physical and financial characteristics of housing. The 2024 ACS questionnaire includes detailed housing questions. For temporarily occupied HUs, a subset of housing data are collected from the occupants. For vacant units, selected housing data are collected from information given by neighbors, determined by observation, or from another

source. This section of the chapter details the concepts associated with some of the housing items.

Units in Structure. All HUs are categorized by the type of structure in which they are located. A structure is a separate building that either has open spaces on all sides or is separated from other structures by dividing walls that extend from ground to roof. In determining the number of units in a structure, all HUs—both occupied and vacant—are counted. Stores and office space are excluded.

Year Structure Built. This question determines when the building in which the sample address is located was first constructed, not when it was remodeled, added to, or converted. The information is collected for both occupied and vacant HUs. Units that are under construction are not considered HUs until they meet the HU definition—that is, when all exterior windows, doors, and final usable floors are in place. This determines the year of construction. For mobile homes, houseboats, and recreational vehicles, the manufacturer’s model year is taken as the year the unit was built.

Year Householder Moved Into Unit. This question is collected only for occupied HUs and refers to the year of the latest move by the householder. If the householder moved back into a HU he or she previously occupied, the year of the last move is reported. If the householder moved from one apartment to another within the same building, the year the householder moved into the present apartment is reported. The intent is to establish the year the householder began the current occupancy of the unit. The year that the householder moved in is not necessarily the same year other members of the household moved in.

Acreage. This question determines a range of the acres on which the house or mobile home is located. A major purpose of this item is to identify farm units. Starting in 2024, if administrative data are available, this question may not be asked.

Agricultural Sales. This refers to the total amount (before taxes and expenses) received from the sale of crops, vegetables, fruits, nuts, livestock and livestock products, and nursery and forest products produced on the property in the 12 months prior to the interview. This item is used to classify HUs as farm or nonfarm residences.

Rooms. The intent of this question is to determine the number of whole rooms in each HU that are used for living purposes. Living rooms, dining rooms, kitchens, bedrooms, finished recreation rooms, enclosed porches suitable for year-round use, and lodger’s rooms are included. Excluded are strip or Pullman kitchens, bathrooms, open porches, balconies, halls or foyers, half rooms, utility rooms, unfinished attics or basements, or other unfinished spaces used for storage. A partially divided room is considered a separate room only if there is a partition from floor to ceiling that extends out at least six inches, but not if the partition consists solely of shelves or cabinets.

Bedrooms. Bedrooms include only rooms designed to be used as bedrooms; that is, the number of rooms the respondent would list as bedrooms if the house, apartment, or mobile home were on the market for sale or rent. Included are all rooms intended for use as bedrooms, even if currently they are being used for another purpose. A HU consisting of only one room is classified as having no bedroom.

Plumbing and Kitchen Facilities. Answers to this question are used to estimate the number of HUs that do not have complete plumbing facilities or do not have complete kitchen facilities. A HU has complete plumbing facilities when it has both hot and cold piped water and a bathtub or shower.⁵⁰ These facilities must be located inside the house, apartment, or mobile home, but not necessarily in the same room. A HU has complete kitchen facilities when it has all three of the following: a sink with piped water, a range or cooktop and oven, and a refrigerator. All kitchen facilities must be located in the house, apartment, or mobile home, but not necessarily in the same room. A HU having only a microwave or portable heating equipment, such as a hot plate or camping stove, is not considered to have complete kitchen facilities.

Telephone Service Available. For an occupied unit to be considered as having telephone service available, there must be a telephone in working order and service available (including a cell phone) in the house, apartment, or mobile home that allows the respondent both to make and receive calls. Households whose service has been discontinued for nonpayment or other reasons are not considered to have telephone service available.

Computer Usage. This question measures usage or ownership of desktops, laptops, smartphones, tablets, or other portable wireless computers, as well as any other types of computers (which have applications that allow them to function like a desktop or laptop computer). Global Positioning System (GPS) devices, digital music players, and devices with limited computing capabilities (such as household appliances) are excluded.

Internet Access. This question determines whether any member of the household has access to the internet at the unit. If yes, it also determines whether that access is provided with or without a subscription service.

Internet Subscription Type. For respondents who indicate that they do access the internet with a subscription service, this question categorizes the type of subscription used. “Cellular data plan for a smartphone or other mobile device” includes any regular or prepaid contract with a mobile phone service provider that includes access to internet services such as email, web access, social networks, or streaming audio or video. “Broadband (high speed) Internet service such as cable, fiber optic, or digital subscriber line (DSL) service installed in this household” usually involves a modem that provides a

⁵⁰ In Puerto Rico, a HU has complete plumbing facilities when it has running water and a bathtub or shower.

wired connection or wireless signal within the house, apartment, or mobile home. “Satellite Internet service installed in this household” usually involves linking a satellite dish to a modem, which provides a wired connection or wireless signal within the house, apartment, or mobile home. “Dial-up Internet service installed in this household” is a type of internet service that uses a regular telephone line to connect to the internet.

Vehicles Available. These data show the number of passenger cars, vans, and pickup or panel trucks of one-ton capacity or less kept at home and available for the use of household members. Vehicles rented or leased for one month or more, company vehicles, and police and government vehicles are included if kept at home and used for nonbusiness purposes. Dismantled or immobile vehicles are excluded, as are vehicles kept at home but used only for business purposes.

House Heating Fuel. House heating fuel information is collected only for occupied HUs. The data show the type of fuel used most to heat the house, apartment, or mobile home.

Selected Monthly Owner Costs. Selected monthly owner costs are the sum of payments for mortgages, deeds of trust, contracts to purchase, or similar debts on the property; real estate taxes; fire, hazard, and flood insurance; utilities (electric, gas, water, and sewer); and fuels (such as oil, coal, kerosene, or wood). These costs include monthly condominium fees, homeowner association (HOA) fees, or mobile home costs.

Supplemental Nutrition Assistance Program Benefit. The Food and Nutrition Service of the U.S. Department of Agriculture (USDA) administers the Supplemental Nutrition Assistance (Food Stamp) Program (SNAP) through state and local welfare offices. This program is the major national income-support program for which all low-income and low-resource households, regardless of household characteristics, are eligible. This question asks if anyone in the households received SNAP benefits at any time during the 12-month period before the ACS interview.

Tenure. All occupied HUs are divided into two categories - owner-occupied and renter-occupied. A HU is owner-occupied if the owner or co-owner lives in the unit, even if it is mortgaged or not fully paid for. All occupied HUs that are not owner-occupied, whether they are rented for cash rent or occupied without payment of rent, are classified as renter-occupied.

Contract Rent. Contract rent is the monthly rent agreed to or contracted for, regardless of any furnishings, utilities, fees, meals, or services that may be included.

Gross Rent. Gross rent is the contract rent plus the estimated average monthly cost of utilities and fuels, if these are paid by the renter.

Value of Property. The survey estimates of value of property are based on the respondent’s estimate of how much the property (house and lot, mobile home and lot, or condominium unit) would sell for. The information is collected for HUs that are owned

or being bought and for vacant HUs that are for sale. If the house or mobile home is owned or being bought, but the land on which it sits is not, the respondent is asked to estimate the combined value of the house or mobile home and the land. For vacant HUs, value is defined as the price asked for the property. This information is obtained from real estate agents, property managers, or neighbors.

Mortgage Status. Mortgage refers to all forms of debt where the property is pledged as security for repayment of the debt.

Mortgage Payment. This item provides the regular monthly amount required to be paid to the lender for the first mortgage on the property.

Detailed Population Information

Detailed population data are collected for all current household members. Some questions are limited to a subset based on age or other responses. The 2024 ACS includes detailed population questions. On the Puerto Rico Community Survey, the questions about the place of birth, residence one year ago (migration), and citizenship differ from those used on the ACS. The definitions below refer specifically to the ACS. This section describes concepts and definitions for the detailed population items.

Place of Birth. Each person is asked whether he or she was born in or outside of the United States. Those born in the United States are then asked to report the name of the state; people born elsewhere are asked to report the name of the country or territory.

Citizenship. The responses to this question are used to determine the U.S. citizen and non-U.S. citizen populations and native and foreign-born populations. Citizens are people who are born in the United States, Puerto Rico, Guam, the U.S. Virgin Islands, or the Northern Marianas; born abroad of a U.S. citizen parent or parents; or are naturalized citizens. Non-citizens are people who do not meet any of the citizenship criteria. People are considered native-born if they are born in the United States, Puerto Rico, Guam, the U.S. Virgin Islands, or the Northern Marianas or if they are born abroad of a U.S. citizen parent or parents. People who do not meet those criteria are considered foreign-born.

Year of Entry. All respondents born outside of the 50 United States and the District of Columbia are asked for the year in which they came to live in the United States. This includes people born in Puerto Rico and U.S. Island Areas, those born abroad of an American (U.S. citizen) parent(s), and foreign-born people. For the Puerto Rico Community Survey, people born outside of Puerto Rico are asked when they came to live in Puerto Rico.

Type of School and School Enrollment. People are classified as enrolled in school if they have attended a regular public or private school or college at any time during the three months prior to the time of the interview. This question includes instructions to “include only nursery or preschool, kindergarten, elementary school, home school, and

schooling which leads to a high school diploma, or a college degree” as a regular school or college. Data are tabulated for people three years and older.

Educational Attainment. Educational attainment data are tabulated for the population 18 years and older. Respondents are classified according to the highest degree or the highest level of school completed. The question includes instructions for people currently enrolled in school to report the level of the previous grade attended or the highest degree received.

Field of Degree. Persons with a bachelor's degree or higher are asked to provide the specific major of this person's bachelor's degree. If a person has more than one bachelor's degree or more than one major, this question asks for the names of all majors for all bachelor's degrees.

Ancestry. Ancestry refers to a person’s ethnic origin or descent, roots or heritage, place of birth, or place of parents’ ancestors before their arrival in the United States. Some ethnic identities, such as “Egyptian” or “Polish,” can be traced to geographic areas outside the United States. Other ethnic identities, such as “Pennsylvania German” or “Cajun,” evolved within the United States.

Language Spoken at Home. Respondents are instructed to mark “Yes” if they sometimes or always speak a language other than English at home and “No” if the language is spoken only at school or is limited to a few expressions or slang. Respondents are asked the name of the non-English language spoken at home. If the person speaks more than one language other than English at home, the person should report the language spoken most often or, if he or she cannot determine the one spoken most often, the language learned first.

Ability to Speak English. Ability to speak English is based on the person’s self-response.

Residence One Year Ago (Migration). Residence one year ago is used in conjunction with the location of the current residence to determine the extent of residential mobility and the resulting redistribution of the population across geographic areas of the country.

Health Insurance. This question measures the insured and uninsured by asking about coverage through an employer, direct purchase from an insurance company, Medicare, Medicaid or other government-assistance health plans, military health care, Veterans Affairs health care, Indian Health Service, or other types of health insurance or coverage plans. Plans that cover only one type of health care (such as dental plans) or plans that only cover a person in case of an accident or disability are not included.

Health Insurance Premiums and Subsidies. These questions capture information about direct purchases of health insurance and whether the residents receive any financial assistance to help pay the health insurance premium.

Disability. Disability is defined as a long-lasting sensory, physical, mental, or emotional condition that makes it difficult for a person to perform activities such as walking, climbing stairs, dressing, bathing, learning, or remembering. It may impede a person from being able to go outside of the home alone or work at a job or business; the definition includes people with severe vision or hearing impairments.

Marital Status. The response categories are “now married,” “widowed,” “divorced,” “separated,” or “never married.” “Now married” includes married persons regardless of whether his or her spouse is living in the household, unless they are separated. If the person's only marriage was annulled, the person should be classified as “never married.” The “divorced” category should be selected only if the person has a divorce decree. Couples who live together (unmarried people, people in common-law marriages) report the marital status they consider the most appropriate.

Marital History. Data are collected on whether the person got married, widowed, or divorced in the past 12 months, the total number of times the person has been married, and the year in which the person last got married. A person is considered divorced in the past 12 months only if the person has received a divorce decree in the past 12 months. Marriages ending in annulment should not be included in the count of total marriages.

Fertility. This question asks if the person has given birth in the previous 12 months.

Grandparents as Caregivers. Data are collected on whether a grandchild lives with a grandparent in the household, whether the grandparent has responsibility for the basic needs of the grandchild, and the duration of that responsibility.

Veteran Status. A “civilian veteran” is a person 18 years old or older who has served (even for a short time) but is not now serving on active duty in the U.S. Army, Navy, Air Force, Marine Corps, or Coast Guard, or who served in the U.S. Merchant Marine during World War II. People who have served in the National Guard or military reserves are classified as veterans only if they were called or ordered to active duty at some point, not counting the four to six months of initial training or yearly summer camps. All other civilians age 18 and older are classified as nonveterans.

Service-Connected Disability Rating. This question determines whether the person has a VA service-connected disability rating, and if yes, identifies the numeric rating. The “0 percent” category should only be selected if the person has a service-connected disability rating of zero; it should not be used to indicate no rating.

Work Status. People age 16 and older who have worked one or more weeks are classified as having “worked in the past 12 months.” All other people age 16 and older are classified as “did not work in the past 12 months.”

Place of Work. Data on place of work refer to the location (street address, city/county, state) at which workers carried out their occupational activities during the reference week.

Means of Transportation to Work. Means of transportation to work refers to the principal mode of travel or type of conveyance that the worker usually used to get from home to work during the reference week.

Number of Carpoolers. The number of persons in the vehicle is used to determine the extent of carpooling and estimate the number of cars, trucks, and vans in each flow.

Time Leaving Home to Go to Work. This item covers the time of day that the respondent usually left home to go to work during the reference week.

Travel Time to Work. This question asks the total number of minutes that it usually took the worker to get from home to work during the reference week.

Labor Force Status. These questions on labor force status are designed to identify: (1) people who worked at any time during the reference week; (2) people on temporary layoff who were available for work; (3) people who did not work during the reference week but who had jobs or businesses from which they were temporarily absent (excluding layoffs); (4) people who did not work but were available during the reference week, and who were looking for work during the last four weeks; and (5) people not in the labor force.

Industry, Occupation, Class of Worker. Information on industry relates to the kind of business conducted by a person's employing organization; occupation describes the kind of work the person does. For employed people, the data refer to the person's job during the previous week. For those who work two or more jobs, the data refer to the job where the person worked the greatest number of hours. For unemployed people, the data refer to their most recent job within the past 5 years. The information on class of worker refers to the same job as a respondent's industry and occupation, and categorizes people according to the type of ownership of the employing organization.

Income. "Total income" is the sum of the amounts reported separately for wage or salary income; net self-employment income; interest, dividends, or net rental or royalty income, or income from estates and trusts; social security or railroad retirement income; Supplemental Security Income; public assistance or welfare payments; retirement, survivor, or disability pensions; and all other income. Income is reported for the past 12 months from the date of the interview. The estimates are inflation-adjusted using the Consumer Price Index.

6.5 Structure of the Group Quarters Questionnaires

The GQ questionnaire includes all of the population items on the HU questionnaire, except for relationship, and there is one person collected per questionnaire. One housing question, Supplemental Nutrition Assistance Program benefit, is asked. Address information is for the GQ facility itself and is collected as part of the automated GQ Facility Questionnaire. The survey information collected from each person selected to be interviewed is entered on a separate questionnaire.

Chapter 6 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
D. Reese, E. Poehler, & S. Clark	11/28/2022	Deleted information on household concepts about business on property and flush toilets, updated the list of internet subscription types, added information about health insurance premium and subsidy and carpoolers. Updated format and content.	3.0
N. Butler & M. Wiland	7/23/2024	Updated definition of “selected monthly owner costs” to include HOA fees. Made other editorial changes.	4.0

Chapter 7: Data Collection and Capture for Housing Unit Addresses

7.1 Overview

The American Community Survey (ACS) divides the overall housing unit (HU) sample into monthly independent samples, which are called panels. The data collection operation for housing unit addresses consists of four modes of response (internet, paper, telephone, and personal visit) and two data collection phases. The first phase focuses on self-response; in the second phase, interviewers contact members of the sampled addresses by telephone or personal visit. The second phase is referred to as the computer-assisted personal interviewing (CAPI) operation.

In the self-response phase, for HUs with mailable addresses, the Census Bureau mails a request to respond via the internet, followed by an option to complete a paper questionnaire and return it by mail. Additionally, respondents can call the Census Bureau’s Telephone Questionnaire Assistance (TQA) line to complete the survey or have questions about it answered. If the Census Bureau does not receive a self-response for a HU, the address may be selected for the CAPI operation. While the first phase focuses on obtaining self-responses, self-responses are accepted from all sampled housing units during both phases.

Data collection for each panel lasts three months in continuously overlapping cycles, so that during any given month, two panels are in the self-response phase, and one panel is in the CAPI phase. Figure 7-1 shows an example of the three-month data collection timeline for three consecutive ACS monthly sample panels.

Figure 7-1: Example of Three Months of ACS HU Data Collection: Response Modes by Sample Panel and Month of the Year

ACS Sample Panel	Month of Data Collection				
	December	January	February	March	April
December Panel	Internet, Paper, and TQA*				
			Personal Visit and Phone		
January Panel	Internet, Paper, and TQA*				
				Personal Visit and Phone	
February Panel	Internet, Paper, and TQA*				
				Personal Visit and Phone	

*Telephone Questionnaire Assistance

For information on ACS response rates, visit [Response Rates | American Community Survey | U.S. Census Bureau](#). For more information on the distribution of response mode by item, visit [People and Households Represented in Each ACS Data Collection Mode](#).

7.2 Self-Response Phase

Self-responses by internet or paper are the least expensive methods of data collection, and the success of the program depends on high levels of self-response. Sample addresses are reviewed to determine whether the available information is sufficient for mailing. The requirement for a “mailable” address in the United States is met if there is either a complete city-style or rural route address. A complete city-style address includes a house number, street name, town or city name, state, and ZIP Code. A complete rural-route address includes a rural-route number, box number, town or city name, state, and ZIP Code. About 99 percent of the 2022 sample addresses in the United States met these criteria and were designated as mailable.

The requirement for a mailable city-style Puerto Rico address is the same as Stateside. Mailing addresses in Puerto Rico generally incorporate the urbanization name (a geographic area roughly equivalent to a neighborhood), which creates a four-line address. Using the urbanization name eliminates the confusion created when street names are repeated in adjacent communities. About 70 percent of the addresses in Puerto Rico were designated mailable in 2022.

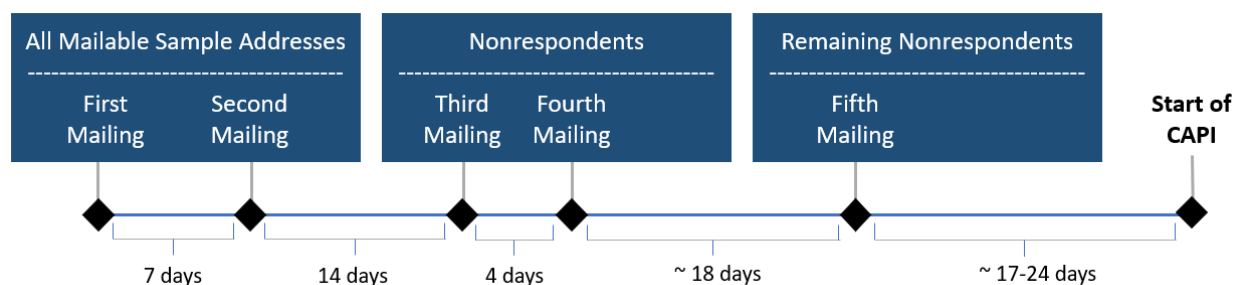
Examples of unmailable addresses include addresses with only physical descriptions of an HU and its location or with post office (P.O.) box addresses and addresses missing place names and zip codes. P.O. box addresses are considered unmailable because of the unknown location of the HU using the P.O. box. Addresses missing zip codes are considered unmailable when the place name is also missing. Unmailable addresses are still included in the sample frame but are ineligible for the self-response phase of data collection. These addresses are eligible for the CAPI phase, where after an initial contact by an interviewer, they can choose to respond by internet.

ACS Self-Response Contact Strategy⁵¹

Because a high level of self-response is critical, the ACS sends up to five mailings to mailable sampled addresses to encourage recipients to respond to the survey via the internet or via a paper questionnaire. Figure 7-2 outlines the ACS self-response contact strategy.

⁵¹ See below for information on the Puerto Rico Community Survey Mail Strategy.

Figure 7-2. ACS Self-Response Contact Strategy



All mailable sample addresses are sent the first two mailings: an initial letter and a reminder letter. A paper questionnaire package and reminder postcard are sent to sample addresses for which a self-response is not received about three weeks after the initial mailing. Finally, households that have not responded are sent another letter with a due date the following month. Details of each mailing are provided below.

Initial Letter. The first mailing is a pressure-sealed letter signed by the director of the Census Bureau. It informs the respondents that their address has been selected for participation in the ACS and asks them to respond online or wait a few weeks for a paper questionnaire. Login information is provided in the letter. The letter explains the purpose of the ACS and how the data are used. Finally, a telephone number is included for respondents if they have questions or need help completing the questionnaire. The letter is mailed on the last Thursday of the month before the self-response month or on the first Thursday of the self-response month if the last day of the month is a Wednesday. The letter is printed at the Census Bureau’s National Processing Center (NPC) using print-on-demand technology to merge letter text, the sampled address, and an associated user ID, which is used to log into the internet survey instrument.

First Reminder Letter. The second mailing is a pressure sealed letter signed by the director of the Census Bureau. The letter is mailed on Thursdays, a week after the initial mail letter, and reminds respondents to complete the survey via the internet. The reminder letter is printed at the NPC using print-on-demand technology to merge text, addresses, and user IDs.

Paper Questionnaire Package. Addresses for which the Census Bureau receives a response to the survey after the second mailing are removed from the address file to create a new mailing universe of nonrespondents. These addresses are sent the third and fourth mailings.⁵² The third mailing is a package containing a paper questionnaire, a follow-up letter, and a postage-paid return envelope. The enclosed letter is signed by the director of the Census Bureau and reminds the household of the importance of the ACS

⁵² Addresses deemed “undeliverable as addressed” (UAA) by the United States Postal Service (USPS) are also removed from the address files for subsequent mailings.

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and asks them to either respond online or return the paper questionnaire. The paper questionnaire package is mailed on Thursdays, about three weeks after the initial mail package.

The NPC assembles and mails the paper questionnaire packages. All the components of the mail packages are printed under contract by outside vendors. NPC quality control staff monitor the work of the contractors and inspect materials to identify samples that do not meet contractual quality standards.

Reminder Postcard. The fourth mailing is a postcard printed on white cardstock. The postcard reminds respondents to fill out and return their completed questionnaire or respond online. The postcard is mailed on Mondays, a few days after the paper questionnaire package. This postcard is also printed in-house at the NPC, using print-on-demand technology to merge text and addresses.

Additional Reminder Letter. After the fourth mailing, addresses for which the Census Bureau has received a response are removed from the address file to create a new mailing universe of nonrespondents. The remaining addresses are sent the fifth mailing. The fifth mailing is a pressure sealed letter signed by the director of the Census Bureau. The letter includes a more urgent message about the response with a due date. The fifth mailing is mailed on Tuesdays, about seven weeks after the initial letter. The letter is printed at the NPC, using print-on-demand technology to merge text displaying the due date, address, and user ID.

The ACS mail contact materials for U.S. addresses are written in English. ACS respondents can request mail contact materials written in Spanish via the telephone questionnaire assistance (TQA) operation. The First Reminder Letter includes the TQA phone number for help in Spanish along with directions to respond online (in Spanish). The Paper Questionnaire Package letter includes the same directions in Spanish along with directions in simplified Chinese, Vietnamese, Russian, and Korean that provide a TQA phone number for help in the corresponding language.

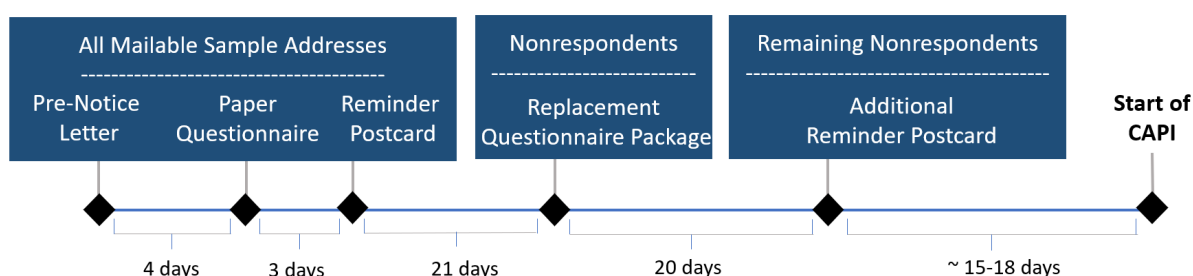
The NPC is responsible for printing the pressure sealed letters and addressing all outgoing mailings. Several months before each sample's mailings, Census Bureau headquarters staff provide an address file to the NPC for use in creating address labels for the first two mailings. An updated address file is provided to the NPC about three days before the mailing of the paper questionnaire package. This file excludes addresses from which a response was received by internet during the first two and a half weeks and addresses identified as "undeliverable as addressed" (UAA) by the U.S. Postal Service (USPS); this usually amounts to about 24 percent of the sampled addresses. An additional updated address file for the final reminder mailing is provided to the NPC two and a half weeks after the paper questionnaire package mailing and excludes addresses from which a response was received via paper or internet.

Most paper and internet responses are received within five weeks after the initial mailing is sent, but the Census Bureau will continue to accept paper or internet responses for three months from the start of each monthly sample. After a specified cutoff date, late returns will not be included in the data set.

Mail Contact Strategy in Puerto Rico

An internet response option is not available for Puerto Rico (PR) households; therefore, a different contact strategy is used in Puerto Rico. Figure 7-3 outlines the Puerto Rico Community Survey (PRCS) mail contact strategy.

Figure 7-3. PRCS Mail Contact Strategy



The PRCS mail contact materials sent to addresses in sample are bilingual, containing messages written in Spanish and English. Sampled addresses in Puerto Rico are sent a pre-notice letter before the paper questionnaire package is sent. This pre-notice letter notifies the respondents that their address has been selected for participation in the PRCS, provides information about the importance of the PRCS, and requests that the paper questionnaire be returned promptly once received. The paper questionnaire package includes a letter, paper questionnaire, and FAQ brochure. The paper questionnaire is in Spanish; respondents can request an English questionnaire by calling TQA. A reminder postcard follows three days later.

Addresses for which the Census Bureau receives a response to the survey after the third mailing are removed from the address file to create a new mailing universe of nonrespondents. These addresses are sent the fourth mailing, a replacement questionnaire package. The timing of the replacement questionnaire package is approximately 3 ½ weeks after the initial mail package for Puerto Rico addresses.

After the fourth mailing, addresses for which the Census Bureau has received a response are removed from the address file to create a new mailing universe of nonrespondents. The remaining addresses are sent the fifth mailing, an additional reminder postcard.

For Puerto Rico, all the components of the mail packages except the pre-notice letter and postcards are printed under contract by outside vendors. As the vendors print the materials, NPC quality control staff monitor the work and inspect materials to identify samples that do not meet

contractual quality standards. The NPC is responsible for printing the letters and postcards, as well as assembling and addressing all outgoing mailings. Several months before each sample's mailings, Census Bureau headquarters staff provide an address file to the NPC for use in creating address labels for the first three mailings. An updated address file is provided to the NPC about three days before the mailing of the replacement mail package. This file excludes Puerto Rico addresses from which a response was received during the first two and a half weeks and addresses identified as UAA by the USPS; these usually amount to about 3 percent of the sample addresses for Puerto Rico. An additional updated address file for the final reminder mailing is provided to the NPC two and a half weeks after replacement mailings, which excludes addresses from which a response was received via paper or internet.

Most paper responses are received within five weeks after the initial mail package is sent, but the Census Bureau will continue to accept paper responses for three months from the start of each monthly sample. After a specified cutoff date, late returns will not be included in the data set.

Internet Response

As mentioned earlier, the initial letter for the ACS includes an invitation to complete the survey online.⁵³ The letter contains a unique code and instructions on how to get to the response website. After logging in, the householder answers security questions and is provided a personal identification number (PIN) that will allow them to exit and return later to complete the survey.

After completing questions about the housing unit and for each person living there, the respondent can review the data and make any needed corrections before submitting the form. Once a respondent submits a form, they cannot access it anymore. The responses received are then processed.

Households at a sample address can complete an internet return at any time during the three months of data collection for the panel to which the address is assigned. If a household has started an internet return but has not completed it, data from those "partial interviews" are processed at the end of the three-month data collection period.

If a respondent starts to answer the survey online and provides an email address but does not complete the survey, an email will be sent to the respondent to remind them to return to the survey to complete their online questionnaire. This email includes a link to the online survey, clear instructions about how to log in, and prominently displays the user identification number. This email is sent only once to a respondent, either before the third mailing or before the fifth mailing, depending on when they started filling out the online survey.

⁵³ PRCS does not have an internet response option.

Paper Questionnaire Response

The 2022 ACS questionnaire contains 48 pages and is a two-color booklet-style form. The paper questionnaires are printed on white paper with colored ink—green for the ACS form and purple for the PRCS form. The cover of the questionnaire includes information in English and Spanish on how to obtain assistance, as well as information on how to respond via the internet. The questionnaire includes questions about the housing unit and the people living in it. The paper questionnaire allocates space for detailed person information for up to five people. Follow-up by telephone is used for households that return their questionnaires by mail and report that six or more people reside in the household (see Failed-Edit Follow-Up below for more information).

Completed paper questionnaires are mailed to the NPC. The check-in unit at the NPC receives mail deliveries two or three times each business day. Each questionnaire contains a unique bar code in the address label area. The paper returns are sent through a laser sorter, where the bar code is scanned; this allows sorting by and within monthly sample, by language, and to separate the ACS and PRCS forms. During this step, the return envelopes are opened mechanically. Staff have three days to check in a form, although they usually check in all forms they receive within one day. Each day, NPC staff transmit a file of the checked-in cases, and headquarters staff update the status of each case in the control file.

After clerks remove the forms from the return envelopes, each page of the form is thoroughly reviewed for issues like damage to the form. The clerks also look for enclosed correspondence, which they forward to headquarters, if necessary. The clerks then scan the bar code on each questionnaire to officially check in the forms and organize them into batches of 50.

Some of the forms are returned to the NPC as UAA by the USPS. UAAs occur for many reasons, including bad or unknown addresses, vacant HUs, or residents' refusals to accept mail delivery. Sample addresses that are UAAs are ineligible for the paper questionnaire package and subsequent mailings. Sample addresses that are UAAs are eligible for the CAPI operations.

Data Capture of Paper Questionnaires

After the paper questionnaires have been checked in and batched into groups of 50, they are scanned. The scanned images continue through the data capture process. Checkboxes on the images are read by optical mark recognition (OMR), and numeric entries are read by optical character recognition (OCR). Any value not read with high confidence by OMR or OCR is presented to a clerk or keyer. After OMR and OCR, the images move to the data entry (keying) unit in the NPC. The keying unit has the goal of keying the responses from the questionnaires within three weeks of receipt. Data keyers enter the information from the forms into a data capture file. Each day, NPC staff transmit a file with the OMR, OCR, and keyed data, and headquarters staff update the status of each case in the control file. The NPC's data keying operation uses stringent quality control procedures to minimize nonsampling errors.

Data keyers move through two stages of quality control verification. When new keyers begin their training, they key four truth deck batches. There is no error rate calculation on the first two batches. If the keyer's field error rate for the third truth deck (the percentage of all keyed data items that are in error) is equal to or less than 1.0 percent, the keyer will be given the fourth truth deck batch to key. If the keyer's field error rate for the fourth truth deck batch is equal or less than 1.0 percent, the keyer is moved to the qualified stage. If the keyer's field error rate for the third or fourth truth deck batch is greater than 1.0 percent, the keyer is given numerous attempts to retry keying the truth deck batches. If the keyer's field error rate is greater than 1.0 percent on the third or fourth batch after numerous attempts, the keyer will not be allowed to key for the ACS.

Once the keyer has cleared training, they are qualified to key and verify batches. If these keyers exceed the production error rate of 1.0 percent, they receive timely feedback. Keyers at all levels are subject to removal from the project and administrative action if they fail to maintain a production error rate of less than 1.0 percent, but most have a much lower rate.

Telephone Questionnaire Assistance (TQA)

Respondents who call the toll-free TQA number reach an interactive voice recognition (IVR) telephone system that provides answers to questions about completing the questionnaire or assists respondents in requesting a questionnaire in another language. The TQA telephone number is listed on the questionnaire, as well as on all the letters, brochures, and postcards. Alternate TQA numbers are listed in the paper questionnaire package for people who need assistance in Spanish, simplified Chinese, Vietnamese, Russian, and Korean.

When respondents call TQA, they enter the IVR system, which provides some basic information on the ACS and directions on using the IVR. Respondents may obtain recorded answers to FAQs or speak directly to an agent during business hours. Respondents can furnish their ACS identification number from any of the mailing pieces, which allows them to hear a customized message about the status of their paper or internet questionnaire. The IVR can indicate whether the NPC has received a completed survey for the sample address and, if not, can state that an ACS interviewer may call or visit. If a respondent chooses to speak directly to an agent, the agent answers the caller's questions and can complete the questionnaire over the telephone if requested. Agents use an automated survey instrument to capture the respondent's answers. Respondents may also contact TQA staff to reset their internet questionnaire if the personal identification number (PIN) provided when the respondent first accessed their questionnaire online is lost and if they are unable to provide answers to their security questions.

Household members from approximately thirteen percent of the sampled addresses called the toll-free number and talked to a TQA agent for assistance in 2022. While the TQA agents can complete the survey over the phone, less than one percent of interviews are completed that way. All calls are logged, and the system can record up to five reasons for each call. Even though

TQA interviews are conducted by telephone, they are treated like paper responses during processing because the call was initiated by the sample household upon receiving the initial mailings.

Failed-Edit Follow-Up

After the data are keyed from paper or the data are returned from the internet, the data files are processed through a computerized edit to check coverage consistency. This edit identifies cases requiring additional information. Cases that fail are eligible for the telephone failed-edit follow-up (FEFU) operation and become part of the FEFU workload if a telephone number for the sample address is available. This operation is designed to improve the final quality of completed surveys.

Cases failing the edit for coverage consistency can take three forms. First, since the ACS paper questionnaire is designed to accommodate detailed answers for households with five or fewer people, a case will fail when a respondent indicates that there are more than five people living in the household or if the reported number of people is greater than the number of people for whom responses are provided. Second, responses that indicate the sample address is vacant or a business are also treated as coverage failures. Third, the number of data-defined fields is less than expected, and the case data will not be retained for downstream processes. A new set of FEFU cases is generated each business day, and Census Bureau contact center staff call respondents to obtain the missing data. The interview period for each FEFU case is three weeks.

7.3 Personal Visit Phase

The last phase of ACS data collection is the CAPI operation, in which Census Bureau field representatives (FRs) call or visit sampled addresses to complete the interview. This phase usually begins on the first day of the third month of data collection for each panel and typically lasts for the entire month.

Two to three weeks after the fifth mailing, responding addresses are removed, and unmailable and UAA addresses are added back in to create the universe of addresses eligible for CAPI. Of this universe, a sample is selected from two categories of cases to be included in the CAPI operation. Mailable addresses with neither a paper or internet response are generally sampled at a rate of 1 in 2, 2 in 5, or 1 in 3 based on the expected rate of completed interviews at the tract level. Unmailable addresses are sampled at a rate of 2 in 3. All eligible addresses in Hawaiian Home Lands, Alaska Native Village Statistical areas, and a subset of American Indian areas are sent to CAPI without CAPI sampling (U.S Census Bureau, 2020). For more information on ACS sampling, see Chapter 4.

Mailable addresses in the CAPI sample are sent a pressure sealed letter, called the CAPI Internet Letter, encouraging them to complete the internet questionnaire to avoid an in-person interview.

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The letter is mailed at the beginning of the CAPI operation. The CAPI Internet Letter is printed in-house at the NPC, using print-on-demand technology to merge text, addresses, and user IDs.

The CAPI operation is conducted by Census Bureau field representatives (FRs) operating from the Census Bureau's six regional offices (ROs). The ROs manage the work for sampled cases within their geographic boundaries. The New York RO is responsible for CAPI data collection in Puerto Rico.

After the databases containing the sample addresses are distributed to the appropriate RO, the addresses are assigned to FRs. The FRs can conduct interviews by telephone or personal visit, using laptop PCs loaded with the survey instrument. The CAPI instrument is available in English and Spanish for both the ACS and PRCS.

If a telephone number is available, the FR will first attempt to call the sample address. Unmailable addresses are an exception because an FR must verify the location of the address in person. The FR will call and confirm that he or she has reached the sample address. If so, the FR uses the automated instrument and attempts to conduct the interview. If an FR cannot reach a resident after calling three to five times at different times of the day during the first few days of the interview period, they must make a personal visit.

Approximately 70 percent of CAPI cases require an FR visit. In addition to trying to obtain an interview, a visit is needed to determine whether the HU exists and to determine the occupancy status. If an HU does not exist at the sample address, that status is documented. If an FR verifies that an HU is vacant, they will interview a knowledgeable respondent, such as the owner, building manager, real estate agent, or a neighbor, and conduct a "vacant interview" to obtain some basic information about the HU. If the HU is currently occupied, the FR will conduct an "occupied" or a "temporarily occupied" interview. An FR conducts a temporarily occupied interview when there are residents living in the HU at the time of the FR's visit, but no resident has been living there or plans to live there for more than two months. Each day, FRs transmit a file with the status of all assigned cases, and headquarters staff update the statuses on the control file.

Collecting data on Tribal Lands is unique because of the sophistication of tribal government's functionality and diverse processes across the different Tribal Lands. For the first visit to a Tribal Land, the RO management staff or Tribal Specialist staff will contact the tribal government and establish a contact person for the initial visit and subsequent visits. Each tribe will have their own protocol for conducting interviews on Tribal Lands.

The FRs are trained to remain polite but persistent when attempting to obtain responses. They are also trained on handling almost any situation that might arise in the field, from responding to a household that claims to have returned its questionnaire by mail or internet to interviewing with a non-English speaking respondent.

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When FRs cannot obtain interviews, they must indicate the reason. Such noninterviews are taken seriously because they have an impact on both sampling and nonsampling error. Noninterviews occur when an eligible respondent cannot be located, is unavailable, or is unwilling to provide the survey information or when the maximum number of attempts allowed across all data collection modes has been reached. Additional noninterviews occur when FRs are unable to confirm the status of a sample HU because of restricted access to an area because of a natural disaster or nonadmission to a gated community during the interview period. Some sample cases will be determined to be ineligible for the survey. These include sample addresses of structures under construction, demolished structures, and nonexistent addresses.

Approximately 2,500-3,000 FRs conduct CAPI interviews for the ACS and PRCS. Interviewers have almost the entire month to complete the monthly CAPI caseload, which averaged approximately 58,000 cases each month in 2022.

In 2023, the ACS began using administrative data to optimize data collection costs and data quality in the CAPI operation by informing contact strategies—when to decrease the number of visits or when to stop interviewing. The ACS does this in two ways. First, a vacancy prediction model is run to identify housing units with high probabilities of being vacant. Interviewers are alerted that, for these cases, the initial contact attempt should be made in person (rather than by phone). A maximum of two attempts are allowed to determine the unit's status and collect the required information. This strategy allows interviewers to identify vacant units more quickly and free up their time to complete interviews for housing units that are more likely to be occupied. Second, for housing units that the vacancy prediction model does not identify, the ACS uses administrative data, historic response data, and survey paradata to develop data- and model-driven business rules to determine cases to stop interviewing before the end of the month. After ten full weekdays and two full weekends for each CAPI month, the models identify the best cases to remove from the workload to minimize adverse effects on data quality. Cases stopped through this process are treated as noninterviews (U.S. Census Bureau, 2023).

FRs participate in a 4-day classroom training session to learn and practice the appropriate interviewing procedures. Each FR is also trained thoroughly on the importance of maintaining high data quality. Since ACS estimates extend to the block group level, and most CAPI cases are subsampled, each potential CAPI interview should be handled with care. Supervisors travel with FRs during their first few work assignments to observe and reinforce the procedures learned in training.

In addition, a sample of FRs is selected each month for reinterview. The primary purpose of the reinterview program is to verify that FRs are conducting interviews correctly. Analysts at the Census Bureau also review survey paradata and response data to identify FRs whose data falls outside of expected parameters and may indicate poor quality.

Data Collection in Remote Alaska

Remote areas of Alaska provide special difficulties for FRs, such as climate, travel, and seasonality of the population. To address some of these challenges, the Census Bureau uses different procedures for ACS interviewing.

For areas of Alaska that the Census Bureau defines as remote, ACS operations are different from operations in the rest of the country. The Census Bureau does not send self-response contact materials to sampled housing units in Remote Alaska. Sampled households in Remote Alaska cannot self-respond via paper questionnaire or by TQA. All interviews for Remote Alaska are conducted using personal visit procedures, and we do not sample for CAPI in Remote Alaska as we do elsewhere. During the personal visits, FRs can provide respondents with information on how to respond to the ACS via the internet.

To allow FRs in Alaska adequate time to resolve some of the transportation and logistical challenges associated with conducting interviews in Remote Alaska areas, the interviewing period is two 6-month interview periods every year. The first starts in January and stops at the end of June. The second starts in July and stops at the end of December. Extending the interview periods into two 6-month periods was found to be more effective in allowing FRs to gain access to remote areas and in finding residents of Native Villages at home who might be away frequently to participate in subsistence activities.

Table 7-1 provides a list of Remote Alaska areas and their associated interview periods.

Table 7-1. Interview Periods by Area in Remote Alaska Areas

Borough name	Interview period for the remote portion of the borough: January–June	Interview period for the remote portion of the borough: July–December
Aleutians East	X	
Aleutians West		X
Bethel		X
Bristol Bay	X	
Chugach	½	½
Copper River	½	½
Denali		X
Dillingham	X	
Haines		X
Hoonah-Angoon	½	½
Kusilvak	½	½
Lake and Peninsula	X	
Nome	½	½
North Slope	X	
Northwest Arctic	½	½
Prince of Wales-Hyder	½	½
Southeast Fairbanks	½	½
Wrangell	½	½
Yakutat	X	
Yukon-Koyukuk		X

Note: X indicates that all of the workload falls in the interview period. ½ indicates that half of the workload falls in the interview period.

7.4 References

- U.S. Census Bureau. 2020. “Accuracy of the Data (2020).” Washington, DC, 2020, https://www2.census.gov/programs-surveys/acs/tech_docs/accuracy/ACS_Accuracy_of_Data_2020.pdf
- U.S. Census Bureau. 2023. “Agility in Action 4.0. Responding and Adapting to Our Evolving Nation.” Washington, DC, 2023. [Agility in Action 4.0: Responding and Adapting to Our Evolving Nation \(census.gov\)](https://www.census.gov/agility-in-action-4.0)

Chapter 7 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
D. Reese & E. Poehler	11/28/2022	Updated content and formatting.	3.0
N. Butler, C. Fisher, & E. Poehler	10/17/2024	Added information about email reminders and the use of administrative data for workload optimization. Additional changes made for clarity.	4.0

Chapter 8: Data Collection and Capture for Group Quarters

8.1 Overview

All living quarters are classified as either housing units (HUs) or group quarters (GQs). An HU is a house, an apartment, a mobile home, a group of rooms, or a single room occupied or intended for occupancy as separate living quarters. Separate living quarters are those in which the occupants live separately from any other people in the building and that are directly accessible from outside the building or through a common hall.

GQs are places where people live or stay in a group living arrangement that is owned or managed by an entity or organization providing housing and/or services for the residents. These services may include custodial or medical care and other types of assistance, and residency is commonly restricted to those receiving these services. People living in GQs usually are not related to each other. GQs include college/university student housing, residential treatment centers, skilled nursing facilities, group homes, military barracks, correctional facilities, workers' group living quarters, Job Corps centers, and emergency and transitional shelters. GQs are defined according to the housing and/or services provided to residents and are identified by census GQ-type codes.

In January 2006, the American Community Survey (ACS) was expanded to include the population living in GQ facilities. A new GQ sample is introduced each month. Data collection for each monthly sample lasts six weeks and does not include a formal nonresponse follow-up operation. The GQ data collection operation is conducted in two phases. First, U.S. Census Bureau field representatives (FRs) conduct interviews with the GQ facility contact person or the administrator of the selected GQ (referred to as the GQ level interview), and second, the FR conducts interviews with a sample of individuals from the facility (referred to as the person- or resident-level interview).

The GQ-level data collection instrument is an automated Group Quarters Facility Questionnaire (GQFQ). Information collected by the FR using the GQFQ during the GQ-level interview is used to determine or verify the type of facility and population size and to draw a random sample of residents to be interviewed. FRs conduct GQ-level data collection at approximately 20,000 individual GQ facilities each year.

During the person-level phase, an FR uses a Computer-Assisted Personal Interviewing (CAPI) automated instrument to collect detailed information for each sampled resident. FRs also have the option to distribute a bilingual (English/Spanish) questionnaire to residents for self-response if unable to complete a CAPI interview. In January 2024, the ACS launched an internet response option for residents living in select GQ types. Data are collected from approximately 150,000 GQ sample residents each year. All methods described in this chapter apply to the ACS GQ

operation in both the United States and Puerto Rico, where the survey is called the Puerto Rico Community Survey (PRCS).⁵⁴

Outside vendors print most GQ data collection materials, such as advance contact letters, questionnaires, and brochures. A questionnaire instruction guide is provided online. Trained quality control staff from the National Processing Center (NPC) monitor the work as the contractors print the materials. The NPC rejects batches of work if they do not meet contractual quality standards. On a monthly basis, the Census Bureau headquarters provides label/address files for GQ materials to the NPC. The NPC receives the files approximately eight weeks before the sample months and is responsible for using these files to assemble GQ and resident-level packages. Each GQ level package contains questionnaire labels, the Frequently Asked Questions (FAQ) brochure, the Survey Package Control List for Special Sworn Status (SSS) Individuals, an Instruction Manual for SSS Individuals, a listing sheet, a Thank You notice, and a copy of the Introductory Letter mailed to the GQ. Each resident-level package includes a questionnaire, a resident-level Introductory Letter, an FAQ Brochure, and a Thank You notice. The NPC delivers both packages and other materials to the regional offices (ROs) two weeks before the start of each monthly survey panel.

8.2 Group Quarters (Facility-Level Phase)

The GQ data collection operation is primarily completed through FR personal interviews. FRs obtain the facility information by conducting a personal visit interview with the GQ contact. Each FR is assigned approximately two sample GQ facilities each month, and interviews are conducted over a period of six weeks.

During the GQ-level interviews, FRs verify sample GQ information such as the name, address, and GQ type. FRs also obtain a roster of residents currently living at the GQ. The GQFQ randomly selects residents for person-level interviews. The information obtained from GQ interviews is transmitted nightly to Census Bureau headquarters through a secure file transfer.

Previsit Mailings

This section details the materials mailed to each GQ facility before the FR makes contact.

GQ Introductory Letter. Approximately two weeks before the FRs begin each monthly GQ assignment, NPC mails an introductory letter to the sampled GQ facility. The letter explains that the FR will visit the facility to collect GQ- and person-level data. It describes the information that the FR will ask during the visit, the uses of the data, the internet address where they can find more information about the ACS, and RO contact

⁵⁴ See sample copies in English and Spanish of [GQ forms and materials](#) used in the GQ data collection on the Census Bureau's website.

information. This letter is printed at the NPC using print-on-demand technology, which merges the letter text and the sample GQ name and address.

There are unique letters for college/university student housing and healthcare facilities administrators because these are some of the most challenging GQ types to interview.

GQ Frequently Asked Questions (FAQ) Brochure. This brochure contains FAQs about the ACS and GQ facilities and is mailed to the sample GQ facility along with the GQ introductory letter. Examples of the FAQs are as follows: “*What is the American Community Survey?*” “*Do I have to answer the questions on the American Community Survey?*” and “*Will the Census Bureau keep my information confidential?*” Similar brochures are sent to sample GQ facilities in Puerto Rico and Remote Alaska.

GQ State and Local Correctional Facilities Letter. FRs may mail another letter to selected correctional facilities after the GQ introductory letter is sent but before calling to schedule an appointment to visit. This letter was developed to assist FRs in gaining access to state and local correctional facilities, although the GQ operation does not require FRs to send the letter. The letter asks for the name and title of a person with the authority to schedule FR visits and coordinate the GQ data collection. It also provides information about the ACS and the dual nature of the FR visit to the facility; it includes a form to return to the RO with the contact’s name, title, and phone number of a designated GQ contact.

Initial Contact with GQ Facility

To conduct the GQ-level interviews for the assigned facility, the FR is instructed to try first to make the initial contact by telephone. If successful in reaching the GQ contact (usually the facility administrator), the FR uses the automated GQFQ, which is available in both English and Spanish, to schedule an appointment to conduct a telephone or personal visit interview to complete the GQ-level data collection phase. The FR will collect information about the facility, such as verifying its name and address.

If the GQ contact refuses to participate, the FR notifies the RO, and the RO staff try to gain the GQ contact’s cooperation. If this attempt at scheduling an appointment is unsuccessful, the FR then visits the GQ facility to try to get the information needed to generate the sample of residents and to conduct the person-level interviews. If still unsuccessful, the RO or FR explains the mandatory nature of the survey, what the FR is attempting to do at the facility, and why. The ACS Group Quarters Branch may also contact the GQ to explain the nature of the survey and try to gain cooperation.

Interviewing the GQ Facility

During the GQ-level data collection phase, the FR updates or verifies the GQ name, mailing and physical address, facility telephone number, contact name(s), and telephone number(s). Using a flashcard, the FR asks the GQ administrator to indicate which GQ-type code best describes the GQ facility. Depending on the size of the facility, either a sample or all of the residents will be interviewed.

After determining that the GQ facility is in scope for GQ data collection, the FR asks for a roster of names and bed locations for everyone living or staying at the sample GQ facility on the day of the visit. This roster is used to generate the sample of residents to be interviewed. If a register is unavailable, the FR creates one using a GQ listing sheet. The facility also has the option to upload an electronic listing of the residents in the online eListing application to generate the subsample of residents. The GQFQ instrument proceeds automatically to the beginning of the sampling component after the FR has entered all required facility information and the GQ contact person verifies that people are living or staying there at the time of the visit. If no residents live or stay at the GQ facility at that time, the FR completes the GQ-level interview to update the GQ information but does not conduct person-level interviews.

The sample of GQ residents to be interviewed is generated from the GQFQ instrument through a systematic sample selection. (See Section 8.3 for information about data collection from individuals.) The FR matches the line numbers generated for the person sample to the register of current residents. A grid up to 15 lines long appears on the GQFQ laptop screen, along with the line number corresponding to the register, a place for the name, the sample person location description, a telephone number, and a GQ control number (assigned by the GQFQ sampling program). To complete the sampling process, the FR enters information into the GQFQ that specifically identifies the location of each sample person.

An interim or final outcome is assigned to each GQ-level interview, and reasons for GQ refusals or noninterviews are also specified. The GQFQ assigns an interim GQ-level interview status reason to allow closure of a case and subsequent reentry. From a list in the GQFQ, the FR selects the appropriate reason for exiting an interview, and the GQFQ assigns an outcome code that reflects the current interview status.

There are several reasons why GQ-level data collection may not be completed, such as the FR being unable to locate a facility, finding that there are no residents living or staying at the sample GQ facility during the data collection period, determining that there are now only housing units at the sample GQ facility, or finding that the facility no longer exists.

All information collected during the GQ-level phase is transmitted nightly from each FR to the Census Bureau through secure electronic file transfer.

8.3 Person-Level Phase

This section describes person-level interviews at sample GQ facilities. During this phase, the FR collects data for up to 15 sample residents at each assigned GQ facility using a CAPI automated instrument.

Person-Level Survey Instruments and Materials

This section details the instruments and materials needed to conduct ACS GQ person-level interviews.

Introductory Letter for the Sample Resident. The FR gives each sampled person an introductory letter at the time of the person-level interview. It provides information about the ACS, describes why it is important that they complete the ACS survey, describes uses of ACS data, stresses the confidentiality of their individual responses, and includes the internet address for the ACS website.

Computer-Assisted Personal Interviewing (CAPI) Questionnaire Instrument (QI). The CAPI QI is the preferred method of data collection. FRs use the CAPI QI to conduct face-to-face or telephone interviews with sample GQ residents. Interviews can be conducted in both English and Spanish. The GQ QI instrument is designed to record detailed population information for one person. It does not include housing questions except for the food stamp benefit question. The QI contains skip patterns based on GQ type. For example, sample residents living in nursing and correctional facilities are not asked the journey-to-work questions.

ACS GQ Paper Questionnaire. The FR distributes a paper GQ questionnaire to residents for self-response when a face-to-face CAPI interview cannot be conducted. This questionnaire is a bilingual, 24-page, two-color, flip-style booklet. Twelve blue pages make up the stateside English language GQ questionnaire; when flipped over, 12 green pages make up the stateside Spanish language version. Like the QI, the GQ paper questionnaire is designed to record detailed population information for one person. It does not include housing questions except for the food stamp benefit question. The paper questionnaire does not have skip patterns based on GQ type.

ACS GQ Internet Data Collection Instrument (GQ-IDC). As mentioned earlier, the ACS added an internet response option for residents at noninstitutional GQs, such as colleges, military barracks, and group homes. FRs offer respondents the option to respond online by leaving a package with the paper questionnaire and letter that includes a unique code and instructions on how to get to the website. If the FR collects the respondent's email address, the FR can email the link and unique code, inviting the respondent to respond online. After logging in, the respondent answers challenge questions and is provided a personal identification number (PIN) that allows them to exit and return later to complete the survey.

GQ residents living in noninstitutional GQs can complete an internet return anytime during the six-week panel. If the respondent has started an internet return but has not completed it, data from those "partial interviews" are processed at the end of the six-week data collection period.

GQ Questionnaire Instruction Guide (online). The questionnaire Instruction Guide is available online to sample residents if needed. This guide provides respondents with detailed information about how to complete the GQ questionnaire. It explains each question with expanded instructions and examples and instructs the respondent about how to mark the checkboxes and record write-in responses.

GQ Frequently Asked Question Brochure. Every sample GQ resident is given a bilingual FAQ brochure. This brochure provides answers to questions about the ACS GQ program.

GQ Return Envelopes. The GQ envelopes are used to return completed paper questionnaires to the FR or sworn GQ contact. These envelopes are not designed for delivery through the U.S. Postal Service.

Completing the GQ CAPI Automated Questionnaire Instrument (QI) or Paper Questionnaire

There are several ways for an FR to obtain a completed interview. The preferred method is for the FR to conduct a face-to-face interview with the sampled resident using the CAPI QI; however, other data collection methods may be necessary. The FR may conduct a telephone interview with the sample resident using CAPI QI, conduct a face-to-face CAPI proxy interview with a relative, guardian, or GQ contact; or leave a paper questionnaire with the resident to complete; or leave the questionnaires with the GQ contact to distribute to sampled residents and collect them when completed. If the questionnaires are left with sample residents to complete, the FR arranges with the resident or GQ contact to return and pick up the completed questionnaire(s) within two days. The FR must ensure that the sample residents are physically and mentally able to understand and complete the questionnaires on their own.

Before a GQ contact or a GQ employee obtains access to the names of the sample residents and their answers to the GQ questionnaire, they must take an oath to maintain the confidentiality of information about GQ residents. By taking this oath, one attains Special Sworn Status (SSS). Generally, an SSS individual is needed when the sample person is not physically or mentally able to answer the questions. An FR must swear in social workers, administrators, or GQ employees under Title 13, United States Code (U.S.C.) if these individuals need to see a sampled resident's responses. In taking the Oath of Nondisclosure, SSS individuals agree to abide by the same rules that apply to other Census Bureau employees regarding safeguarding Title 13 respondent information and other protected materials and acknowledge that they are subject to the same penalties for unauthorized disclosure. Relatives or legal guardians do not need to be

sworn as SSS individuals. If the sample person gives a GQ employee permission to answer questions or help to answer on their behalf, the GQ employee does not need to be sworn in.

Questionnaire Review

When a CAPI interview is conducted, the QI contains automated edit checks within the instrument to ensure the quality of the interview and to determine the final outcome (completed, sufficient partial, or insufficient partial interview). When paper questionnaires are used for self-response, the edit screen in the CAPI instrument is used by FRs to verify that all responses are legible and that the write-in entries and check boxes contain appropriate responses according to the skip patterns on the questionnaire. The FRs also determine whether the self-response questionnaire is a completed, sufficient, partial, or incomplete interview. The FR records the final outcome code for each self-response paper questionnaire on the Census Use Only page on the questionnaire.

An interview is considered complete when all or most of the questions have been answered, and a sufficient partial when enough questions have been answered to define the basic characteristics of the sample person. A case is classified as a noninterview when the answers do not meet the criteria of a complete or sufficient partial interview.

The FR conducts a GQ-level assignment review. This review is necessary to ensure that all CAPI interviews have been conducted and that all self-response questionnaires dropped off have been accounted for.

FRs ship paper questionnaires to the RO on a flow basis throughout each 6-week data collection period. The ROs conduct a final review of the questionnaires before sending completed questionnaires to NPC for keying. Nightly, CAPI interviews are transmitted from laptops to the Decennial Information Technology Division (DITD) processing staff.

8.4 Check-In and Data Capture

CAPI QI interview data are transmitted nightly via automated procedures to Census Bureau headquarters. Based on the final outcome code recorded for each paper questionnaire, the RO separates blank questionnaires from those with data. Only questionnaires that contain completed or sufficient partial data are shipped each week to NPC for check-in and keying. The forms are sorted according to the sample month and location (United States or Puerto Rico).

Check-In

The NPC check-in staff are given three days to check in a form, although they usually check in all forms they receive within one day. The check-in process results in batches of 50 questionnaires for data capture.

The NPC accepts completed questionnaires shipped from the RO on a weekly basis for a period of six weeks from the start of the sample month. Each RO closes out the sample month GQ assignments and accounts for all questionnaires. The NPC completes the sample month check-in within seven days of receipt of the final shipment from each RO. Each questionnaire contains a unique bar code that enables tracking of that questionnaire. The forms for the United States and Puerto Rico contain slightly different formatting and are checked in, batched, and keyed in separate batches.

Clerks review each page of every returned ACS GQ questionnaire. They look for correspondence, which they forward to headquarters if necessary. They then wand or hand key each bar code to officially check in and batch the form simultaneously, retain the English or Spanish pages of the questionnaire, and organize the forms into batches of 50 questionnaires.

Data Capture

After the questionnaires have been checked in and batched, they are prepped for scanning using the integrated Computer Assisted Data Entry (iCADE) system. After the scanning processes are completed, the images move to the keying unit for further processing using Key-From-Image (KFI) technology. The keying staff at NPC key, verify, and classify the data from the images. Upon completion of the keying process, the data files are transmitted to Census Bureau headquarters at the end of the day. Keyers have approximately two months to complete the keying for a given interview panel.

8.5 Special Procedures

Some exceptions to the data collection procedures are necessary to collect data efficiently from all GQ facilities, such as those in remote geographic locations or those with GQ security requirements.

Reduced Months of Data Collection at College Dorms

Fewer students live in dorms during the summer months (May-August) compared with other months of the year, which causes a significant increase in the number of noninterviews during this period. To reduce data collection costs and account for periods when dorms are primarily vacant, beginning in January 2013, the ACS decided to only collect data at college dorms in the non-summer months of January-April and September-December of each data collection year.

Biannual Data Collection in Remote Alaska

FRs collect data at sample GQ facilities in Remote Alaska during two separate periods each survey year; they visit a sample of GQ facilities from January through mid-July and from July through mid-January. This exception is needed because of difficulties accessing these areas at

certain times of the year. Chapter 7, Section 7.4 provides additional information about data collection in Remote Alaska.

Survey Period and Security Restrictions in Federal Correctional Facilities

Person-level data collection for the Federal Bureau of Prisons (BOP) operation is conducted over four months (September through December) for selected federal prisons and detention centers. The BOP provides the Census Bureau with a file containing all federal prisons and detention centers and a complete roster list of inmates for each federal facility. The Census Bureau updates the GQ-level information and generates the person-level samples for these GQ facilities.

Chapter 8 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
D. Reese & E. Poehler	11/28/2022	Updated content and formatting.	3.0
N. Butler	8/16/2024	Added information about internet data collection for residents in select GQ types.	4.0

Chapter 9: Language Assistance Program

9.1 Overview

The American Community Survey (ACS) language assistance program includes a set of methods and procedures designed to assist sample households with limited English proficiency in completing the ACS interview. The ACS program provides language assistance in many forms, including translated instruments and other survey materials, bilingual interviewers, and multiple language support by telephone. Providing language assistance is one of many ways the ACS can improve survey quality by reducing levels of survey nonresponse, the potential for nonresponse bias, and introducing response errors. Language support can help individuals with limited English skills to understand the survey questions, their rights as respondents, and the importance of the ACS.

The ACS language assistance program includes several tools to support each mode of data collection—mail, internet, telephone, and personal visits. Staff developed these tools based on research that assessed the current performance of the ACS for non-English speakers.

McGovern (2004) found that, despite the limited availability of mail questionnaires in languages other than English, the ACS successfully interviewed non-English speakers by telephone and personal visit follow-up. She also found that the level of item nonresponse for households speaking languages other than English was consistent with the low levels of item nonresponse in English-speaking households. These results led to a focus on improving the quality of data collected in the personal visit data collection modes. The language program includes assistance in many languages during the personal visit nonresponse follow-up stages, as well as some assistance in other languages during the mail and internet phases.

This chapter provides details on the current language assistance program. It begins with an overview of the language support, translation, and pretesting guidelines. It then discusses methods for each of the four data collection modes. The chapter closes with a discussion of associated research and evaluation activities.

9.2 Background

The ACS prioritizes developing and testing tools to improve the quality of data collected from limited English-speaking households. Many households speak very little English at home. Almost five percent of households that responded to the 2022 ACS were limited English-speaking households. The ACS language assistance program is critical in reaching populations that do not understand the questions on the survey or may not be able to respond at all.

9.3 Guidelines

The U.S. Census Bureau does not require the translation of all survey instruments or materials. Each census and survey determine the appropriate set of translated materials and language assistance options needed to ensure high-quality survey results. The Census Bureau does require that surveys and censuses follow specific guidelines when they translate data collection instruments, respondent letters, and other respondent materials.

In 2004, the Census Bureau released language support translation and pretesting guidelines. These state that data collection instruments translated from a source language into a target language should be reliable, complete, accurate, and culturally appropriate. Reliable translations convey the intended meaning of the original text. Complete translations should neither add new information nor omit information already provided in the source document. An accurate translation is free of both grammatical and spelling errors. Cultural appropriateness considers the target population's culture when developing the text for translation. In addition to meeting these criteria, translated Census Bureau data collection instruments and related materials should have semantic, conceptual, and normative equivalence. The Census Bureau guidelines recommend using a translation team approach to ensure equivalence. The language support guidelines include recommended practices for preparing, translating, and revising materials and ensuring sound documentation (U.S. Census Bureau, 2004). The ACS uses these Census Bureau guidelines to prepare data collection instruments, advance letters, and other respondent communications.

9.4 Mail and Internet Data Collection

The mailings (requesting a response via the internet) and the internet instrument are available in English or Spanish. The Census Bureau currently mails English-language ACS questionnaires to each nonresponding address unless a householder requests a Spanish questionnaire. The cover of the questionnaires contains a message written in the other language requesting people to call the toll-free assistance number if they need help. About 0.4 percent of internet and mail responses use a Spanish questionnaire.

In Puerto Rico, questionnaires are mailed in Spanish unless a householder requests English materials. The cover of the questionnaires contains a message written in the other language requesting that people who prefer to complete the survey in that language call an assistance number to obtain assistance or to request the appropriate form.

The ACS provides telephone questionnaire assistance for about 15 languages. There are specific numbers for respondents to receive general assistance in Spanish, simplified Chinese, Russian, Korean, and Vietnamese. Spanish is the only additional language in which the interview can be conducted over the phone.

9.5 Personal Visit Follow Up

The regional offices that conduct the computer-assisted personal interviewing (CAPI) nonresponse follow-up operations make every effort to hire bilingual staff. Language workloads are monitored and generally stable over time. Regional offices successfully meet the language needs of the population in their regions by hiring field representatives with the necessary language skills.

The regional offices train CAPI interviewers to search for interpreters within the sample household to assist in data collection. The regional offices maintain a list of multilingual field staff available to assist in the language preferred by a household respondent. Interviewers use a flashcard to identify the specific language spoken when they cannot communicate with a particular household. CAPI interviewers can also provide translated versions of some informational materials to respondents who speak Spanish, Chinese, Russian, Korean, Vietnamese, Polish, Portuguese, French, Haitian-Creole, or Arabic. These materials include an introductory letter, two brochures explaining the survey, and a bookmark that thanks the respondent for their participation.

The ACS CAPI survey instruments currently are available in both English and Spanish. Interviewers can conduct interviews in additional languages if they have that capability. Because a translated instrument is not available in languages other than English and Spanish, interviewers translate the English version during the interview and record the results on the English instrument. The standard classroom interviewer training for all ACS interviewers includes this language assistance training. The training is designed to improve the consistency of these procedures and to remind interviewers of the importance of collecting complete data for all households.

Bilingual interviews currently provide support in more than 30 languages. Interviewer language capabilities include English, Spanish, Portuguese, Chinese, Russian, French, Polish, Korean, Vietnamese, German, Japanese, Arabic, Haitian Creole, Italian, Navajo, Tagalog, Greek, and Urdu.

The CAPI instruments collect important data on language-related issues, including the frequency of the use of interpreters and the Spanish instrument, allowing the Census Bureau to monitor how interviewers complete survey interviews. The instruments record how often interviewers conduct translations on their own into different languages. For example, Griffin (2006) found that in 2005, more than 86 percent of all CAPI interviews with Spanish-speaking households were conducted by a bilingual (Spanish/English) interviewer.

Additional data collected allow managers to identify CAPI cases that the contact centers and the regional offices did not complete because of language barriers. A profile of this information by language highlights those languages needing greater support. For example, Fish (2010) found

that from 2006 to 2008, some regional offices' total language CAPI workloads experienced moderate changes, while others' total language workloads remained stable. These changes were primarily driven by an increase or decrease in the regional offices' English and/or Spanish language workloads. This research also demonstrated that estimated language workloads and the estimated linguistically isolated language workloads aligned well with the available language assistance resources. Regional offices have hired field representatives with the necessary language skills to accommodate their unique linguistically isolated language workloads, allowing the ACS to conduct interviews in over 30 languages every year.

9.6 Group Quarters

Chapter 8 describes the data collection methodology for people living in group quarters (GQ) facilities. Five instruments are used in GQ data collection— a CAPI instrument to conduct interviews with sample GQ residents, a paper survey questionnaire, an online application for uploading facility roster information, and an automated instrument for collecting administrative information from each facility. Beginning in January 2024, the ACS added an internet response option for residents living in select GQ types. The Census Bureau designed and field-tested a bilingual (English/Spanish) GQ questionnaire in 2005. Interviewers used these questionnaires to conduct interviews with a small sample of GQ residents. An interviewer debriefing found that the interviewers had no problems with these questionnaires and, as a result, the GQ data collection currently uses this form. The Census Bureau will hire bilingual interviewers to conduct interviews with non-English speakers in Puerto Rican GQ facilities. The Group Quarters Facility Questionnaire is available in both English and Spanish.

9.7 Research and Evaluation

The ACS established early research and development priorities for the language assistance program. Of critical importance was a benchmarking of the effectiveness of current methods. McGovern (2004) and Griffin and Broadwater (2005) assessed the potential for nonresponse bias because of language barriers. In addition, ACS staff created a website on quality measures, including annual information about the effect of language barriers on survey nonresponse.⁵⁵ These evaluations and the website both show that current methods result in very low levels of noninterviews caused by the interviewer's inability to speak the respondent's language. These nonresponse levels remain low because of special efforts in the field to use interpreters and other means to conduct these interviews. McGovern (2004) also assessed item-level nonresponse. She found that the mail returns received from non-English speakers were nearly as complete as those from English speakers and that the interviews conducted by telephone and personal visits with

⁵⁵ Quality measures can be found at: Sample Size and Data Quality: [Sample Size and Data Quality | American Community Survey | U.S. Census Bureau](#). Language barrier detail can be found at: Response Rates and Reasons for Noninterviews (in percent) for Housing Units: [Response Rates | American Community Survey | U.S. Census Bureau](#)

non-English speakers were as complete as those from English speakers. The Census Bureau continues to monitor unit nonresponse because of language barriers.

Language barriers can result in measurement errors when respondents do not understand the questions or when interviewers incorrectly translate a survey question. Limited-English-speaking households may face language barriers when responding to the ACS. Executive Order 13166, Improving Access to Services for Persons with Limited English Proficiency (LEP), directs each federal agency to “examine the services it provides and develop and implement a system by which LEP persons can meaningfully access those services consistent with, and without unduly burdening, the fundamental mission of the agency.” The Census Bureau continues to evaluate, research, and update Spanish survey translations within data collection instruments to improve survey data quality. Providing translations enables Spanish-speaking households to respond on their own and reduces overall survey nonresponse and the cost of in-person interviewing with bilingual field representatives or interpreters. In addition, using professional translators to ensure translations are clear and accurate improves item nonresponse rates and reduces measurement error, thus improving data quality.

9.8 References

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Chapter 9 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
D. Reese	11/28/2022	Updated and approved final content and formatting.	3.0
N. Butler, M. Wiland, C. Fisher	9/6/2024	Clarified who can serve as interpreters and updated GQ data collection to add internet response option.	4.0

Chapter 10: Data Preparation and Processing for Housing Units and Group Quarters

10.1 Overview

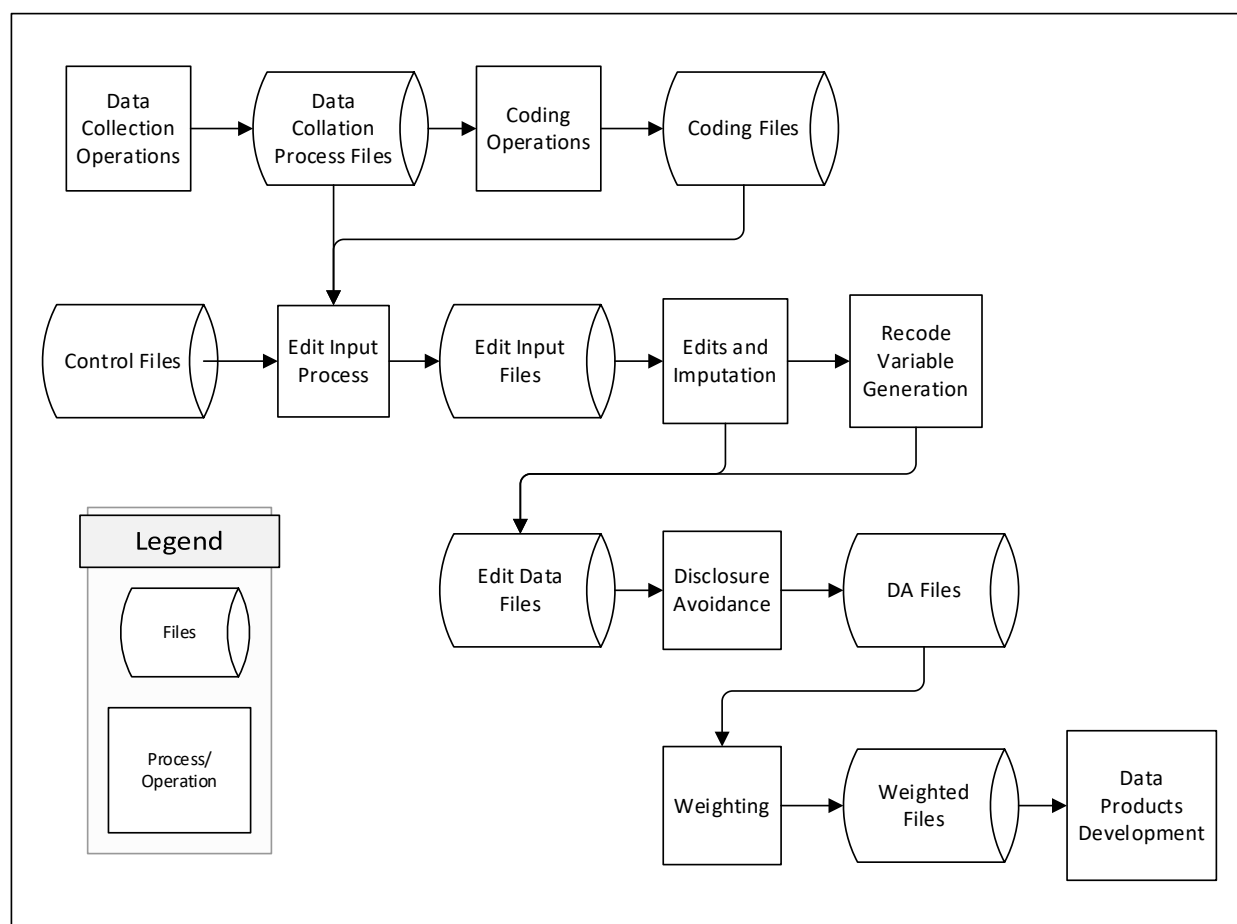
Data preparation and processing are critical steps in the survey process, particularly in terms of improving data quality. It is typical for developers of a large ongoing survey, such as the American Community Survey (ACS), to develop stringent procedures and rules to guide these processes and ensure that they are done consistently and accurately. This chapter discusses the actions taken during ACS data preparation and processing, provides the reader with an understanding of the various stages involved in readying the data for dissemination, and describes the steps taken to produce high-quality data.

The main purpose of data preparation and processing is to take the response data gathered from each survey collection mode to the point where they can be used to produce survey estimates. Data returning from the field typically arrive in various stages of completion, from a completed interview where all data items are addressed to one where most or all data items are left blank. There can be inconsistencies within the interviews, such that one response contradicts another. There may also be multiple interviews returned from the same household. These returns could contain different answers to the same question.

Upon arrival at the U.S. Census Bureau, all data undergo data preparation, where responses from different modes are captured, normalized, collated, and accumulated, forming Data Collation Process files (DCP files). The write-in entries from these files are then subject to monthly coding operations. Once all response data for the year have been processed, a series of steps are taken to produce Edit Input Files. These are created by merging operational status information (such as whether the unit is vacant, occupied, or nonexistent) for each housing unit (HU) and group quarters persons within a facility (GQ) with the files that include the response data. These combined data then undergo several processing steps before they are ready for use in data products.

Figure 10-1 depicts the overall flow of data as they pass from data collection operations through data preparation and processing and into data products development. While there are no set definitions of data preparation versus data processing, all activities leading to the creation of the Edit Input Files are considered data preparation activities, while those that follow are considered data processing activities.

Figure 10-1. American Community Survey (ACS) Data Preparation and Processing



10.2 Data Preparation

The ACS control file is integral to data preparation and processing because it provides a single database for all units in the sample. The control file includes detailed information documenting operational outcomes for every ACS sample case. For the self-response operations, it documents the receipt and check-in date of questionnaires returned by mail or the date completed on the internet. The status of data capture for questionnaires and the results of the Failed-Edit Follow-up (FEFU) operation are also recorded in this file. Chapter 7 provides a detailed discussion of self-response data collection, computer-assisted personal interviewing (CAPI), and telephone questionnaire assistance (TQA) operations.

For CAPI operations, the ACS control file stores information on whether a unit was determined to be occupied or vacant. Data preparation, which joins together each case's control file information with the raw, unedited response data, involves three activities: creation and processing of response data files, coding, and creation of edit input files.

Creation and Preparation of Data Capture Files

Many processing procedures are necessary to prepare the ACS data for tabulation. In this section, we examine each data preparation procedure separately. These procedures occur daily or monthly, depending on the file type (control or data capture) and the data collection mode (internet, mail, TQA, or CAPI). The processing that produces the final input files for data products is conducted on a yearly basis.

Daily Data Processing

The HU data are collected on a continual basis throughout the year by internet, mail, TQA, and CAPI. Sampled households first are mailed a request to complete their form on the internet and then are subsequently mailed the ACS questionnaire. A sample of households that do not complete their form by self-response is sent to the field for CAPI interviews, together with a sample of cases that could not be mailed. Figure 10-2 shows the monthly processing of HU response data. Each day (as shown by the dotted lines in Figure 10-2), the status of each sample case is updated in the ACS control file (large cylinder in Figure 10-2) based on data from data collection and capture operations (rectangle boxes in Figure 10-2). While the control file does not record response data, it does indicate when cases are completed to avoid additional survey completion attempts in another mode.

The creation and processing of the data depends on the mode of data collection. Data from questionnaires received by mail and internet are processed daily. Incoming data received by mail or internet (parallelograms in Figure 10-2) are run through a computerized process that checks for sufficient responses and for large households that require follow-up. Cases failing the process are sent to the FEFU operation. For more information about the FEFU operation, see Chapter 7.

The TQA operation uses a computer-assisted telephone interviewing (CATI) instrument to collect data. Responses from the various collection operations are captured in data files, shown as cylinders in Figure 10-2.

Figure 10-2. Daily Processing of Housing Unit Data

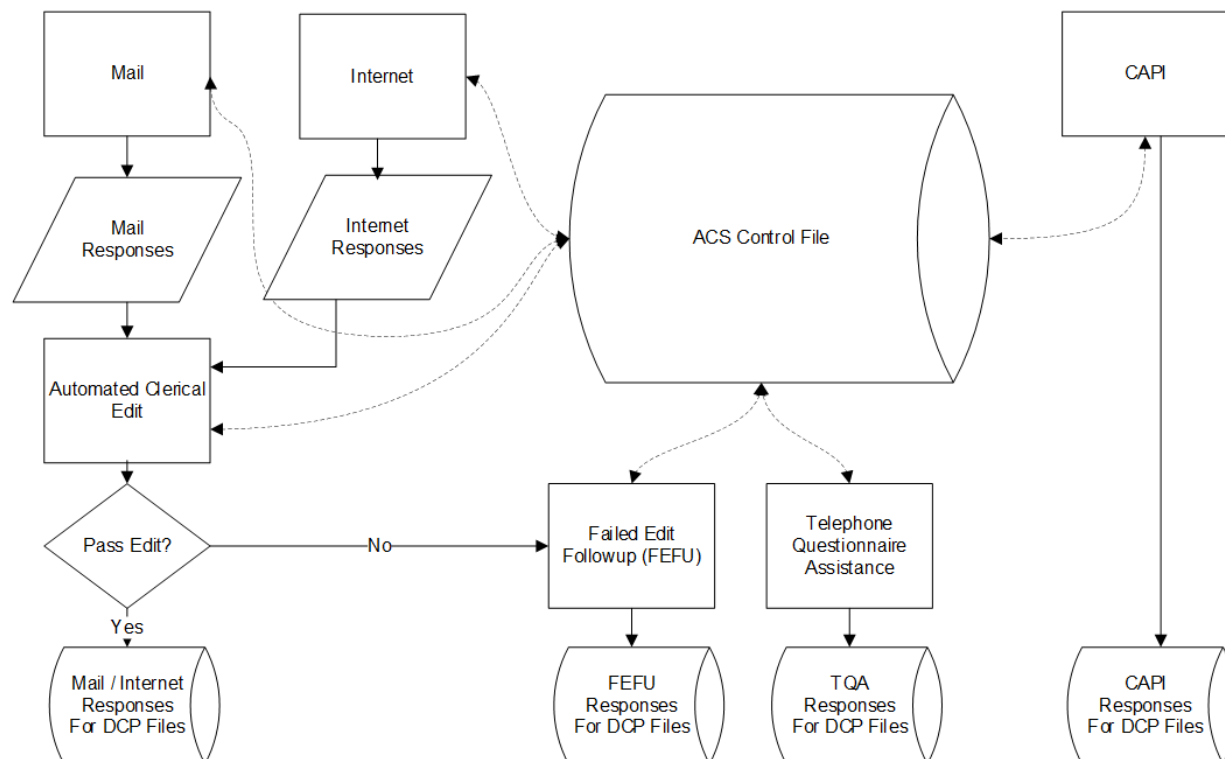


Figure note: DCP = Data collation process

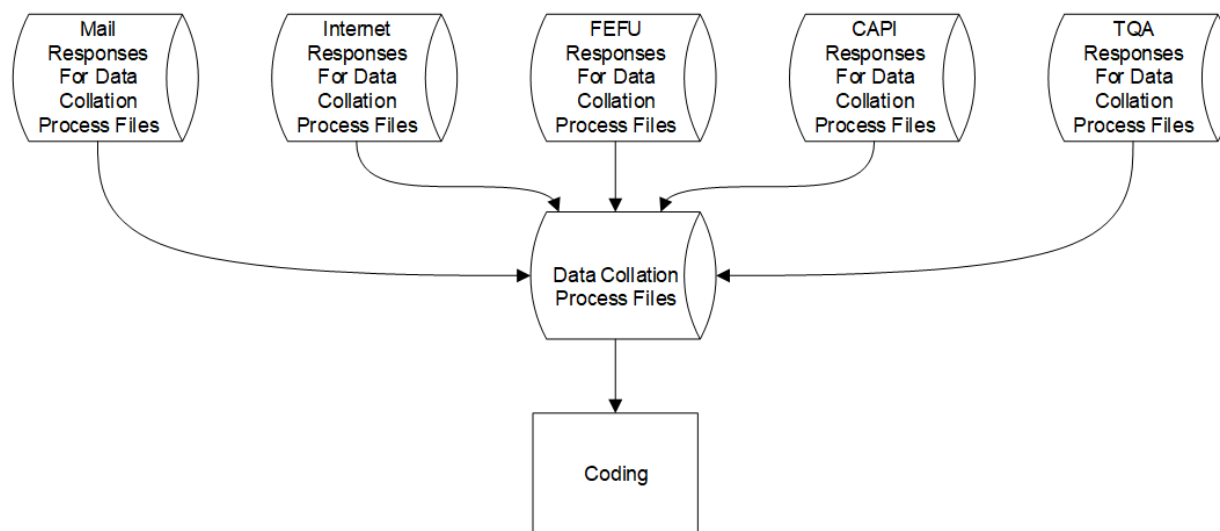
TQA and FEFU telephone operations are conducted at Census Bureau contact centers. Data collected through telephone and personal interviews are entered into a software instrument using BLAISE (a Computer-Assisted Interviewing (CAI) system and survey processing tool created by Statistics Netherlands). Operational data are transmitted to Census Bureau headquarters daily to update the control file with the status of each case. For data collected via the CAPI mode, Census Bureau field representatives (FRs) enter the ACS data directly into a laptop during a personal visit or phone call to the sample address. The FR transmits completed cases from the laptop to headquarters using an encrypted internet connection. The control file also is updated with the status of the case. Like HU data, status information for GQs is transmitted to headquarters each day for use in updating the control file. The GQ data are collected by internet and via paper forms that are sent to the National Processing Center on a flow basis for response data processing or by CAPI.

Monthly Data Processing

At the end of each month, the data collation process compiles all of the internet, mail, FEFU, CAPI, and TQA data collected during the past month and creates the data collation process files (DCP files) (see Figure 10-3). These represent all data collected during the previous month, regardless of the sample month for which the HU or GQ was chosen. The FEFU files include

both cases successfully completed and those for which the required number of attempts have been made without successful resolution.

Figure 10-3. Monthly Data Collation Process File Additions



During the data collation process, responses are reviewed. Those containing responses of "Don't Know," "Refused," and "Illegal" values are identified. Rules may cause some illegal values to be changed to legal ones. Once this response data has gone through this initial data cleaning, the next step is processing the questions that require coding.

10.3 Coding

The ACS questionnaire includes a set of questions that offer the possibility of write-in responses, each requiring coding to assign a numeric value to the response to make it machine-readable. Part of preparing newly received data for entry into the DCP involves identifying these write-in responses and placing them in a series of files that serve as input to the coding operations. The DCP includes HU and GQ data files, as well as a separate file for each write-in type. The HU and GQ write-ins are stored together.

During the coding phase for write-in responses, fields with write-in values are translated into a prescribed list of valid codes. All ACS coding is first automated (i.e., using a series of computer programs to assign codes) and then clerically coded (coded by a person) when needed. The write-ins are organized into three types of coding: write-in coding, industry and occupation coding, and geocoding. The items sent to coding are illustrated below in Table 10-1, followed by descriptions of each type of coding.

Table 10-1. Types of Coding

Item	Type of Coding
Race	Write-in Coding
Hispanic Origin	Write-in Coding
Ancestry	Write-in Coding
Language	Write-in Coding
Health Insurance	Write-in Coding
Field of Degree	Write-in Coding
Computer Types	Write-in Coding
Internet Service	Write-in Coding
Industry	Industry & Occupation
Occupation	Industry & Occupation
Place of Birth	Geocoding
Migration	Geocoding
Place of Work	Geocoding

Write-in Coding

The first type of coding involves the most items—write-in coding. Write-in items are those that allow respondents to write in some response other than the categories listed.

If a respondent checks the “White,” “Black or African American,” “American Indian or Alaska Native,” “Other Asian,” “Other Pacific Islander,” or “Some Other Race” boxes, he or she is asked to provide a write-in response. Figure 10-4 illustrates write-in coding processes.

All write-in coded items go through an automated process for the first pass of coding called Autocoding. The written responses are matched to a data dictionary. The data dictionary contains a list of the most common responses, with a code attached to each.⁵⁶ The coding program attempts to match the response to an entry in the dictionary to assign a code. For example, the question of language spoken in the home is automatically coded to one of over 1,000 language categories. These categories were developed from a master code list of 55,000 language names and variations. If the respondent lists more than one non-English language, only the first language is coded.

However, not all cases can be assigned a code using the automated coding program. Responses that do not match the data dictionary must be sent to clerical coding. Trained human coders look at each case and assign a code.

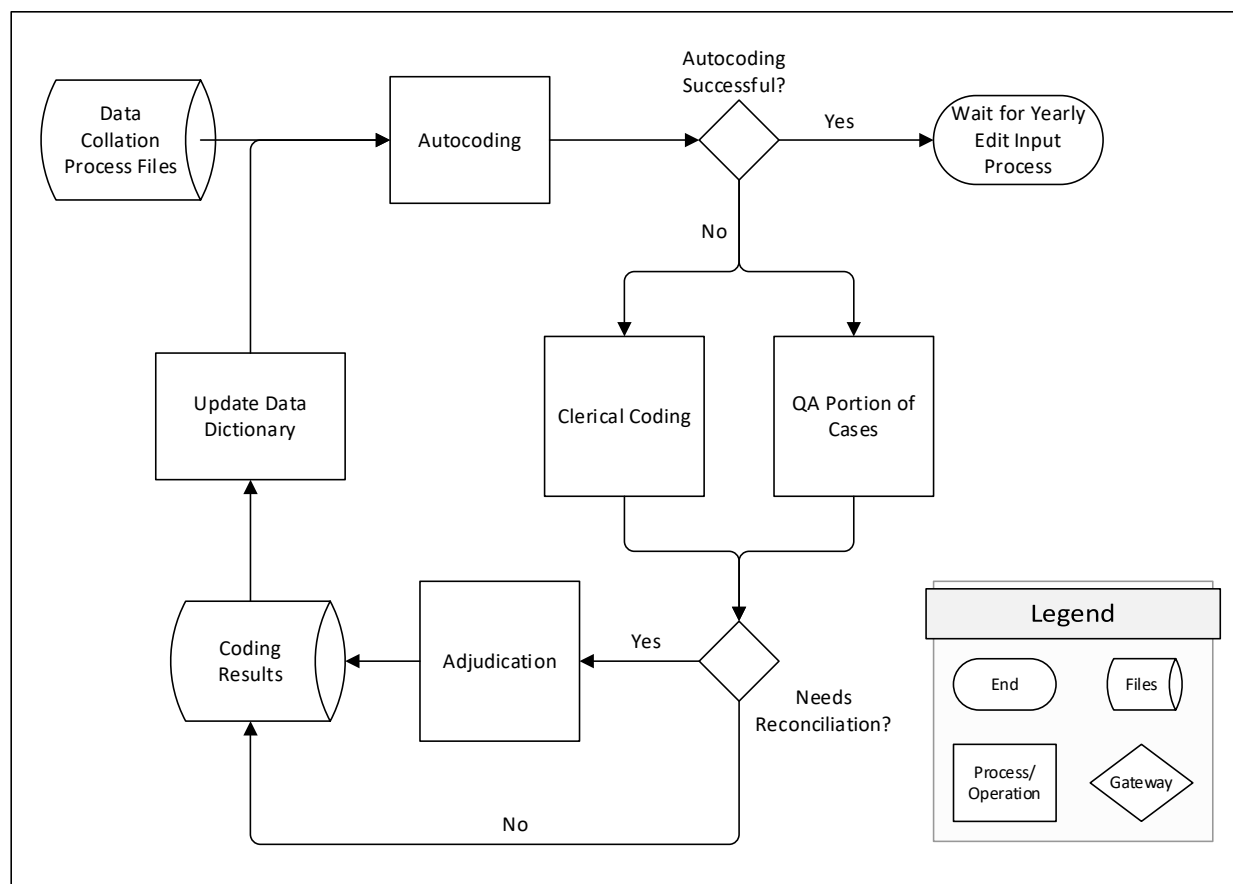
To ensure that clerical coding is accurate, a percentage of the write-in coded items are sent through the quality assurance (QA) process. The algorithm is specified according to the number

⁵⁶ Alternate spellings are also included in the data dictionary along with misspellings previously used by respondents that were clerically coded.

of returns in a batch. Batches of 1,000 randomly selected cases are sent to two QA coders who independently assign codes. If the codes they assign do not match one another, or the codes assigned by the automated coding program or clerical coder do not match, the case is sent to adjudication.

Adjudicator coders are experienced coders with additional training and resources. The adjudicating coder decides the proper code, and the case is considered complete.

Figure 10-4. Write-in Coding



Industry and Occupation Coding

The second type of coding is industry and occupation coding, which is slightly different from write-in coding. The ACS collects information concerning many aspects of the respondents' work, including commute time and mode of transportation to work, salary, and type of organization employing the household members. The ACS also asks about industry and occupation to give a clear picture of the kind of work in which the resident population is engaged. Industry information relates to the person's employing organization and the kind of business it conducts. Occupation is the work the person does for that organization. The industry and occupation questions are included in a series of questions regarding employment, as shown in Figures 10-5 and 10-6.

Figure 10-5. ACS Industry Questions, #42b-42c

b. What was the name of this person's employer, business, agency, or branch of the Armed Forces?

c. What kind of business or industry was this?
Include the main activity, product, or service provided at the location where employed. (For example: elementary school, residential construction)

Figure 10-6. ACS Occupation Questions, #42e-42f

e. What was this person's main occupation?
(For example: 4th grade teacher, entry-level plumber)

f. Describe this person's most important activities or duties. *(For example: instruct and evaluate students and create lesson plans, assemble and install pipe sections and review building plans for work details)*

Two additional supporting questions are asked to aid in coding the industry and occupation write-in questions—one before the industry question (class of worker) and one before the occupation question (industry type); see Figures 10-7 and 10-8.

Figure 10-7. ACS Class of Worker Question, #42a

a. Which one of the following best describes this person's employment last week or the most recent employment in the past 5 years?
Mark (X) ONE box.

PRIVATE SECTOR EMPLOYEE

☐ **For-profit** company or organization

☐ **Non-profit** organization (including tax-exempt and charitable organizations)

GOVERNMENT EMPLOYEE

☐ **Local government** (for example: city or county school district)

☐ **State government** (including state colleges/universities)

☐ **Active duty** U.S. Armed Forces or Commissioned Corps

☐ **Federal government** civilian employee

SELF-EMPLOYED OR OTHER

☐ **Owner of non-incorporated** business, professional practice, or farm

☐ **Owner of incorporated** business, professional practice, or farm

☐ Worked **without pay** in a **for-profit** family business or farm for 15 hours or more per week

Figure 10-8. ACS Industry Type Question, #42d

d. Was this mainly – Mark (X) ONE box.

☐ manufacturing?

☐ wholesale trade?

☐ retail trade?

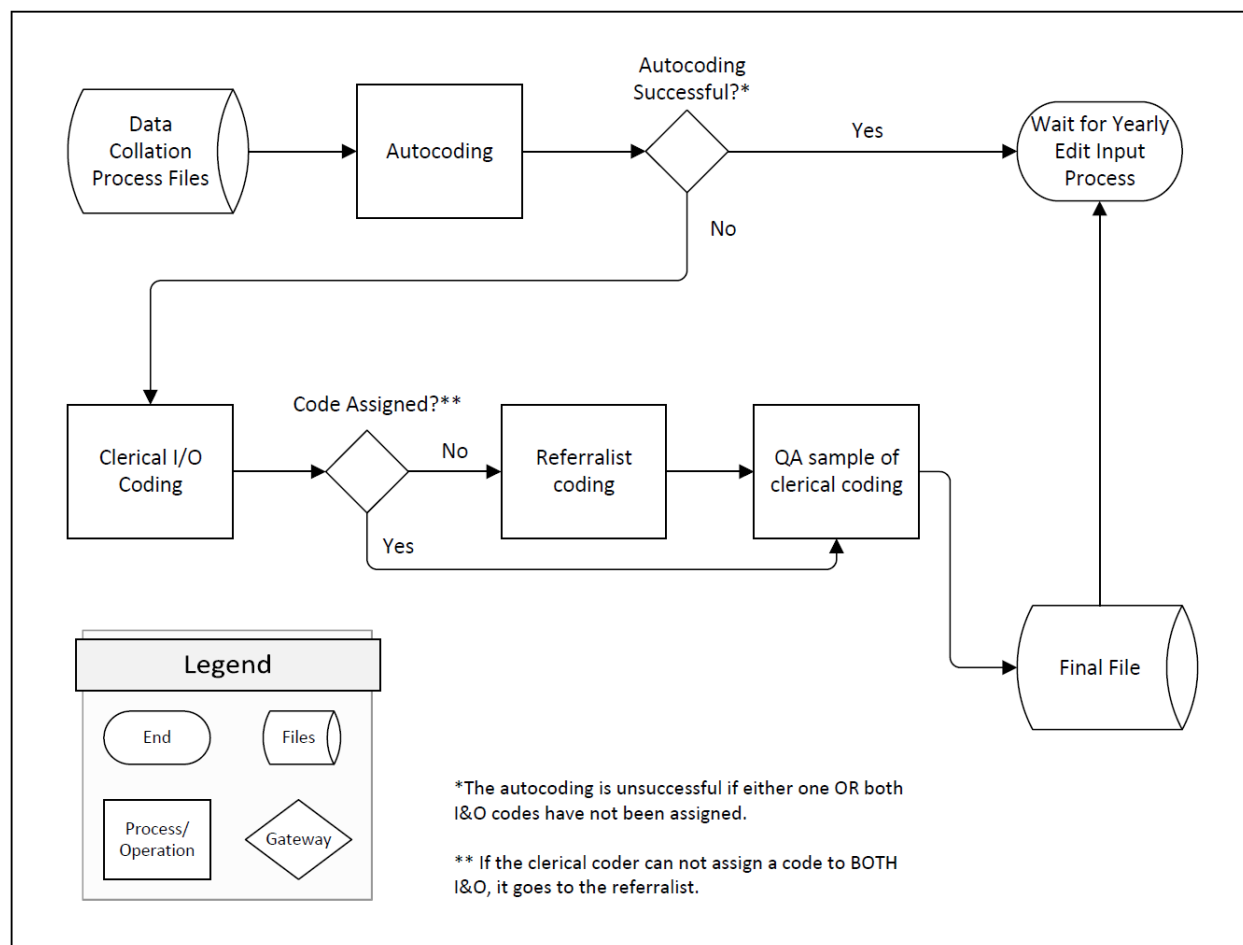
☐ other (agriculture, construction, service, government, etc.)?

From questions 42a-42f, monthly processing converts industry and occupation write-in responses to a 4-digit code category. Before 2012, specialized industry and occupation clerical coders assigned all codes.

Since 2012, as a first step in the industry and occupation coding process, the ACS uses an autocoder to assign industry and occupation codes. Instead of a straight string match, the industry and occupation autocoder is a set of logistic regression models, data dictionaries, and consistency edits (“hardcodes”) developed from around two million clerk-coded records, including group quarters and Spanish records (Thompson, et al., 2012). In September 2023, these data dictionaries were updated with data from the 2019 to 2021 ACS industry and occupation write-ins. The autocoder assigns an industry or occupation code if the quality score, based on agreement with clerk-coded records, is sufficiently high. If one or both of industry or occupation remain unassigned, these residual records are then assigned a code by specialized industry and occupation clerical coders. When clerical coders are unable to assign a code, the case is sent to

an expert, or coding referralist, for a decision. Once these cases are assigned both an industry and an occupation code, they are placed in the general pool of completed responses. Both the autocoder and the clerical coding have independent quality check processes. Figure 10-9 illustrates the industry and occupation coding process.

Figure 10-9. Industry and Occupation (I/O) Coding



Both the automated and clerical industry and occupation coding use the Census Industry or Occupation Classification System to code responses. The Census Industry Code list is based on the North American Industry Classification System (NAICS) and the Standard Occupational Classification (SOC) for the Census Occupation Code list. The Census Classification System's 4-digit code categories can be bridged directly to the NAICS and SOC for comparisons. However, to make meaningful comparisons over time, data users need to apply conversion rates to published industry and occupation statistics and microdata that use previous code lists to make equivalent comparisons. The degree of correspondence between the systems may vary, depending on the level of specificity of responses collected on the ACS. For instance, some 4-digit Census industry codes correspond to a 2-digit NAICS industry sector, while others correspond to a 6-digit NAICS U.S. industry. Similarly, some 4-digit Census occupation codes

correspond to a 3-digit SOC minor occupation group, while others correspond to a 6-digit SOC detailed occupation.

Standardized procedures and additional resources are maintained to aid in the assigning of industry and occupation code lists. Both the autocoder and clerical coders are given access to additional responses, such as education level, age, and geographic location. Clerks may also use an alphabetical index of industries and occupations. If the name of the company or business for which a person works is available, clerical coders can look up the name on a reference listing of employers and their industries. The Census Bureau has used many versions of this reference list, often referred to as the Employer Name List (ENL). Some ENLs have been developed from public publications while others have used previously coded records. There are also dedicated clerks who code group quarters records and Spanish records. Finally, the coding referralists have access to even more resources, including access to state registries and use of the internet for finding more information about the response.

Both the autocoder and clerical coders are subjected to regular quality checks. The autocoder quality control (QC) process is currently being revised. Reviews of the autocoder software and the associated dictionaries are conducted when major changes are implemented in the occupation or industry classifications.

The clerk quality assurance (QA) process begins with a sample of each coder's records being independently assigned by a different clerk who does not see the assigned code. The sampling rate is dynamic, based on the number of records a clerk codes in a month. The sample must also meet a minimum sample size. After the samples are re-coded, the codes then are reconciled to determine which is correct. Coders are required to maintain a high monthly agreement rate and a minimum production rate to remain qualified to code. A coding supervisor oversees the QA process.

Geocoding

The third type of coding that ACS uses is geocoding. This is the process of assigning a standardized code to geographic data. Place-of-birth, migration, and place-of-work responses require coding of a geographic location. These variables can be as localized as a street address or as general as a country of origin (Boertlein, 2007b).

The first category is place-of-birth coding, a means of coding responses to a U.S. state, the District of Columbia, Puerto Rico, a specific U.S. Island Area, or a foreign country where the respondents were born (Boertlein, 2007b). These data are gathered through a two-part question on the ACS asking in what state the person was born (if in the United States) or country (if outside the United States).

The second category of geocoding, migration coding, again requires matching the write-in responses of state, county, city, inside/outside city limits, and ZIP code given by the respondent to geocoding reference files and attaching geographic codes to those responses. A series of three questions collects these data, which are shown in Figure 10-10. If the person moved from a foreign country, they are asked to provide the country name.

Figure 10-10. ACS Migration Questions

15 a. Did this person live in this house or apartment 1 year ago?

☐ Person is under 1 year old → *SKIP to question 16*

☐ Yes, this house → *SKIP to question 16*

☐ No, outside the United States and Puerto Rico – *Print name of foreign country, or U.S. Virgin Islands, Guam, etc., below; then SKIP to question 16*

☐ No, different house in the United States or Puerto Rico

b. Where did this person live 1 year ago?

Address (Number and street name)

Name of city, town, or post office

Name of U.S. county or municipio in Puerto Rico

Name of U.S. state or Puerto Rico

ZIP Code

First, respondents are asked if they lived at this address a year ago; if the respondent answers no, there are several follow-up questions, such as the name of the city, county, state, and ZIP code of the previous home. The goal is to code the residence one year ago to the block level, if possible; if not possible, to code it to the place level.

The final category of geocoding is place-of-work coding. The place-of-work questions are shown in Figure 10-11.

Figure 10-11. ACS Place-of-Work Questions

30 a. **LAST WEEK, did this person work for pay at a job (or business)?**

☐ Yes → *SKIP to question 31*

☐ No – Did not work (or retired)

b. **LAST WEEK, did this person do ANY work for pay, even for as little as one hour?**

☐ Yes

☐ No → *SKIP to question 36a*

31 **At what location did this person work LAST WEEK?** *If this person worked at more than one location, print where he or she worked most last week.*

a. **Address (Number and street name)**

If the exact address is not known, give a description of the location such as the building name or the nearest street or intersection.

b. **Name of city, town, or post office**

c. **Is the work location inside the limits of that city or town?**

☐ Yes

☐ No, outside the city/town limits

d. **Name of county**

e. **Name of U.S. state or foreign country**

f. **ZIP Code**

The ACS questionnaire first establishes whether the respondent worked in the previous week. If this question is answered “Yes,” follow-up questions regarding the physical location of this work are asked.

The place-of-work coding requires matching the write-in responses of structure number and street name address, place, inside/outside city limits, county, state/foreign country, and ZIP code to reference files and attaching geographic codes to those responses. If the street address location

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information provided by the respondent is inadequate for geocoding, the employer's name often provides the necessary additional information. The inside/outside city limits indicator and ZIP code responses are used in the coding operations but are not a part of the final outgoing geographic codes. The goal is to code the place of work to the block level if possible; if not, code it to the place level.

The State and Foreign Country File (SFCF) is the main reference file used for geocoding. The SFCF contains two key pieces of information for geocoding. They are:

- The names and abbreviations of each state, the District of Columbia, Puerto Rico, and the U.S. Island Areas.
- The official names, alternate names, and abbreviations of foreign countries and selected foreign city, state, county, and regional names.

Other reference files (such as a military installation list and City Reference File) are available and used in instances where “the respondent's information is either inconsistent with the instructions or is incomplete” (Boertlein, 2007b).

Responses do not have to match a reference file entry exactly to meet the requirements for a correct geocode. The coding algorithm for this automated geocoding allows for equivocations, such as using Soundex values of letters (for example, m=n, f=ph) and reversing consecutive letter combinations (ie=ei). Each equivocation is assigned a numeric value, or confidence level, with exact matches receiving the best score or highest confidence (Boertlein, 2007b). A preference is given for matches consistent with any check boxes marked and/or response boxes filled. The responses must match a reference file entry with a relatively high level of confidence for the automated match to be accepted. Soundex values are used for most types of geocoding and are generally effective in producing matches for given responses. Table 10-2 summarizes the properties of the geocoding workload by category of codes that were assigned a code automatically as of 2022.

Table 10-2. Percentage of Geocoding Cases with Automated Matched Coding

Characteristic	Percentage of Cases Assigned a Code Through Automated Geocoding
Place of Birth (State Level)	99.9 Percent
Migration (Place Level)	99.0 Percent
Migration (Block Level)	71.1 Percent
Place of Work (Place Level)	99.3 Percent
Place of Work (Block level)	60.1 Percent

Source: U.S. Census Bureau, American Community Survey, 2022

The remaining responses that are not assigned a code through the automated system are processed in computer-assisted clerical coding (CACC) operations. The CACC coding is separated, with one operation coding to place-level and one to block-level responses.

Both the place-and block-level CACC operations involve long-term, specially trained clerks who use additional reference materials to code responses that cannot be resolved using the standard reference files and procedures. Clerks use interactive computer systems to search for and select reference file entries that best match the responses, and the computer program then assigns the codes associated with that geographic entity. The CACC operations are also generally effective at assigning codes.

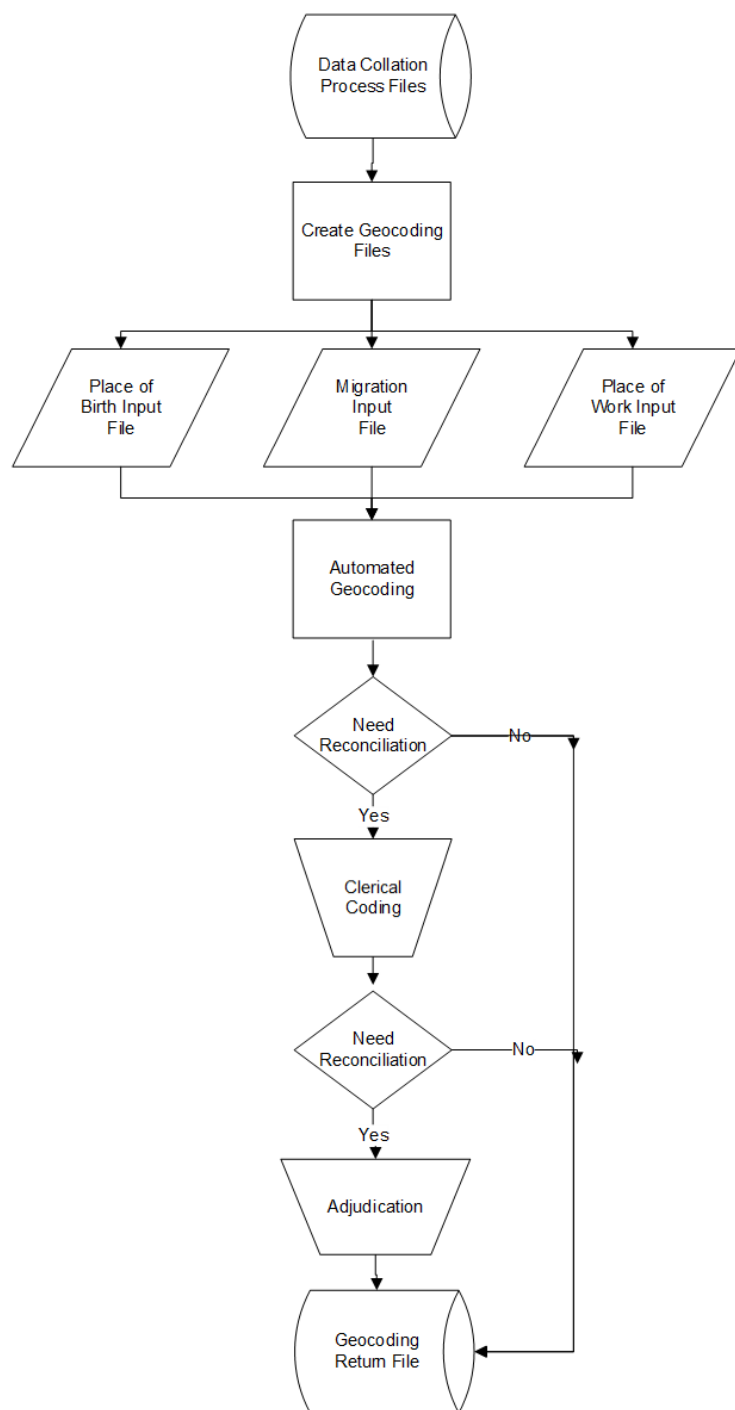
All three geocoding items—place of birth, migration, and place of work—require QA to ensure that the most accurate code has been assigned. The first step of assigning a geocode, the automated coding system, currently does not have a QA step. In both the 1990 and 2000 Decennial Censuses, the automated coding system had an error rate of less than 2.4 percent of all cases (Boertlein, 2007a); since then, the automated coder software has undergone revisions and has been shown to have an even lower error rate.

Among the place-of-birth, migration, and place-of-work cases that were not assigned geocodes by the automated coding system and that subsequently are sent to CACC, 5 percent will be sent to three independent clerical coders. If two out of three coders agree on a match, the third coder is assigned an error for the case. Coders must maintain an error rate of less than five percent per month (Boertlein, 2007a).

For place of work block-level coding, the QA protocol is slightly different. Block-level coders must maintain an error rate at or below 10 percent to continue coding. These coders also are expected to have an uncodable rate of 35 percent or less. If block-level coders do not maintain these levels, 100 percent of their work is reviewed for accuracy, and additional training may be provided (Boertlein, 2007a).

The QA system for ACS geocoding also includes feedback to the coders. Those with high error rates or high uncodable rates, as well as those who have low production rates or make consistent errors, may be offered additional training or general feedback on how to improve. Figure 10-12 illustrates automatic geocoding.

Figure 10-12. Geocoding



10.4 Preparation for Creating Select Files and Edit Input Files

The final data preparation operation involves creating Select Files and Edit Input Files for data processing. To create these files, several preparatory steps must be followed. By the end of the year, the response data stored in the DCP files will have been updated 12 times and will become a principal source for the edit-input process. Coding input files are created from the DCP files of write-in entries. Edit Input Files combine data from the DCP files and the returned coding files, and operational information for each case is merged with the ACS control file. The resulting file includes housing and person data. Vacant units are included, as they may have some housing data.

Creation of the Select and Edit Input Files involves carefully examining several components of the data, each described in more detail below. First, the response type and number of people in the household unit are assessed to determine inconsistencies. Second, the return is examined to establish if there are enough data to count the return as complete, and third, any duplicate returns undergo a process of selection to assess which return will be used.

Response Type and Number of People in the HU

Each HU is assigned a response type that describes its status as occupied, temporarily occupied, vacant, delete, or noninterview. Deleted HUs are units that are determined to be nonexistent, demolished, or commercial units and therefore are out of scope for the ACS.

While this type of classification already exists in the DCP, it can be changed from “occupied” to “vacant” or even to “noninterview” under certain circumstances, depending on the final number of persons in the HU, in combination with other variables. In general, if the return indicates that the HU is not occupied and that there are no people listed with data, the record and number of people (which equals 0) are left as is. If the HU is listed as occupied, but the number of persons for whom data are reported is 0, it is considered vacant.

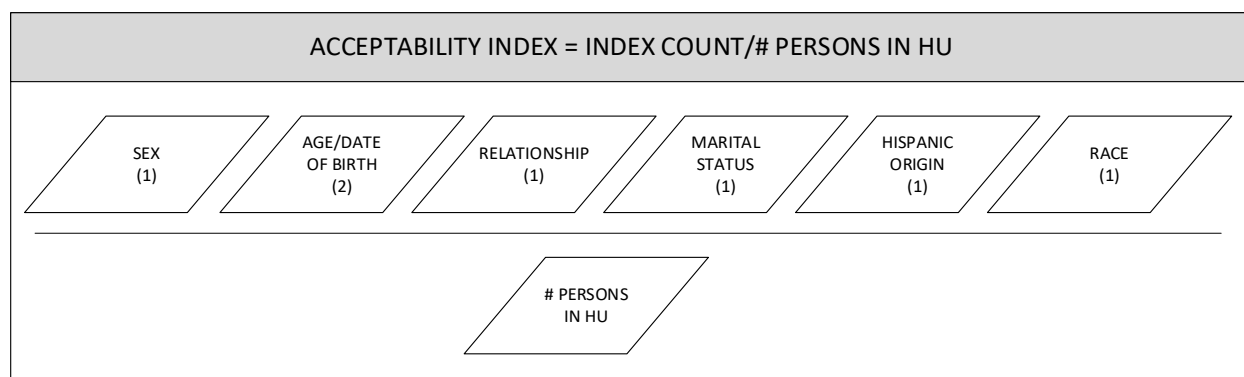
The data are also examined to determine the total number of people living in the HU, which is not always a straightforward process. For example, on a mail return, the count of people on the cover of the form may not match the number of people reported inside. Several steps are taken to reconcile the numbers, but generally, the largest number listed is used (Powers, 2023b).

Determining if a Return Is Acceptable

The acceptability index is a data quality measure used to determine if the data collected from an occupied HU or a GQ are complete enough to include a person record. Figure 10-13 illustrates the acceptability index. For the acceptability index, each GQ person is treated as a single person household. Six basic demographic questions plus marital status are examined for answers, although relationship is excluded for GQ persons. One point is given for each question answered for a total of seven possible points (six for GQ) that could be assigned to each person in the

household. A person with a response to either age or date of birth scores two points because, given one, the other can be derived or assigned. The total number of points is then divided by the total number of household members. (This is equal to one for GQ). For the interview to be accepted, there must be an average of 2.5 responses per person in the household. Household records or GQ persons that do not meet this acceptability index are classified as noninterviews and will not be included in further data processing (Powers, 2023a and 2023b). These cases will be accounted for in the weighting process, as described in Chapter 11.

Figure 10-13. Acceptability Index



If the Acceptability Index is greater than 2.5, the person record is accepted as a complete return. If the Acceptability Index is less than 2.5, the person record is not accepted as a complete return.

Unduplicating Multiple Returns

Once the universe of acceptable interviews is determined, the HU data are reviewed to unduplicate multiple returns for a single HU. There are several reasons why more than one response can exist for an HU. A household might return two self-responses, one to the request to complete by internet and a second response using the paper questionnaire. Depending on the timing, a household might return an internet response or mailed form but may also be interviewed in CAPI before the internet or mail form is logged in as returned.

If more than one return exists for an HU, a quality index is used to select one as the final return. This index is calculated as the percentage of items with responses out of the total number of items that should have been completed. The index considers responses to both population and housing items. The mode of each return is also considered in the decision regarding which of two returns to accept, with preference generally given to mail/internet returns (Powers, 2023b).

After the resolution of multiple returns, each sample case is assigned a value for three critical variables—data collection mode, month of interview, and case status. The month in which data were collected from each sample case is determined and then used to define the universe of cases to be used in the production of survey estimates. For example, a record in sample in December

2018, but for which data was collected in January 2019, was included in the 2019 ACS data products because the data collected were associated with the 2019 ACS data collection.

10.5 Creating the Select Files and Edit Input Files

Select Files

Select Files are the series of paradata files that identify the status of all cases included in processing for a given year, whether they responded or not. These files include the case status, the interview month, and the data collection mode for all cases. The largest select file, also called the Omnibus Select File, contains every available case from 14 months of sample—the current (selected) year and November and December of the previous year.⁵⁷ This file identifies the response status of all cases: occupied, vacant, noninterview, non-housing unit, subsampled out at the CAPI stage⁵⁸, and other.

This file is then reduced to the Final Accepted Select File, which includes only occupied housing units, vacant units, and GQ persons to be included in tabulations for the current year. That is, returns that were tabulated in the prior year or that will be tabulated in the next year are excluded. The final screening removes returns from vacant boats because they are not included in the ACS estimation universe.

Edit Input Files

The next step is the creation of the files that will be the input to the editing and imputation process, generally called the Edit Input Files. There are three such files: the Housing Edit Input File, the HU Person Edit Input File, and the GQ Person Edit Input File.

The Housing Edit Input file is created by first merging the DCP housing data files with the codes for computer type and internet service. This file is then merged with the Final Accepted Select File. Then, there are several “clean-up” steps. Date variables are modified into the proper format, variables are given the prefix “U,” followed by the variable name to indicate they are unedited variables, and answers that are “Don’t Know” and “Refused” are set as missing blank values for the edit process.

The Person Edit Input Files are created by first merging the DCP person data files with the codes for place of birth, migration place, Hispanic origin, race, ancestry, language, field of degree, place of work, health insurance, and current or most recent industry and occupation. These files are then merged with the Final Accepted Select File to create a file with all person information

⁵⁷ November and December of the previous year are included to account for late year returns coming in during the beginning of the next calendar year and processed with the new calendar year.

⁵⁸ See Chapter 7 for a full discussion of subsampling in the ACS.

for all accepted HUs and a file with all GQ person information. As was done for the housing items, the person items are set with a “U” in front of the variable name to indicate that they are unedited variables, and “Don’t Know” and “Refused” answers are set as missing blank values for the edit process. Additionally, various name flags are set to identify people with Spanish surnames and those with “non-name” first names, such as “female” or “boy.” When complete, the Edit Input Files encompass the information from the DCP housing and person files but only for the unduplicated response records with data collected during the calendar year.

10.6 Editing and Imputation

The edit and imputation process begins once the Edit Input Files have been generated and verified. The main steps in this process are:

- Editing and imputation
- Generating recoded variables
- Reviewing edit results
- Creating input files for data products

Editing

As editing and imputation begin, the data file still contains blanks and inconsistencies. When data are missing, it is standard practice to use a statistical procedure called imputation to fill in missing responses. Filling in missing data provides a complete dataset, making analysis of the data both feasible and less complex for users. Imputation can be defined as the placement of one or more estimated answers into a field of a data record that previously had no data or had incorrect or implausible data (Groves et al., 2004). Imputed items are flagged so that analysts understand the source of these data.

As mentioned, the blanks come from blanked-out invalid responses and missing data on internet returns or mail questionnaires that were not corrected during FEFU, as well as from CAPI cases with answers of “Refusal” or “Don’t Know.” The files also include the codes from the eleven topics that allow for open-ended responses. As a preliminary step, data are separated by state because the HU editing and imputation operations are completed on a state-by-state basis, whereas GQ editing and imputation operations are completed at the national level by GQ type.

Edit and imputation rules are designed to ensure that the final edited data are as consistent and complete as possible and are ready for tabulation. The first step is to address those internally inconsistent responses that were not resolved during data preparation. The editing process looks at internally contradictory responses and attempts to resolve them. Examples of contradictory responses are:

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- A person is reported as having been born in Puerto Rico but not a United States citizen.
- A young child answers the questions on wage or salary income.
- A person under the age of 15 reports being married.
- A male responds to the fertility question (Diskin, 2007).

Subject matter experts at the Census Bureau develop rules to handle these types of responses. Applying such edit rules helps maintain data quality when contradictory responses exist. Some edits are more complex than others. For example, joint economic edits examine the combination of multiple variables related to a person's employment, such as most recent job activity, industry, type of work, and income. This approach maximizes information that can be used to impute any economic-related missing variables. As noted by Alexander et al. (1997):

Editing the ACS data to identify for obviously erroneous values and imputing reasonable values when data were missing involved a complex set of procedures. Demographers and economists familiar with each specific topic developed the specific procedures for different sets of data, such as marital status, education, or income. The documentation of the procedures is over 1,000 pages long, so only a very general discussion will be given here.

As Alexander et al. (1997) note, edit checks encompass range and consistency. They also provide justification for the edit rules:

The consistency edit for fertility deletes response from anyone identified as Male or under age 15. In setting a cutoff like this, a decision must be made based on the data about which categories have more 'false positives' than 'true positives.' The consistency edit for housing value involves a joint examination of value, property taxes, and other variables. When the combination of variables is improbable for a particular area, several variables may be modified to give a plausible combination with values as close as possible to the original.

Another edit step relates to the income components reported by respondents for the previous 12 months. Because of general price-level increases, answers from a survey taken in January 2018 are not directly comparable to those of December 2018 because the value of the dollar changed during this period. Consumer Price Index (CPI) indexes are used to adjust these income components for inflation. For example, a household interviewed in March 2021 reports their income for the preceding 12 months—March 2020 through February 2021. This reported income is adjusted to the reference year by multiplying it by the 2021 (January–December 2021) CPI and dividing by the average CPI for March 2020– February 2021.

Imputation

There are two principal imputation methods to deal with missing or inconsistent data—assignment and allocation. Assignment involves looking at other data reported by the respondent to fill in missing responses. For example, when determining sex, if a person reports giving birth to children in the past 12 months, this would indicate that the person is female. This approach also uses data as reported by other people in the household to fill in a blank or inconsistent field. For example, if the reference person and the spouse are both citizens by birth, a child with a blank response to citizenship is assumed also to be a citizen. Assigned values are expected to have a high probability of correctness. Assignments are tallied as part of the edit output.

Certain values, such as a person's educational attainment, cannot be ascertained from other responses on the questionnaire, so responses from similar housing units and people are used to fill in needed data. This commonly used approach of imputation is known as hot-deck allocation, which uses a statistical method to supply responses for missing or inconsistent data from responding HUs or people in the sample who are similar.

Hot-deck allocation is conducted using a hot-deck matrix that contains the data for prospective donors and is called upon when a recipient needs data because a response is inconsistent or blank. For each question or item, subject matter analysts develop detailed specification outlines for how the hot-deck matrices for that item are to be structured in the editing system. Classification variables for an item are used to determine categories of “donors” (referred to as cells) in the hot deck. These donors are records of other HUs or people in the ACS sample with complete and consistent data for the item(s) of interest. One or more cells constitute the matrix used for allocating one or more items. For example, for the place-of-work question some blanks remain after write-in coding is conducted. Codes are allocated from a similar person based on other variables such as age, sex, education, and number of weeks worked. If all items are blank, they are filled in using data allocated from another case, or donor, whose responses are used to fill in the missing items for the current case, the “recipient.”

Some hot-deck matrices are simple and contain only one cell, while others may have thousands. For example, in editing the housing item known as tenure (which identifies whether the housing unit is owned or rented), a simple hot deck of three cells is used, where the cells represent responses from single-family buildings, multi-unit buildings, and cases where a value for the question on type of building is not reported. Alternatively, dozens of different matrices are defined with thousands of cells specified in the joint economic edit, where many factors are used to categorize donors for these cells, including sex, age, industry, occupation, hours and weeks worked, wages, and self-employment income.

Sorting variables are used to order the input data before processing to determine the best matches for hot-deck allocation. In the ACS, the variables used for this purpose are mainly geographic, such as state, county, census tract, census block, and basic street address. This sequence is used

because it has been shown that housing and population characteristics are often more similar within a given geographic area. The sorting variables for place of work edit, for example, are used to combine similar people together by industry groupings, means of transportation to work, minutes to work, state of residence, county of residence, and the state in which the person works.

For each cell in the hot deck, up to four donors (e.g., other ACS records with housing or population data) are stored at any one time. The hot-deck cells are given starting values determined in advance to be the most likely for particular categories. Known as cold-deck values, they are used as donor values only in rare instances where there are no donors.

Procedures are employed to replace these starting values with actual donors from cases with similar characteristics in the current data file. This step is referred to as hot-deck warming.

The edit and imputation programs look at the housing and person variables according to a predetermined hierarchy. For this reason, each item in a response record is edited and imputed in an order delineated by this hierarchy, which includes the basic person characteristics of sex, age, and relationship, followed by most of the detailed person characteristics, and then all the housing items. Finally, the remainder of the detailed person items, such as migration and place of work, are imputed. For HUs, the edit and imputation process is performed for each state separately, with the exception of the place of work item, which is done at the national level. For GQ facilities, the data are processed nationally by GQ type, with facilities of the same type (e.g., nursing homes, prisons) edited and imputed together.

In the edit and imputation system, a flag is associated with each variable to indicate whether it was changed and, if so, the nature of the change. These flags support the subject matter analysts in their review of the data and provide the basis for the calculation of allocation rates. Allocation rates measure the proportion of values that required hot-deck allocation and are an important measure of data quality. The rates for all variables are provided in the quality measures section on the ACS Web site. Chapter 15 also provides more information about these quality measures.

Generating Recoded Variables

New variables are created during data processing. These recoded variables, or recodes, are calculated based on the response data. Recoding is usually done to make commonly used, complex variables user-friendly and to reduce errors that could occur when users incorrectly recode their own data. There are many recodes for both housing and person data, enabling users to understand the characteristics of an area's people, employment, income, transportation, and other important categories.

Data users' ease and convenience are primary reasons for creating recoded variables. For example, one recode variable is "Presence of Persons 60 and Over." While the ACS also provides more precise age ranges for all people in a given county or state, having a recoded

variable that will give the number and percentages of households in a region with one or more people age 60 or over in a household provides a useful statistic for policymakers planning for current and future social needs or interpreting social and economic characteristics to plan and analyze programs and policies (U.S. Census Bureau, 2021). Other recodes represent more complicated composites of multiple variables. For example, the ACS does not specifically ask about poverty status. However, a poverty-status recode is created using information about respondents' income and household composition or GQ type.”

Reviewing Edit Results

The review process involves a review of the editing process and a reasonableness review. After editing and imputation are complete, Census Bureau subject matter analysts review the resulting data files. The files contain both unedited and edited data and the accompanying imputation flag variables that indicate which missing, inconsistent, or incomplete items have been filled by imputation methods. Subject matter analysts first compare the unedited and edited data to see that the edit process worked as intended. The subject matter analysts also undertake their own analyses, looking for problems or inconsistencies in the data from their perspectives. If year-to-year changes do not appear to be reasonable, they institute a more comprehensive review to reexamine and resolve the issues. Allocation rates from the current year are compared with those of previous years to check for notable differences. A review is conducted by variable, and results on unweighted data are compared across years to see if there are substantial differences. Analysts also examine unusual individual cases that were changed during editing to ensure accuracy.

The analysts use a number of special reports for comparisons based on the edit outputs and multiple years of survey data. These reports and data are used to help isolate problems in specifications or processing. They include detailed information on imputation rates for all data items, performance metrics for each hot-deck matrix, and counts of tallies that represent the number of times certain programmed logic checks were executed during editing. Finally, analysts verify that all recodes have been correctly applied. If editing problems are discovered in the data during this review process, it is often necessary to rerun the programs and repeat the review.

Creating Input Files for Weighting and Disclosure Avoidance

Once the subject matter analysts have approved data within the edited files, and their associated recodes, the files are ready to serve as inputs to the weighting operation and disclosure review. If errors attributable to editing problems are detected during the creation of data products, it may be necessary to repeat the editing and review processes. Further details on the weighting operation can be found in Chapter 12; information on disclosure avoidance techniques is discussed in Chapter 13.

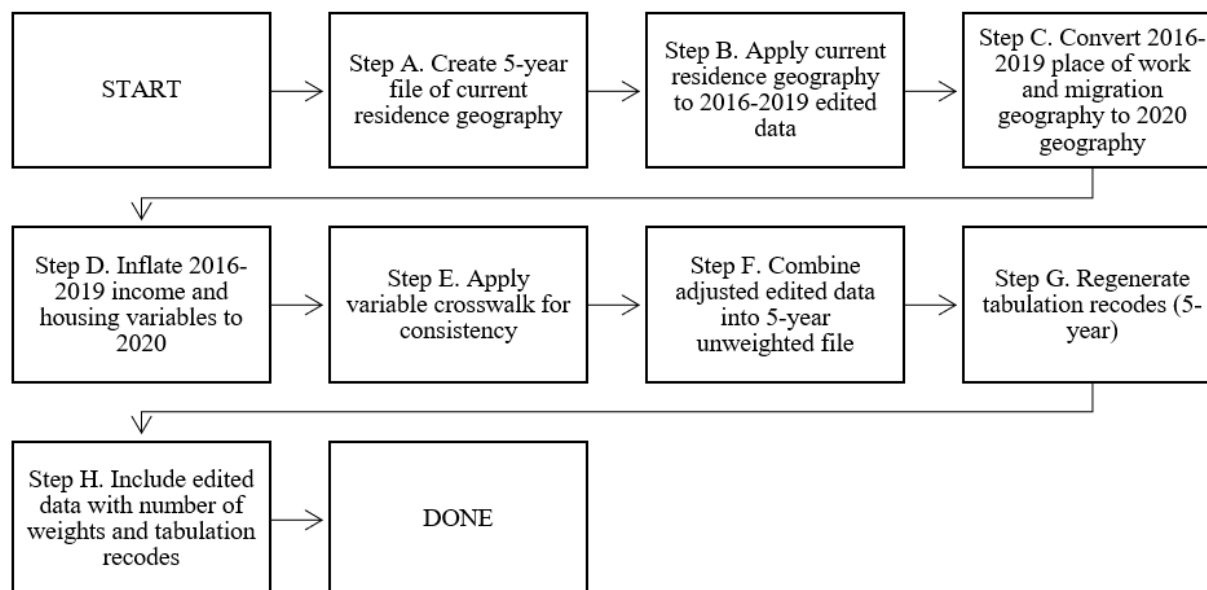
10.7 Multiyear Data Processing

ACS multiyear estimates were published for the first time in 2008 based on the 3-year combined file from the 2005 ACS, 2006 ACS, and 2007 ACS. To do this, multiyear edited data (or microdata) were used as the basis for producing the 3-year ACS tabulated estimates for the multiyear period. The 3-year estimates are not published anymore, but the 5-year estimates have been published since 2010 (on the 2005–2009 data). This discussion will focus on the 2018–2022 5-year files and describe the steps to implement multiyear data processing.

A number of steps must be applied to the previous years' final edited data to make them consistent for multiyear processing. The first step is to update the current residence geography for 2018–2022 data to 2022 geography. The most complex step in the process pertains to how the vintage of geography in the “Place of Work” and “Migration” variables and recodes are updated to bring them up to the current year (2022). This step is required because the 2018 edited data for these variables and recodes are in the 2018 vintage geography, while the edited data for the remaining individual years, 2017, 2018, and 2019 edited data are in their respective year's vintage geography. The geocodes in these variables and recodes from prior years need to be converted to current geography. This transformation is accomplished using a matching process to multiyear geographic bridge files to update these variables to 2022 geography. Inflation adjustments must also be applied to monetary income and housing variables and recodes to inflate them to a constant reference year of 2022 for the 2018–2022 5-year edited file. Yet another step is needed to deal with variable changes across years so that a consistent 5-year file may be created. A crosswalk table for the multiyear process attempts to map values of variables that changed across years into a consistent format. For example, because the race and Hispanic Origin coding changed for 2020, the existing codes in 2018 and 2019 1-year microdata had to be crosswalked into the new coding scheme when creating the 2018–2022 5-year file. Similarly, because the response categories changed for the relationship to householder question in 2019, the existing data for 2016 through 2018 1-year microdata had to be crosswalked forward to match the new categories when creating the 2018–2022 5-year file.

When all of these modifications have been applied to the prior years' data, these data are combined with the 2022 data into an unweighted multiyear edited dataset. Tabulation recodes are then created from this file, and outputs of that process are joined with the 5-year weights and edited data to create the multiyear weighted and edited file. At this point, the 5-year ACS edited and weighted data file will be suitable for input to the data products system. See Figure 10-14 for a flowchart showing high-level process flow.

Figure 10-14. Multiyear Edited Data Process



10.8 References

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Chapter 10 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
E. Poehler	11/28/2022	Updated content and formatting.	3.0
K. Dawson	8/30/2024	Updated figures to be clearer.	4.0

Chapter 11: Weighting and Estimation

11.1 Overview

In general, the Census Bureau produces and publishes estimates for the same set of statistical, legal, and administrative entities as the previously published Census long form: the nation, states, American Indian and Alaska Native (AIAN) areas, counties (*municipios* in Puerto Rico), minor civil divisions (MCDs), incorporated places, and census tracts, among others (see Chapter 14, “Data Dissemination”). The Census Bureau publishes up to three sets of estimates for a geographic area depending on its total population.

- For all statistical, legal, and administrative entities, including census tracts, block groups, and small incorporated places, such as cities and towns, the Census Bureau publishes 5-year estimates based on data collected during the 60 months of the five most recent calendar years.
- For geographic entities with populations of at least 20,000, the Census Bureau will also publish 1-year Supplemental Estimates based on data collected during the 12 months of the most recent calendar year. These tables are generally less detailed and more robust than the standard 1-year tables.
- For geographic entities with populations of at least 65,000, the Census Bureau also publishes single-year estimates based on data collected during the 12 months of the most recent calendar year.

The basic estimation approach is a ratio estimation procedure that assigns two sets of weights: a weight to each sample person record, both household and group quarters (GQ) persons, and a weight to each sample housing unit (HU) record. As with most household surveys, weights are used to bring the characteristics of the sample more into agreement with those of the full population by compensating for differences in sampling rates across areas, differences between the full sample and the interviewed sample, and differences between the sample and independent estimates of basic demographic characteristics (Alexander, Dahl, & Weidman, 1997).

In particular, the ACS uses ratio estimation to take advantage of independent population estimates by sex, age, race, and Hispanic origin, as well as estimates of total HUs produced by the Population Estimates Program (PEP) of the Census Bureau. This increases the precision of the estimates and corrects for under- or over-coverage by geography and demographic detail. This method also produces ACS estimates consistent with the population estimates by these characteristics and the estimates of total HUs for each county in the United States.

For any given geographic area, a characteristic total is estimated by summing the weights assigned to the people, households, families, or HUs possessing the characteristic. Estimates of population characteristics are based on the person weight. Estimates of family, household, and HU characteristics are based on the HU weight.

This chapter describes the weighting methodology used for the 2022 data year, although much of the methodology has been unchanged for many years. Sections 11.2–11.6 describe the single-year weighting and estimation methodology for calculating person weights for the GQ person records as implemented for the 2011 ACS forward. This weighting for GQ persons is done independently of the weighting for HUs. Sections 11.7–11.10 describe the single-year weighting methodology for calculating HU weights and person weights for the household sample records for the 2021 ACS forward. The weighting for household persons makes use of the GQ person weights so that the household and GQ person weights can be combined to produce estimates of the total population. While the methodology for the multiyear weighting is largely the same as the single-year weighting methodology, Section 11.11 outlines where the 5-year weighting methodology differs from the single-year methodology.

11.2 ACS GQ Person Weighting—Overview

Since the 2006 data collection year, estimates from the ACS have included data from both people living in HUs and GQs. The weighting and estimation methodology for GQs significantly changed in 2011. Readers interested in the methodology used before 2011 should reference the 12/2010 revision of this chapter posted on the ACS website. The new methodology was designed to address a significant limitation of the current sample design and the previous weighting methodology. Because of constraints on both sample size and budget, the sample design was optimized at the state level rather than the small area level, as is the case for the HU sample. In addition, the lack of independent GQ population estimates at the substate level led to the decision to optimize the weighting at the state level as well, to support the GQ products that are released at that level. The trade-off, however, was increased substate variation in both the estimate of the total GQ population and the characteristics of that population.

As a result of this variation, there were many counties and tracts that did not have GQ representation even with the 5-year estimates (Asiala, Beaghen, & Navarro, 2011). This variation was substantial enough to impact the estimates of the characteristics of the total resident population for the substate areas, including counties (Beaghen & Stern, 2009).

To address this limitation, a new GQ estimation methodology was developed and implemented with the 2011 data products. At its core is a mass imputation procedure whereby whole person records taken from the interviewed sample are copied (i.e., imputed) into not-in-sample GQs. By doing so, the GQ estimates better reflect the substate distribution of the GQs present on the sampling frame and reduce the variability in the substate estimates.

This estimation methodology has four basic components:

- Construct enhanced GQ imputation frame
- Select donors for whole person record imputation into select not-in-sample GQs
- Weighting

- Construct the post-imputation microdata

Each component is described in detail in the subsequent sections.

11.3 ACS GQ Person Weighting—Construct Imputation Frame

The goal of the enhanced GQ imputation frame is to start with the sampling frame for the given year (see Chapter 3 for more details) and update that frame with all information regarding the frame that is collected during the year. Most updates that are available come from sample cases that were fielded after the creation of the sampling frame. These updates include the number of persons residing in the GQ, GQ type, and identification of nonexistent or out-of-scope GQ facilities.

If only the size of the sampled facilities were updated on the enhanced frame, then the imputation into the not-in-sample facilities would not reflect the trends observed in the in-sample facilities. For example, if GQs that were in sample for a particular major type are tending to be larger than expected the same trend is expected to occur in the not-in-sample GQs for the same major type.

For this reason, the expected populations of the not-in-sample GQs are adjusted using the empirical relationship between the observed and expected population for the in-sample GQs. This adjustment is calculated within cells defined by major GQ type (see Table 11-1) by size class (small GQ, large GQ but not in sample with certainty, large GQ in sample with certainty). Note that, in addition to the standard seven major GQ types used for sampling and weighting, Federal Prisons are separated from other Correctional Institutions for the imputation because of differences in data sources and data collection methods. In the final weighting methodology, only the seven standard major types are used.

Table 11-1. Major GQ Type

Major GQ type	Definition	Institutional/Noninstitutional
0	Correctional Institutions—Federal Prisons	Institutional
1	Correctional Institutions—Other	Institutional
2	Juvenile Detention Facilities	Institutional
3	Nursing Homes	Institutional
4	Other Long-Term Care Facilities	Institutional
5	College Dormitories	Noninstitutional
6	Military Facilities	Noninstitutional
7	Other Noninstitutional Facilities	Noninstitutional

To improve the imputation, a flag is set on the enhanced frame to identify single-sex facilities. A facility is designated as a single-sex facility using either the Federal Bureau of Prisons demographics file, historical ACS sample interview data, or the most recent census for facilities with no historical ACS sample interview data. If the sex distribution for the GQ is at least 90 percent male or 90 percent female, then it is identified as a single-sex GQ, and only persons of that sex are imputed into that facility. All other GQs do not take sex into account when imputing records into the facility. For more information on creating the enhanced frame, see the detailed computer specifications (Jordan, 2023a).

11.4 ACS GQ Person Weighting—Select Donors for Imputation

The overarching goal of the imputation procedure is for the substate GQ estimates to better reflect the distribution present on the frame. To accomplish this, this goal is separated into two objectives:

- To establish representation of county by major type in the tabulations for each combination that exists on the frame for the 1- and 5-year data.
- To establish representation of tract by major type in the tabulations for each combination that exists on the frame for the 5-year data.

To accomplish these two objectives, while providing some limits on the degree of imputation required, the imputation is targeted toward two groups:

- All not-in-sample GQs that have an expected population of greater than 15 persons are selected to receive imputed whole person records.

- A subset of the not-in-sample GQs that have an expected population of 15 or fewer persons likewise are selected as necessary to achieve the two objectives stated above.

The larger GQs are selected with certainty to ensure a base distribution of the GQ estimates in the broadest set of geographic areas. Since these GQs contain the largest proportion of the GQ population, targeting these GQs to receive imputed records has the greatest visibility and impact on the estimates. The smaller GQs are selected only as needed to achieve the stated objectives for areas that do not meet those objectives after accounting for the sample GQs and the imputation into the not-in-sample larger GQs. Thus, if there is a tract by major type combination that exists on the enhanced frame that is comprised of entirely small GQs and no interviewed sample exists, then one small GQ is selected at random to represent the set of small GQs that exist for that combination.

Once the GQs are selected for imputation, the number of imputed person records to allocate to each GQ is determined. For the larger GQs, the number of imputed GQ person records is calculated as the larger of 2.5 percent of the expected population or one. For the smaller GQs, the number of imputed person records is the larger of 20 percent of the expected population or one.

Once the subset of not-in-sample GQs has been selected and the number of GQ imputed records to be assigned to the GQ has been computed, donors from the interviewed sample are selected. The selection process is implemented through an expanding search algorithm that first searches for a donor within county of the same specific GQ type. The specific types are a more detailed breakdown of the seven major types into more than 30 specific types. For example, the major type for correctional institutions is further classified into federal prisons, state prisons, jails, and half-way houses. If a donor is not found, the search expands to within county but of the same major GQ type. If a donor is still not found, the geographic region is expanded, and the process repeats until a donor is found. The levels of search are as follows:

- Within a geographic level, the search is first within the same specific type and then within the same major type
- Geographic levels expand as necessary in the following order: county, state, division, region, nation

To guard against the excessive reuse of donors, a particular donor is limited to being used three times within a single tract and five times within a single county. For more information on selecting donors, see the detailed computer specifications (Jordan, 2023c).

11.5 ACS GQ Person Weighting—Base Weights, Constraints, and Controls

The GQ weighting makes no distinction between the sampled and imputed GQ person records. The weighting has three basic steps: assigning an initial weight that reflects the observed

combined sampled/imputed rate, an adjustment of those weights to match substate totals from the enhanced frame, and a coverage adjustment at the state level.

Base Weights

The base weights (BW) for GQ persons are defined so that the sum of the base weights is equal to the domain that they represent. That domain differs depending on whether the GQ is small or large. Large GQs are self-representing, and thus, the sum of the base weights for the persons in that GQ is equal to the actual or adjusted expected population of the GQ. The base weights for all persons in the GQ are defined to be equal, and hence, for the i -th person in the GQ, BW is defined as follows:

$$\begin{aligned} BW_i &= \text{Actual or adjusted expected population, } N_p, \text{ of the GQ} \\ &\div \\ &\quad \text{Total number of sampled or imputed GQ person records, } n_p \\ &= \frac{N_p}{n_p} \end{aligned}$$

For the small GQs, the domain that the sum of the base weights is to represent is the total GQ population residing in small GQs for the tract by major type combination. Thus, the definition of BW is adjusted to account for the potential random selection of the small GQ with sampled or imputed data from the set of all small GQs in the tract by major type combination:

$$\begin{aligned} BW_i &= (\text{Number of small GQs, } N_{GQ}, \text{ on the frame for the tract by major type combination}) \\ &\div \\ &\quad \text{Number of small GQs, } n_{gq}, \text{ with either sampled or imputed GQ person records}) \\ &\times \\ &\quad (\text{Actual or adjusted expected population, } N_p, \text{ of the GQ}) \\ &\div \\ &\quad \text{Total number of sampled or imputed GQ person records, } n_p) \\ &= \frac{N_{GQ}}{n_{gq}} \times \frac{N_p}{n_p} \end{aligned}$$

Note that, as defined, the base weights also account for nonresponse within the GQ and within the tract (for small GQs).

Tract-level Constraint

The next steps are a series of constraints to ensure that the weighted totals of the sample and imputed records match the frame totals of the adjusted population. One reason why the sum of the initial weights may not match the frame totals is the fact that the base weights of the small GQs reflect the equal probability selection of the small GQs within a tract (for the imputed GQs).

While in expectation, the sum of the base weights may match the frame totals at the tract level, there may be a small deviance between the two because the first factor in the base weight calculation does not account for the population totals of the small GQs.

The tract-level constraint is thus defined as follows:

$$\begin{aligned}
 TRCON_{tg} &= \text{Sum of adjusted GQ population, } ADJEXPPOP_j, \text{ for all GQs, } j, \text{ on the enhanced frame within} \\
 &\quad \text{the tract } t \text{ and major type } g \\
 &\quad \div \\
 &\quad \text{Sum of base weights for all GQ person records, } i, \text{ sampled or imputed in tract } t \text{ and} \\
 &\quad \text{major type } g \\
 &= \frac{\sum_{GQ=j} ADJEXPPOP_j}{\sum_{Person=i} BW_i}
 \end{aligned}$$

The weight after the tract-level constraint, $WTRCON$, is achieved by multiplying the constraint factor by the base weight:

$$WTRCON_i = BW_i \times TRCON_{t(i)g(i)}$$

County-level Constraint

A second source of deviance between the weighted totals and the frame counts are ungeocoded GQs on the frame. These GQs do not have the census block codes required for tabulation but do have a county code assigned to them. For this reason, ungeocoded GQs are ineligible for imputation (they are still eligible for sampling, however, where they can be geocoded during data collection). To maintain consistency with the frame, the population total of all ungeocoded GQs on the frame are distributed to the geocoded GQs within county and major type via the county-level constraint. Note that in 2011 when the methodology was developed, the issue of ungeocoded records was relatively small because of the robustness of the sampling frame that was built from the 2010 Census. In future years, new updates to the frame that cannot be geocoded through automated means may make this constraint more important. The county-level constraint is defined as follows:

$$\begin{aligned}
 CTYCON_{cg} &= \text{Sum of adjusted GQ population for all GQs, } j, \text{ on the enhanced frame within the} \\
 &\quad \text{county } c \text{ and major type } g \\
 &\quad \div \\
 &\quad \text{Sum of the weight after the tract-level constraint for all GQ person records, } i, \\
 &\quad \text{sampled or imputed in county } c \text{ and major type } g \\
 &= \frac{\sum_{GQ=j} ADJEXPPOP_j}{\sum_{Person=i} WTRCON_i}
 \end{aligned}$$

The weight after the county-level constraint, $WCTYCON$, is achieved by multiplying the constraint factor by the weight after the tract-level constraint:

$$WCTYCON_i = WTRCON_i \times CTYCON_{c(i)g(i)}$$

State-level Constraint

The last constraint is designed to be a safety net in case there exists an ungeocoded GQ in a county where there are no geocoded GQs of the same major type. In that case, the population of that GQ is spread over all GQs of the same major type within the state. In practice, this is a relatively rare situation, and the constraint is very close to one.

The state-level constraint is defined as follows:

$$\begin{aligned} STCON_{sg} &= \text{Sum of adjusted GQ population for all GQs, } j, \text{ on the enhanced frame within the state } s \text{ and major type } g \\ &\div \\ &\quad \text{Sum of weight after the county-level constraint for all GQ person records, } i, \text{ sampled or imputed in state } s \text{ and major type } g \\ &= \frac{\sum_{GQ=j} ADJEXP_j}{\sum_{Person=i} WCTYCON_i} \end{aligned}$$

The weight after the state-level constraint, $WSTCON$, is achieved by multiplying the constraint factor by the weight after the county-level constraint:

$$WSTCON_i = WCTYCON_i \times STCON_{s(i)g(i)}$$

GQ Post-Stratification Adjustment to Controls

The final step in the GQ person weighting process is to apply the GQ Person Post-Stratification Factor ($GQPPSF$). The post-stratification cells are defined within state by GQ major type. This is consistent with the nature of the PEP GQ population estimates that are updated and maintained by major type. Using state as the level of geography for the post-stratification allows the GQ distribution on the frame to drive the substate distribution of the estimates.

All sample interviewed and imputed persons are placed in their appropriate cells. The $GQPPSF$ for each cell is then calculated:

$$\begin{aligned} GQPPSF_{sg} &= \text{PEP GQ population estimate for state } s \text{ and major type } g \\ &\div \\ &\quad \text{Sum of weight after the state-level constraint for GQ person records that are either interviewed sample or imputed in state } s \text{ and major type } g \\ &= \frac{GQPOP_{sg}}{\sum_{Person=i} WSTCON_i} \end{aligned}$$

where

$$GQPOP_{sg} = \text{PEP GQ population estimate for state } s \text{ and major type } g$$

The weight after post-stratification, $WGQPPSF$, is achieved by multiplying the post-stratification factor by the weight after the GQ state constraint adjustment:

$$WGQPPSF_i = WSTCON_i \times GQPPSF_{s(i)g(i)}$$

These weights are then rounded to form the final GQ person weights. For more information on creating the GQ person weights, see the detailed computer specifications (Lamas, 2023).

11.6 ACS GQ Person Weighting—Post-Imputation Microdata

The final person-level microdata are assembled by concatenating the sample interview microdata with the imputed records. The microdata for each imputed record is created by joining the geographic information of the GQ selected for imputation with the edited response information from the donor. For geographically-tied characteristics, some adjustments are necessary to preserve certain data relationships. For example, if the donor listed the same county for their residence one year ago as their current county of residence, the microdata for the imputed record is adjusted so that the same relationship is true for the donee record as was true for the donor record. Similar procedures are performed to preserve analogous relationships for place of work and journey to work. These steps help maintain the integrity of these characteristics for the imputed person records so that the estimates formed from the sampled and imputed records are not adversely impacted. For more information on creating the post-imputation microdata, see the detailed computer specifications (Jordan, 2023b).

11.7 ACS HU Weighting—Overview

The single-year weighting is implemented in three stages. In the first stage, weights are computed to account for differential selection probabilities based on the sampling rates used to select the HU sample. In the second stage, the weights of responding HUs are adjusted to account for nonresponding HUs. In the third stage, weights are controlled so that the weighted estimates of HUs and persons by age, sex, race, and Hispanic origin conform to estimates from the PEP of the Census Bureau at a specific point in time. The estimation methodology is implemented by “weighting area,” either a county or a group of less populous counties. Note that this section reflects the methodology as implemented after the telephone mode of data collection was discontinued. Readers interested in the methodology used before 2017 should reference the 01/2014 revision of this chapter posted on the ACS website.

11.8 ACS HU Weighting—Probability of Selection

The first stage of weighting involves three steps. In the first step, each HU is assigned a basic sampling weight that accounts for the sampling probabilities in both the first and second phases of sample selection. Chapter 4 provides more details on the sampling. In the second step, these sampling weights are adjusted to account for late mail or internet returns. In the third step, the weights are adjusted to reduce variability in the monthly weighted totals.

Sampling Weight

The first step is to compute the basic sampling weight for the HU based on the inverse of the probability of selection. This sampling weight is computed as a multiplication of the base weight (*BW*) and a Computer-Assisted Personal Interviewing (CAPI) subsampling factor (*SSF*). The *BW* for an HU is calculated as the inverse of the final overall first-phase sampling rate, which, for 2022, ranges from approximately 0.5 percent to 15 percent. HUs sent to CAPI are eligible to be subsampled (second-phase sampling) at rates generally ranging from 1-in-3 to 2-in-3 except for areas in remote Alaska and select American Indian areas which have a 100 percent CAPI sampling rate (see Chapter 4 for further details). Records that were not eligible for the CAPI subsample receive a CAPI *SSF* of 1.0. Those selected for the CAPI subsample, including those that returned a late mail or internet return, are assigned a CAPI *SSF* equal to the inverse of their (second phase) subsampling rate. This is a change from years before 2020 when late mail or internet return returns had been assigned a CAPI *SSF* equal to 1.0 even if selected for CAPI. Those not selected for the CAPI subsample receive a factor of 0.0. There are HUs that were eligible for CAPI subsampling but were not selected for CAPI but still returned a late mail or internet return; these records receive a CAPI *SSF* of 1.0. This portion of the *SSF* assignment is unchanged from 2019.

The CAPI *SSF* is then used to calculate a new weight for every HU, the weight after the CAPI subsampling factor (*WSSF*). It is equal to the *BW* times the *SSF*. After each of the subsequent weighting steps, a new weight is calculated as the product of the new factor and the weight following the previous step. Table 11-2 summarizes the computation of the *WSSF* by weighting step and the sample disposition of HUs. Additional information can be found in the detailed computer specifications for the HU weighting (Albright, 2023).

Table 11-2. Computation of the Weight after CAPI Subsampling Factor (*WSSF*)

Weighting step	Sample Disposition for mail/internet respondent	Sample Disposition for CAPI sampled units	Sample Disposition for CAPI non-sampled units	Sample Disposition for CAPI eligible, non-sampled, but ultimately mail/internet respondent
Base Weight (<i>BW</i>)	$1 \div \text{OSR}$	$1 \div \text{OSR}$	$1 \div \text{OSR}$	$1 \div \text{OSR}$
CAPI Subsampling Factor (<i>SSF</i>)	1	$1 \div \text{SSR}$	0	1
Weight after Subsampling Factor (<i>WSSF</i>)	$1 \div \text{OSR}$	$(1 \div \text{OSR}) \times (1 \div \text{SSR})$	0	$1 \div \text{OSR}$

Notes: OSR = Original Sampling Rate for sample record; SSR = CAPI Subsampling Rate for sample record

Adjustment for Late Mail/Internet Returns

Even though the late mail or internet returns were not selected for the CAPI subsample, they still receive a non-zero *WSSF* weight since they did complete a return. This results in the total weight for all HUs after *WSSF* exceeding the total weight after the *BW*. The sample at this point can be broken up into two groups, eligible for CAPI subsampling and not eligible for CAPI subsampling. Since the late mail or internet returns are in the eligible for CAPI group, that is the part of the sample where the excess weight is contained. To correct for this, an *SSF* correction factor (*SSFCORR*) is calculated for the CAPI subsampled records as follows:

$$\begin{aligned}
 SSFCORR &= \frac{\text{(Total weight after CAPI subsampling adjustment factor of all HUs selected for CAPI subsample)} \\
 &\quad - \text{(Total weight after CAPI subsampling adjustment factor of all HUs not selected for CAPI subsample)}}{\text{Total weight after CAPI subsampling adjustment factor of all HUs selected for CAPI subsample}} \\
 &= \frac{\sum_{i \in \text{Selected for CAPI}} WSSF_i - \sum_{i \in \text{Not Selected for CAPI}} WSSF_i}{\sum_{i \in \text{Selected for CAPI}} WSSF_i}
 \end{aligned}$$

where

$$WSSF_i = \text{weight after the CAPI subsampling factor for HU } i$$

This adjustment factor is computed within each of the 2,175 ACS single-year weighting areas (either a county or a group of less populous counties). The index for weighting area is suppressed in this and all other formulas for weighting adjustment factors.

Certain types of HUs are excluded from the *SSFCORR* calculation as they are eligible for CAPI subsampling, but there is no mechanism where there can be a late mail or internet return. These include areas in remote Alaska and select American Indian areas that have a 100 percent CAPI sampling rate, as well as all HUs that were unmailable (see Chapter 4 for further details). These records all receive an *SSFCORR* of 1.0. All HUs that were not eligible for CAPI subsampling received an *SSFCORR* of 1.0. Table 11-3 summarizes the computation of the calculation of the weight after *SSF* correction factor.

Table 11-3. Computation of the Weight after SSF Correction Factor (*WSSFCORR*)

Interview status	<i>WSSFCORR_i</i>
Not eligible for CAPI subsampling	<i>WSSF_i</i>
Remote Alaska or American Indian areas Unmailable HUs	<i>WSSF_i</i>
Not Selected for CAPI subsampling	<i>WSSF_i</i>
Selected for CAPI subsampling	<i>WSSF_i × SSFCORR</i>

Variation in the Monthly Sample Factor

The goal of ACS estimation is to represent the characteristics of a geographic area across the specified period. For single-year estimates, this period is 12 months, and for 5-year estimates, it is 60 months. The annual sample is allocated into 12 monthly samples. The monthly sample becomes a basis for the operations of the ACS data collection, preparation, and processing, including weighting and estimation.

The data for HUs assigned to any sample month can be collected at any time during a 3-month period. For example, the households in the January sample month can have their data collected in January, February, or March. Each HU in a sample belongs to a tabulation month (the month the interview is completed). This is either the month the processing center checked in the completed mail questionnaire, the month the internet questionnaire is submitted, or the month the interview is completed by CAPI.

Because of seasonal variations in response patterns, the number of HUs in tabulation months may vary, thereby over-representing some months and under-representing other months in the single- and multiyear estimates. For the ACS to represent the time period across the entire year equitably, an even distribution of HU weights by month is desirable. To smooth out the total weight for all sample months, a variation in monthly response factor (*VMS*) is calculated for each month, *m*, as:

$$\begin{aligned}
 VMS_m &= \frac{\text{Total base weights of all HUs in that sample month}}{\text{Total weight after WSSF correction factor of all HUs interviewed in that sample month}} \\
 &= \frac{\sum_{i \in \text{Sample Month}(m)} BW_i}{\sum_{i \in \text{Interview Month}(m)} WSSFCORR_i}
 \end{aligned}$$

where

$$\begin{aligned}
 BW_i &= \text{base weight for sample HU } i, \\
 WSSFCORR_i &= \text{weight after the } WSSF \text{ correction factor for interviewed HU } i.
 \end{aligned}$$

Table 11-4a and Table 11-4b illustrate the computation of the *VMS* adjustment factor within a particular county for the numerator and denominator respectively. In this example, the total *BW* for each sample month is 100 (as shown in Table 11-4a). The total *WSSF* weight across modes within each month varies from 90 to 115 (as shown in the first line of Table 11-4b). The *VMS* factors are then computed by month as the ratio of the total *BW* to the total *WSSF* (as shown in the final line of Table 11-4b).

Table 11-4a. Example of Computation of Variation in Monthly Response Factor (*VMS*)—Numerator: Sum of Base Weight Across Sample Month

Numerator	March	April	May	June	July
Total base weight (<i>BW</i>) across released samples	100	100	100	100	100

Table 11-4b. Example of Computation of Variation in Monthly Response Factor (*VMS*)—Denominator: Sum of Weight After CAPI Subsampling Across Interview Month

Denominator and Components	March	April	May	June	July
Denominator: Total Weight after CAPI Subsampling (<i>WSSF</i>) across modes	115	95	90	100	105
Total weight of mail returns from three panels	55 Jan-Mar sample	45 Feb-Apr sample	40 Mar-May sample	45 Apr-Jun sample	50 May-Jul sample
Total weight of internet returns from three panels	30 Jan-Mar sample	25 Feb-Apr sample	30 Mar-May sample	30 Apr-Jun sample	25 May-Jul sample
Total weight of CAPI sample	30 Jan sample	25 Feb sample	20 Mar sample	25 Apr sample	30 May sample
<i>VMS</i> Adjustment Factor (numerator from 11-3a divided by denominator above)	$100 \div 115$	$100 \div 95$	$100 \div 90$	$100 \div 100$	$100 \div 105$

The weight after the variation of monthly response adjustment (*WVMS*) is the product of the weight after the *WSSF* correction factor (*WSSFCORR*) and the variation of the monthly response factor (*VMS*). When the *VMS* factor is applied, the total weight across all HUs tabulated in a sample month is equal to the total base weight of all HUs selected in that month's sample. The result is that each month contributes approximately 1/12 to the total single-year estimates. In other words, the single-year estimates of ACS characteristics are a 12-month average without over- or under-representing any single month because of variation in monthly response.

11.9 ACS HU Weighting—Noninterview Adjustment

The noninterview adjustment changed for the 2017 data year moving forward. Readers who are interested in the methodology used before 2017 should reference the 01/2014 revision of this chapter posted on the ACS website. The new methodology simplifies the adjustments done in prior years without impacting data quality (Gutentag, Castro, & Asiala, 2018).

During data collection, nothing new is learned about the HU or person characteristics of noninterviewed HUs, so only characteristics known at the time of sampling can be used in adjusting for them. In other surveys and censuses, characteristics that have been shown to be related to HU response include census tract and building type grouped into single- versus multi-unit structure (Weidman, Alexander, Diffendal, & Love, 1995). It is expected that many other characteristics may be correlated with these characteristics given the local nature of the adjustment. The noninterview adjustment step is applied to all HUs interviewed by any mode—mail, internet, or CAPI.

Note that vacant units and ineligible units, such as deletes, are excluded from the noninterview adjustment.⁵⁹ The weight corresponding to these HUs remains unchanged during this stage of the weighting process since it is assumed that all vacant units and deletes are properly identified in the field and, therefore, are not eligible for the noninterview adjustment. The weighting adjustment is carried out only for the occupied, temporarily occupied (those HUs that are occupied but whose occupants do not meet the ACS residency criteria), and noninterviewed HUs. After completion of the adjustment to the weights of the interviewed HUs, the noninterviewed HUs can be dropped from subsequent weighting steps; their assigned weights are equal to 0.

Calculation of the Noninterview Adjustment Factor

In this step, all HUs are placed into adjustment cells based on the cross-classification of building type (single- versus multi-unit structures) and census tract. If a cell contains fewer than 10 interviewed HUs, it is collapsed with an adjoining tract until the collapsed cell meets the minimum size of 10.⁶⁰ Cells with zero noninterviews are not collapsed, regardless of size, unless they are forced to collapse with a neighboring cell that fails the size criterion. The noninterview adjustment factor (*NIF*) for each eligible cell, *c*, is:

⁵⁹ Deletes or out-of-scope addresses fall into three categories: (1) addresses of living quarters that have been demolished, condemned, or are uninhabitable because they are open to the elements; (2) addresses that do not exist; and (3) addresses that identify commercial establishments, units being used permanently for storage, or living arrangements known as group quarters.

⁶⁰ Data are sorted by the weighting area, building type, and tract. Within a building type, a tract that has 10 or more responses is put in its own tract. A tract that has no nonresponses and some responses (even though the total is fewer than 10) is put in its own tract. A tract that has nonresponses and fewer than 10 responses is collapsed with the next tract. If the final tract needs to be collapsed, it is collapsed with the previous tract.

$$\begin{aligned}
 NIF_c &= \text{Total HU weight in cell } c \text{ after variation in monthly response adjustment factor of} \\
 &\quad \text{interviewed occupied and temporarily occupied HUs and noninterviewed HUs} \\
 &\quad \div \\
 &\quad \text{Total HU weight after variation in monthly response adjustment factor of} \\
 &\quad \text{interviewed occupied and temporarily occupied HUs} \\
 &= \frac{\sum_{i \in \text{Interviews}(c)} WVMS_i + \sum_{i \in \text{Noninterviews}(c)} WVMS_i}{\sum_{i \in \text{Interviews}(c)} WVMS_i}
 \end{aligned}$$

where

$$WVMS_i = \text{Adjusted HU weight after the variation in monthly response adjustment for HU } i$$

All occupied and temporarily occupied interviewed HUs are adjusted by this noninterview factor. Vacant and deleted HUs are assigned a factor of 1.0, and noninterviews are assigned a factor of 0.0. The computation of the weight after the noninterview adjustment factor is summarized in Table 11-5 below.

Table 11-5. Computation of the Weight after the Noninterview Adjustment (*WNIF*)

Interview status	<i>WNIF_i</i>
Occupied or temporarily occupied HU	$WVMS_i \times NIF_{c(i)}$
Vacant or deleted HU	$WVMS_i$
Noninterviewed HU	0

11.10 ACS HU Weighting—Housing Unit and Population Controls

This stage of weighting forces the ACS total HU and person weights to conform to estimates from the Census Bureau PEP. The PEP of the Census Bureau annually produces estimates of population by sex, age, race, and Hispanic origin, as well as total HUs for each county in the United States as of July 1. They also produce annually updated estimates of the total population for incorporated places and minor civil divisions (MCDs) as of July 1. The ACS estimates are based on a probability sample and may vary from their true population values due to sampling and nonsampling error (see Chapters 12 and 14). In addition, it can be seen from the formulas for the adjustment factors in the previous two sections that the ACS estimates also vary based on the combination of interviewed and noninterviewed HUs in each tabulation month. As part of the process of calculating person weights for the ACS, estimates of totals by sex, age, race, and Hispanic origin are controlled to be equal to population estimates by weighting area. There are two reasons for this: (1) to reduce the variability of the ACS HU and person estimates, and (2) to reduce bias due to under-coverage of HUs and the people within them in household surveys. The bias that results from missing these HUs and people is partially corrected by using these controls (Alexander, Dahl, & Weidman, 1997).

The assignment of final weights involves the calculation of three factors based on the HU and population controls. The first adjustment involves the independent HU estimates. A second and separate adjustment relies on the independent population estimates. The final adjustment is implemented to achieve consistency between the ACS estimates of occupied HUs and householders.

Models for PEP Estimates of HUs and Population

The Census Bureau produces estimates of total HUs for states and counties as of July 1 on an annual basis. The estimates are computed based on a model:

$$HU2X = HU20 + (NC2X + NM2X) - HL2X$$

where the suffix “X” indicates the year of the housing unit estimates, and HU2X = Estimated 202X HUs

HU20	=	Geographically updated 2020 Census HUs
NC2X	=	Estimated new residential construction, April 1, 2020 to July 1, 202X
NM2X	=	Estimated new residential mobile home placements, April 1, 2020 to July 1, 202X
HL2X	=	Estimated residential housing loss, April 1, 2020 to July 1, 202X.

More detailed background on the current methodology used for the HU estimates can be found on the Census Bureau’s website (U.S. Census Bureau, 2023a).

The Census Bureau also produces population estimates as of July 1 on an annual basis. Those estimates are computed based on the following simplified model:

$$P1 = P0 + B - D + NDM + NIM + NMM$$

where

P1	=	population at the end of the period (current estimate year)
P0	=	population at the beginning of the period (previous estimate year)
B	=	births during the period
D	=	deaths during the period
NDM	=	net domestic migration during the period
NIM	=	net international migration during the period
NMM	=	net military movement during the period

In practice, the model is considerably more complex in leveraging the best information available from multiple sources. More detailed background on the current methodology used for the HU estimates can be found on the Census Bureau’s website (U.S. Census Bureau, 2023b).

Production of the population estimates for Puerto Rico is limited to sex-age distribution by *municipio* and does not contain any race or Hispanic origin detail. For this reason, estimates of totals by *municipio*, sex, and age for the PRCS are controlled to be equal to the population estimates. Currently, there are no HU estimates available from the PEP for Puerto Rico and there are also no subcounty estimates available.

Creation of the Subcounty Control Areas

The subcounty control areas are formed to give both MCDs and incorporated places the benefit of using subcounty controls. To achieve this balance, the basic units for forming the subcounty areas are the county/MCD/place intersections or parts where the “balance of county” is also considered as another fundamental subcounty area. Note that outside of the strong and weak MCD states (U.S. Census Bureau, 2023c) for which the PEP produce total population estimates this defaults to simply the county/place parts. These subcounty areas are then combined until all subcounty areas within a county have a total population of 24,000 or greater. If it is not possible to partition a county into two or more subcounty areas of this size, then the subcounty area is simply coexistent with the county.

Calculation of Housing Unit Post-Stratification Factor

Note that both HU and population estimates used as controls have a reference date of July 1 which means that the 12-month average of ACS characteristics is controlled to the population with the reference date of July 1. If person weights are controlled to the population estimates as of that date, it is logical that HUs also are controlled to those estimates to achieve a consistent relationship between the two totals.

The housing unit post-stratification factor (*HPF*) is employed to adjust the estimated number of ACS HUs by subcounty area within a weighting area to agree with the PEP estimates. For the subcounty area, *sa*, within a weighting area, this factor is:

$$\begin{aligned}
 HPF_{sa} &= \text{PEP HU estimate for the subcounty area, } sa \\
 &\div \\
 &\quad \text{Total HU weight after the noninterview adjustment across all interviewed occupied,} \\
 &\quad \text{interviewed temporarily occupied, and vacant HUs } i \text{ for the subcounty area, } sa, \\
 &= \frac{HU_{sa}}{\sum_{i \in \text{Occupied and Vacant Interviews (sa)}} WNIF_i}
 \end{aligned}$$

Note that if the PEP HU subcounty estimates are summed across all subcounty areas within a county, the total is consistent with the PEP county-level HU estimates. The denominator of the *HPF* formula aggregates the adjusted HU weight after the noninterview factor adjustment (*WNIF*) across 12 months for the interviewed occupied, temporarily occupied, and vacant HUs. All HUs are then adjusted by this HU post-stratification factor. Therefore, $WHPF = WNIF \times HPF$, where *WHPF* is the adjusted HU weight after the HU post-stratification factor adjustment.

Calculation of Person Weights

The next step in the weighting process is to assign weights to persons via a three-dimensional raking-ratio estimation procedure. This is done so that (1) the estimate of total population for the subcounty areas conforms to the population estimates; (2) the combined estimates of spouses and unmarried partners conform to the combined estimate of married-couple and unmarried-partner households and the estimate of householders conforms to the estimate of occupied HUs; and (3) the estimates for certain demographic groups are equal to their population estimates.

The population estimates used for the household person weighting are derived from the PEP estimates of total resident population by first starting with the PEP total and then subtracting the corresponding ACS GQ estimate for that same population. For example, the control total used for county household population is derived by subtracting the ACS GQ estimate of total GQ population from the PEP estimate of total resident population. By doing so, the ACS estimate of total resident population (formed by summing the household and GQ population) conforms to the PEP estimate for the same population. This procedure is also used to derive the controls for subcounty areas and demographics.

Each person in an interviewed occupied HU is assigned an initial person weight equal to the HU weight after the HU post-stratification factor is applied (*WHPF*). Next, there are three steps of ratio adjustment. The first step uses one cell per subcounty control area defined within the weighting area. The second step uses four cells to classify persons by spousal relationship, householder, and non-householder. The third step uses up to 156 cells defined by race/Hispanic origin, sex, and age. The steps are defined as follows:

Step 1: Subcounty Population Controls. All persons are assigned to one subcounty area within the weighting area. The marginal totals (i.e., the single-dimension control totals for a raking matrix) are equal to the derived household population control totals for the subcounty area, as described above.

Step 2: Spouse / Unmarried Partner and Householders. All persons are placed into one of four cells:

1. Persons who are the primary person in a two-partner relationship—all householders in a married-couple or unmarried-partner household,
2. Persons who are the secondary person in a two-partner relationship—all spouses or unmarried partners in those same households, or
3. Persons who are a householder but do not fit into the first cell, or
4. Balance of population—all persons not fitting into the first three cells.

The marginals for the first two columns of cells are both equal to the estimate of married-couple plus unmarried-partner households using the *WHPF* weight. The marginal for the third column is the estimate of occupied HUs using the *WHPF* weight minus the marginal for the first column.

In this manner, the estimate of households, equal to the first column plus the third column of cells, is controlled to the estimate of occupied HUs. The marginal for the fourth column is equal to the derived household population estimate minus the sum of the marginals used for the other three columns of cells. In this manner, the estimate of total household population is controlled to the derived population estimates.

Step 3: Race-Hispanic Origin/Sex/Age. The third step assigns all persons to one of up to 156 cells: six classifications of race-Hispanic origin by sex by 13 age groups. The marginals for these rows at the weighting area level come from the derived population estimates (PEP total resident population minus the ACS GQ estimate for the same characteristic cross-classification). Some weighting areas do not have sufficient sample to support all 156 cells; in these cases, some collapsing is necessary. This collapsing is done before the raking and remains fixed for all iterations of the raking.

Race and Hispanic origin are combined to define six unique race-ethnicity groups consistent with those used in weighting the 2000 Census long form. These groups are created by crossing “non-Hispanic” with the five major single-race groups, plus the group of all Hispanics regardless of race. The race-ethnicity groups are:

1. Non-Hispanic White
2. Non-Hispanic Black
3. Non-Hispanic American Indian and Alaskan Native (AIAN)
4. Non-Hispanic Asian
5. Non-Hispanic Native Hawaiian or Pacific Islander (NHPI)
6. Hispanic

The assignment of a single major race to a person can be complicated because people can identify themselves as being of multiple races. People responding either with multiple races or “Other Race” are included in one of the six race-ethnicity groups for estimation purposes only. This reduces the likelihood of creating many sparse cells which could cause extreme adjustments while grouping race-ethnicity groups of similar coverage rates together. Subsequent ACS tabulations are based on the full set of responses to the race question.

Initial estimates of population totals are obtained from the ACS sample for each of the weighting area/race-ethnicity groups. These estimates are calculated based on the initial person weight of *WHPF*. The demographic control totals are derived from the Census Bureau’s PEP total resident population estimate minus the ACS GQ estimate for each weighting area/race-ethnicity group. These derived population estimates are used as control totals in the remaining steps of the ACS household person weighting.

The initial sample and population estimates for each weighting race-ethnicity group are tested against a set of criteria that require a minimum of 10 sample people and a ratio of the population control to the initial sample estimate that is between 1/3.5 and 3.5. This is done to reduce the

effect of large weights on the variance of the estimates. If there are weighting race-ethnicity groups that do not satisfy these requirements, they are collapsed until all groups satisfy the collapsing criteria. Collapsing decisions are made following the specified order shown below.

1. If the requirements are not met for Hispanics, then Hispanics are collapsed with the largest non-Hispanic non-White group.
2. If the requirements are not met for any non-Hispanic non-White group, it is collapsed with the largest (prior to collapsing) non-Hispanic non-White group.
3. If the largest collapsed non-Hispanic non-White group still does not meet the requirements, it is collapsed with the surviving non-Hispanic non-White groups in the following order until the requirements are met: Black, American Indian and Alaskan Native, Asian, and Native Hawaiian or Pacific Islander.
4. If all non-Hispanic non-White groups have been collapsed together the collapsed group still does not meet the requirements, it is collapsed with the non-Hispanic White group.
5. If the requirements are not met for the non-Hispanic White group, then it is collapsed with the largest non-Hispanic non-White group.
6. If the requirements are not met when all non-Hispanic race groups are combined then all weighting race-ethnicity groups are collapsed together and the collapsing is complete.

Within each collapsed weighting race-ethnicity group, the persons are placed in sex-age cells formed by crossing sex by the following 13 age categories: 0–4, 5–14, 15–17, 18–19, 20–24, 25–29, 30–34, 35–44, 45–49, 50–54, 55–64, 65–74, and 75+ years. If necessary, these cells are also collapsed to meet the requirements of the same sample size and a ratio between (1/3.5) and 3.5. The goals of the collapsing scheme are to keep children ages 0–17 together whenever possible by first collapsing across sex within the first three age categories. In addition, the collapsing rules keep men ages 18–54, women ages 18–54, and people 55+ in separate groups by collapsing across age.

The initial sample cell estimates are then scaled and rescaled via iterative proportional fitting, or raking, so that the sum in each row or column consecutively agrees with the row or column household estimate (Steps 1 & 2) or population estimate (Step 3). This procedure is iterated a fixed number of times, and final person weights are assigned by applying an adjustment factor to the initial weights.

The scaling and rescaling between rows and columns is referred to as an iteration of raking. An iteration of raking consists of the following three steps. (The weighting matrix is included to facilitate the discussion below.) For clarity, the three-step process has been split into two tables, Table 11-6 and Table 11-7.

Table 11-6. Steps 1 and 2 of the Weighting Matrix

Step 2: Cell	Step 1 Cell: Subcounty Area #1	...	Step 1 Cell: Subcounty Area #N	Step 2: Control
Householder in two-partner relationship				Survey estimate of married-couple and unmarried-partner households
Spouse / unmarried partner in two-partner relationship				Survey estimate of married-couple and unmarried-partner households
Householder not in two-partner relationship				Survey estimate of all single-headed households
Balance of population				Derived total household population estimate minus the sum of the other three controls
Step 1: Control	Derived household population estimate for Subcounty Area #1	...	Derived household population estimate for Subcounty Area #N	Derived total household population estimate for weighting area

Step 1. At this step, the initial person weights are adjusted to make the sum of the weights of all household persons equal to the derived household population controls for the defined subcounty control area.

Step 2. The Step 1 adjusted person weights are adjusted to make both the sum of the weights of householders in married-couple or unmarried-partner households and the sum of the weights of their spouses or unmarried partners equal to the survey estimate of married-couple and unmarried-partner households. In addition, the weights are adjusted so that the sum of the weights of householders not in a two-partner relationship is equal to the survey estimate of other single-headed households. The survey estimate is calculated using the HU weight after the HU post-stratification factor adjustment for both constraints. Lastly, the weights of all other persons are adjusted to make the sum of all person weights equal to the derived household population estimates.

Step 3. The Step 2 adjusted person weights are adjusted a third time by the ratio of the population estimates of race-Hispanic origin/age/sex groups to the sum of the Step 2 weights for sample people in each of the demographic groups described previously.

The three steps of ratio adjustment are repeated in the order given above until the predefined stopping criterion is met. The stopping criterion is a function of the difference between Step 2 constraints and their corresponding marginals calculated using the Step 3 weights. Internal research has shown that Step 1 converges quickly and that the driver for convergence of the

weights being within tolerance of the constraints for each dimension is Step 2 (they will be equal for Step 3 since that is the last step). The raking for a weighting area is stopped when each of the marginals for Step 2 is within 0.01 percent of the constraint if fewer than 20 iterations have been run. If the weighting area has not converged before 20 iterations, the stopping criterion is relaxed to 0.1 percent, and the raking continues. If the threshold continues not to converge, a maximum iteration limit of 40 is enforced. Approximately 95 percent of the weighting areas converge within 20 or fewer iterations. The weights obtained from Step 3 of the final iteration are the final person weights, *WPPSF* (the weight after the person post-stratification).

For review (and historical comparison) purposes, we define the person post-stratification factor (*PPSF*) as the net adjustment that results from the entire ratio-raking estimation process. It is calculated as follows:

$$\begin{aligned} PPSF_i &= \text{final person weight after raking} \div \text{the initial person weight} \\ &= \frac{WPPSF_i}{WHPF_i} \end{aligned}$$

The factor is calculated and applied to each person so that their weights become the product of their initial weights and the factor.

Table 11-7. Steps 2 and 3 of the Weighting Matrix

Step 3: Cells	Step 2 Cell: Householder in two-partner relationship	Step 2 Cell: Spouse / unmarried partner in two- partner relationship	Step 2 Cell: Householder not in two-partner relationship	Step 2 Cell: Balance of population	Step 3: Controls
Non-Hispanic White by 26 age-sex cells					Derived household population estimate for demographic cell
Non-Hispanic Black by 26 age-sex cells					...
Non-Hispanic AIAN by 26 age-sex cells					...
Non-Hispanic Asian by 26 age-sex cells					...
Non-Hispanic NHPI by 26 age-sex cells					...
Hispanic by 26 age-sex cells					Derived household population estimate for demographic cell
Step 2: Controls	Survey estimate of married-couple and unmarried-partner households	Survey estimate of married-couple and unmarried-partner households	Survey estimate of all single-headed households	Derived population estimate minus the sum of the other three controls	Derived total household population estimate for weighting area

Calculation of Final Housing Unit Factors

Before the calculation of person weights, each HU has a single weight which is independent of the characteristics of the persons residing in the HU. After the calculation of person weights, a new HU weight is computed that takes into account the characteristics of the householder in the HU. In each interviewed occupied HU, the householder is identified as one of the persons who rents or owns the HU. Adjustment of the HU weight to account for the householder characteristics is done by assigning a householder factor (*HHF*) for an HU equal to the person post-stratification factor (*PPSF*) of the householder. Their *PPSF*s give an indication of relative coverage for households whose householders have the same demographic characteristics. The

HHF adjustment uses this information to adjust for the resultant bias. Note that because the total weight for occupied units was constrained in the person raking methodology, this adjustment impacts the distribution of the weight of the housing unit weights to reflect the relative coverage rates of different householders, but it generally does not perform any further adjustments for overall coverage. The adjustment for overall coverage for all housing units was addressed in the *HPF* adjustment. Vacant HUs are given an *HHF* of 1.0 because they have no householders.

The adjusted HU weight accounting for householder characteristics is computed as a multiplication of the adjusted HU weight after the HU post-stratification factor adjustment (*WHPF*) with the householder factor (*HHF*). Therefore, $WHHF = WHPF \times HHF$, where *WHHF* is the adjusted HU weight after the householder factor adjustment. The HU weight after the householder factor adjustment becomes the final HU weight.

The ACS weighting procedure results in two separate sets of weights: one for HUs and one for persons residing within HUs. However, since the housing unit weight is equal to the person weight of the householder, the survey produces logically consistent estimates of occupied housing units, households, and householders. With this weighting procedure, the survey estimate of total HUs differs slightly from the PEP total housing unit estimates but is typically within a tenth of a percent at the county level.

11.11 Multiyear Estimation Methodology

The multiyear estimation methodology involves reweighting the data for each sample address in the 5-year period and is not just a simple average of the 1-year estimates. The weighting methodology for the multiyear estimation is very similar to the methodology used for the single-year weighting. Thus, only the differences between the single- and multiyear weighting are described in this section.

Pooling the Data

The data for all sample addresses over the multiyear period are pooled together into one file. The single-year base weights are then adjusted by the reciprocal of the number of years in the period so that each year contributes its proportional share to the multiyear estimates. For the 5-year weighting, the base weights are thus all divided by five.

The interview month assigned to each address is also recoded so that all the data from the entire period appears as though it came from a one-year period. For example, in the 2018–2022 5-year weighting, all addresses that were originally assigned an interview month of January 2018, 2019, 2020, 2021, or 2022 are assigned the common interview month of January. Thus, when the weighting is performed, those records are all treated as though they come from the same month for the *VMS* adjustment. This should better preserve the seasonal trends that may be present in the population as captured by the ACS.

Geography

The geography for all sample addresses in the period is updated into the common geography of the final year. This allows the data to be tabulated in a consistent, constant geography that is the most recent and likely most relevant to data users. When tabulating estimates for an area, all interviews from the period that are considered to be inside the boundaries of that area in the final year of the period are included in the estimates regardless of if they were considered to be inside the boundaries for that area at the time of interview. As a by-product of this methodology, the ACS is also able to publish multiyear estimates for newly created places or counties that did not exist when the interviews for the addresses in that place or county were collected.

Derivation of the Multiyear Controls

Since the multiyear estimate is an estimate for the period, the controls are not those of a particular year but are the average of the annual independent population estimates over the period. The Population Estimates Program refreshes its entire time series of estimates going back to the previous census each year using the most current data and methodology. Each of these time series is considered a “vintage.” To use the best available population estimates as controls, the ACS multiyear weighting uses the population estimates of the most recent vintage for all years in the period to derive the multiyear controls.

These derived estimates are created for the housing unit, group quarters population, and total population for use as controls in the multiyear weighting. The derived county-level housing unit estimates are the simple average across all years in the period. Since the average is typically not an integer, the result is rounded to form the final estimate. Likewise, the derived group quarters population estimates for state by major type group are the simple average across all years in the period. Those averages are then control rounded so that the rounded state average estimate is within 1 of the unrounded estimate. Finally, the derived total population estimates by race, ethnicity, age, and sex are averaged across all years in the period and control rounded to form the final derived estimates. This is done before the collapsing of the estimates into the 156 cells per weighting area needed for the demographic dimension of the household person weighting as described in the single-year person weighting section.

The weighting areas used for the multiyear estimation are generally smaller than those used for the single-year estimation. They are still formed by complete counties or aggregations of counties, and they must meet a threshold of 400 unweighted person interviews at the time of their formation. In addition, for the five-year estimation, the weighting area must have a minimum population of 2,500.

However, since there is no publication threshold for the five-year data product, there are counties that are not their own weighting area, and therefore, greater differences between the ACS and PEP estimates of total population may exist, particularly for counties smaller than 2,500 in total

population. For the formation of the subcounty control areas, the single-year threshold of 24,000 in total population is reduced for the five-year weighting to 2,500.

Model-assisted Estimation

Once the data are pooled and put into the geography of the final year, they are weighted using the single-year weighting methodology through the *NIF* adjustment. It is after this adjustment that the only multiyear-specific weighting step is implemented, the model-assisted estimation procedure. An earlier research project (Starsinic, 2005) compared the variances of ACS tract-level estimates formed from the 1999–2001 ACS to the variances of the 2000 Census long-form estimates. The results of that research showed that the variances of the ACS tract-level estimates were higher in relation to the long form than was expected based on sample size alone. The primary source of that increased variance was attributed to the lack of ACS subcounty controls at the tract-level or lower as was used for the long form.

Several options were explored on how the ACS might improve our estimates of variance for subcounty estimates. One option considered was to use the ACS sampling frame counts as subcounty controls. Other options explored ways to create subcounty population controls, including tract-level population controls. The final approach that was chosen introduces a model-assisted estimation step into the multiyear weighting that makes use of both the sampling frame counts and administrative records to reduce the level of variance in the subcounty estimates (Fay, 2006). An important feature of the model-assisted estimation procedure is that the administrative record data are not used directly to produce ACS estimates. The administrative record data are only used to help reduce the level of variance. This is an important property of this step as the coverage for administrative records for small areas such as tracts can vary significantly year-to-year. Had we pursued using the administrative record directly to impact the estimates, this variability in coverage could have led to unpredictable impacts on the quality of the ACS data. By using the administrative record data for variance reduction only, the impact of this coverage on the ACS is principally on the degree of variance reduction that is achieved and not on the directly on estimates themselves. The published ACS estimates are still formed from weighted totals of the ACS survey data.

The model-assisted estimation step is calculated at the at the tract level for the ACS 5-year data. The entire model-assisted estimation process is summarized in these steps.

1. Create frame counts for tracts that contain at least 300 housing unit addresses.
2. Link the administrative records to the ACS sampling frame (the Master Address File or MAF) and drop administrative records that cannot be linked.
3. Form unweighted geographic totals of the linked administrative record characteristics.
4. Apply the *WNIF* weights at the housing-unit level to the linked administrative records that fall into the ACS sample. The weighted estimates at this step represent (essentially) unbiased estimates of the unweighted totals in Step 3.

5. Using generalized regression estimation, fit a model to calibrate the ACS weights so that the weighted totals from the linked ACS records match the unweighted totals from Step 3 and so that the weighted ACS estimate of HUs match the frame totals in Step 1. The categories of the variables considered in the regression are collapsed or removed as necessary to fit a good model.
6. Proceed with the remaining steps of the ACS weighting starting with the *HPF* adjustments, including the person weighting using the derived multiyear controls as described in the preceding section.

Frame Counts: The base weights (*BW*), which reflect the sampling probabilities of selection, should sum to the count of records on the sampling frame at the county and, generally, the subcounty level. However, after the weighting steps that follow the base weight assignment through the noninterview adjustment, the weighted subcounty distribution of the interviewed sample cases can deviate from the original frame distribution. This can impact both the subcounty estimates and the variances on those estimates. The use of the frame counts reestablishes the original subcounty distribution of housing unit addresses on the frame in the weighted sample. This control to the frame counts is the simplest model and is used if a model with administrative record data cannot be estimated. Otherwise, it is one part of the entire calibration performed in this step.

Link Administrative Records to Frame: The administrative record data used for this step is created from linking two primary files maintained by the Economic Reimbursable Division at the Census Bureau. The first file includes person characteristics and has been created from a combination of Social Security and census information. The second file uses administrative records to identify all possible addresses of the persons on the first file. A merged file is then created which contains only the age, sex, race, and Hispanic origin of each person and an identifier that links that person to the best address available in the MAF via a Master Address File ID (MAFID). No other characteristics or publicly identifiable information are present on the file. This file is updated annually to account for new births, death information, and for updated address information.

Administrative Universe Counts: For each MAFID, it is possible to create household demographic totals of people by age/sex and race/ethnicity from the merged administrative records for each address that is matched to the MAF. The age/sex totals are calculated within seven categories:

1. All persons ages 0–17
2. All persons ages 18–29
3. Males ages 30–44
4. Females ages 30–44
5. Males ages 45–64

6. Females ages 45–64
7. All persons ages 65 and older

The race/ethnicity totals are calculated within four categories:

1. All Hispanics regardless of race
2. All non-Hispanic blacks
3. All non-Hispanic whites
4. All non-Hispanics other races

These household-level totals can then be used to create unweighted tract-, place- and MCD-level administrative record universe totals using the geography associated with the address.

Weighted Administrative Sample Counts: The administrative records that match to the sampling frame can also be linked to the actual ACS sample records themselves. Using the *WNIF* weights, the records that match to the ACS sample can then be used to create weighted administrative record totals for the same geographic areas. Since the ACS sample weights should reflect the frame counts, these weighted administrative record totals should be an unbiased estimate of the unweighted matched universe totals.

Applying GREG Estimation: Using generalized regression estimation (or GREG), the ACS weights are first calibrated so that the weighted administrative record totals match the unweighted universe counts for the seven age/sex categories. Two conditions are checked: is the regression equation solvable and are all of the resulting weights greater than 0.5. If either condition fails, then the age/sex categories are collapsed and the regression is attempted again.

Two levels of collapsing are attempted:

Collapsing across age/sex categories into three categories: all persons ages 0–17, all persons ages 18–44, and all persons 45 and older.

Collapse all categories into a single cell of total administrative persons.

If the condition still fails after the second level of collapsing, then the administrative record data are not used.

If the regression passes using at least the single cell of total administrative persons, then an attempt is made to add the race/ethnicity covariates to the model. First, a collapsing procedure is run that tests which race/ethnicity categories can be used. The criteria for including a race/ethnicity category in the regression is that both the administrative records universe count for the category being tested and the total for all other categories must be greater than 300 persons. This procedure is carried out first for the largest race/ethnicity category not including the non-Hispanic white category, then the next largest such category, and finally the last remaining category other than non-Hispanic white.

As an example, if the largest category other than non-Hispanic white was the Hispanic category, then the first test would be if 1) the Hispanic category had a universe count which was greater than 300 and 2) the other three categories combined had a universe count greater than 300. If it passes, the Hispanic category is flagged to be included and the remaining categories are tested. If the next largest category is non-Hispanic black, it is tested to determine if its universe count is greater than 300 and if the balance, now only the non-Hispanic other races and non-Hispanic white, is greater than 300. If it passes, then the procedure moves on to test the smallest category other than non-Hispanic white. In this example, that is the non-Hispanic other race category. If a similar test on that category fails (or on any previous attempt) then the race collapsing is complete and the covariates for each race/ethnicity category that passed are added to the model. The regression is then attempted including both the age/sex and race/ethnicity covariates. The same conditions used in the age/sex category collapsing are applied to the new attempt. If the regression passes both conditions, then the covariate matrix is considered final. If the regression fails either condition, then the smallest race/ethnicity category is not included in the model and the regression is attempted again. This process continues until either the regression passes or all race/ethnicity covariates have been removed. The final result of this step is the creation of the GREG Weighting Factor (*GWTF*) for each ACS record, which captures the calibration performed in the regression. A summary of the impact of the *GWTF* is given in Table 11-8.

Table 11-8. Impact of GREG Weighting Factor Adjustment

Interview Status	Frame and Model Criteria Status	ACS Record Matches to Administrative Data	Impact of <i>GWTF</i>
Noninterview or non-responding HU not sampled for CAPI	Not applicable	Not applicable	No impact (factor set to 1)
Interview or field-determined ineligible HU	Tract fails both frame count and model criteria	Not applicable	No impact (factor set to 1)
Interview or field-determined ineligible HU	Tract passes the frame count criterion but fails model criteria	Not applicable	Calibrates weights to frame counts for the tract
Interview or field-determined ineligible HU	Tract passes both the frame count and model criteria	Does not match	Calibrates weights to frame counts for the tract
Interview or field-determined ineligible HU	Tract passes both the frame count and model criteria	Does match	Calibrates weights to frame counts for the tract and Calibrates weighted admin data to admin universe counts for tract

This factor is then applied to the WNIF weights to create the Weight after the GREG Weighting Factor (*GWTF*). The computation of this weight is summarized in Table 11-9.

Table 11-9. Computation of the Weight After the GREG Weighting Factor (*GWTF*)

Interview Status	<i>GWTF_j</i>
Interview or field-determined ineligible housing unit	$WNIF_j \times GWTF_j$
All others	0

After this step is complete, the multiyear weighting mirrors the single-year weighting, picking up again at the *HPF* step.

Other Multiyear Estimation Steps

In addition to the adjustments to the single-year weighting methodology for weighting the multiyear data, there are other steps involved in the multiyear estimation that are not weighting related. These include standardizing definitions of variables, updating the geography for place of work and migration characteristics, and the adjustment of income, value, and other dollar amounts for inflation over the period. The details of these adjustments are given in Chapter 10.

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Chapter 11 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
M. Asiala & E. Castro	11/28/2022	Content and formatting edits. Approval of Final Edits	3.0
M. Asiala & E. Castro	8/16/2024	Added section on adjustments for late mail and internet returned. Content and formatting edits. Approval of Final Edits	4.0

Chapter 12: Variance Estimation

12.1 Overview

Sampling error is the uncertainty associated with an estimate based on data gathered from a sample of the population rather than the full population. Note that sample-based estimates will vary depending on the particular sample selected from the population. Measures of the magnitude of sampling error, such as the variance and the standard error (the square root of the variance), reflect the variation in the estimates over all possible samples that could have been selected from the population using the same sampling methodology.

The American Community Survey (ACS) is committed to providing its users with measures of sampling error along with each published estimate. To accomplish this, all published ACS estimates are accompanied by 90 percent margins of error based on ACS direct variance estimates. Due to the complexity of the sampling design and the weighting adjustments performed on the ACS sample, unbiased design-based variance estimators do not exist. As a consequence, the direct variance estimates are computed using a replication method that repeats the estimation procedures independently several times. The variance of the full sample is then estimated using the variability across the resulting replicate estimates. Note that the methodology for variance estimation used for the ACS has been relatively stable over its history. For reference, this chapter reflects the methodology used for the 2022 data products released in 2023.

Although the variance estimates calculated using this procedure are not completely unbiased, the current method produces variances that are accurate enough for analysis of the ACS data.

For Public Use Microdata Sample (PUMS) data users, replicate weights are provided to approximate standard errors for the PUMS-tabulated estimates. Design factors are also provided with the PUMS data, so PUMS data users can compute standard errors of their statistics using either the replication method or the design factor method.

12.2 Variance Estimation for Housing Unit and Person Estimates

Unbiased estimates of variances for ACS estimates do not exist because of the systematic sample design, as well as the ratio adjustments used in estimation. As an alternative, ACS implements a replication method for variance estimation. An advantage of this method is that the variance estimates can be computed without consideration of the form of the statistics or the complexity of the sampling or weighting procedures, such as those being used by the ACS.

The ACS employs the Successive Differences Replication (SDR) method (Wolter, 1984; Fay & Train, 1995; Judkins, 1990) to produce variance estimates. It has been the method used to calculate ACS estimates of variances since the start of the survey. The SDR was designed to be used with systematic samples for which the sort order of the sample is informative, as in the case

of the ACS’s geographic sort. Applications of this method were developed to produce estimates of variances for the Current Population Survey (U.S. Census Bureau, 2006) and the Census 2000 Long Form estimates (Gbur & Fairchild, 2002).

In the SDR method, the first step in creating a variance estimate is constructing the replicate factors. Replicate base weights are then calculated by multiplying the base weight for each housing unit (HU) by the factors. The weighting process is then rerun, using each set of replicate base weights in turn, to create final replicate weights. Replicate estimates are created by using the same estimation method as the original estimate but applying each set of replicate weights instead of the original weights. Finally, the replicate and original estimates are used to compute the variance estimate based on the variability between the replicate estimates and the full sample estimate.

The following steps produce the ACS direct variance estimates:

1. Compute replicate factors.
2. Compute replicate weights.
3. Compute variance estimates.

Replicate Factors

Computation of replicate factors begins with selecting a Hadamard matrix of order R (a multiple of 4), where R is the number of replicates. A Hadamard matrix H is a k -by- k matrix with all entries either 1 or -1 , such that $H'H = kI$ (that is, the columns are orthogonal). For the ACS, the number of replicates is 80 ($R = 80$). Each of the 80 columns represents one replicate.

Next, a pair of rows in the Hadamard matrix is assigned to each sampling cluster. (For the HU sample, this is the HU; for the group quarters sample, it is the “hit.”⁶¹) An algorithm assigns two rows of an 80×80 Hadamard matrix to each cluster. The ACS uses a repeating sequence of 780 pairs of rows in the Hadamard matrix to assign rows to each record in a particular sort order (Navarro, 2001a). The assignment of Hadamard matrix rows repeats every 780 records until all records receive a pair of rows from the Hadamard matrix. The first row of the matrix, in which every cell is always equal to one, is not used.

The replicate factor for each record is then determined from these two rows of the 80×80 Hadamard matrix. For record i ($i = 1, \dots, n$, where n is the sample size) and replicate r ($r = 1, \dots, 80$), the replicate factor is computed as:

$$f_{i,r} = 1 + 2^{-1.5}a_{R1(i),r} - 2^{-1.5}a_{R2(i),r}$$

⁶¹ A “hit” is either an entire small group quarters facility, or a group of 10 people in a large group quarters facility. For more information, see Chapter 4 of the Design and Methodology Report.

where $R1(i)$ and $R2(i)$ are, respectively, the first and second row of the Hadamard matrix assigned to the i -th HU, and $a_{R1(i),r}$ and $a_{R2(i),r}$ are respectively the matrix elements (either 1 or -1) from the Hadamard matrix in rows $R1i$ and $R2i$ and column r . Note that the formula for $f_{i,r}$ yields replicate factors that can take one of three approximate values: 1.7, 1.0, or 0.3. That is:

- If $a_{R1(i),r} = +1$ and $a_{R2(i),r} = +1$, the replicate factor is 1.
- If $a_{R1(i),r} = -1$ and $a_{R2(i),r} = -1$, the replicate factor is 1.
- If $a_{R1(i),r} = +1$ and $a_{R2(i),r} = -1$, the replicate factor is approximately 1.7.
- If $a_{R1(i),r} = -1$ and $a_{R2(i),r} = +1$, the replicate factor is approximately 0.3.

The expectation is that 50 percent of replicate factors will be 1, and the other 50 percent will be evenly split between 1.7 and 0.3 (Gunlicks, 1996).

The following example demonstrates the computation of replicate factors for a sample of size five, using a Hadamard matrix, H , of order four:

$$H = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

Table 12-1 presents an example of a two-row assignment developed from this matrix and the values of replicate factors for each sample unit.

Table 12-1. Example of Two-Row Hadamard Matrix Assignment and Replicate Factors

Case (i)	Row Pair (R1, R2)	Matrix elements Replicate #1 ($a_{R1,1}, a_{R2,1}$)	Matrix elements Replicate #2 ($a_{R1,1}, a_{R2,1}$)	Matrix elements Replicate #3 ($a_{R1,1}, a_{R2,1}$)	Matrix elements Replicate #4 ($a_{R1,1}, a_{R2,1}$)	f_1	f_2	f_3	f_4
1	2, 3	(+1, +1)	(-1, +1)	(+1, -1)	(-1, -1)	1	0.3	1.7	1
2	3, 4	(+1, +1)	(+1, -1)	(-1, -1)	(-1, +1)	1	1.7	1	0.3
3	4, 2	(+1, +1)	(-1, -1)	(-1, +1)	(+1, -1)	1	1	0.3	1.7
4	2, 3	(+1, +1)	(-1, +1)	(+1, -1)	(-1, -1)	1	0.3	1.7	1
5	3, 4	(+1, +1)	(+1, -1)	(-1, -1)	(-1, +1)	1	1.7	1	0.3

Note that row 1 is not used, and the index i is not shown in Table 12-1 for clarity. For the third case ($i = 3$), rows four and two of the Hadamard matrix are used to calculate the replicate factors. For the second replicate ($r = 2$), the replicate factor, $f_{i,j}$, is computed using the values in the second column of rows four (-1) and two (-1) as follows:

$$f_{3,2} = 1 + 2^{-1.5}a_{4,2} - 2^{-1.5}a_{2,2} = 1 + (2^{-1.5} \times -1) - (2^{-1.5} \times -1) = 1$$

Replicate Weights

Replicate weights are produced in a way similar to that used to produce full sample final weights. All of the weighting adjustment processes performed on the full sample final survey weights (such as applying noninterview adjustments and population controls) also are carried out for each replicate weight. However, collapsing patterns are retained from the full sample weighting and are not determined again for each set of replicate weights.

Before applying the weighting steps explained in Chapter 11, the replicate base weight (RBW) for replicate r is computed by multiplying the full sample base weight (BW— see Chapter 11 for the computation of this weight) by the replicate factor $f_{i,r}$; that is, $RBW_{i,r} = BW_i \times f_{i,r}$, where $RBW_{i,r}$ is the replicate base weight for the i -th HU and the r -th replicate ($r = 1, \dots, 80$).

One can elaborate on the previous example of the replicate construction using five cases and four replicates: Suppose the full sample BW values are given under the second column of the following table (Table 12-2). Then, the replicate base weight values are given in columns 7–10.

Table 12-2. Example of Computation of Replicate Base Weight Factor (RBW)

Case # (i)	BW _i	f _{i,1}	f _{i,2}	f _{i,3}	f _{i,4}	RBW _{i,1}	RBW _{i,2}	RBW _{i,3}	RBW _{i,4}
1	100	0.3	1	1	1.7	29	100	100	171
2	120	1.7	1	0.3	1	205	120	35	120
3	80	1	1	1	0.3	80	80	80	23
4	120	0.3	1	1	1.7	35	120	120	205
5	110	1.7	1	0.3	1	188	110	32	110

The rest of the weighting process (Chapter 11) then is applied to each replicate weight $RBW_{i,r}$ (starting from the adjustment for Computer-Assisted Personal Interviewing (CAPI) subsampling) and proceeding to the population control adjustment or raking). Basically, the weighting adjustment process is repeated independently 80 times and the $RBW_{i,r}$ is used in place of $BW_{i,r}$ (as in Chapter 11).

By the end of this process, 80 final replicate weights for each HU and person record are produced.

Variance Estimates

Given the replicate weights, the computation of variance for any ACS estimate is straightforward. Suppose that θ is an ACS estimate of any type of statistic, such as mean, total, or proportion. Let θ_0 denote the estimate computed based on the full sample weight, and $\theta_1, \theta_2, \dots, \theta_{80}$ denote the estimates computed based on the replicate weights. The variance of θ_0 , $v(\theta_0)$,

is estimated as the sum of squared differences between each replicate estimate θ_r ($r = 1, \dots, 80$) and the full sample estimate θ_0 . The formula is as follows⁶²:

$$v(\theta_0) = \frac{4}{80} \sum_{r=1}^{80} (\theta_r - \theta_0)^2$$

This equation holds for count estimates as well as any other types of estimates, including percentages, ratios, and medians.

Special Cases for Variance Estimates—Controlled Estimates

There are certain cases, however, where this formula does not apply. The first and most important case is for estimates that are “controlled” to population totals and have their standard errors set to zero. These are estimates that are forced to equal independent population estimates during the weighting process’s raking step, for example, total population and collapsed age, sex, and Hispanic origin estimates for weighting areas. Although race is included in the raking procedure, race group estimates are not controlled; the categories used in the weighting process (see Chapter 11) do not match the published tabulation groups because of multiple race responses and the “Some Other Race” category. Information on the final collapsing of the person post-stratification cells is passed from the weighting to the variance estimation process in order to identify estimates that are controlled. This identification is done independently for all weighting areas and then is applied to the geographic areas used for tabulation. Standard errors for those estimates are set to zero, and published margins of error are set to “*****” (with an appropriate accompanying footnote).

Special Cases for Variance Estimates—Zero Count Estimates

Another special case deals with zero-estimated counts of people, households, or HUs. A direct application of the replicate variance formula leads to an estimated standard error of zero for a zero-estimated count. However, there may be people, households, or HUs with that characteristic in that area that were not selected to be in the ACS sample, but a different sample might have selected them, so an estimated standard error of zero is inaccurate. For these cases, a model-based estimation of standard error was implemented.

The model-based method requires census counts and ACS data. For ACS data collected in a census year, the ACS zero-estimated counts (for characteristics included in the 100 percent

⁶² A general replication-based variance formula can be expressed as

$$v(\theta_0) = \sum_{r=1}^n c_r (\theta_r - \theta_0)^2$$

where c_r is the multiplier related to the r -th replicate determined by the replication method. For the SDR method, the value of c_r is $4 / R$, where R is the number of replicates (Fay & Train, 1995)

census (“short form”) count) can be checked against the corresponding census estimates. At least 90 percent of the census counts for the ACS zero-estimated counts should be within a 90 percent confidence interval based on our modeled standard error. Let the variance of the estimate be modeled as some multiple (K) of the average final weight (for a state or the nation). That is:

$$v(0) = K \times (\text{average weight})$$

Then, set the 90 percent upper bound for the zero estimate equal to the census count:

$$\begin{aligned} \text{Upper confidence bound} &= 0 + 1.645 \times SE(0) \\ &= 1.645 \times \sqrt{K \times (\text{average weight})} \\ &= \text{census count} \end{aligned}$$

Solving for K yields:

$$K = \left(\frac{\text{census count}}{1.645} \right)^2 \frac{1}{(\text{average weight})}$$

The first K was computed for all ACS zero-estimated counts from 2000 which were matched to the 2000 Census short-form counts, and then the 90th percentile of those K s was determined. Based on the 2000 Census data, we used a value for K of 400 (Navarro, 2001b) from 2001 through the 2010 data products.

When the results of the 2010 Census became available in 2011, work was done to update the K value. The new value for K was determined to be 223 and was used for the first time with the 2011 data products. In 2012, additional research was conducted. As a result, for the 1-year data products, the k value continues to be 223, but for the 5-year data products one of six possible values (4, 8, 10, 14, 18, or 22) is assigned based on the population size of the geographic area.

For publication, the standard error (SE) of the zero count estimate is computed as:

$$SE(0) = \sqrt{K \times (\text{average weight})}$$

Where:

- $K = 400$ for ACS data products from 2001 to 2010
- 223 for ACS data products for 2011
- 4, 8, 10, 14, 18, 22 or 223 for ACS data products starting with 2012

The average weights are the maximum of the average housing unit and average person final weights. They are calculated at the state and national level for each ACS single-year or multiyear data release. Estimates for geographic areas wholly contained within a state use that state’s average weight. Estimates for geographic areas that cross state boundaries use the national average weight.

Special Cases for Variance Estimates—Zero Percent Estimates

Finally, a method must be used to create an approximate standard error for both ACS zero percent and 100 percent estimates. In both cases, as with the zero counts above, all replicates are equal, and the replicate formula produces a zero standard error. To preserve the desirable property $SE(p) = SE(1-p)$, we want to use the same approximation for zero percent and 100 percent, and we will present the zero percent case for this discussion. (Note that we do not produce approximate standard errors for other zero estimates, such as ratios or medians.)

Let Z be the unweighted count of respondents with a given characteristic. If Z is modeled as a Poisson variable with parameter λ , what is a likely value for λ if $Z = 0$? We could find the value λ_0 such that the probability of $\lambda \leq \lambda_0$ is 0.9, given $Z = 0$ (sort of a 90% upper confidence bound).

$$P(\lambda \leq \lambda_0 | Z = 0) = \int_0^{\lambda_0} \frac{\lambda^0 e^{-\lambda}}{0!} = \int_0^{\lambda_0} e^{-\lambda} = -e^{-\lambda_0} - (-e^0) = 1 - e^{-\lambda_0} = 0.9$$

$$e^{(-\lambda_0)} = 0.1 \Rightarrow \lambda_0 = -\ln(0.1) \approx 2.3$$

Z can also be represented as X / AW , where X is the weighted estimate of the numerator, and AW is the average weight, as described in the previous section.

$$\text{Upper Bound } (E(Z)) = \text{Upper Bound } (E(X / AW)) = 2.3$$

$$\Rightarrow \text{Upper Bound } (E(X)) = 2.3 \times AW$$

$$\Rightarrow \text{Upper Bound } (E(X/Y)) = 2.3 \times AW / Y$$

This gives an “upper bound” value for the proportion that we can use to calculate a corresponding “upper bound” value for its standard error.

$$R = \frac{2.3 \times AW}{Y}$$

$$SE(R) \approx \sqrt{\frac{pq}{n}} = \sqrt{\frac{R(1-R)}{\# \text{ of unweighted cases for } Y}} \approx \sqrt{\frac{R(1-R)}{Y/AW}} = \sqrt{\frac{AW}{Y} R(1-R)}$$

To be conservative, if $R > 0.5$ (which will only happen if Y is fairly small), then 0.5 can be used instead of R , since $SE(R)$ is a maximum for a given AW and Y when $R = 0.5$.

Finite Population Correction Factor for Multiyear Estimates

The 2000 Census long form, as noted above, used the same SDR variance estimation methodology as the ACS currently does. The long form methodology also included a finite population correction (FPC) factor in its calculation. The overall sampling rate for ACS 1-year

published areas is not large enough for an FPC to have much impact on the estimated variances. This is because while the ACS samples areas at up to 10 percent of the housing units per year, these small areas do not meet the publication threshold for the 1-year products. However, with the ACS 5-year products, all areas are published, including areas where up to 50 percent of housing units may have been in sample over the 5-year period. Applying an FPC factor to multi-year ACS replicate estimates enables a more accurate estimate of the variance, particularly for small areas. The FPC factor is applied in the creation of the replicate factors:

$$f_{i,r}^* = 1 + (2^{-1.5}a_{R1i,r} - 2^{-1.5}a_{R2i,r}) \times FPC$$

where

$$FPC = \sqrt{1 - n/N}$$

Generically, n is the unweighted sample size, and N is the unweighted universe size. The ACS uses two separate FPC factors: one for HUs responding by mail or internet, and a second for HUs responding via personal visit follow-up.

The FPC is typically applied as a multiplicative factor “outside” the variance formula. However, under certain simplifying assumptions, the variance using the replicate factors after applying the FPC factor is equal to the original variance multiplied by the FPC factor. This method allows a direct application of the FPC to each housing unit’s or person’s set of replicate weights and a seamless incorporation into the current variance production methodology rather than having to keep track of multiplicative factors when tabulating across areas of different sampling rates.

The adjusted replicate factors are used to create replicate base weights, and ultimately final replicate weights. It is expected that the improvement in the variance estimate will carry through the weighting and will be seen when the final weights are used.

The ACS FPC factor could be applied at any geographic level. Since the ACS sampling rates are determined at the small area level (mainly census tracts and governmental units), a low level of geography was desirable. At higher levels, the high sampling rates in specific blocks would likely be masked by the lower rates in surrounding blocks. For that reason, the factors are applied at the census tract level.

Group quarters persons do not have an FPC factor applied to their replicate factors.

Design Factor Adjustment for Group Quarters Imputation

Beginning with the 2011 ACS data products, a new imputation-based methodology was incorporated into processing (see the description in Chapter 11). Group Quarters (GQ) person records were imputed into not-in-sample GQs in order to improve estimates for the characteristics of the GQ population for substate areas. While the ACS does not publish GQ

characteristic data for substate areas (apart from our Public Use Microdata Sample files), the methodology does significantly improve sub-state characteristic estimates for the total population, especially in areas where the GQ population is a significant proportion of the total population (e.g., small areas, college towns, etc.)

The number of these wholly imputed person records has generally ranged between 80 percent and 90 percent of the number of interviewed GQ persons, so estimates based on this methodology could contain a sizable amount of imputation error. This imputation error would not be picked up by the normal replicate variance formula, leading to an underestimate of standard errors.

An adjustment was needed to the production replicate weight variance methodology to account for the non-negligible amount of additional variation being introduced by the new technique. However, it needed to be simple enough to be implemented within the structure of the current estimate and variance tabulation system. Creating a set of design factors to inflate the basic replicate variance was determined to be the best approach.

To that end, 40 separate sets of imputed GQ data were created, with the full ACS weighting applied to each set. Estimates selected from ACS subject table S2601A, “Characteristics of the Group Quarters Population in the United States,” were created from each of the 40 sets at the state and county level. The total variance of the estimates, including from the imputation, was calculated using a formula from Rubin (1996):

$$Var(\bar{\theta}) = \frac{1}{m} \sum_{j=1}^m Var(\theta_j) + \frac{m+1}{m} \sum_{j=1}^m \frac{1}{m-1} (\theta_j - \bar{\theta})^2$$

where the first term is the average “within imputation” variance (the standard ACS variance), and the second term is the “between imputation” variance that is not captured by the standard method.

Preliminary design factors were created by estimate characteristic group, major GQ type, and state by fitting a no-intercept regression model with the independent variable as the “within imputation” variance and the dependent variable as the total variance in the above formula.

Variability in the preliminary design factors was much more pronounced between major GQ types than between characteristics, so (conservative) “overall” design factors were calculated to create a single design factor for each state by major GQ type.

These design factors were applied to the replicate factors, similarly to the multiyear finite population correction factor described above, although with the intention of increasing the variance rather than decreasing it.

$$f_{i,r,s,g}^* = 1 + (2^{-1.5}a_{R1i,r} - 2^{-1.5}a_{R2i,r}) \times DF_{s,g}$$

where $DF_{s,g}$ is the design factor for state s and GQ major type g . The design factors are capped at 1.4, as a value larger than the square root of 2 would lead to negative replicate factors.

For more information regarding this methodology, see Asiala and Castro (2012).

12.3 Margin of Error and Confidence Interval

Once the standard errors have been computed, margins of error (MOE) are produced for each estimate. These are the measures of overall sampling error presented along with each published ACS estimate. All published ACS margins of error presented in the ACS data products are based on a 90 percent confidence level, which is the Census Bureau’s standard (U.S. Census Bureau, 2023b). A margin of error contains two components: the standard error of the estimate and a multiplication factor based on a chosen confidence level. For the 90 percent confidence level, the value of the multiplication factor used by the ACS is 1.645.

The margin of error of an estimate θ can be computed as:

$$\text{Margin of error}(\theta) = 1.645 \times SE(\theta)$$

where $SE(\theta)$ is the standard error of the estimate θ . Given this margin of error, the 90 percent confidence interval can be computed as:

$$\theta \pm \text{Margin of error}(\theta)$$

That is, the lower bound of the confidence interval is $[\theta - \text{margin of error}(\theta)]$, and the upper bound of the confidence interval is $[\theta + \text{margin of error}(\theta)]$. Roughly speaking, this interval is a range that will contain the “full population value” of the estimated characteristic with a known probability.

Users are cautioned to consider “logical” boundaries when creating confidence bounds from the margins of error. For example, a small population estimate may have a calculated lower bound of less than zero. A negative number of people does not make sense, so the lower bound should be set to zero instead. Likewise, bounds for percentages should not go below zero percent or above 100 percent. For other characteristics, like income, negative values may be legitimate.

Given the confidence bounds, a margin of error can be computed as the difference between an estimate and its upper or lower confidence bounds:

$$\text{Margin of error} = \max(\text{upper bound} - \text{estimate}, \text{estimate} - \text{lower bound})$$

Using the margin of error (as published or calculated from the bounds), the standard error is obtained as follows:

$$\text{Standard error}(\theta) = \text{Margin of error}(\theta) \div 1.645$$

For ranking tables and comparison profiles, the ACS provides an indicator as to whether two estimates, Est_1 and Est_2 , are statistically significantly different at the 90 percent confidence level. That determination is made by initially calculating:

$$Z = \frac{Est_1 - Est_2}{\sqrt{SE(Est_1)^2 + SE(Est_2)^2}}$$

If $Z < -1.645$ or $Z > 1.645$, the difference between the estimates is significant at the 90 percent level. Determinations of statistical significance are made using unrounded values of the standard errors, so users may not be able to achieve the same result using the standard errors derived from the rounded estimates and margins of error as published. Only pairwise tests are used to determine significance in the ranking tables; no multiple comparison methods are used.

12.4 Variance Replicate Tables

Beginning with the 2010–2014 5-year data, the ACS has published the Variance Replicate Estimate Table package, which allows advanced users to calculate variances, standard errors, and margins of error using replicate estimates, as described in section 12.2 above.

The table package provides augmented versions of over 100 detailed ACS tables published on data.census.gov for a limited set of geographic summary levels. For this limited set of tables, the standard errors and eighty variance replicate estimates are provided in addition to the original published estimates and margins of error for each characteristic. This additional information allows data users to compute variances using the replicate methodology similar to the one used by the ACS during its production. The benefit of using this set of expanded tables is that it allows advanced data users to calculate MOEs for their own defined characteristics without using approximation formulas. This includes forming their own aggregate estimates across geographies or aggregating categories within a table, as well as ratios, proportions, and percentages.

12.5 Variance Estimation for the PUMS

The Census Bureau cannot possibly predict all combinations of estimates and geography that may interest data users. Data users can download PUMS files and tabulate the data to create their own estimates. Because the ACS PUMS contains only a subset of the full ACS sample, estimates from the ACS PUMS file will often differ from the published ACS estimates based on the full ACS sample.

Users of the ACS PUMS files can compute the estimated variances of their statistics using one of two options: (1) the replication method using replicate weights released with the PUMS data and

(2) the design factor method. In general, the replication method produces significantly more accurate variance estimates and is strongly recommended over the design factor method whenever practical.

PUMS Replicate Variances

Direct variance estimates based on the SDR formula described in Section 12.2 above can be implemented for the replicate method. Users can tabulate 80 replicate estimates in addition to their desired estimate by using the provided 80 replicate weights and then apply the variance formula:

$$v(\theta_0) = \frac{4}{80} \sum_{r=1}^{80} (\theta_r - \theta_0)^2$$

PUMS Design Factor Variances

Similar to methods used to calculate standard errors for PUMS data from the 2000 Census, the ACS PUMS provides tables of design factors for various topics such as age for persons or tenure for HUs. The ACS PUMS design factors are published at national and state levels and are updated periodically, but not necessarily on an annual basis. The design factor approach was developed based on a model that uses a standard error from a simple random sample as the base and then inflates it to account for an increase in the variance caused by the complex sample design. Standard errors for almost all counts and proportions of persons, households, and HUs are approximated using design factors. For 1-year ACS PUMS files beginning in 2005,

$$SE(Y) = DF \times \sqrt{99 \times Y \times \left(1 - \frac{Y}{N}\right)}$$

for a total, and

$$SE(p) = DF \times \sqrt{\frac{99}{B} \times p \times (100 - p)}$$

for a percent, where

Y = the estimate of a total or count

p = the estimate of a percent

DF = the appropriate design factor based on the topic of the estimate

N = the total for the geographic area of interest (if the estimate is of HUs, the number of HUs is used; if the estimate is of families or households, the number of households is used; otherwise, the number of persons is used as N).

B = the denominator (base) of the percent.

The value 99 in the formula is the value of the 1-year PUMS FPC factor, which is computed as $(100 - f) / f$, where f (given as a percent) is the sampling rate for the PUMS data. Since the PUMS is approximately a 1 percent sample of HUs, $(100 - f) / f = (100 - 1) / 1 = 99$.

For 5-year PUMS files, beginning with 2005–2009, the 5 years’ worth of data represent approximately a 5 percent sample of HUs. So, the 5-year PUMS FPC is $(100 - f) / f = (100 - 5) / 5 = 95 / 5 = 19$, which can be substituted for 99 in the above formulas.

The design factor (DF) is the ratio of the standard error of an estimated parameter (computed under the replication method described in Section 12.2) to the standard error based on a simple random sample of the same size. The DF reflects the effect of the actual sample design and estimation procedures used for the ACS. The DF for each topic was computed by modeling the relationship between the standard error under the replication method (RSE) with the standard error based on a simple random sample ($SRSSE$); that is, $RSE = DF \times SRSSE$, where the $SRSSE$ is computed as follows:

$$SRSSE(Y) = DF \times \sqrt{39 \times Y \times \left(1 - \frac{Y}{N}\right)}$$

The value 39 in the formula above is the FPC factor based on an approximate sampling fraction of 2.5 percent in the ACS; that is, $(100 - 2.5) / 2.5 = 97.5 / 2.5 = 39$.

The value of DF is obtained by fitting the no-intercept regression model $RSE = DF \times SRSSE$ using standard errors (RSE , $SRSSE$) for various published table estimates at the national and state levels. The values of DFs by topic can be obtained from the “PUMS Accuracy of the Data” statement that is published with each PUMS file. For example, 2022 1-year PUMS DFs can be found in the “PUMS Accuracy of the Data (2022)” (U.S. Census Bureau, 2023a). The documentation also provides examples on how to use the design factors to compute standard errors for the estimates of totals, means, medians, proportions or percentages, ratios, sums, and differences.

The topics for the PUMS design factors are, for the most part, the same ones that were available for the 2000 Census PUMS. We recommend to users that, in using the design factor approach, if the estimate is a combination of two or more characteristics, the largest DF for this combination of characteristics is used. The only exceptions to this are items crossed with race or Hispanic origin; for these items, the largest DF is used after removing the race or Hispanic origin DFs from consideration.

12.6 References

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Chapter 12 Revision History

Author	Date	Brief Description of Change	Version #
C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
M. Asiala & E. Poehler	11/28/2022	Updated content and formatting	3.0
M. Asiala	8/16/2024	Updates dates and references; no substantive changes from version 3.0	4.0

Chapter 13: Preparation and Review of Data Products

13.1 Overview

This chapter discusses the data products derived from the American Community Survey (ACS). ACS data products include the tables, reports, and files that contain estimates of social, economic, housing, and demographic characteristics. These products cover geographic areas within the United States and Puerto Rico.

Most surveys of the population provide sufficient samples to support the release of data products only for the nation, the states, and perhaps select sub-state geographic areas. The ACS is a very large nationwide survey that collects data continuously every year and designed to provide data products for many types of geographic areas, including smaller geographic areas such as places, townships, and census tracts.

The first step in the preparation of a data product is defining the topics and characteristics it will cover. Once the initial characteristics are determined, they must be reviewed by the Census Bureau Disclosure Review Board (DRB) to ensure that individual responses and the identity of individual respondents will be kept confidential. Based on this review, the specifications of the products may be revised. The DRB also may require that the microdata files be altered in certain ways and may impose restrictions on the publication of estimates for certain geographic areas based on the size of the ACS sample or the population size of such areas. These activities are collectively referred to as disclosure avoidance.

The actual processing of the data products cannot begin until all response records for a given year or years are edited and imputed in the data preparation and processing phases, the final weights are determined, and disclosure avoidance techniques are applied. Using the weights, the sample data are tabulated for a wide variety of characteristics according to the predetermined content. These tabulations are done for selected geographic areas for the 1-year products and for small geographic areas down to the census tract and block group levels for 5-year products. The Public Use Microdata Sample (PUMS) data files are created by different processes because the data are a subset of the full sample data.

After the estimates are produced, Census Bureau subject matter analysts review them for reasonableness. When the estimates have passed the final review, they are released to the public. PUMS data files are reviewed in a separate process.

While the 2005 ACS sample was limited to the housing unit (HU) population for the United States and Puerto Rico, starting in sample year 2006, the ACS was expanded to include the group quarters (GQ) population. This step made the 2006 ACS and Puerto Rico Community Survey (PRCS), published in 2007, representative of the total resident population of the United States and Puerto Rico.

In 2008, the Census Bureau, for the first time, released products based on three years of ACS sample data collected from 2005 through 2007 for areas of 20,000 people or greater. However, the 3-year product was subsequently discontinued after the release of the 2011-2013 ACS 3-year estimates. In 2010, the Census Bureau released the first products based on five years of consecutive ACS samples, 2005 through 2009. Since several years of sample form the basis of these multiyear products, reliable estimates can be released for much smaller geographic areas than is possible for products based on single-year data. Starting with the 2014 data year, the Census Bureau began releasing 1-year Supplemental Estimates, which are simplified versions of popular ACS Detailed Tables for geographic areas with at least 20,000 people.

In addition to data products regularly released to the public, the Census Bureau releases other data products developed through a fee-based custom tabulations program. Government agencies, private organizations and businesses, or individual data users can request custom tabulations of data. As is the case for regular data products, custom tabulation requests are reviewed and approved by the DRB to assure protection of the confidentiality of individual responses. Section 13.7 provides information on methods used by the DRB to protect census data, including data from the ACS.

Chapter 14 describes the dissemination of the data products discussed in this chapter, the ACS data release schedule, and the locations of various products and supporting documents on the Census Bureau's website.

13.2 Geography

The Census Bureau strives to provide ACS estimates for the geographic areas that are most important to data users. The two main types of geographic areas for which ACS estimates are provided include legal areas and statistical areas.

Legal areas are defined specifically by law and include state, local, and tribal government units (e.g., Alaska Native Regional Corporation [ANRC] areas, states, counties, minor civil divisions [MCDs], and incorporated places), as well as specially defined administrative areas, such as congressional districts and state legislative districts.

Statistical areas are defined to tabulate and provide census data for areas not specifically defined by law. Some statistical areas are defined directly by the Census Bureau and state, regional, or local authorities (e.g., census tracts, block groups, and census designated places [CDPs]), while other statistical areas, such as Core Based Statistical Area (CBSA) geographies (e.g., Metropolitan Statistical Area, Micropolitan Statistical Area, Combined Statistical Area [CSA], and Metropolitan Division), are defined and maintained by the Office of Management and Budget (OMB).

“Geographic Summary Level” is a term used by the Census Bureau to designate the different geographic levels or types of geographic areas for which data are summarized. Geographic components describe subsets within summary levels and are most commonly used to identify population ranges within a summary level. See [Geography & ACS](#) for more information about the [types of geographic areas](#) for which the Census Bureau publishes ACS estimates and other resources.

The ACS 1-year estimates are published annually for recognized legal, administrative, or statistical areas with populations of 65,000 or more. The population sizes are based on the latest ACS population estimates. If a geographic area met the 1-year threshold for the previous period but dropped below it for the current period, data will continue to be published for that area if the population does not drop more than five percent. The ACS 5-year estimates, based on five successive years of ACS samples, are published for all legal, administrative, and statistical areas down to the block group level, regardless of population size. The release of ACS estimates is also subject to confidentiality restrictions and data quality filtering, which are covered in more detail in Section 13.7.

The PRCS provides estimates for legal, administrative, and statistical areas in Puerto Rico. The data release and threshold rules, confidentiality restrictions, and data quality filtering for the U.S. resident population also apply to the publication of data for the PRCS.

Many areas for which ACS estimates are produced undergo periodic boundary changes because of annexation, boundary correction, deannexation, consolidation with other areas, or redistricting. Each year, the Census Bureau’s Geography Division, working with state and local governments, conducts the Boundary and Annexation Survey (BAS) to collect updated boundary information.⁶³ In addition to the BAS, there are a variety of update programs underway in the Geography Division at any given time. In addition, the Redistricting Data Program periodically collects state legislative and congressional district boundaries when they are changed by the states through redistricting.⁶⁴

Twice a year, the Geography Division stops all update programs and performs an activity called benchmarking. At this time a ‘snapshot’ is taken of the Master Address File (MAF) Topologically Integrated Geographic and Encoding Reference (TIGER) database, or MAF/TIGER database (MTdb). All of the address and geographic units used in the ACS sample and data products come from these Geography Division biannual benchmarks conducted in December and July.

⁶³ See [About BAS](#) for more information.

⁶⁴ See [Redistricting Data Program](#) for more information.

In the case of multiyear estimates, the boundaries of the legal entity for which the estimate is produced are those of the final year of data collection for that multiyear estimate. For example, geographic units for which a 2016–2020 estimate is produced are those in effect in 2020.

13.3 Defining the Data Products

The initial set of data products for the ACS was developed by starting with the 2000 Census Summary File tables and gathering information on public data needs through several Federal Register Notices.

The ACS Data Products Planning Working Group (DPPWG) and ACS Portfolio Management Governance Board (PMGB) have played a critical role in establishing more efficient processes for considering, reviewing, and approving proposals for new and modified data products. These groups have helped to allocate resources appropriately to develop new products, to consider innovative product design approaches suggested by both internal and external groups, and to revise existing products to make them more useful. They also help to ensure that technical constraints are fully assessed against technical requirements and that potential product innovations align with future goals. These groups also help develop data products when new topics are introduced to the ACS or new topics using existing data are identified.

The ACS Data Users Group (DUG) was formed in early 2013 to provide regional planners, demographers, community leaders, and other data users with an externally managed forum for exchanging information on understanding and using ACS data. The group includes an online community and helpful resources, such as links to ACS reference materials and webinars. Membership in the group is free and is open to all interested ACS data users. Since 2013, membership in the ACS Data Users Group has grown to over 5,000 members. More information is available on ACS DUG activities at: <https://acsdatacommunity.prb.org/>.

In 2015, the ACS Data Product Redesign Group (DPRG) was established to provide the Census Bureau with proactive feedback on the current suite of data products in terms of usage and gaps, and to provide feedback on changes or revisions to data products and data dissemination methods. The feedback from the DPRG facilitated the decision to release the ACS 1-year Supplemental Estimates and the ACS 5-year Variance Replicate Tables. In 2021, the former ACS Data Products Redesign Group (DPRG) merged with the ACS Data Users Group Steering Committee to form one ACS Data Users Group Steering Committee.

In partnership with the U.S. Census Bureau and the Population Reference Bureau (PRB), the ACS DUG is led by a steering committee representing a broad spectrum of ACS data users. The DUG Steering Committee's main responsibilities are improving data users' understanding of the value and utility of ACS data. In addition, the DUG Steering Committee informs PRB and Census Bureau staff about the concerns of ACS data users and provides feedback on ACS-related activities, data products, dissemination channels, and/or methodology. The DUG Steering

Committee members also help plan ACS Data Users Conferences with PRB and the Census Bureau, which are held biennially to broaden understanding about the importance of the ACS. The first ACS Data Users Conference was held in 2014.

13.4 Description of Tabular Data Products

The ACS releases new data every year through a variety of data tables and other datasets that are accessible with different data tools. (See Chapter 14 for how we release these products to the public.)

Data for the ACS are collected from a sample of housing units (HUs), as well as the GQ population, and are used to produce estimates of the actual figures that would have been obtained by interviewing the entire population. The data tables contain the estimates from the survey responses. Each estimate is created using the sample weights from respondent records that meet certain criteria. For example, the ACS estimate of people under the age of 18 in Chicago for a given year is calculated by adding the weights from all respondent records from interviews completed in that year in Chicago with residents under 18 years old.

This section provides a description of each tabular product. Each product described is available as 1-year period estimates; unless otherwise indicated, they are also available as 5-year estimates. The data products described below contain all estimates planned for release each year, including those from multiple years of data. Data release rules described in Section 13.7 prevent certain 1-year period estimates from being released if they do not meet ACS requirements for statistical reliability. All ACS estimates within tabular data products are accompanied by a margin of error at a 90 percent confidence level. Unless specifically mentioned, these ACS data products are available at data.census.gov.

Detailed Tables

The Detailed Tables provide basic distributions of characteristics. They are the foundation upon which other data products are built. They include social, economic, housing, and demographic characteristics and provide 1- or 5-year period estimates for the nation and states, as well as for counties, towns, and other small geographic entities such as census tracts and block groups, the latter of which are only available as 5-year estimates.

The Census Bureau's initial goal in defining ACS data products was to maintain a high degree of comparability between ACS Detailed Tables and Census 2000 Summary File 3 sample-based data products. In addition, characteristics not measured in the 2000 Census tables were included in the ACS base tables. For example, the 2022 ACS data products included more than 1,400 Detailed Tables that cover a wide variety of social, economic, housing, and demographic characteristics.

Data Profiles

There are four Data Profiles, one each for social, economic, housing, and demographic characteristics, and they contain a summary of statistics for each topic in the ACS. For a given geographic area, the Data Profiles include distributions for such characteristics as sex, age, type of household, race and Hispanic origin, school enrollment, educational attainment, disability status, veteran status, language spoken at home, ancestry, income, poverty, physical housing characteristics, occupancy and owner/renter status, and housing value. The profiles provide both point estimates and percentage estimates of characteristics.

Comparison Profiles

Beginning with the 2007 ACS, the Census Bureau published a 1-year Comparison Profile that compares the sample year's estimates with those of each of the four previous years. For example, the 2019 Comparison Profile compared 2019 ACS 1-year estimates with those of the 2015, 2016, 2017, and 2018 ACS 1-year estimates. These profile reports include the results of a statistical significance test for each previous year's estimate compared to the current year. This test result indicates whether the previous year's estimate is significantly different (at a 90 percent confidence level) from the current year's. Beginning with the 2011–2015 ACS 5-year release, the 5-year Comparison Profiles show data side-by-side from two different non-overlapping 5-year periods, indicating where there is a statistically significant difference between the two sets of estimates. For example, the 2018–2022 Comparison Profile compares estimates from 2013–2017 with those from the 2011–2015 period.

Narrative Profiles

Sourced from the Data Profiles, the 5-year Narrative Profiles allow users to explore a narrative and graphical presentation of the statistics for their own communities. This descriptive report describes a geographic area using custom text and graphics across 18 topics. This product is available using the 5-year data and can be generated for the Nation, States (including Puerto Rico), Counties, County Subdivisions, Places, Metropolitan/Micropolitan Statistical Areas, ZIP Code Tabulation Areas (ZCTAs), Census Tracts, and American Indian/Alaska Native /Hawaiian Home Land (AIA/ANA/HHL) Areas.⁶⁵

Subject Tables

Subject Tables cover frequently requested information on a single topic for a single geographic area. However, Subject Tables contain more detail than the ACS Data Profiles. In general, a Subject Table contains distributions for a few key universes, such as the race groups and people in various age groups, which are relevant to the topic of the table. The estimates for these

⁶⁵ The [5-year Narrative Profiles](#) are available through the ACS website starting with the 2011-2015 ACS data.

universes are displayed as whole numbers. The distribution that follows is displayed in percentages. For example, Subject Table S1501 on educational attainment provides the estimates for two different age groups—18 to 24 years old and 25 years and older— as a whole number. For each age group, these estimates are followed by the percentages of people in different educational attainment categories (high school graduate, college undergraduate degree, etc.). Subject Tables also contain other measures, such as medians, and they include the imputation rates for relevant characteristics. About 70 topic-specific Subject Tables are released each year.

Selected Population Profiles

Selected Population Profiles (SPPs) provide certain characteristics from the Data Profiles for a specific race, Hispanic origin, ancestry, and place of birth group. SPPs are provided every year for 1-year estimates. There are two thresholds used for SPPs. First, the SPPs are released for geographic areas with a total population of at least 500,000. Before the 2007 ACS, SPPs were produced for geographic areas with a total population of at least 1,000,000. Second, we look at the size of the population group of interest. The SPPs are released for population groups with an estimated population of 65,000 or more within a given geographic area.

Selected Population Tables and American Indians and Alaska Native Tables

The ACS Selected Population Tables and American Indian and Alaska Native (AIAN) Tables include a set of data products comparable to the Census 2000 Summary File 4/AIAN products. Representing a collection of Detailed Tables and Data Profiles, this product, created using ACS 5-year estimates, provides data iterated by detailed race, ethnicity, tribal, and ancestry population groups. This product was first released in May 2012 and included estimates for the period 2006 to 2010. The Census Bureau produces the Selected Population Tables and American Indian and Alaska Native Tables approximately every five years.

Report Production Series and Analytical Applications

Subject matter analysts at the Census Bureau create analytical data products based on recent ACS data. The analytical products include reports, infographics, and data visualizations. These products provide data users with insightful analysis on the current state of the nation. Some subject matter areas produce special tabulations of the data for their topics. The Census Bureau also produces methodological analysis products based on ACS collection and processing research.⁶⁶

13.5 Public Use Microdata Sample (PUMS)

The PUMS files comprise individual records that contain information collected about each person and housing unit (HU). PUMS files are extracts from the microdata for which disclosure

⁶⁶ Information on these various analytic products can be found in the [ACS Library](#) website.

avoidance techniques are applied to protect confidential information about households or individuals. These extracts cover the same characteristics contained in the full microdata sample files. The only geography other than region, division, and state shown on a PUMS file is the Public Use Microdata Area (PUMA). PUMAs are special non-overlapping areas that partition a state; each PUMA contains a population of about 100,000 or more. State governments draw the PUMA boundaries based on the Census Bureau’s criteria and guidelines at the time of the Census.⁶⁷ PUMAs are redefined after each decennial census, and the new PUMAs are introduced into the ACS in years ending in “2,” so ACS PUMS records through 2021 have the 2010 PUMAs, and the ACS records from 2022 on have the 2020 PUMAs.

The Census Bureau releases 1-year and 5-year PUMS files. The multiyear PUMS files combine annual PUMS files to create larger samples in each PUMA, covering a longer period of time. The 5-year PUMS files include all of the records on the individual 1-year PUMS files, with items like income and geography brought forward to the latest year. This allows users to create estimates that are more statistically reliable.

13.6 Multiyear Estimates

While the types of data products for the multiyear estimates are almost entirely identical to those used for the 1-year estimates, there are several distinctive features of the multiyear estimates that data users must bear in mind.⁶⁸

First, the geographic boundaries that are used for multiyear estimates are based on the same set of geographic boundaries as the single-year estimates from the last year in the estimate period. For example, the 2018-2022 ACS 5-year estimates and 2022 ACS 1-year estimates were tabulated on the same vintage of geographic boundaries. If a geographic area has gained or lost territory over time, it can affect the user’s interpretation of estimates for that geographic area when making comparisons over time. To learn about the specific vintage of geographies used in a particular reference year of the ACS, please visit the Census Bureau’s [Geography Boundaries by Year](#) website.

Secondly, for multiyear period estimates based on monetary characteristics (for example, median earnings), inflation factors are applied to the data to create estimates that reflect the dollar values in the final year of the multiyear period.

Finally, changes to the questionnaire may occur from time to time. Whenever possible, changes are crosswalked forward to make the data comparable across time, as described in Chapter 10.

⁶⁷ PDF-format maps of PUMA boundaries are available from the Census Bureau’s [Geography Reference Maps](#) website.

⁶⁸ References to multiyear data primarily refer to ACS 5-year estimates. ACS 3-year estimates were discontinued after the release of the 2011-2013 ACS 3-year data.

However, when crosswalking is not possible, changes to a question can result in the inability to produce certain estimates for a multiyear period containing the year in which the question was changed. In addition, if a new question is introduced, estimates of characteristics related to the new question will not be available until they have been included for the entire multiyear period.

13.7 The Creation of Data Products

The subject matter analysts in the Census Bureau’s Social, Economic, and Housing Statistics Division (SEHSD) and Population Division (POP) provide the specifications for ACS data products. These specifications include the logic used to calculate every estimate in each data product and the exact textual description associated with each estimate. The ACS Data Products Planning Workgroup (DPPWG), which serves as a Change Control Board, approves specification changes. After the review and approval of the table specifications by the DRB, the specifications are validated and verified through a series of simulation and system tests. The tabulation starts when the weighted and edited microdata become available (see Chapters 10 and 11). ACS data products are generated by various automated tabulation systems, beginning with the 1-year period estimates for the Detailed Tables.

13.8 Data Release Rules

Various kinds of restrictions are applied to ACS data to limit the disclosure of information about individual respondents and to limit the production of ACS estimates with unacceptable statistical reliability.

Population Thresholds for ACS Data

Population thresholds restrict the availability of ACS 1-year estimates to populations of 65,000 or more. The Census Bureau started to release a subset of simplified Detailed Tables called the ACS 1-year Supplemental Estimates with a population threshold of 20,000 or more beginning with the 2014 ACS data. There is no population threshold for the 5-year estimates. Therefore, areas or groups of all population sizes are eligible for 5-year estimates.⁶⁹

Other population threshold rules apply in cases where a drop in population size changes the qualification of an area for types of ACS estimates for which it was previously qualified. Consider, for example, the case of a geographic area that met the 1-year or 1-year Supplemental Estimates population thresholds for a previous period but dropped below it for the current period. The Census Bureau would like to preserve the continuity of data for such areas. To give such an area a reasonable chance to retain access to 1-year or 1-year Supplemental Estimates, the Census Bureau applies a data release rule that provides for the continued production of 1-year or 1-year

⁶⁹ The Census Bureau produced 3-year estimates for populations of 20,000 or more from the 2005-2007 ACS 3-year period through the 2011-2013 ACS 3-year period.

Supplemental Estimates unless the population drops more than five percent below the specific period threshold for 1-year or 1-year Supplemental Estimates.

Data Quality Filtering

Another kind of data release rule, data quality filtering, applies to ACS 1-year and 1-year Supplemental Estimates. Every Detailed Table consists of a series of estimates. Each estimate is subject to sampling variability that can be summarized by its standard error. If more than half of the estimates in the table are not statistically different from zero (at a 90 percent confidence level), then the table fails to meet the rule's requirements and is not published. Dividing the standard error by the estimate yields the coefficient of variation (CV) for each estimate. (If the estimate is 0, a CV of 100 percent is assigned.) For each table at a given geographic area, CVs are calculated for each table's detailed estimates (excluding totals and subtotals), and the median CV value is determined. If the median CV value for the table is less than or equal to 61 percent (in other words, the median of the ratio of the margin of error to the estimate is less than one), the table passes for that geographic area and is published; if it is greater than 61 percent, the table fails and is not published.

Data quality filtering is not applied to 5-year estimates.

Disclosure Review Board Rules

Beyond the population thresholds and data quality filtering applied to ACS tables, the Disclosure Review Board (DRB) establishes additional rules that protect against disclosing information about individual respondents. These rules describe, for example, how medians or other quantiles must be calculated, what requirements apply to medians, means, or aggregates, the thresholds for tables involving specific kinds of geographic areas or tables with more than 100 data cells, and thresholds for tables of unweighted counts of people and housing units.

The DRB uses data swapping as a disclosure limitation technique for the ACS. Data swapping involves the swap, or interchange, of a small percentage of household records between pairs of households in different geographic regions. The selection process for deciding which households should be swapped is highly targeted to affect the records with the most disclosure risk. Pairs of households that are swapped match on a minimal set of demographic variables, and the household pairs, in most cases, are located in the same census tract. The methodology for swapping is confidential to further protect the confidentiality of respondents. All data products (tables and microdata) are created from the swapped data files.

For PUMS data, the following techniques are employed in addition to swapping:

- Top-coding is a method of disclosure avoidance in which all cases in or above a certain percentage of the distribution are placed into a single category.
- Geographic population thresholds prohibit the disclosure of data for individuals or HUs for geographic units with population counts below a specified level.
- Age perturbation (modifying the age of household members) is required for large households containing 10 people or more because of concerns about confidentiality.
- Detail for categorical variables is collapsed if the number of occurrences in each category does not meet a specified national minimum threshold.
- Partial synthesis is used to protect the confidentiality of housing units and persons at risk for disclosure based on certain characteristics.

13.9 Data Review and Acceptance

After the editing, imputation, data products generation, disclosure avoidance, and application of the release rules are complete, subject matter analysts perform a final review of the ACS data and estimates before release. This final data review and acceptance process helps to ensure that there are no obvious errors or other data anomalies.

Each year, the subject matter analysts review the ACS estimates following a multistep review process and provide clearance before estimates are released to the public. Because of the short time available to review such a large amount of data, the program has developed two tools: the Automated Review Tool (ART) and the Visual Automated Review Tool (VisART), in addition to various supplementary single- and multiyear review tools, to facilitate the process. These tools enable subject matter analysts to detect statistically significant differences in estimates from one year to the next using several statistical tests. ART features predesigned reports and includes functions allowing for ad hoc, user-defined queries for hundreds of single-year estimates for national, state, county, and place geographies. VisART includes a scatterplot, mapping functionality, and the ability to create custom analyst queries with multiple variables. This tool has been used for the review of 5-year estimates as well as portions of the 1-year review.

13.10 Custom Data Products

The Census Bureau offers a wide variety of general-purpose data products from the ACS designed to meet the needs of most data users. They contain predefined sets of data for standard census geographic areas. For users whose data needs are not met by the general-purpose products, the Census Bureau offers customized tabulations on a cost-reimbursable basis through the ACS custom tabulation program. Custom tabulations are created by tabulating data from

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ACS edited and weighted data files. These projects vary in size, complexity, and cost, depending on the needs of the sponsoring client.

In many cases, requests for ACS custom tabulations from federal agencies with statutory requirements for information have led to innovative approaches to providing ACS data. For example, the Census Bureau fulfilled a Department of Justice tabulation request for ACS estimates to meet Section 203 requirements of the Voting Rights Act by providing modeled ACS estimates.

Each custom tabulation request is reviewed in advance by the DRB to ensure that confidentiality is protected. The requestor may be required to modify the original request to meet disclosure avoidance requirements.⁷⁰

⁷⁰ See [American Community Survey Custom Tabulations](#) for more detailed information.

Chapter 13 Revision History

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C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
N. Walters & E. Poehler	11/28/2022	Updates to content and formatting.	3.0
N. Walters	9/20/2024	Updates made for clarity.	4.0

Chapter 14: Data Dissemination

14.1 Overview

This chapter provides a description of the release schedule and dissemination channels for the American Community Survey (ACS) data products, including the 1-year estimates, 1-year Supplemental Estimates, 5-year estimates, and Public Use Microdata Sample (PUMS) files.

The primary data dissemination systems for ACS data products are:

- Standard ACS 1-year, 1-year Supplemental Estimates, and 5-year tabulated data products are available using the Census Bureau’s main data dissemination environment, data.census.gov, and using the Census Bureau’s [Application Programming Interface \(API\)](#).
- The ACS Summary File for Detailed Tables, along with supporting documentation, is available on the [ACS Summary File webpage](#).
- ACS 1-year Geographic Comparison Tables, 1-year Ranking Tables, and 5-year Narrative Profiles are available through the [ACS data tables page](#).
- ACS PUMS files are available for download through the [ACS PUMS webpage](#).
- In addition, the ACS Public Use Microdata Sample (PUMS) are available for use using the [Microdata Analysis Tool \(MDAT\)](#).
- ACS 5-year Variance Replicate Estimates (VRE) Tables are available for a subset of Detailed Tables through the [VRE Tables page](#).

14.2 Release Schedule

The Census Bureau typically releases the ACS 1-year estimates in September; ACS 1-year PUMS files and ACS 1-year Supplemental Estimates in October; ACS 5-year estimates in December; and ACS 5-year PUMS files and 5-year Variance Replicate Estimates in January of the following year. Table 14-1 shows the general ACS data release schedule.

Table 14-1. ACS Data Release Schedule

Planned Release Month	Data Products	Lowest Level of Geography
September	1-Year Detailed Tables, Data Profiles, Comparison Profiles, Selected Population Profiles, and Subject Tables, Summary File	Places, County Subdivisions (where available) Geographies of 65,000+ population
October	1-Year Public Use Microdata Sample (PUMS) File	Public Use Microdata Area (PUMA)
October	1-Year Supplemental Estimate Tables	Places, County Subdivisions (where available) Geographies of 20,000+ population
December	5-Year Detailed Tables	Block Groups
December	5-Year Data Profiles	Census Tracts
December	5-Year Comparison Profiles	Places/County Subdivisions
December	5-Year Subject Tables	Census Tracts
December	5-Year Summary File	Block Group
December	5-Year Narrative Profiles	Census Tracts
January	5-Year Public Use Microdata Sample (PUMS) File	Public Use Microdata Area (PUMA)
January	5-Year Variance Replicate Estimate Tables	Block Group

Starting in 2010, the information on social, economic, housing, and demographic characteristics previously available only once a decade is available annually for all areas through the ACS 5-year estimates. Table 14-2 displays a history of the ACS data releases.

Table 14-2. ACS Data Release History by Estimate Type, Population Threshold, and Year of Data Collection

Year of data release	Year of Data Collection for 1-year Estimates (population threshold: 65,000+)	Year of Data Collection for 1-year Supplemental Estimates (population threshold: 20,000+)	Year of Data Collection for 3-year Estimates (population threshold: 20,000+)*	Year of Data Collection for 5-year Estimates (population threshold: All areas)
2006	2005	NA	NA	NA
2007	2006	NA	NA	NA
2008	2007	NA	2005–2007	NA
2009	2008	NA	2006–2008	NA
2010	2009	NA	2007–2009	2005–2009
2011	2010	NA	2008–2010	2006–2010
2012	2011	NA	2009–2011	2007–2011
2013	2012	NA	2010–2012	2008–2012
2014	2013	NA	2011–2013	2009–2013
2015	2014	NA	NA	2010–2014

Year of data release	Year of Data Collection for 1-year Estimates (population threshold: 65,000+)	Year of Data Collection for 1-year Supplemental Estimates (population threshold: 20,000+)	Year of Data Collection for 3-year Estimates (population threshold: 20,000+)*	Year of Data Collection for 5-year Estimates (population threshold: All areas)
2016	2015	2014/2015	NA	2011–2015
2017	2016	2016	NA	2012–2016
2018	2017	2017	NA	2013–2017
2019	2018	2018	NA	2014–2018
2020	2019	2019	NA	2015–2019
2021	2020 Experimental Data [†]	NA	NA	2016–2020
2022	2021	2021	NA	2017–2021
2023	2022	2022	NA	2018–2022

Table Notes: NA = Not available.

* The Census Bureau produced ACS 3-year estimates starting in 2008, but that series was discontinued after the release of the 2011–2013 estimates. Five-year estimates are available for areas as small as census tracts and block groups.

[†] Because of the impact of the COVID-19 pandemic, the Census Bureau changed the 2020 ACS release schedule. Instead of providing the standard 1-year data products, the Census Bureau released experimental estimates from the 1-year data. This includes a limited number of data tables for the nation, states, and the District of Columbia. The 2020 ACS 1-year experimental estimates are posted on the [2020 ACS 1-Year Experimental Data Tables](#) page; they are not available on data.census.gov.

14.3 Accessing ACS Data and Supporting Documents

Data.census.gov

In 2019, [data.census.gov](#) was launched as the new platform and an integral part of [www.census.gov](#) for accessing data and digital content from the U.S. Census Bureau. One of the goals of the new platform was to streamline duplicative data access tools and capabilities by incorporating the best aspects of each tool into a single platform. The 2018 ACS 1-year estimates were the first ACS data release to be disseminated via data.census.gov and marked the end of any releases into the former data dissemination platform, American FactFinder (AFF). AFF was officially retired in March 2020. ACS data available on data.census.gov include all 1- and 5-year releases since the 2010 1-year and 2006–2010 5-year. Older releases and the 3-year products are available via FTP and the API.

Data.census.gov is the primary online tool for accessing ACS data products, which include Detailed Tables, Data Profiles, Comparison Profiles, Subject Tables, 1-year Supplemental Estimates, and Selected Population Profiles (1-year data only). In addition, microdata access through [MDAT](#) allows access to the PUMS files.

Other ACS Data Tools

In addition to data.census.gov, the Census Bureau’s primary tool for accessing data from the ACS, there are many other ways to access ACS estimates. Popular tools include:

- QuickFacts-Provides quick, easy access to facts about people, business, and geography
- My Congressional District-Interactive tool that provides the latest demographic and economic statistics for every Congressional District
- My Tribal Area-Interactive tool that provides the latest demographic and economic statistics for tribal areas
- My Community Explorer-Interactive map-based tool that highlights demographic and socioeconomic data that measures inequality
- OnTheMap for Emergency Management-Census Bureau’s data for disasters, natural hazards, and weather events
- Census Business Builder-Interactive map that allows small business owners and regional planners a way to easily navigate to and use key demographic and economic data

TIGER⁷¹/Line Shapefiles with Selected Demographic and Economic Data-TIGER/Line Shapefiles pre-joined with data in geodatabase format for mapping ACS data with Geographic Information System (GIS) software. Visit the [ACS Data Tools webpage](#) for a complete list of our data tools, along with additional information, to help you access the information you need from the ACS.

ACS Website

The ACS website contains a wealth of information, documentation, and research papers about the ACS program.⁷² The site also hosts many supporting documents related to data products, such as subject definitions, code lists, geographic information, comparison guidance, and the latest information for each data release. Documentation on the accuracy of the data is also included, providing information about the sample design, confidentiality, sampling error, nonsampling error, and estimation methodology. Quality measures showing sample sizes, coverage rates, response rates, and allocation rates are provided. The User and Errata Notes

⁷¹ TIGER stands for Topologically Integrated Geographic Encoding and Referencing. TIGER/Line® is a registered trademark of the Census Bureau.

⁷² The ACS website can be found at: <https://www.census.gov/programs-surveys/acs/>.

section lists updates made to the data. The geography section gives a brief explanation of the Census Bureau’s geographic hierarchy, common terms, and specific geographic areas presented.

The Guidance for Data Users section contains important information to help data users access and use ACS data. The section includes information such as how to compare ACS data over time, when to use 1-year or 5-year estimates, how to determine which ACS data table or tool to use, and how to determine which ACS estimates are statistically different from one another. The Guidance for Data Users section also contains a set of 14 ACS data user handbooks tailored to specific data user audiences or tools. The handbooks include information on how to access and use the data, including case studies featuring real-world applications of ACS data.

File Transfer Protocol (FTP) Site

The FTP site hosts the ACS Summary Files and PUMS files, and it is intended for advanced data users.⁷³ The data and technical documentation can be downloaded into Excel, PDF, or text files. Users of the FTP site can import the files into the spreadsheet/database software of their choice for data analysis and table presentation. Documentation describing the layout of the site in the README file is available in the main directory on the FTP server.

Application Programming Interface (API)

The Census Data Application Programming Interface (API) is a data service that enables researchers and software developers to use Census Bureau data within their programs and applications. The Census Data API presents data in a standardized way. Standardizing the API query language across multiple datasets reduces the learning curve for researchers and developers.

The API includes many Census Bureau datasets, including the ACS 1-year, 1-year Supplemental Estimates, 3-year, and 5-year datasets.⁷⁴

Developers can build programs using statistical software, create mobile and Web apps, and develop interactive data visualizations that display data directly in an app or browser window in an easily consumable format. Examples of Census Bureau applications that draw ACS data directly from the Census Data API include My Tribal Area and My Congressional District.⁷⁵

The Census Data API is also used widely by organizations outside of the Census Bureau. For example, the Pennsylvania State Data Center uses the API to produce a series of dashboards

⁷³ Data users can access the [ACS Summary Files](#) website and [PUMS files](#) website or can access the files directly on the [FTP site](#).

⁷⁴ The API can be accessed at: [Available APIs \(census.gov\)](#).

⁷⁵ Data users can access the [My Tribal Area](#) website and the [My Congressional District](#) website.

based on ACS estimates.⁷⁶ Cleargov.com uses the Census Data API and other sources to produce online profiles of cities and towns for state and local governments.⁷⁷

Federal Statistical Research Data Centers

The Census Bureau partners with various universities, research institutions, and agencies to form a nationwide network of secure Federal Statistical Research Data Centers (FSRDC). At these secure centers, qualified researchers with approved projects (i.e., projects that benefit Census Bureau programs and demonstrate scientific merit) can perform statistical analysis on selected internal microdata from the Census Bureau (including ACS) and other statistical agencies.⁷⁸

⁷⁶ For more information see the [Pennsylvania State Data Center \(PaSDC\) Dashboards](#).

⁷⁷ Cleargov.com can be accessed at: <https://www.cleargov.com>.

⁷⁸ More information about the FSRDCs is available at: <https://www.census.gov/fsrdc>.

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C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
E. Poehler	11/28/2022	Updated content and formatting.	3.0
N. Walters	9/23/2024	Updated descriptions of dissemination systems.	4.0

Chapter 15: Improving Data Quality by Reducing Nonsampling Error

15.1 Overview

Total survey error includes two components: sampling error and nonsampling error. Chapters 4 and 11 provide information about the steps the Census Bureau takes to reduce sampling error in the American Community Survey (ACS). As with all surveys, the quality of ACS data reflects how well data collection, processing, and quality control procedures address potential sources of nonsampling error. Groves (1989) identified four primary sources of nonsampling error:

- **Coverage Error:** The failure to give some units in the target population any chance of selection into the sample or giving units more than one chance of selection.
- **Nonresponse Error:** The failure to collect complete data from all units in the sample.
- **Measurement Error:** The inaccuracy in responses recorded on survey instruments, arising from:
 - The effects of interviewers on the respondents' answers to survey questions.
 - Respondents' inability to answer questions, lack of requisite effort to obtain the correct answer, or other psychological or cognitive factors.
 - Faulty wording of survey questions.
 - Data collection mode effects.
- **Processing Error:** Errors introduced after the data are collected, including:
 - Data capture errors.
 - Errors arising during coding and classification of data.
 - Errors arising during editing and item imputation of data.

This chapter identifies the operations and procedures designed to reduce these sources of nonsampling error and thus improve the quality of ACS estimates. It also includes information about ACS Quality Measures which help the Census Bureau monitor some types of nonsampling error. Finally, it describes the annual review process used to demonstrate that ACS data meet the Census Bureau's Statistical Quality Standards. The ACS releases the survey estimates, as well as the [Quality Measures](#), at the same time each year so that users can consider data quality in conjunction with the survey estimates.

15.2 Coverage Error

All surveys experience some degree of coverage error. It can take the form of undercoverage or overcoverage. Undercoverage occurs when units in the target population do not have a chance of

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selection into the sample, for example, when the address frame is missing certain housing units or when respondents erroneously exclude some people from a household roster. Overcoverage occurs when units or people have multiple chances of selection, for example, when the address frame lists a housing unit more than once. In general, coverage error will affect survey estimates if the characteristics of the individuals or units erroneously excluded or included differ from the characteristics of those correctly listed.

The ACS uses the Census Bureau's Master Address File (MAF) as its sampling frame and includes procedures for reducing coverage error in the MAF. Twice a year, the Census Bureau receives the U.S. Postal Service (USPS) Delivery Sequence File (DSF) that consists of the addresses, including a house number and street name, rather than a rural route or post office box. The Geography Division uses this file to update the city-style addresses (street number and name, city, state, and ZIP code) on the MAF. See Chapter 3 for further details.

The Census Bureau does several things to address within-household person-level coverage error. In the electronic data collection instruments (everything but the paper questionnaire), after the basic roster is collected, the survey asks several questions to identify other people who live there who might have been missed or people who might have been erroneously included. For respondents who use the paper questionnaire, the ACS conducts a telephone follow-up operation when there is conflicting information about the total number of household members to confirm that the survey includes the appropriate persons.

The Census Bureau compares the ACS estimate that is created using weights that account for probabilities of selection and nonresponse to independent estimates coming from the Census Bureau's Population Estimates Program. The main goal of this process is to adjust the weights so that estimates match or come as close to the Population Estimates Program estimates as possible. This also allows us to measure survey undercoverage or overcoverage. If the ACS estimate before poststratification is less than the independent estimate, then it is an indicator of undercoverage. Conversely, if the ACS estimate before poststratification is more than the independent estimate, then it is an indicator of overcoverage.

The ACS monitors coverage error by producing housing unit-level and person-level coverage rates for various geographic levels and groups, as shown in Table 1.

Table 15-1. Group and Geographic Level that Coverage Rates are Calculated

Group	Nation	Each State
Housing Units	X	X*
Group Quarters Population	X	
All Persons	X	X
Male and Female	X	X
White Non-Hispanic, Black Non-Hispanic, American Indian and Alaska Native Non-Hispanic, Asian Non-Hispanic, Native Hawaiian and Other Pacific Islander Non-Hispanic, Hispanic	X	

* Housing unit coverage rates are not published for Puerto Rico because independent housing unit estimates are not available.

Coverage rates are published on the [ACS Sample Size and Data Quality](#) webpage and on data.census.gov in tables [B98011](#), [B98012](#), [B98013](#), and [B98014](#).

15.3 Nonresponse Error

There are two forms of nonresponse error: unit nonresponse and item nonresponse. Unit nonresponse results from the failure to collect data for an eligible sample unit or obtain the minimum required data. Item nonresponse occurs when respondents do not report individual data items, or provide data considered invalid or inconsistent with other answers.

Bias results from systematic differences between nonrespondents and respondents so survey practitioners strive to increase both unit and item response to reduce the potential for bias introduced into survey estimates. Nonresponse bias may impact the accuracy of survey estimates, but the magnitude of this impact is difficult to measure.

The ACS reduces the potential for nonresponse bias by reducing the amount of unit and item nonresponse through procedures and processes listed below. For more details, see Chapters 7 and 8.

- ACS data are collected using four modes: internet, paper, telephone, and personal visits. This provides the option for each household to respond in the way they are most comfortable. Respondents are provided a telephone number to call to get help or to complete the survey.
- The housing unit mailing protocol is based on a strategy shown in research studies to obtain a high response rate (Risley and Berkley, 2020). Both the online survey and paper questionnaire have been tested to make them user-friendly for respondents and reflect accepted principles of survey design. The ACS program regularly tests modified mailing materials and contact strategies to improve self-response.

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- Response to the ACS is mandated by law, and information about the mandatory requirement to respond is provided in most materials and reinforced in communication with respondents in all stages of data collection.
- The ACS includes a telephone failed-edit follow-up (FEFU) interview with self-respondents whose responses do not meet quality standards or are identified as vacant units.
- The ACS uses Census Bureau professional interviewers trained in refusal conversion methods for the computer-assisted personal interviewing (CAPI) follow-up. Data for people in group quarters (GQs) is mostly collected by professional interviewers when possible.
- The internet and CAPI instruments are available in Spanish, a Spanish paper questionnaire is available, and bilingual interviewers provide language support in other languages.

Measuring Unit Nonresponse

The Census Bureau presents survey response and nonresponse rates as part of the ACS Quality Measures. The survey response rate is the percent of units eligible to be interviewed that completed an interview by the end of data collection. The ACS Quality Measures provide separate rates for GQ persons and housing units. For the housing unit (HU) response rate, the numerator is a weighted estimate of the number of cases interviewed after all modes of data collection. The denominator is the weighted estimate of cases eligible to be interviewed.

For the GQ person response rate, the numerator is a weighted estimate of all interviewed GQ persons after a personal visit. The denominator is the weighted estimate of the total number of persons eligible to be interviewed in GQs.

The ACS Quality Measures also include the weighted estimates of nonresponse rates broken down by the reason for nonresponse. These reasons include refusal, unable to locate the sample unit, temporarily absent during the interview period, language problem, insufficient data (not enough data collected to consider it a response), and other (such as “sample address not accessible” or “death in the family”). For the GQ rates, there are two additional reasons for noninterviews: whole GQ refusal and whole GQ other (such as unable to locate the GQ). For HUs, there are two additional reasons for noninterviews: no one home during the data collection period and maximum contact attempts were reached.

ACS response rates are published on the [ACS Sample Size and Data Quality](#) webpage and on data.census.gov in tables [B98021](#) and [B98022](#).

Measuring Item Nonresponse

The ACS Quality Measures provide information about item nonresponse. When respondents do not report individual data items, or provide data considered invalid or inconsistent with other answers, the Census Bureau imputes the necessary data. The imputation methods use either rules to determine acceptable answers (referred to as “assignment”) or answers from similar people or HUs (“allocation”). Assignment involves logical imputation, in which a response to one question implies the value of a missing response to another question. For example, a response about the school grade or level a person is currently attending can often be used to assign a value to the highest degree or level of school completed. Allocation involves using statistical procedures to impute for missing values.

The ACS Quality Measures include allocation rates as a measure of the extent to which item nonresponse required imputation across over 120 person and HU characteristics, as well as overall person and HU rates.

Item allocation rates are published on the [ACS Sample Size and Data Quality](#) webpage, which provides the rates for all housing units or people eligible for the question. Those tables are published at the national and state levels.

The [B99 series of tables](#) on [data.census.gov](#) provide the number of people or housing units allocated and not allocated for an item for all geographic levels at which the item is published. These allocation counts correspond to the universe of data published on data.census.gov, which may differ from those eligible for the question.

For example, the ACS Sample Size and Data Quality web page gives the allocation rates for Educational Attainment for the population 3 years or older, the universe of people eligible for the question. [Table B99151 on data.census.gov](#) provides the number of people allocated and not allocated among those 25 years of age or older, the population for the Educational Attainment table on data.census.gov.

15.4 Measurement Error

Measurement error is defined as the difference between the recorded answer and the true answer. Measurement error may occur in any mode of data collection and can be caused by vague or ambiguous questions misinterpreted by respondents; questions that respondents cannot answer or questions where respondents deliberately falsify answers for social desirability reasons (see Tourangeau and Yan (2007) for information on social desirability); or interviewer characteristics or actions such as the tone used in reading questions, the paraphrasing of questions, or leading respondents to certain answers. In 2012, the Census Bureau conducted a reinterview with a subsample of ACS respondents to attempt to quantify the amount of inconsistency in reporting

between the original interview and reinterview. This study found minimal evidence of measurement error for most ACS questions (Murphy, 2012).

The ACS minimizes measurement error in several ways, some of which also help to reduce nonresponse.

- ACS pretests new or modified survey questions in all modes before introducing them into the ACS as mandated in the Census Bureau Statistical Quality Standard “Developing Data Collection Instruments and Supporting Materials” (Census Bureau, 2022).
- The ACS survey instruments are designed to consider the features of each data collection mode. Methods that make it easier for the respondent to understand the questions also increase the chances that the individual will respond to the questionnaire.
 - The ACS uses a questionnaire design on the internet and paper form that reflects accepted principles of survey design to improve respondent friendliness and navigation (Tancreto et al., 2012).
 - The internet questionnaire provides help via a hyperlink for respondents who need additional information on how to interpret and respond to specific questions and the ability to review/edit answers at the end of the survey.
 - The internet and CAPI instruments include functionality that helps achieve valid responses for some questions. For example, the instruments check for values outside of the expected range to ensure that the reported answer reflects an appropriate response.
 - The telephone instrument breaks up questions with long or complicated response categories into separate questions.
 - The CAPI instrument includes flashcards to present answer choices to the respondent.
- The internet and CAPI instruments are available in Spanish, and bilingual interviewers provide language support in other languages.
- Respondents may call Telephone Questionnaire Assistance (TQA) to speak with trained interviewers to answer general ACS questions and questions regarding specific items or even complete the survey over the telephone.
- Training for the interviewing staff includes instruction on reading the questions as worded and answering respondent questions. The training also encompasses extensive role-playing opportunities. All interviewers receive a manual that explains each question in detail and provides detailed responses to questions often asked by respondents.

- Reinterviews are conducted with CAPI respondents to evaluate data collected by filed representatives to identify and minimize falsification of data. Response data are also regularly monitored to identify procedural errors and improve data quality.
- Performance measures are regularly monitored; these metrics unit and item nonresponse rates by survey mode, internet instrument response breakoffs, help screen usage, and mail form processing.

15.5 Processing Error

The final component of nonsampling error is processing error—error introduced in the post-data collection process of creating published data. For example, a processing error may occur when keying the data from the mail questionnaires. Another example is the miscoding of write-in responses, either clerically or by automated methods. The degree to which imputed data differ from the truth also reflects processing error—specifically imputation error.

A number of practices are in place to control processing error (more details are discussed in Chapters 7 and 10). For example:

- Data processing programs have several internal validation checks to ensure that they are implemented and functioning properly.
- Data capture of paper questionnaires includes a quality control procedure involving a double keying of a sample of fields and adjudication by a third keyer.
- Coding of write-in responses involves a multi-step process. An automated system is used first to code the write-ins. Write-ins that the automated system cannot code are coded by clerical coding staff. Clerical coding includes a quality control procedure involving double coding a sample of the cases and adjudication by a third coder.
- Procedures for selecting one interview or return from multiple returns for an address rely on reviewing the quality of data derived from each response and selecting the return with the most complete data.
- After completion of all phases of data collection (internet, mail, and CAPI), questionnaires with insufficient data do not continue in the survey processing; instead, they receive a noninterview code and are accounted for in the weighting process.
- Edit and imputation rules reflect the combined efforts and knowledge of subject matter experts, as well as experts in processing, and include evaluation and subsequent improvements as the survey continues to progress.
- Subject matter and survey experts extensively review the data and tables, comparing results with previous years' data and other data sources.

15.6 Census Bureau Statistical Quality Standards

In 2012, the ACS program started conducting a new, formal assessment to demonstrate that its data products meet the Census Bureau's quality standards (see Census Bureau, 2022). Because the ACS covers a very broad set of topics, the Census Bureau chose a selection of seventeen critical core 5-year estimates reflecting social, demographic, economic, and housing characteristics to gauge the quality of the survey's products. This assessment uses several of the ACS Quality Measures associated with nonsampling error, including coverage rates, unit response rates, and item response rates (Sections 15.2 and 15.3 of this document explain most of these nonsampling errors). We also derived a combined nonsampling error rate from these three measures. Given the ACS's goal to produce estimates for small areas, counties were chosen to assess quality for smaller geographies large enough to result in stable quality measures. The assessment calculated these four rates at the county level, summarizing those results as medians. To meet the quality standard, the median county-level unit response rate must equal or exceed 60 percent, the median county-level coverage rates must equal or exceed 70 percent, and the median county-level item response rates for each of the 17 estimates must equal or exceed 70 percent. The median county-level combined measure of nonsampling error (the product of a coverage rate, the unit response rate, and the item response rate) must equal or exceed 50 percent for each of the 17 estimates.

The assessment also includes a measure of sampling error that does not come from the ACS Quality Measures. Sampling error in survey estimates arises because of the use of probability sampling. The ACS was designed to produce reliable tract-level estimates based on 5-year data. To determine a measure of sampling error, the assessment includes the median tract-level 5-year coefficient of variation (CV) for each of the 17 estimates. The CV equals the standard error (SE) of a weighted estimate divided by the weighted estimate itself. Most estimates must have a median CV of less than or equal to 30 percent to meet the quality standard.

The ACS met the Census Bureau's quality standards each year from 2012 to 2019 but did not meet the sampling error standard in 2020 because of the effects of the COVID-19 pandemic on the ACS. For more information, see the User Note [“Increased Margins of Error in the 5-Year Estimates Containing Data Collected in 2020.”](#) Additionally, one of the 17 key characteristics failed to meet the minimum 70 percent item response rate in 2022. For more information, see the User Note [“Item Allocation Rate for Total Income.”](#)

15.7 References

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Chapter 15 Revision History

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C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
D. Raglin & E. Poehler	11/28/2022	Updated content and resolved comments and formatting issues	3.0
D. Raglin	8/30/2024	Updated content for clarity.	4.0

Chapter 16: Research and Evaluation

The short- and long-range planning of the American Community Survey (ACS) includes research and evaluation (R&E) activities in all aspects of the ACS program. This chapter describes the process for identifying, prioritizing, and implementing R&E activities.

The Census Bureau conducts research on an ongoing basis to improve the ACS, its operations, and the resulting data products. Research projects for the ACS are those that determine a new or revised (alternative) method or procedure for an operation, process, or system. Evaluation projects for the ACS are those that analyze, interpret, assess, and synthesize the effectiveness of the methods or procedures of the ACS program and their impact on key survey attributes, such as data quality and coverage, using data collected from ACS operations, processes, systems, and auxiliary data collections (including debriefings and focus groups). The program has also developed a series of internal performance and data quality measures that it uses to monitor the survey on an ongoing basis.

R&E projects include experiments (quantitative and qualitative studies), simulations (including alternative weighting methodologies, alternative filtering methods, and variance estimation), feasibility and cost/benefit studies, quality assessments and measurements of error, and characteristics of populations living in housing units and group quarters.

R&E projects originate from a variety of sources, including Census Bureau researchers, the National Academy of Sciences, and stakeholders. Research ideas may be designed to improve response rates, enhance data quality, or identify cost savings and stakeholder (e.g., federal sponsors or external experts) suggestions.

The ACS R&E Workgroup governs ACS R&E projects. Workgroup members include staff in the American Community Survey Office as well as Census Bureau subject matter experts, statisticians, and survey methodologists. Each proposed R&E project is evaluated by the workgroup based on several criteria, including alignment with strategic goals of the Census Bureau and ACS program, benefits to stakeholders and data users, potential for follow-on investments, and feasibility.

Researchers conduct the R&E projects with oversight and guidance from the R&E Workgroup and ACS senior-level managers. Once a work request is approved, the researchers develop a Research and Evaluation Analysis Plan (REAP). This plan outlines the history, motivations, objectives, and research questions related to the project. It also explains, in detail, the research questions and the methodology and metrics that will be used. Design assumptions and limitations are also included in the REAP. R&E Workgroup members with expertise in the area of the research, known as critical reviewers, are assigned to review each REAP and to provide feedback and guidance on the research questions, methodology, and limitations. After feedback is incorporated, the critical reviewers approve the REAP, and research begins.

Once the research is complete, findings are documented in a written report following the methodology outlined in the REAP. After the report is drafted, it undergoes review and is presented to the ACS R&E Workgroup and ACS senior-level managers before being finalized. The final R&E reports are published in the [ACS R&E memoranda series](#). R&E results are also regularly presented at professional forums like the American Association of Public Opinion Research (AAPOR) conference and the Joint Statistical Meetings (JSM).

R&E projects have provided a great benefit to the ACS program. For example, over the last few years, several projects have been designed to improve self-response through better mail materials, which have positively influenced the response rates once implemented into production. The periodic Content Tests, which test new and revised content for the survey (see Chapter 5 for details), are overseen through the R&E process. A report on the devices used to respond to the ACS shows a dramatic shift toward mobile devices. Research into administrative records use on the survey led to the first-ever ACS production usage in 2024 for the acreage topic.

The Census Bureau regularly publishes an overview report outlining planned research and evaluation activities as well as summaries (see [Agility in Action Reports](#)).

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C. Rosol	1/30/2014	Final draft (2014 edition) to be published	2.0
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D. Raglin	8/30/2024	Minor, non-substantive changes.	4.0

Appendix: Glossary

Term/Acronym	Description
AAPOR	American Association of Public Opinion Research
ACS	American Community Survey
ACSO	American Community Survey Office
Address Canvassing	A process for updating addresses and map databases to ensure they are accurate and current. As of 2020, address canvassing is conducted in two ways: in-office address canvassing and in-field address canvassing. In-office address canvassing operation uses imagery to detect housing change or stability compared to the Master Address File. For in-field address canvassing, address listing staff (address listers) visit specific geographic areas where in-office methods could not assure a complete and accurate address list. Listers identify every place where people could live or stay and verify, correct, add, and delete address records, as appropriate.
AFF	American FactFinder
AHS	American Housing Survey
AIA	American Indian Areas
AIAN	American Indian and Alaska Native
AIANA	American Indian and Alaska Native Areas
AIANHH	American Indian Areas /Alaska Native Areas /Hawaiian Home Lands
AIR	American Indian Reservations
ANA	Alaska Native Areas
ANRC	Alaska Native Regional Corporation
API	Application Programming Interface
ART	Automated Review Tool
ASA	American Statistical Association
AW	Average weight
Automated Coding	A process of coding write-in responses using digital means. Written-in responses are keyed into digital data and then matched to a data dictionary. The data dictionary contains a list of the most common responses, with a code attached to each. The coding programs attempt to match the keyed response to an entry in the dictionary to assign a code.
BAS	Boundary and Annexation Survey: An annual survey of all active, functioning legal governments across the United States and its territories. These governments can update boundaries and features for states, counties or county equivalents, minor civil divisions, consolidated cities, census-designated places, and federally recognized tribes with a reservation or off-reservation trust land.
BOP	Bureau of Prisons
BR	Base rate

Term/Acronym	Description
BW	Base weight
C2SS	Census 2000 Supplementary Survey
CACC	Computer-Assisted Clerical Coding
CAI	Computer-Assisted Interviewing
CAPI	Computer-Assisted Personal Interviewing
CATI	Computer-Assisted Telephone Interviewing
CAUS	Community Address Updating System
CBSA	Core-Based Statistical Area. Metropolitan and Micropolitan Statistical Areas are collectively referred to as Core-Based Statistical Areas.
CDPs	Census Designated Places
Clerical Coding	A process of coding write-in responses by trained human coders.
CNSTAT	Committee on National Statistics
COVID / COVID-19	Coronavirus disease 2019
Coverage Error	See Overcoverage and Undercoverage.
CPI	Consumer Price Index
CPS	Current Population Survey
CQR	Count Question Resolution
CSA	Combined Statistical Area
CSAC	Bureau of Census Scientific Advisory Committee
CV	Coefficient of variation
DAAL	Demographic Area Address Listing
DCP	Data Collation Process
DF	Design Factor
DITD	Decennial Information Technology Division
DOC	Department of Commerce
DOD	Department of Defense
DPPWG	ACS Data Products Planning Working Group
DPRG	ACS Data Product Redesign Group
DRB	Disclosure Review Board
DSF	Delivery Sequence File
DSL	Digital Subscriber Line
DUG	Data Users Group
EDS	Excluded from Delivery Statistics
EIA	Energy Information Administration
ENL	Employer Name List
FAQs	Frequently Asked Questions
FEFU	Failed-edit Follow-up
FPC	Finite Population Correction
FR(s)	Field Representative(s)
FSRDC	Federal Statistical Research Data Centers
FTP	File Transfer Protocol
GAO	Government Accountability Office

Term/Acronym	Description
GIS	Geographic Information System
GPS	Global Positioning System
GQs	Group Quarters
GQFQ	Group Quarters Facility Questionnaire
GQ-IDC	Group Quarters Internet Data Collection
GQMOS	Group Quarters Measure of Size
GQPPSF	GQ Person Post-Stratification Factor
GREG	Generalized Regression Estimation
GWTF	GREG Weighting Factor
HHF	Householder Factor
HOA	Homeowners association
HPF	Housing Unit Post-Stratification Factor
HUs	Housing Units
I&O	Industry and Occupation
iCADE	Integrated Computer Assisted Data Entry system
ICF	World Health Organization's International Classification of Functioning, Disability, and Health
ICSP-SACS	Interagency Council on Statistical Policy – Subcommittee on the American Community Survey
IVR	Interactive Voice Recognition
JSM	Joint Statistical Meetings
KFI	Key-from-Image
LACS	Locatable Address Conversion Service
LEP	Limited English Proficiency
LiMA	Listing and Mapping Application
LUCA	Local Update of Census Addresses
MAF	Master Address File
MAFID	Master Address File Identification number
MCDs	Minor Civil Divisions
MDAT	Microdata Analysis Tool
MENA	Middle Eastern or North African
MOE	Margin of Error
MOS	Measure of Size
MTdb	MAF/TIGER database
NAC	Census Bureau National Advisory Committee on Racial, Ethnic, and Other Populations
NAICS	North American Industry Classification System
NCES	National Center for Education Statistics
NHPI	Native Hawaiian or Pacific Islander
NIF	Housing Unit Noninterview Adjustment Factor
NPC	National Processing Center
OCR	Optical Character Recognition
OLQ	Other Living Quarters
OMB	Office of Management and Budget

Term/Acronym	Description
OMR	Optical Mark Recognition
OSA	Office of Strategic Alliances
Overcoverage	The extent to which error occurs because elements (e.g., addresses) that do not belong to the target population are included in a sample or frame
PEP	Population Estimates Program
PMGB	ACS Portfolio Management Governance Board
PO	Post Office
POP	Population Division
PPSF	Housing Unit Person Post-Stratification Factor
PR	Puerto Rico
PRB	Population Reference Bureau
PRCS	Puerto Rico Community Survey
PUMA	Public Use Microdata Area
PUMS	Public Use Microdata Sample
QA	Quality Assurance
QC	Quality Control
QI	Questionnaire Instrument
R&E	Research & Evaluation
RBW	Replicate Base Weight
REAP	Research & Evaluation Analysis Plan
RO	Regional Office
RR	Rural route
RSE	Replication method standard error
SAS	SAS (previously “Statistical Analysis System”) is statistical software developed by SAS Institute
SDR	Successive Differences Replication
SE	Standard Error
SEHSD	Social, Economic, and Housing Statistics Division
SEMOS	Smallest Entity Measure of Size
SFCF	State and Foreign Country File
SIPP	Survey of Income and Program Participation
SNAP	Supplemental Nutrition Assistance Program
SOC	Standard Occupational Classification
SPs	Special Places (facilities such as prisons, hotels, migrant farm camps, and universities)
SRSSE	Simple Random Sample Standard Error
SSF	CAPI Subsampling Factor
SSFCORR	CAPI Subsampling Correction Factor
SSS	Special Sworn Status
TDD	Telecommunications Device for the Deaf
ACS Test Counties / Test Sites	Testing to demonstrate the feasibility of conducting the ACS occurred from 1996 to 2004. The testing included 36 counties organized into 31 test sites. Most of these were single-county

Term/Acronym	Description
	sites; others consisted of multiple, contiguous counties (such as the ACS test site of Vilas and Oneida Counties, WI).
TIGER	Topologically Integrated Geographic Encoding and Referencing
TMOS	Tract Measure of Size
TQA	Telephone Questionnaire Assistance
UAA	Undeliverable as Addressed
Undercoverage	The extent to which error occurs because of the exclusion of elements (e.g., persons in a census) belonging to the target population from a sample or frame
U.S.	United States
U.S.C.	United States Code
USDA	U.S. Department of Agriculture
USPS	United States Postal Service
VA	Veterans Affairs
VisART	Visual Automated Review Tool
VMS	Housing Unit Variation in Monthly Sample Factor
WCTYCON	GQ Weight after the County-level Constraint
WGQPPSF	GQ Weight after the GQ Person Post-Stratification Factor
WGWTF	HU Weight after the GREG Weighting Factor
WHHF	HU Weight after the Householder Factor
WHPF	HU Weight after the Housing Unit Post-Stratification Factor
WNIF	HU Weight after Housing Unit Noninterview Adjustment
WPPSF	HU Person Weight after the Person Post-Stratification Factor
WSSF	HU Weight after the CAPI Subsampling Factor
WSSFCORR	HU Weight after the CAPI Subsampling Correction Factor
WSTCON	GQ Weight after the State-level Constraint
WTRCON	GQ Weight after the Tract-level Constraint
WVMS	HU Weight after the Variation in Monthly Sample Factor
ZCTA	ZIP Code Tabulation Area

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